THE MINIMAL VOLUME OF A STABLE SURFACE

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Abstract. We prove that the minimal volume of klt stable surfaces is $1/48983$, which is the surface case of a conjecture of Totaro. Moreover, up to isomorphism the surface of Alexeev and W. Liu is the only klt stable surface of volume $1/48983$. As an application, we show that every normal stable surface has at least two linearly independent m-canonical sections for every integer $m \geq 887$. We also obtain birationality of $|m(K_X + \Delta)|$ for every stable surface pair (X, Δ) with reduced boundary and every integer $m \geq 1882$.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

A klt stable surface is a projective surface X with klt singularities such that K_X is ample, and the volume of X is K_X^2. More generally, the volume of a projective variety X with klt singularities and ample canonical divisor is $K_X^{\dim X}$. A stable surface is a projective surface with semi-log canonical singularities [Kol13b, Definition–Lemma 5.10] and ample canonical divisor. Kollár raised the problem of the smallest volumes of log surfaces of general type [Kol94, Section 6].

By the DCC theorem of Alexeev [Ale94, Theorem 8.2], the volumes of klt stable surfaces form a DCC set. Thus they attain a minimum. In any fixed dimension d, the volumes of projective varieties of dimension d with klt singularities and ample canonical divisor form a DCC set by [HMX14, Theorem 1.3], so they also attain a minimum. The DCC theorem does not determine the minimum. An explicit lower bound is due to Alexeev and Mori [AM04, Theorem 4.8]. As

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written out in [AL19c, Section 10], it implies that any klt stable surface X satisfies

$$K_X^2 \geq \frac{1}{42 \cdot 84^{128 \cdot 42^5 + 168}},$$

and the right-hand side is about $10^{-3.22 \cdot 10^{10}}$.

Alexeev and W. Liu [AL19c, Theorems 1.4 and 6.1] constructed a projective surface with klt singularities, ample canonical divisor, and Picard number two, whose volume is

$$\frac{1}{48983} = \frac{1}{11 \cdot 61 \cdot 73}.$$

In [AL19c, Section 10] they wrote that the minimum might be close to, or equal to, $1/48983$. Totaro observed that the surface of Alexeev and W. Liu is a hypersurface X_{438} of degree 438 in $\mathbb{P}(11, 61, 146, 219)$ (see Theorem 1.5 for the equation) [Tot24, Introduction and the second paragraph after Conjecture 2.2].

Totaro constructed such a hypersurface in any dimension $d \geq 2$, improving on the examples of Totaro and Wang [TW23, Theorem 0.1]. Let $s_0 = 2$ and $s_{i+1} = s_0 s_1 \cdots s_i + 1$ for $i \geq 0$ be the Sylvester sequence. For an integer $d \geq 2$, set

$$a_{d+1} = \begin{cases} \frac{1}{4}(s_d^2 - s_d + 2) & \text{if } d \text{ is even,} \\ \frac{1}{4}(s_d^2 - 3s_d + 4) & \text{if } d \text{ is odd,} \end{cases}$$

$$a_d = (s_d - 2)a_{d+1} + s_d - 1, \qquad e = (s_d - 1)(1 + a_d + a_{d+1}),$$

and $a_i = e/s_i$ for $0 \leq i \leq d-1$. Totaro [Tot24, Theorem 2.1] proves that there exists a hypersurface of degree e in $\mathbb{P}(a_0, \ldots, a_{d+1})$ that is well-formed and klt with canonical divisor $\mathcal{O}(1)$, and he writes its equation explicitly right after the theorem. For $d = 2$, we have $s_2 = 7$, $a_3 = 11$, $a_2 = 5 \cdot 11 + 6 = 61$, $e = 6 \cdot 73 = 438$, $a_1 = 146$, and $a_0 = 219$. Thus for $d = 2$ the hypersurface of [Tot24, Theorem 2.1] is X_{438}, up to the order of the weights. Totaro conjectured the following.

Conjecture 1.1 ([Tot24, Conjecture 2.2])**.** *For any integer $d \geq 2$, the hypersurface of dimension d in* [Tot24, Theorem 2.1] *has the minimum volume among all projective varieties of dimension d with klt singularities and ample canonical divisor.*

The following results on small volumes of surfaces and pairs related to klt stable surfaces are known, in chronological order of their first arXiv versions.

(1) Alexeev and W. Liu constructed a projective lc surface pair with non-zero reduced boundary and volume $1/462$ [AL19c, Section 5], and a projective surface with ample canonical divisor, Picard number one, and $K_X^2 = 1/6351$ [AL19c, Theorem 8.2].
(2) The minimum of $\mathrm{vol}(K_X)$ over projective lc surfaces X of general type with positive geometric genus $h^0(X, K_X) > 0$ is $1/143$ [Liu26, Theorem 1.1(ii) and (iii)]. Moreover, the minimal resolution of a klt stable surface X with $K_X^2 < 1/143$ is a rational surface [Liu26, Corollary 6.12]. We use the last result as Theorem 2.7.
(3) Any accumulation point of the volumes K_X^2 of projective lc surfaces X with ample K_X is the volume $(K_X + B)^2$ of a projective lc surface pair (X, B) that is not klt, where B is a reduced divisor and $K_X + B$ is ample [AL19b, Theorem 1.1]. The smallest accumulation point is at least $1/86436 = 1/(7^2 \cdot 42^2)$ [AL19b, Theorem 1.7(i) and (iii)].
(4) Among the surfaces with ample canonical divisor and Picard number one of [AL19a], whose construction starts with four lines in $\mathbb{P}^2$, the smallest volume is $1/6351$ [AL19a, Introduction].
(5) The lower bound $1/462$ for $(K_X + S)^2$ of Theorem 1.2(2) is [LS23, Theorem 1.4]. It is attained by the pair of [AL19c, Section 5], and it answers a question of Kollár [Kol94], [Kol13a, Paragraph 40].

(6) The minimum $1/825$ of Theorem 1.3(1) for projective lc surfaces with ample canonical divisor that are not klt is [LL23, Theorem 1.1]. Moreover, $1/825$ is the smallest accumulation point of the volumes of projective lc surfaces with ample canonical divisor [LL23, Corollary 1.2].
(7) The minimum $1/6351$ of Theorem 1.4 for stable surfaces of Picard number one is [LL26, Theorem 1.1].
(8) For projective lc surface pairs (X, B), where B is a non-zero reduced divisor, such that $K_X + B$ is ample and B contains a zero-dimensional lc center of (X, B), the minimum of $(K_X + B)^2$ is $1/42$, and the pair attaining the minimum is unique up to isomorphism [LS26, Theorem 1.3].

In dimension at least three, no optimal lower bound for the volume had been proved as of 2023 [LS23, Introduction], although Esser and Totaro described the pair of volume $1/462$ as a hypersurface and generalized it to all dimensions [ET24, Introduction]. For surfaces, [LS23, Introduction, paragraph after Example 1.5] asks for the smallest volume of klt projective surfaces without boundary and writes that it is expected to be $1/48983$, and [LS26, Introduction, second paragraph] writes that the minimal volume of normal stable surfaces is unknown.

We recall the main results of [LS23], [LL23], and [LL26].

Theorem 1.2 ([LS23, Theorems 1.1 and 1.4])**.**

(1) Let (X, bS) be a projective klt surface pair, where b is a real number and $S \geq 0$ is a non-zero Weil divisor. Assume that $K_X + bS \sim_{\mathbb{R}} 0$. Then $b \leq 10/11$ or $b = 12/13$.
(2) Let (X, S) be a projective lc surface pair, where $S \geq 0$ is a non-zero Weil divisor. Assume that $K_X + S$ is ample. Then $(K_X + S)^2 \geq 1/462$, and if $(K_X + S)^2 = 1/462$, then S is a smooth rational curve.

Theorem 1.3 ([LL23, Theorems 1.1 and 3.4, Corollary 3.6])**.**

(1) Let X be a projective lc surface. Assume that K_X is ample and X is not klt. Then $K_X^2 \geq 1/825$. Moreover, up to isomorphism there exists exactly one such surface X with $K_X^2 = 1/825$.
(2) Let (X, S) be a projective lc surface pair, where $S \neq 0$ is a reduced Weil divisor. Assume that $K_X + S$ is ample and $(K_X + S)^2 = 1/462$. Then (X, S) is isomorphic to the pair of [LL23, Example 3.2].
(3) Let X be a stable surface. Assume that X is not normal. Then $K_X^2 > 1/462$.

Theorem 1.4 ([LL26, Theorem 1.1])**.** *Let X be a stable surface with $\rho(X) = 1$. Then $K_X^2 \geq 1/6351$. Moreover, if $K_X^2 = 1/6351$, then X is klt and is unique up to isomorphism.*

Theorem 1.2(1) is the main result of [LS23]. The proof in this paper does not apply it directly, but uses its analogue for pseudo-effective thresholds, which are equal to $12/13$ or at most $10/11$ (Theorem 2.17 [LL23, Theorem 2.6]), and the fact that log canonical thresholds are equal to 1 or at most $5/6$ (Theorem 2.16 [LS23, Lemma 3.6]). Theorem 1.2(2) and items (1) and (2) of Theorem 1.3 enter the proof in parts (b) and (d) of the sketch of the proof.

By Theorem 1.4, any klt stable surface X with $\rho(X) = 1$ satisfies $K_X^2 \geq 1/6351 > 1/48983$, so the inequality of Conjecture 1.1 for $d = 2$ holds strictly for surfaces of Picard number one. However, X_{438} has Picard number two [AL19c, Theorem 6.1], and this paper removes the assumption $\rho(X) = 1$. The proof does not use Theorem 1.4 itself, but results of [LL26]: divisorial valuations computing the minimal log discrepancy of a surface at a point [LL26, Section 4], and the classification of klt Calabi–Yau surface pairs of Picard number one [LL26, Theorem 3.1]. As in [LL26, Section 1.5], we divide the problem at the value $5/46$ of the minimal log discrepancy $\mathrm{mld}(X)$ of X (see Definition 2.2): we treat $\mathrm{mld}(X) < 5/46$ and $\mathrm{mld}(X) \geq 5/46$ separately.

We now explain where the method of [LL26] uses the assumption $\rho(X) = 1$. First, let $f : Z \to X$ be the extraction of a special divisorial valuation E over a point of X [LL26, Definitions 4.2 and 4.3]. One such E computes the minimal log discrepancy of X at that point

[LL26, Proposition 4.6]. If $\rho(X)=1$, $K_X^2 \le 1/6351$, and K_Z+tE is not nef for any $0 \le t < 5/6$, then by [LL26, Construction-Theorem 4.12], $\rho(Z)=2$, the K_Z-MMP with scaling of E contracts a single curve onto a surface T with $\rho(T)=1$, K_T+E_T is ample, where E_T is the image of E, and there exists a number p such that K_T+pE_T is numerically trivial. Thus (T, pE_T) is a klt Calabi–Yau surface pair of Picard number one, which is the input of [LL26, Theorem 3.1]. When $\rho(X) \ge 2$, we have $\rho(Z)=\rho(X)+1 \ge 3$, and the reduction to a Calabi–Yau pair of Picard number one is not available.

Second, under the assumptions of [LL26, Construction-Theorem 4.12], [LL26, Theorem 6.3] shows that there exist finitely many possibilities for the singularities of X when the minimal log discrepancy of X is at least $5/46$. The assumption $\rho(X)=1$ enters [LL26, Theorem 6.3] through two results of [LL26]. Since $K_X^2 \le 1/6351 < 1/143$, X is rational by Theorem 2.7, so by [LL26, Theorem 2.5], a bound of Bogomolov type for klt rational surfaces of Picard number one due to Keel and McKernan [KM99, Corollary 9.2], X has at most six singular points. Let $\pi : \widetilde{X} \to X$ be the minimal resolution, with exceptional curves C_i, and write $K_{\widetilde{X}} + \sum_i b_i C_i = \pi^* K_X$. By [LL26, Theorem 2.17], which follows [LS23, Lemma 4.3], when X is rational and $\rho(X)=1$ the sum over the singular points x of X of the invariant of [LS23, Notation 4.2], which is the number of the C_i over x minus the sum of $b_i(K_{\widetilde{X}} \cdot C_i)$ over these C_i, equals $9-K_X^2$. For rational X with $\rho(X) \ge 2$ the sum equals $10-\rho(X)-K_X^2$ [LL26, Theorem 2.17], and the bound of [LL26, Theorem 2.5] is not available. Without the assumption $\rho(X)=1$, the orbifold Bogomolov–Miyaoka–Yau inequality [HK08, Theorem 3.1] (klt surface singularities are quotient singularities [KM98, Proposition 4.18]) only implies

$$\sum_x \Big(1-\frac{1}{|G_x|}\Big) \le 2+\rho(X)-\frac{K_X^2}{3}$$

for a klt stable surface X with $K_X^2 < 1/143$, where x runs over the singular points of X and G_x is the local fundamental group of X at x. Indeed, $e(X)=2+\rho(X)$ by Corollary 2.8. The right-hand side grows with $\rho(X)$, whereas for $\rho(X)=1$ the sum is less than 3.

The goal of this paper is to determine the minimal volume of klt stable surfaces and the surfaces attaining it, that is, to prove Conjecture 1.1 for $d=2$ together with the uniqueness of the minimizer.

Main results. The main result of this paper determines the minimal volume of klt stable surfaces.

Theorem 1.5. *Let X be a klt stable surface. Then $K_X^2 \ge 1/48983$. Moreover, let*

$$X_{438} := (w^2+z^3+xy^7+x^{21}yz=0) \subset \mathbb{P}(11,61,146,219),$$

where x,y,z,w are the coordinates of weights $11,61,146,219$ respectively. Then X_{438} has klt singularities, $K_{X_{438}}$ is ample, $K_{X_{438}}^2 = 1/48983$, and

$$\mathcal{O}_{X_{438}}(K_{X_{438}}) \cong \mathcal{O}_{X_{438}}(1).$$

Furthermore, $K_X^2 = 1/48983$ if and only if $X \cong X_{438}$.

As an application of the volume bound, we obtain uniform effective pluricanonical nonvanishing for normal stable surfaces. Theorem 11.1 shows that, for every such surface X,

$$P_m(X) := h^0(X, \mathcal{O}_X(mK_X)) \ge 2 \qquad \text{for every integer } m \ge 887,$$

where $\mathcal{O}_X(mK_X)$ is interpreted reflexively. If X is not klt, the bound improves to $m \ge 116$. The first pluricanonical pencil on X_{438} occurs in degree 292, so the optimal uniform bound for the first pencil on normal stable surfaces lies between 292 and 887.

Combining the volume bounds with Hisamatsu's log-pluricanonical birationality estimate [His03, Theorem 1.1], and using doubling to treat reduced boundaries, we also obtain effective birationality for stable surface pairs. For every connected stable surface pair (X,Δ) with reduced

boundary, $|m(K_X+\Delta)|$ defines a rational map birational onto its image for every $m \geq 1882$. If $\Delta \neq 0$ or X is non-normal, the bound improves to $m \geq 275$ (Corollary 11.9).

The proof of Theorem 1.5 uses two finite computations. The first is the proof that the explicit system of linear constraints in integer variables of Definition 9.15 has no integer solution (Proposition 9.21), which is a proof file checked by a computer. The proof file is a derivation in the VIPR format of [CGS16], which is the ancillary file `large_mld_proof.vipr.xz`.

The proof file was checked by the exact rational checker `viprchk` [CGS16, Section 5]. The system contained in the proof file coincides with an independent regeneration of the system from the formulas in Section 9, made with exact rational arithmetic by the ancillary script `regenerate_and_compare.py`.

The ancillary file `README.txt` describes how the proof file was produced and explains how to repeat the check, and the file `SHA256SUMS` lists the hashes of the ancillary files. The version of `viprchk` and its input and output adapter are described in Subsection 9.6. Apart from `check_tables.py` and `tables.json`, the ancillary files are used only for Proposition 9.21, whose proof is the proof file. The rows of the tables of Appendix A are exact identities or exact values, which can be checked with integer arithmetic.

The second computation shows that the entries of Table 9, the headers of Tables 10 and 11, and the entries of Table 11 run through all the parameter tuples allowed by the conditions in the proofs of Theorem 9.10 and Proposition 9.9. It is made by the ancillary script `check_tables.py`, which lists the parameter tuples from the conditions, compares them with the tables, and recomputes the rows from the definitions with exact rational arithmetic. Any other step of the proof is proved in the text.

Sketch of the proof. By Lemma 2.6, there exists a minimizer X (Definition 2.3), and $K_X^2 \leq K_{X_{438}}^2 = 1/48983$ (Lemma 3.1). It suffices to show that $X \cong X_{438}$ (Section 10). As in [LL26, Section 1.5], the proof splits at $\operatorname{mld}(X) = 5/46$.

(1) For Theorem 9.1, suppose that $\operatorname{mld}(X) \geq 5/46$ and $K_X^2 \leq 1/48983$. Then the local costs of (a) below satisfy $C_x > 0$ (Theorem 9.10) and $\sum_x C_x < 151733$ (Proposition 9.11), so finitely many types of singular points remain (Theorem 9.14), and X yields an integer solution of the system Σ of Definition 9.15 in 1396 integer variables (Proposition 9.18), which has none (Proposition 9.21).

(2) For Theorem 8.1, suppose that $\operatorname{mld}(X) < 5/46$, and let p and λ be the numbers of Set-up 7.2 and Construction-Theorem 7.3. If $p \geq 115/134$, then $X \cong X_{438}$ (Theorem 6.2). If $p < 115/134$, then $\lambda < 0$ (Construction-Theorem 7.3(2)), and if $X \ncong X_{438}$, then X is not a minimizer (Propositions 8.8 and 8.10 and Subsection 8.4).

(3) If $K_X^2 \leq 1/48983$, then $P_m(X) := h^0(X, \mathcal{O}_X(mK_X))$ vanishes for $1 \leq m \leq 17$ unless $X \cong X_{438}$ (Proposition 4.14), and $P_m(X) \leq 1$ for $1 \leq m \leq 156$ (Proposition 5.1). Step (1) uses both, and step (2) uses the first through Theorem 6.2.

The key new steps are the local costs of (a) below, for step (1), and, for $\lambda < 0$, Lemma 8.6: a klt surface Y with K_Y big such that, on a common resolution, the pullback of K_X minus the pullback of K_Y is effective (we say that K_X dominates K_Y) and non-zero has $\operatorname{vol}(K_Y) < K_X^2$ by the Hodge index theorem. Thus step (2) classifies minimizers only. The other links use known sharp bounds and [LL26, Theorem 3.1], together with the constructions of (d) below.

(a) Local costs. Since $K_X^2 < 1/143$, the minimal resolution of X is rational [Liu26, Corollary 6.12]. Eliminating $\rho(X)$ from the Noether formula and the orbifold Bogomolov–Miyaoka–Yau inequality leaves only the bound $\sum_x \mu_x \leq 12 - \frac{4}{3}K_X^2$ on local terms μ_x (Lemma 9.4), which, unlike [LL26, Theorem 2.5], does not bound the number of singular points, since $\mu_x < 0$ for germs such as $\frac{1}{18}(1,1)$. By Riemann–Roch, since $P_2(X) = \cdots = P_9(X) = 0$ (Proposition 4.4(1)), the local corrections $c_x(m)$ satisfy $\sum_x c_x(m) = -1 - \binom{m}{2}K_X^2$ for $2 \leq m \leq 9$. Thus a positive combination C_x of μ_x and the $c_x(m)$, called the cost of $X \ni x$ (Definition 9.2), satisfies $\sum_x C_x < 151733$ (Proposition 9.11), while $C_x > 0$ at any germ with $\operatorname{mld}(X \ni x) \geq 5/46$ (Theorem 9.10).

Therefore, in step (1), each singular point of X is of one of the 1368 types of the set J of Definition 9.15, so the singular points enter only through the numbers of points of each type, which with $P_{18}(X), \dots, P_{45}(X)$ are the variables of Σ.

(b) The case $\lambda \geq 0$. Let $f : Z \to X$ extract a special divisorial valuation E computing $\mathrm{mld}(X)$ (Definition 2.13, [LL26, Section 4]), let $g : Z \to T$ be the ample model of $K_Z + nE$ at the nef threshold n, and let $\lambda := (K_T + E_T) \cdot E_T$, where $E_T := g_*E$. If $\lambda > 0$, then $K_T + E_T$ is ample, so $(K_T + E_T)^2 \geq 1/462$ by Theorem 1.2(2). If $\lambda = 0$, then contracting E_T yields a projective lc surface with ample canonical divisor that is not klt, so $(K_T + E_T)^2 \geq 1/825$ by Theorem 1.3(1).

We use only the inequality $\mathrm{mld}(X) < 5/46$, which with either bound implies $p > 20/23 > 115/134$ for the pseudo-effective threshold p, as in [LL26, Proposition 5.1]. Thus Theorem 6.2 implies $X \cong X_{438}$ when $\lambda \geq 0$.

(c) Comparison with a third surface. If $\lambda < 0$, then E_T contracts to a klt stable surface X' (Construction-Theorem 7.3(3)). We show that X is not a minimizer unless $X \cong X_{438}$: we construct a klt surface Y over X' such that, on common resolutions, K_X dominates K_Y with non-zero difference and K_Y dominates $K_T + tE_T$ for some $t > 115/134$ (Lemma 8.7). The second condition implies that K_Y is big, since $p < 115/134$ unless $X \cong X_{438}$ (Theorem 6.2), and the first then implies $\mathrm{vol}(K_Y) < K_X^2$ (Lemma 8.6).

The work is to construct Y. By Definition-Proposition 8.2, the morphism $X \to X'$ contracts curves whose strict transforms on the minimal resolution of X are (-1)-curves. Their strict transforms on Z intersect E in three configurations: at a smooth point of Z, where $t \geq 7/8$ (Subsection 8.2), at a point where E and the strict transform are the images of the coordinate axes of a cyclic quotient of $\mathbb{C}^2$, where $t > 63/73$ (Subsection 8.3), and through a fork, where $t > 7/8$ (Subsection 8.4), apart from one case with $t = 6/7 < 115/134$, which the inequality $K_X^2 \geq q(b-n)^2$ of Theorem 2.14(3) excludes, and the case where the singularity of X at $f(E)$ is of type D-II (Definition 2.9), which is excluded without constructing Y.

(d) Pluricanonical sections. For a non-zero element of $H^0(X, \mathcal{O}_X(mK_X))$ with divisor of zeros D, let B be the reduced sum of the exceptional divisors and the strict transform of D on a log resolution W of $(X, \mathrm{Supp}\, D)$. The log canonical model of (W, B) is a pair with non-zero reduced boundary and volume at most $(m+1)^2 K_X^2$ (Construction-Proposition 4.2), so sharp bounds for pairs become lower bounds for K_X^2. For $1 \leq m \leq 17$, the volume is less than $1/42$, so by [LS26, Theorem 1.3] the boundary contains no zero-dimensional lc center, and it is a smooth rational curve (Construction-Proposition 4.2(2)). Then Theorem 2.17, by which the pseudo-effective threshold of the boundary is $12/13$ or at most $10/11$, and integrality leave only the degrees 11, 13, and 15, and degree 15 reduces to degree 5 by Construction-Proposition 4.2(3).

Degree 11 cannot be excluded, since $P_{11}(X_{438}) = 1$. There the log canonical model is the unique pair of volume $1/462$ (Theorem 1.3(2)), and forms built from its coordinates extend to elements of degrees 61 and 146 of the canonical ring (Theorem 4.10), since the curves of X contracted on it are toric (Lemma 4.7).

The elements x, y, z of degrees 11, 61, and 146 and a square root of $-(z^3 + xy^7 + x^{21}yz)$ built likewise show that the rational map to $\mathbb{P}(11, 61, 146)$ is not birational. Since its degree is at most $11 \cdot 61 \cdot 146 \cdot K_X^2 \leq 2$ (Lemma 4.9), it has degree two, and X and X_{438} are both the normalization of $\mathbb{P}(11, 61, 146)$ in one quadratic extension of $\mathbb{C}(\mathbb{P}(11, 61, 146))$ (Theorem 4.10). Degree 13 is excluded analogously (Proposition 4.13). If $P_m(X) \geq 2$ with $m \leq 156$, then the moving part M of $|\lfloor m\pi^* K_X \rfloor|$, where π is the minimal resolution of X, has $M^2 = 0$, and the π-exceptional curves E with $E \cdot M > 0$ yield $K_X^2 \geq 1/(m(2m+1))$, which contradicts $K_X^2 \leq 1/48983$ (Proposition 5.1).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce the types of quotient singularities and the toric description of a cyclic quotient singularity along a smooth curve, special divisorial valuations and their extractions [LL26, Section 4], the gap theorems for lc thresholds [LS23, Lemma 3.6] and for pseudo-effective thresholds [LL23, Theorem 2.6], the classification [LL26, Theorem 3.1] of klt Calabi–Yau surface

pairs of Picard number one whose boundary is b times a non-zero Weil divisor with $b \geq 6/7+1/938$, and the rationality theorem [Liu26, Corollary 6.12].

Section 3 proves the properties of X_{438} that we use. Section 4 proves the vanishing of $P_m(X)$ in step (3) of the sketch of the proof, and Section 5 proves the upper bound for $P_m(X)$ in that step.

Section 6 shows that a klt stable surface X with $K_X^2 \leq 1/48983$ is isomorphic to X_{438} if there exists a birational morphism $f : Z \to X$ from a klt surface Z contracting exactly one curve E, with rational normalization and with log discrepancy less than 1, such that $K_Z + tE$ is pseudo-effective only for $t \geq 115/134$. Section 7, for a klt stable surface X with $K_X^2 \leq 1/48983$ and $\mathrm{mld}(X) < 5/46$, fixes the morphisms $f : Z \to X$ and $g : Z \to T$ and the number λ of the sketch of the proof. Section 8 shows that $\lambda < 0$ unless $X \cong X_{438}$, treats the case $\lambda < 0$, and ends with the proof that a klt stable surface of minimal volume with $\mathrm{mld}(X) < 5/46$ is isomorphic to X_{438}.

Section 9 proves that $K_X^2 > 1/48983$ for any klt stable surface X with $\mathrm{mld}(X) \geq 5/46$, through the system of Definition 9.15. Section 10 proves Theorem 1.5, and Section 11 gives the applications to pluricanonical nonvanishing and effective birationality. Appendix A contains the tables of the finite checks in Section 9.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and klt stable surfaces.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any pair (X, B), we say that (X, B) is *lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc center* of (X, B) is the center on X of a prime divisor E over X with $a(E, X, B) = 0$.

Definition 2.2. A *surface germ* $X \ni x$ consists of a surface X and a closed point $x \in X$. Let $X \ni x$ be a surface germ with klt singularities. The *minimal log discrepancy* of $X \ni x$ is

$$\mathrm{mld}(X \ni x) := \inf\{a(E, X, 0) \mid E \text{ is a prime divisor over } X \text{ with } \mathrm{center}_X E = x\},$$

and for a surface X with klt singularities we set

$$\mathrm{mld}(X) := \inf\{\mathrm{mld}(X \ni x) \mid x \in X \text{ is a closed point}\}.$$

Definition 2.3. A *klt stable surface* is a projective surface X with klt singularities such that K_X is ample. For a klt stable surface X and an integer $m \geq 0$, we set $R(X)_m := H^0(X, \mathcal{O}_X(mK_X))$, where $\mathcal{O}_X(mK_X)$ is the reflexive sheaf of rank one, $P_m(X) := \dim R(X)_m$, and $R(X) :=$

$\bigoplus_{m\geq 0} R(X)_m$. We say that a klt stable surface X is a *minimizer* if $K_X^2 \leq K_Y^2$ for any klt stable surface Y.

2.2. **Minimal model program and volumes.**

Theorem 2.4 ([Fuj12, Theorem 1.1])**.** *Let (X, B) be a projective lc surface pair. Then there exists a sequence of extremal contractions*

$$(X, B) = (X_0, B_0) \to (X_1, B_1) \to \cdots \to (X_k, B_k)$$

of the $(K_X + B)$-MMP such that (X_i, B_i) is lc for any i, and if $K_X + B$ is pseudo-effective, then $K_{X_k} + B_k$ is semi-ample.

For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on a normal projective surface X, $\operatorname{vol}(D)$ is the volume of D [Laz04, Definition 2.2.31 and Remark 2.2.39], which depends only on the numerical class of D [Laz04, Proposition 2.2.41].

Lemma 2.5. *Let X be a normal projective surface.*

1. *$\operatorname{vol}(D + D') \geq \operatorname{vol}(D)$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D and any pseudo-effective $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D' on X.*
2. *Let $h : X' \to X$ be a birational morphism from a $\mathbb{Q}$-factorial normal projective surface X', D a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, and $E \geq 0$ an h-exceptional $\mathbb{Q}$-divisor on X'. Then $\operatorname{vol}(h^*D + E) = \operatorname{vol}(D)$.*
3. *If X is smooth and $D = P + N$ is the Zariski decomposition* [Zar62, Theorem 7.7], [Laz04, Theorem 2.3.19] *of a big $\mathbb{Q}$-divisor D on X, then $\operatorname{vol}(D) = P^2$.*

Proof. (1) First assume that $D' \geq 0$. Choose a positive integer a such that aD and aD' are Cartier. For any positive integer m, multiplication by the section of $\mathcal{O}_X(maD')$ with divisor of zeros maD' is an injection

$$H^0(X, \mathcal{O}_X(maD)) \hookrightarrow H^0(X, \mathcal{O}_X(ma(D + D'))).$$

Thus

$$\operatorname{vol}(D + D') = a^{-2} \operatorname{vol}(a(D + D')) \geq a^{-2} \operatorname{vol}(aD) = \operatorname{vol}(D)$$

by [Laz04, Remark 2.2.39].

In general, fix a very ample Cartier divisor A on X and a rational number $\epsilon > 0$. Then $D' + \epsilon A$ is big, since the big cone is the interior of the pseudo-effective cone [Laz04, Theorem 2.2.26]. Choose a positive integer c such that $c(D' + \epsilon A)$ is Cartier. By [Laz04, Corollary 2.2.7], there exist a positive integer l and a Cartier divisor $N \geq 0$ such that $lc(D' + \epsilon A) \sim A + N$. Since A is very ample, $A \sim H$ for a Cartier divisor $H \geq 0$. Thus $D' + \epsilon A$ is numerically equivalent to the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\frac{1}{lc}(H + N) \geq 0$, and the first case together with [Laz04, Proposition 2.2.41] implies

$$\operatorname{vol}(D + D' + \epsilon A) \geq \operatorname{vol}(D).$$

Letting $\epsilon \to 0$ through rational numbers, the continuity of the volume [Laz04, Theorem 2.2.44] implies $\operatorname{vol}(D + D') \geq \operatorname{vol}(D)$.

(2) Since X' is $\mathbb{Q}$-factorial, $h^*D + E$ is $\mathbb{Q}$-Cartier. Choose a positive integer a such that aD and aE are Cartier, and let m be a positive integer. Since $maE \geq 0$ is h-exceptional, $h(\operatorname{Supp} E)$ is a finite set. Thus a rational function ϕ on X with $\operatorname{div}(\phi) + maE \geq 0$ over an open subset $V \subset X$ is regular on $V \backslash h(\operatorname{Supp} E)$ (since X' is normal and $h_*\mathcal{O}_{X'} = \mathcal{O}_X$, as h is birational, X is normal, and $h_*\mathcal{O}_{X'}$ is coherent [Har77, Theorem III.8.8]), hence on V since X is normal. Therefore $h_*\mathcal{O}_{X'}(maE) = \mathcal{O}_X$, and the projection formula implies

$$h_*\mathcal{O}_{X'}(mah^*D + maE) = \mathcal{O}_X(maD) \otimes h_*\mathcal{O}_{X'}(maE) = \mathcal{O}_X(maD).$$

Hence

$$H^0(X', \mathcal{O}_{X'}(ma(h^*D + E))) = H^0(X, \mathcal{O}_X(maD))$$

for any positive integer m, and $\operatorname{vol}(h^*D + E) = \operatorname{vol}(D)$ by [Laz04, Remark 2.2.39]. For $E = 0$ this is [Laz04, Proposition 2.2.43].

(3) Choose a positive integer a such that aD is a divisor. Then $aD = aP + aN$ is the Zariski decomposition of aD, and

$$\operatorname{vol}(D) = a^{-2}\operatorname{vol}(aD) = a^{-2}(aP)^2 = P^2$$

by [Laz04, Remark 2.2.39, Theorem 2.3.19, Proposition 2.3.21 and Corollary 2.3.22]. □

2.3. **Minimizers and rationality.**

Lemma 2.6. *The set $\{K_X^2 \mid X$ is a klt stable surface$\}$ has a minimum, which is at most $1/48983$. In particular, there exists a klt stable surface that is a minimizer.*

Proof. Let $\mathcal{V} := \{K_X^2 \mid X \text{ is a klt stable surface}\}$, and let X be a klt stable surface. Since X is normal and klt, X is semi-log canonical and K_X is log canonical on the minimal resolution of X [Ale94, Definitions 1.5, 1.7 and 1.9], so K_X is MR semi-log canonical in the sense of [Ale94, Definition 1.12]. Since K_X is ample, the DCC theorem of Alexeev [Ale94, Theorem 8.2], applied to the pairs $(X, 0)$ and the DCC set $\{0\}$, implies that $\mathcal{V}$ is a DCC set.

By Lemma 3.1, X_{438} is a klt surface,

$$\mathcal{O}_{X_{438}}(K_{X_{438}}) \cong \mathcal{O}_{X_{438}}(1)$$

is ample, and $K_{X_{438}}^2 = 1/48983$, so $1/48983 \in \mathcal{V}$.

Hence $\mathcal{V}$ is a non-empty DCC set of real numbers containing $1/48983$. A non-empty DCC set of real numbers has a minimum, since otherwise it would contain a strictly decreasing infinite sequence. Thus $v_{\min} := \min \mathcal{V}$ exists and $v_{\min} \le 1/48983$, and any klt stable surface X with $K_X^2 = v_{\min}$ is a minimizer. □

Theorem 2.7 ([Liu26, Corollary 6.12])**.** *Let X be a klt stable surface with $K_X^2 < 1/143$. Then the minimal resolution of X is a rational surface.*

Theorem 2.7 is the case of [Liu26, Corollary 6.12] where the boundary is zero: a klt stable surface X is a projective log canonical surface of general type, and the volume of K_X is K_X^2 since K_X is ample.

Corollary 2.8. *Let X be a klt stable surface with $K_X^2 < 1/143$, let $\pi : \widetilde{X} \to X$ be the minimal resolution, and let L be the number of π-exceptional curves. Then $\chi(\mathcal{O}_X) = 1$, $e(X) = 2 + \rho(X)$, where $e(X)$ denotes the topological Euler number of X, and $\rho(\widetilde{X}) = 10 - K_{\widetilde{X}}^2 = \rho(X) + L$.*

Proof. By Theorem 2.7, $\widetilde{X}$ is a rational surface. Since X is klt, X has rational singularities [KM98, Theorem 5.22], so $\chi(\mathcal{O}_X) = \chi(\mathcal{O}_{\widetilde{X}})$. Since $\widetilde{X}$ is a smooth rational surface, $q(\widetilde{X}) = p_g(\widetilde{X}) = 0$ [Har77, Theorem V.5.5 and Proposition V.3.4]. Thus

$$\chi(\mathcal{O}_X) = \chi(\mathcal{O}_{\widetilde{X}}) = 1 - q(\widetilde{X}) + p_g(\widetilde{X}) = 1.$$

By Noether's formula [BHPV04, Theorem I.5.5, formula (4)], where the second Chern class of $\widetilde{X}$ is $e(\widetilde{X})$ [BHPV04, Section I.4],

$$e(\widetilde{X}) = 12\chi(\mathcal{O}_{\widetilde{X}}) - K_{\widetilde{X}}^2 = 12 - K_{\widetilde{X}}^2.$$

By [Har77, Corollary V.5.4, Theorem V.5.8, Example V.5.8.2, and Remark V.5.8.4], a sequence of contractions of (-1)-curves maps $\widetilde{X}$ onto $\mathbb{P}^2$ or onto a $\mathbb{P}^1$-bundle over $\mathbb{P}^1$. The blow-up of a point of a smooth projective surface increases the Picard number by one and decreases the self-intersection of the canonical divisor by one [Har77, Propositions V.3.2 and V.3.3]. Since $\rho(\mathbb{P}^2) = 1$, $K_{\mathbb{P}^2}^2 = 9$, and a $\mathbb{P}^1$-bundle Y over $\mathbb{P}^1$ satisfies $\rho(Y) = 2$ [Har77, Proposition V.2.3] and $K_Y^2 = 8$ [Har77, Corollary V.2.11], we obtain $\rho(\widetilde{X}) = 10 - K_{\widetilde{X}}^2$. Thus

$$e(\widetilde{X}) = 2 + \rho(\widetilde{X}).$$

For any closed point $x \in X$, let ℓ_x be the number of π-exceptional prime divisors over x. Then $\ell_x = 0$ for all but finitely many x, and the π-exceptional curves over x form a tree of ℓ_x smooth

rational curves meeting transversally [KM98, Theorem 4.7], whose topological Euler number is $2\ell_x - (\ell_x - 1) = \ell_x + 1$ when $\ell_x > 0$. By the additivity of the topological Euler number over $X \setminus \operatorname{Sing} X$ and the fibers of π over $\operatorname{Sing} X$,

$$e(X) = e(\widetilde{X}) - \sum_x \ell_x.$$

Since X is $\mathbb{Q}$-factorial [KM98, Proposition 4.11], for any Weil divisor C on $\widetilde{X}$ the $\mathbb{Q}$-divisor $C - \pi^*\pi_* C$ is π-exceptional, so the numerical classes of pullbacks of divisors on X and of the π-exceptional curves span the space of numerical classes on $\widetilde{X}$. By the projection formula, π^* is injective on numerical classes and $\pi^* D \cdot G = 0$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on X and any π-exceptional curve G. Since the intersection matrix of the π-exceptional curves is negative definite [KM98, Lemma 3.40], their numerical classes are linearly independent modulo π^* of the numerical classes on X. Hence

$$\rho(\widetilde{X}) = \rho(X) + \sum_x \ell_x.$$

Thus

$$e(X) = e(\widetilde{X}) - \sum_x \ell_x = 2 + \rho(\widetilde{X}) - \sum_x \ell_x = 2 + \rho(X).$$

Since $\sum_x \ell_x = L$, this proves the statement. □

2.4. **Quotient singularities.** A klt surface germ $X \ni x$ that is not smooth is a quotient singularity, the exceptional curves of the minimal resolution of $X \ni x$ are smooth rational curves, and their dual graph is a chain or a tree with exactly one vertex of degree three [KM98, Proposition 4.18 and Theorem 4.7]. The quotient surface singularities were classified by Brieskorn [Bri68, Satz 2.11], and we use the form of the classification recalled in [HK08, Section 5, Table 1].

Definition 2.9 ([LL26, Definition–Theorem 2.9 and Definition 4.1])**.** Let $X \ni x$ be a klt surface germ that is not smooth, and $\pi : \widetilde{X} \to X$ the minimal resolution of $X \ni x$.

(1) For coprime integers $r > q \geq 1$, the *Hirzebruch–Jung chain* $[e_1, \dots, e_\ell]$ of r/q is defined by the integers $e_i \geq 2$ with

$$\frac{r}{q} = e_1 - 1/(e_2 - 1/(\cdots - 1/e_\ell)).$$

If the dual graph of π is a chain $E_1, \dots, E_\ell$ with $E_i^2 = -e_i$ and $[e_1, \dots, e_\ell]$ is the Hirzebruch–Jung chain of r/q, we say that $X \ni x$ is a *cyclic quotient singularity* of *type* $\frac{1}{r}(1, q)$ and of *order* r.

(2) If the dual graph of π has a vertex E_0 of degree three, we call E_0 the *fork*. The three arms of the graph at E_0, read from E_0 outwards, are the Hirzebruch–Jung chains of r_i/q_i for $i = 1, 2, 3$, ordered so that $r_1 \leq r_2 \leq r_3$, and $q_i \leq q_j$ if $i < j$ and $r_i = r_j$. We say that $X \ni x$ is of type $[e_0; (r_1, q_1), (r_2, q_2), (r_3, q_3)]$, where $e_0 := -E_0^2$. We say that $X \ni x$ is of type *D-I* if $(r_2, q_2) = (2, 1)$ and $e_0 \geq 3$, of type *D-II* if $(r_2, q_2) = (2, 1)$ and $e_0 = 2$, of type *E-I* if $r_2 = 3$ and $e_0 \geq 3$, and of type *E-II* if $r_2 = 3$ and $e_0 = 2$.

Lemma 2.10. *Let $X \ni x$ be a klt surface germ of type $[e_0; (r_1, q_1), (r_2, q_2), (r_3, q_3)]$. Let $\sigma := \sum_{i=1}^3 1/r_i - 1$ and $\eta := e_0 - \sum_{i=1}^3 q_i/r_i$. Then $\sigma > 0$, $\eta > 0$, and (r_1, r_2, r_3) is one of $(2, 2, r)$ with $r \geq 2$, $(2, 3, 3)$, $(2, 3, 4)$, and $(2, 3, 5)$. The fork has log discrepancy $a_0 := \sigma/\eta$, and the curve on the i-th arm intersecting the fork has log discrepancy $(q_i a_0 + 1)/r_i$. The order of the local fundamental group of $X \ni x$ is $4\eta/\sigma^2$.*

Proof. Let $\pi : \widetilde{X} \to X$ be the minimal resolution of $X \ni x$, E_0 the fork, and M the intersection matrix of the π-exceptional curves. For $i = 1, 2, 3$, let $E_{i,1}, \dots, E_{i,\ell_i}$ be the curves of the i-th arm, numbered from E_0 outwards, and let $e_{i,j} := -E_{i,j}^2$. Then $[e_{i,1}, \dots, e_{i,\ell_i}]$ is the Hirzebruch–Jung

chain of r_i/q_i. In particular $r_i > q_i \geq 1$, so $r_i \geq 2$. Since the π-exceptional curves are smooth rational curves with simple normal crossings [KM98, Theorem 4.7], we have

$$K_{\widetilde{X}} + (1-c_0)E_0 + \sum_{i=1}^{3}\sum_{j=1}^{\ell_i}(1-c_{i,j})E_{i,j} = \pi^* K_X, \tag{2.1}$$

where $c_0 := a(E_0, X, 0)$ and $c_{i,j} := a(E_{i,j}, X, 0)$, and $c_0 > 0$ because $X \ni x$ is klt. For a π-exceptional curve G with $G^2 = -e$ and δ neighbors in the dual graph, $K_{\widetilde{X}} \cdot G = e - 2$ by adjunction, so the intersection of (2.1) with G shows that $ec_G - \sum_{G'} c_{G'} = 2 - \delta$, where c_G is the log discrepancy of G and G' runs over the neighbors of G. With $c_{i,0} := c_0$ and $c_{i,\ell_i+1} := 1$, we obtain

$$e_0 c_0 - \sum_{i=1}^{3} c_{i,1} = -1, \qquad e_{i,j}c_{i,j} = c_{i,j-1} + c_{i,j+1} \quad (1 \leq j \leq \ell_i). \tag{2.2}$$

Fix i. For $0 \leq j \leq \ell_i - 1$ let $\mu_{i,j}$ be the determinant of minus the intersection matrix of $E_{i,j+1}, \dots, E_{i,\ell_i}$, and let $\mu_{i,\ell_i} := 1$ and $\mu_{i,\ell_i+1} := 0$. The Laplace expansion along the first row shows that $\mu_{i,j-1} = e_{i,j}\mu_{i,j} - \mu_{i,j+1}$ for $1 \leq j \leq \ell_i$. We have $\mu_{i,j} > 0$ for $0 \leq j \leq \ell_i$, since minus the intersection matrix of any set of π-exceptional curves is positive definite [KM98, Lemma 3.40]. Thus descending induction on j shows that $\gcd(\mu_{i,j-1}, \mu_{i,j}) = 1$ and

$$\frac{\mu_{i,j-1}}{\mu_{i,j}} = e_{i,j} - 1/(e_{i,j+1} - 1/(\cdots - 1/e_{i,\ell_i}))$$

for $1 \leq j \leq \ell_i$. For $j = 1$ the right-hand side is r_i/q_i, so $\mu_{i,0} = r_i$ and $\mu_{i,1} = q_i$. Let $\nu_{i,0} := 0$, $\nu_{i,1} := 1$, and $\nu_{i,j+1} := e_{i,j}\nu_{i,j} - \nu_{i,j-1}$ for $1 \leq j \leq \ell_i$. Since $x_j = \mu_{i,j}$ and $x_j = \nu_{i,j}$ both satisfy $x_{j+1} = e_{i,j}x_j - x_{j-1}$ for $1 \leq j \leq \ell_i$, the number $\mu_{i,j}\nu_{i,j+1} - \mu_{i,j+1}\nu_{i,j}$ does not depend on j. For $j = 0$ it is r_i, and for $j = \ell_i$ it is ν_{i,ℓ_i+1}. Thus $\nu_{i,\ell_i+1} = r_i$, and the numbers $(c_0\mu_{i,j} + \nu_{i,j})/r_i$ satisfy the second equation of (2.2), with the value c_0 at $j = 0$ and the value 1 at $j = \ell_i + 1$. For fixed c_0, the second equation of (2.2) is a linear system in $c_{i,1}, \dots, c_{i,\ell_i}$ whose matrix is minus the intersection matrix of the i-th arm, of determinant $\mu_{i,0} = r_i \neq 0$. Hence $c_{i,j} = (c_0\mu_{i,j} + \nu_{i,j})/r_i$ for any j, and in particular

$$c_{i,1} = \frac{q_i c_0 + 1}{r_i}.$$

By the first equation of (2.2), $e_0c_0 - \sum_i (q_ic_0 + 1)/r_i = -1$, that is, $\eta c_0 = \sigma$.

The Laplace expansion of $\det(-M)$ along the row of E_0 shows that

$$\det(-M) = e_0\mu_{1,0}\mu_{2,0}\mu_{3,0} - \sum_{i=1}^{3}\mu_{i,1}\prod_{j\neq i}\mu_{j,0} = e_0 r_1 r_2 r_3 - \sum_{i=1}^{3} q_i \prod_{j\neq i} r_j = r_1 r_2 r_3 \eta.$$

Since $-M$ is positive definite [KM98, Lemma 3.40], $\eta > 0$. Hence $\sigma = \eta c_0 > 0$, $c_0 = \sigma/\eta = a_0$, and $c_{i,1} = (q_i a_0 + 1)/r_i$. Since $\sigma > 0$, we have $1/r_1 + 1/r_2 + 1/r_3 > 1$ with $2 \leq r_1 \leq r_2 \leq r_3$. If $r_1 \geq 3$, then $\sum_i 1/r_i \leq 1$, so $r_1 = 2$ and $(r_1, q_1) = (2, 1)$. Then $1/r_2 + 1/r_3 > 1/2$. If $r_2 \geq 4$, or if $r_2 = 3$ and $r_3 \geq 6$, then $1/r_2 + 1/r_3 \leq 1/2$. Thus (r_1, r_2, r_3) is $(2, 2, r)$ with $r \geq 2$, $(2, 3, 3)$, $(2, 3, 4)$, or $(2, 3, 5)$.

By [KM98, Proposition 4.18], $X \ni x$ is analytically isomorphic to the germ of $\mathbb{C}^2/G$ at the image of 0, where G is a finite group acting on $(0 \in \mathbb{C}^2)$ freely in codimension one. By Cartan's linearization lemma [Car57, Section 4, pp. 97–98], after a holomorphic change of coordinates G acts linearly, so we may assume that $G \subset GL(2, \mathbb{C})$. An element $g \neq 1$ of G does not have 1 as an eigenvalue, since otherwise g is diagonalizable with eigenvalues 1 and $\lambda \neq 1$ and fixes a line pointwise. Thus g fixes only 0, G acts freely on the simply connected space $\mathbb{C}^2\backslash\{0\}$, and the local fundamental group of $X \ni x$ is G. The order $|G|$ is listed in [HK08, Section 5, Table 1]. If $(r_1, r_2, r_3) = (2, 2, r_3)$, then $|G| = 4dr_3$ with $d := (e_0 - 1)r_3 - q_3$. Here $\sigma = 1/r_3$ and

$\eta = e_0 - 1 - q_3/r_3 = d/r_3$, so

$$\frac{4\eta}{\sigma^2} = 4 \cdot \frac{d}{r_3} \cdot r_3^2 = 4dr_3.$$

If (r_1, r_2, r_3) is $(2,3,3)$, $(2,3,4)$, or $(2,3,5)$, then $|G|$ is $24m$, $48m$, or $120m$ respectively, where m is the integer listed in each row of [HK08, Section 5, Table 1] as a function of b, with $b = e_0$ and the dual graph $\langle b; s_1, t_1; s_2, t_2; s_3, t_3\rangle$ read as $[e_0; (r_1, q_1), (r_2, q_2), (r_3, q_3)]$. In any row, $m = (e_0 - 2)/\sigma + c$, where $\sigma = 1/6$ and $c = 9 - 2(q_2 + q_3)$ if $(r_1, r_2, r_3) = (2,3,3)$, $\sigma = 1/12$ and $c = 18 - 4q_2 - 3q_3$ if $(r_1, r_2, r_3) = (2,3,4)$, and $\sigma = 1/30$ and $c = 45 - 10q_2 - 6q_3$ if $(r_1, r_2, r_3) = (2,3,5)$. Since $r_1 = 2$, $q_1 = 1$, and $r_2 = 3$, we have $\sigma c = 3/2 - q_2/3 - q_3/r_3$ in any case, so $\eta = e_0 - 2 + \sigma c$ and $m = \eta/\sigma$. Hence

$$\frac{4\eta}{\sigma^2} = 144\eta = 24m, \qquad \frac{4\eta}{\sigma^2} = 576\eta = 48m, \qquad \frac{4\eta}{\sigma^2} = 3600\eta = 120m$$

respectively. Thus $|G| = 4\eta/\sigma^2$ in any case. □

2.5. **Toric germs and adjunction.**

Definition-Lemma 2.11. Let $r \geq 1$ and $0 \leq Q \leq r-1$ be integers with $\gcd(Q, r) = 1$, let $N \subset \mathbb{Q}^2$ be the lattice generated by $\mathbb{Z}^2$ and $(Q, 1)/r$, let $X \ni x$ be the germ at the torus fixed point of the affine toric surface of the cone spanned by $(1,0)$ and $(0,1)$ in N, and let C and C' be the curves of $(1,0)$ and $(0,1)$. For integers j, k with $j \equiv Qk \bmod r$ write $v_{(j,k)} := (j,k)/r \in N$, and set $\det_r((j,k),(j',k')) := (jk' - j'k)/r$. Then $X \ni x$ is smooth if $r = 1$, and a cyclic quotient singularity of order r if $r \geq 2$. If $r \geq 2$, then the minimal resolution of $X \ni x$ is toric, and the chain of the minimal resolution read from the strict transform of C is the Hirzebruch–Jung chain of r/Q. Moreover, the following hold.

(1) Let $0 \leq b, b' < 1$ be rational numbers. If $j, k \geq 0$ and $v_{(j,k)}$ is primitive in N, then the toric divisor of $v_{(j,k)}$ has log discrepancy $((1-b)j + (1-b')k)/r$ with respect to $(X, bC + b'C')$.

(2) Let $v_{w_1}, \ldots, v_{w_l}$ be primitive vectors of N in the interior of the cone, ordered from $(1,0)$ to $(0,1)$, let $Y \to X$ be the proper birational toric morphism given by the subdivision of the cone by the rays of $v_{w_1}, \ldots, v_{w_l}$, let $G_1, \ldots, G_l$ be the exceptional curves of $Y \to X$, G_i corresponding to v_{w_i}, and let $w_0 := (r, 0)$ and $w_{l+1} := (0, r)$. Let G_0 and G_{l+1} be the strict transforms of C and C'. Then $\det_r(w_{i-1}, w_i)$ is a positive integer for $1 \leq i \leq l+1$, and for $1 \leq i \leq l$,

$$G_i^2 = -\frac{\det_r(w_{i-1}, w_{i+1})}{\det_r(w_{i-1}, w_i)\det_r(w_i, w_{i+1})}, \quad G_i \cdot G_{i-1} = \frac{1}{\det_r(w_{i-1}, w_i)}, \quad G_i \cdot G_{i+1} = \frac{1}{\det_r(w_i, w_{i+1})}.$$

(3) Let μ_r act on $\mathbb{A}^2$ with coordinates u, v with weights $(Q, 1)$. Then $X \ni x$ is isomorphic to the germ of $\mathbb{A}^2/\mu_r$ at the image of the origin, so that C and C' are the images of $(u = 0)$ and $(v = 0)$.

Proof. If $r = 1$, then $Q = 0$, $N = \mathbb{Z}^2$, and X is the affine plane. Suppose that $r \geq 2$, so that $1 \leq Q \leq r-1$. Since N is generated by $\mathbb{Z}^2$ and $f_1 := v_{(Q,1)}$, and $(0,1) = rf_1 - Qf_2$ for $f_2 := (1,0)$, the vectors f_1, f_2 form a basis of N. In the basis f_1, f_2 the cone is $\mathrm{Cone}(f_2, rf_1 - Qf_2)$, which is the normal form $\mathrm{Cone}(e_2, de_1 - ke_2)$ of [CLS11, Proposition 10.1.1] with $e_1 = f_1$, $e_2 = f_2$, $d = r$, and $k = Q$. By [CLS11, Theorems 10.2.3, 10.2.5 and 10.4.4], the toric resolution of $X \ni x$ given by the normal form is a chain of exceptional curves which, read from the ray of C, is the Hirzebruch–Jung chain of r/Q. Since all entries of a Hirzebruch–Jung chain are at least 2, the toric resolution is the minimal resolution, and $X \ni x$ is a cyclic quotient singularity of type $\frac{1}{r}(1, Q)$ and of order r by Definition 2.9(1).

(1) Let $M := \mathrm{Hom}(N, \mathbb{Z})$ and $M_{\mathbb{Q}} := M \otimes \mathbb{Q}$. The vectors $(1,0)$ and $(0,1)$ are primitive in N, as $(1,0) = f_2$ and $(0,1) = rf_1 - Qf_2$ with $\gcd(r, Q) = 1$ (for $r = 1$, $N = \mathbb{Z}^2$). Let $u \in M_{\mathbb{Q}}$ be the linear form with $\langle u, (1,0)\rangle = 1 - b$ and $\langle u, (0,1)\rangle = 1 - b'$, and let m be a positive integer with $mu \in M$. Let $v := v_{(j,k)}$, let $h : Y' \to X$ be the toric morphism given by the subdivision of

the cone by the ray of v (h is the identity if v is $(1,0)$ or $(0,1)$), and let D_v, $C_{Y'}$, and $C'_{Y'}$ be the toric divisors of v, $(1,0)$, and $(0,1)$ on Y'. By [CLS11, Theorem 8.2.3], $K_X = -C - C'$ and $K_{Y'} = -C_{Y'} - C'_{Y'} - D_v$ (with D_v omitted if h is the identity). By [CLS11, Proposition 4.1.2],

$$m(K_X + bC + b'C') = -m(1-b)C - m(1-b')C' = -\operatorname{div}_X(\chi^{mu}),$$

and

$$h^* \operatorname{div}_X(\chi^{mu}) = \operatorname{div}_{Y'}(\chi^{mu}) = m\big((1-b)C_{Y'} + (1-b')C'_{Y'} + \langle u, v\rangle D_v\big).$$

Thus

$$K_{Y'} + bC_{Y'} + b'C'_{Y'} + (1 - \langle u, v\rangle)D_v = h^*(K_X + bC + b'C'),$$

so the log discrepancy of D_v with respect to $(X, bC + b'C')$ is $\langle u, v\rangle = ((1-b)j + (1-b')k)/r$ (if h is the identity, the claim is immediate from the definition of $a(D_v, X, bC + b'C')$).

(2) The lattice N contains $\mathbb{Z}^2$ with index r, so the parallelogram spanned by a basis of N has area $1/r$. For $w = (j,k)$ and $w' = (j',k')$ with $v_w, v_{w'} \in N$, the index of $\mathbb{Z}v_w + \mathbb{Z}v_{w'}$ in N is

$$r \cdot \left|\det\left(\frac{w}{r}, \frac{w'}{r}\right)\right| = \frac{|jk' - j'k|}{r} = |\det_r(w, w')|.$$

Since $v_{w_0} = (1,0)$, $v_{w_1}, \dots, v_{w_l}$, $v_{w_{l+1}} = (0,1)$ are ordered counterclockwise in the first quadrant, $d_i := \det_r(w_{i-1}, w_i) > 0$, so d_i is a positive integer for $1 \le i \le l+1$. The cone spanned by the primitive vectors $v_{w_{i-1}}$ and v_{w_i} defines the cyclic quotient $\mathbb{C}^2/\mu_{d_i}$, acting freely outside 0, and G_{i-1} and G_i are the images of the coordinate axes [CLS11, Propositions 10.1.1 and 10.1.2]. The pullbacks of G_{i-1} and G_i by the quotient morphism of degree d_i are the coordinate axes, which intersect transversally at 0. Since $G_{i-1} \cap G_i$ is the torus fixed point of $\operatorname{Cone}(v_{w_{i-1}}, v_{w_i})$, the projection formula implies that $G_{i-1} \cdot G_i = 1/d_i$.

For $1 \le i \le l$, let $u \in M_{\mathbb{Q}}$ be the linear form $\langle u, v\rangle := \det(v_{w_{i-1}}, v)$, so that $\langle u, v_w\rangle = \det_r(w_{i-1}, w)/r$ for any $w = (j,k)$ with $j \equiv Qk \bmod r$, and let m be a positive integer with $mu \in M$. By [CLS11, Proposition 4.1.2],

$$\operatorname{div}_Y(\chi^{mu}) = m\sum_\rho \langle u, v_\rho\rangle D_\rho,$$

where ρ runs over the rays of the fan of Y, v_ρ is the primitive vector of ρ, and D_ρ is the toric divisor of ρ. Since G_i is a complete curve, $\operatorname{div}_Y(\chi^{mu}) \cdot G_i = 0$. Among the divisors D_ρ, only G_{i-1}, G_i, and G_{i+1} intersect G_i, and $\langle u, v_{w_{i-1}}\rangle = 0$. Hence

$$\frac{\det_r(w_{i-1}, w_i)}{r} G_i^2 + \frac{\det_r(w_{i-1}, w_{i+1})}{r} \cdot \frac{1}{\det_r(w_i, w_{i+1})} = 0,$$

which is the formula for G_i^2.

(3) Since N is generated by $(1,0)$, $(0,1)$, and $(Q,1)/r$, we have $M = \{(a,b) \in \mathbb{Z}^2 \mid aQ + b \equiv 0 \bmod r\}$. Thus the map sending $\chi^{(a,b)}$ to $u^a v^b$ is an isomorphism from the coordinate ring of the affine toric surface, which is spanned by the characters $\chi^{(a,b)}$ with $(a,b) \in M$ and $a, b \ge 0$, onto the ring of μ_r-invariants of $\mathbb{C}[u,v]$. By [CLS11, Proposition 4.1.2], $\chi^{(a,b)}$ has order a along C and order b along C'. Since the quotient map $\mathbb{A}^2 \to \mathbb{A}^2/\mu_r$ is étale in codimension one, C and C' are the images of $(u = 0)$ and $(v = 0)$. □

Definition-Lemma 2.12 ([Kol13b, Theorem 3.36], [LL26, Proposition 2.14])**.** Let (X, S) be a projective lc surface pair such that S is a smooth curve and (X, S) is plt near S. Then any singular point y of X contained in S is a cyclic quotient singularity, and

$$(K_X + S) \cdot S = 2g(S) - 2 + \sum_y \left(1 - \frac{1}{r_y}\right),$$

where y runs over the singular points of X contained in S and r_y is the order of $X \ni y$. Moreover, for any such y, there exists a unique integer Q_y such that the germ $(X \ni y, S)$ is analytically isomorphic to the germ of Definition-Lemma 2.11 with $r = r_y$ and $Q = Q_y$, where S corresponds to C, and the Hirzebruch–Jung chain of r_y/Q_y is the chain of the minimal resolution of $X \ni y$

read from the strict transform of S. In addition, $S^2 - \sum_y Q_y/r_y$ is an integer. More generally, if S is a reduced curve on a normal surface X and $y \in S$ is a singular point of X at which (X, S) is plt, then S is smooth at y, $X \ni y$ is a cyclic quotient singularity of order r_y, there exists a unique integer Q_y such that $(X \ni y, S)$ is analytically isomorphic to the germ of Definition-Lemma 2.11 with $r = r_y$ and $Q = Q_y$, where S corresponds to C, and the Hirzebruch–Jung chain of r_y/Q_y is the chain of the minimal resolution of $X \ni y$ read from the strict transform of S.

Proof. **Step 1.** In this step, we find the local form of $(X \ni y, S)$ at a singular point y of X at which (X, S) is plt. Let $y \in S$ be a singular point of X at which (X, S) is plt. By [Kol+92, Proposition 16.6, (16.6.3)] (cf. [KM98, Theorem 4.15(3)] for the chain of exceptional curves), the germ $(X \ni y, S)$ is analytically isomorphic to $((x = 0) \subset \mathbb{C}^2/\mu_r)$, where μ_r acts with weights $(1, q)$, $r \geq 2$, and $\gcd(q, r) = 1$. Let $N := \mathbb{Z}^2 + \mathbb{Z}(1, q)/r$, and let $N' := \mathbb{Z}^2$, generated by the ray generators $(1, 0)$ and $(0, 1)$. By [CLS11, Example 1.3.20], the affine toric surface of the first quadrant in N is $\mathbb{C}^2/(N/N')$. By the formula (10.1.3) of [CLS11], the class of $(1, q)/r$, a generator of $N/N' \cong \mu_r$, multiplies $x := \chi^{(1,0)}$ by $\zeta := e^{2\pi i/r}$ and $\chi^{(0,1)}$ by ζ^q. The curve $(x = 0)$ is the toric divisor of $(1, 0)$, since $\operatorname{div}(\chi^{(1,0)})$ on $\mathbb{C}^2$ is the divisor of the ray of $(1, 0)$ [CLS11, Proposition 4.1.2]. Let $1 \leq Q \leq r - 1$ be the integer with $Qq \equiv 1 \bmod r$. Then $Q(1, q)/r - (Q, 1)/r = (0, (Qq - 1)/r) \in \mathbb{Z}^2$, and $\gcd(Q, r) = 1$, so

$$\mathbb{Z}^2 + \mathbb{Z}\frac{(1, q)}{r} = \mathbb{Z}^2 + \mathbb{Z}\frac{(Q, 1)}{r}.$$

Thus $(X \ni y, S)$ is analytically isomorphic to the germ of Definition-Lemma 2.11 with the integers r and Q above, where S corresponds to C. By Definition-Lemma 2.11, $X \ni y$ is a cyclic quotient singularity of order r, that is, $r = r_y$. The minimal resolution of a surface germ depends only on the analytic isomorphism class of the germ, so we compute it at y in the germ of Definition-Lemma 2.11.

Step 2. In this step, we prove the adjunction formula. By [Kol13b, Theorem 3.36],

$$(K_X + S) \cdot S = \deg K_S + \deg \operatorname{Diff}_S(0),$$

where the different $\operatorname{Diff}_S(0)$ is a $\mathbb{Q}$-divisor on S whose coefficient at a point $y \in S$ is 0 if X is smooth at y, and is $1 - 1/m$ if X is singular at y and (X, S) is plt at y, where μ_m is the group of the local form of **Step 1** [Kol+92, Proposition 16.6 and Corollary 16.7]. By **Step 1**, $m = r_y$ at the singular points of X on S. As $\deg K_S = 2g(S) - 2$, we obtain the adjunction formula.

Step 3. In this step, we construct the integer Q_y and prove that $S^2 - \sum_y Q_y/r_y$ is an integer. By **Step 1**, any singular point y of X contained in S is a cyclic quotient singularity, and $(X \ni y, S)$ is analytically isomorphic to the germ of Definition-Lemma 2.11 with $r = r_y$ and some $Q = Q_y$, where S corresponds to C. By Definition-Lemma 2.11, the chain of the minimal resolution of $X \ni y$ read from the strict transform $\widetilde{S}$ of S is the Hirzebruch–Jung chain of r_y/Q_y. Since a Hirzebruch–Jung chain determines r_y/Q_y, the integer Q_y is unique. **Step 1** and the first paragraph of **Step 3** use only the germ $(X \ni y, S)$, and a reduced curve is smooth at a point at which the pair is plt [KM98, Proposition 5.51], so they also prove the last sentence of the statement.

Thus Q_y is the integer $q_y(S)$ of [LL26, Definition 2.12], and the statement on $S^2 - \sum_y Q_y/r_y$ is [LL26, Proposition 2.14]. □

2.6. **Special divisorial valuations.**

Definition 2.13 ([LL26, Definitions 4.2 and 4.3])**.** Let $X \ni x$ be a klt surface germ that is not Du Val, and $\pi : \widetilde{X} \to X$ the minimal resolution of $X \ni x$. A *special divisorial valuation* over $X \ni x$ is a π-exceptional prime divisor E such that one of the following holds:

(1) $X \ni x$ is a cyclic quotient singularity and $E^2 \leq -3$,
(2) $X \ni x$ is of type D-I, E-I, or E-II, and E is the fork, or

(3) $X \ni x$ is of type D-II, and E is the curve with $E^2 \leq -3$ on the arm of (r_3, q_3) that is closest to the fork.

If X is a projective surface with klt singularities and E is a special divisorial valuation over $X \ni x$, the *extraction* of E is the projective birational morphism $Z \to X$ from a normal surface Z whose exceptional locus is the strict transform of E. It exists by Lemma 2.15(1), since $a(E, X, 0) < 1$ [KM98, Corollary 4.3 and Definition 4.4], and it is unique up to isomorphism over X by Lemma 2.15(2).

In the next theorem, (1) is [LL26, Proposition 4.6], (2) is [LL26, Proposition 4.4], and (3) is [LL26, Lemmas 4.8 and 4.9]. None of them assumes that $\rho(X) = 1$.

Theorem 2.14 ([LL26, Propositions 4.4 and 4.6, Lemmas 4.8 and 4.9])**.** *Let X be a klt stable surface and $x \in X$ a point such that $X \ni x$ is not Du Val. Let E be a special divisorial valuation over $X \ni x$, and let $f : Z \to X$ be the extraction of E, where E also denotes the exceptional curve of f. Let $b := 1 - a(E, X, 0)$, so that $K_Z + bE = f^*K_X$, let $q := -(E^2)_Z$, the self-intersection of E on Z, and let $n := \inf\{t \geq 0 \mid K_Z + tE \text{ is nef}\}$. Then the following hold.*

(1) There exists a special divisorial valuation E_0 over $X \ni x$ with $a(E_0, X, 0) = \mathrm{mld}(X \ni x)$.
(2) $q \geq 7/30$ if $X \ni x$ is of type E-II, and $q > 1$ otherwise.
(3) $b > n$ and $K_X^2 \geq q(b - n)^2$.

Lemma 2.15 (cf. [BCHM10, Corollary 1.4.3])**.** *Let X be a normal projective surface and $E_1, \dots, E_k$ prime divisors over X that are exceptional over X. Then the following hold.*

(1) If X is klt and $a(E_i, X, 0) < 1$ for any i, then there exists a projective birational morphism $Y \to X$ from a $\mathbb{Q}$-factorial klt projective surface Y whose exceptional divisors are exactly $E_1, \dots, E_k$.
(2) Let $Y \to X$ and $Y' \to X$ be projective birational morphisms from normal surfaces Y and Y' whose exceptional divisors are exactly $E_1, \dots, E_k$. Then there exists an isomorphism $Y \to Y'$ over X.

Proof. (1) By [BCHM10, Corollary 1.4.3], applied to the klt pair $(X, 0)$ and the divisors $E_1, \dots, E_k$, there exists a proper birational morphism $g : Y \to X$ from a $\mathbb{Q}$-factorial normal surface Y whose exceptional divisors are exactly $E_1, \dots, E_k$ (in [BCHM10], Y is a log terminal model over X). The matrix $((E_i \cdot E_j))_{i,j}$ is negative definite: for a non-zero $\mathbb{R}$-divisor $D = \sum_i x_i E_i$ and a resolution $h : V \to Y$, $D^2 = (h^*D)^2 < 0$ by [KM98, Lemma 3.40] applied to the resolution $g \circ h$ of X. Thus there exist rational numbers $c_1, \dots, c_k$ such that $F := \sum_i c_i E_i$ satisfies $F \cdot E_j = -1$ for $1 \leq j \leq k$. Since Y is $\mathbb{Q}$-factorial, $-mF$ is Cartier for some positive integer m, and $-mF \cdot E_j = m > 0$ for any j. As the curves contracted by g are $E_1, \dots, E_k$, $-mF$ is g-ample by [KM98, Theorem 1.42]. Hence g is projective, and Y is projective, since X is projective [Har77, Exercise II.4.9]. Since X is klt, $0 < a(E_i, X, 0) < 1$ for any i, and

$$K_Y + \sum_{i=1}^{k} (1 - a(E_i, X, 0))E_i = g^*K_X.$$

Thus $a(E, Y, \sum_i (1 - a(E_i, X, 0))E_i) = a(E, X, 0)$ for any prime divisor E over X [KM98, Lemma 2.30], so the pair $(Y, \sum_i (1 - a(E_i, X, 0))E_i)$ is klt, and Y is klt.

(2) Let V be a resolution of Y and Y' with morphisms $h : V \to Y$ and $h' : V \to Y'$ over X. A prime divisor on V is h-exceptional, or equivalently h'-exceptional, if and only if it is neither the strict transform of a prime divisor on X nor one of $E_1, \dots, E_k$. Thus a curve on V is contracted by h if and only if it is contracted by h'. For any $y \in Y$, $h^{-1}(y)$ is connected by Zariski's main theorem, and any curve in $h^{-1}(y)$ is contracted by h', so $h'(h^{-1}(y))$ is a point. Thus the image Γ of V in $Y \times_X Y'$ is proper and birational over Y with finite fibers, and $\Gamma \to Y$ is an isomorphism by Zariski's main theorem, since Y is normal. Hence $h' = \phi \circ h$ for a morphism $\phi : Y \to Y'$ over X. Exchanging the roles of h and h', $h = \psi \circ h'$ for a morphism $\psi : Y' \to Y$ over X. The

morphisms $\psi \circ \phi$ and $\phi \circ \psi$ are the identity on dense open subsets, so they are the identity, and ϕ is an isomorphism over X. □

2.7. **Gap theorems and Calabi–Yau pairs of Picard number one.** The following gap of log canonical thresholds is due to Kuwata [Kuw99] and Prokhorov [Pro02, Corollary 3.3] (cf. [LS23, Lemma 3.6]).

Theorem 2.16 ([LS23, Lemma 3.6]). *Let X be a klt surface and $D \geq 0$ a non-zero $\mathbb{Q}$-Cartier Weil divisor on X. Let*

$$\operatorname{lct}(X; D) := \sup\{t \geq 0 \mid (X, tD) \text{ is lc}\}.$$

Then $\operatorname{lct}(X; D) = 1$ or $\operatorname{lct}(X; D) \leq 5/6$.

The next theorem is [LL23, Theorem 2.6] with $A = 0$. The condition in [LL23, Definition 2.5] that (X, tB) is lc does not change the infimum defining $\operatorname{pet}(X; B)$, since (X, tB) is lc for $0 \leq t \leq 1$ [KM98, Corollary 2.35(1)] and $K_X + B$ is pseudo-effective.

Theorem 2.17 ([LL23, Theorem 2.6]). *Let (X, B) be a projective lc surface pair such that $B \neq 0$ is a reduced $\mathbb{Q}$-Cartier Weil divisor and $K_X + B$ is big and nef. Let*

$$\operatorname{pet}(X; B) := \inf\{t \geq 0 \mid K_X + tB \text{ is pseudo-effective}\}.$$

Then $\operatorname{pet}(X; B) = 12/13$ or $\operatorname{pet}(X; B) \leq 10/11$. If $\operatorname{pet}(X; B) = 12/13$, then any singular point of X contained in B is a cyclic quotient singularity of order at most 5.

The next theorem is [LS23, Lemma 3.8(1)–(3)] in the case of [LS23, Set-up 3.2] with $T = 0$ and one boundary coefficient b. The statement for $b > 5/6$ is [LS23, Lemma 3.7] applied to (Y, bS_Y), which satisfies [LS23, Set-up 3.2] with $T = 0$.

Theorem 2.18 ([LS23, Lemmas 3.7 and 3.8]). *Let (X, bS) be a projective klt surface pair with $K_X + bS \equiv 0$, where $2/3 < b < 1$ and $S \geq 0$ is a non-zero Weil divisor. Then there exists a birational morphism $f : X \to Y$ to a projective surface Y with $\rho(Y) = 1$ such that $S_Y := f_*S \neq 0$, (Y, bS_Y) is klt, $K_X + bS = f^*(K_Y + bS_Y)$, and $K_Y + bS_Y \sim_{\mathbb{R}} 0$. If moreover $b > 5/6$, then (Y, S_Y) is lc.*

Theorem 2.19 ([LL26, Theorem 3.1]). *Let (X, bS) be a projective klt surface pair, where $S \geq 0$ is a non-zero Weil divisor, $b \geq 6/7 + 1/938$, $K_X + bS \equiv 0$, and $\rho(X) = 1$. Then one of the following holds.*

1. *S is not a smooth rational curve, and (X, S) is isomorphic to one of the pairs in cases 4–8, 10–14, and 16 of* [Abe97, Table, pp. 9–14].
2. *$b = 12/13$, and*

$$(X, bS) \cong \left(\mathbb{P}(3, 4, 5), \frac{12}{13}(x^3y + y^2z + z^2x = 0)\right).$$

3. *S is a smooth rational curve, $b = 10/11$, (X, S) is plt, X has exactly three singular points, they are contained in S, and they are cyclic quotient singularities of types $\frac{1}{2}(1, 1)$, $\frac{1}{5}(1, 4)$, and $\frac{1}{7}(1, 3)$ with $Q_y = 1$, 4, and 3 in Definition-Lemma 2.12, respectively.*
4. *S is a smooth rational curve, $b = 15/17$, (X, S) is plt, X has exactly three singular points, they are contained in S, and they are cyclic quotient singularities of types $\frac{1}{3}(1, 2)$, $\frac{1}{5}(1, 4)$, and $\frac{1}{7}(1, 2)$ with $Q_y = 2$, 4, and 2 in Definition-Lemma 2.12, respectively.*

In (3) and (4), [LL26, Theorem 3.1] does not state that (X, S) is plt, although the integers $q_x(S)$ are defined only at points where (X, S) is plt [LL26, Definition 2.12]. [LL26, Theorem 3.1] states $q_x(S)$ only at the point of order 7. At the points of orders 2, 3, and 5, the chain of the minimal resolution is a palindrome, so Q_y is determined by the type. We deduce that (X, S) is plt from the other conclusions.

Proof that (X, S) is plt in Theorem 2.19(3) and (4). Since (X, bS) is klt, X is $\mathbb{Q}$-factorial [KM98, Proposition 4.11], so X is klt. We have $\operatorname{lct}(X; S) \geq b > 5/6$, so $\operatorname{lct}(X; S) = 1$ by Theorem 2.16, and (X, S) is lc, since the log discrepancies of (X, tS) are affine functions of t. Suppose that (X, S) is not plt at a point $y \in S$. Then X is singular at y, since S is smooth. By [KM98, Theorem 4.15(2)], the dual graph of the minimal resolution of $X \ni y$ together with the strict transform $\widetilde{S}$ of S is $\widetilde{S} - E_1 - \cdots - E_k$, with two (-2)-curves attached to E_k. If $k \geq 2$, then E_k has three neighbors among the exceptional curves, and $X \ni y$ is not a cyclic quotient singularity. If $k = 1$, then the exceptional curves form the chain $[2, c, 2]$ with $c := -E_1^2 \geq 2$, and

$$2 - 1/(c - 1/2) = \frac{4(c-1)}{2c-1}, \qquad \gcd(4(c-1), 2c-1) = 1,$$

so $X \ni y$ is a cyclic quotient singularity of order $4(c-1)$. In (3) and (4), the singular points of X are the three cyclic quotient singularities of orders 2, 5, 7, respectively 3, 5, 7, none of which is divisible by 4. This is a contradiction, so (X, S) is plt. □

3. The hypersurface X_{438}

In this section, we prove the properties of the surface X_{438} of Theorem 1.5 that the rest of the paper uses. Lemma 3.1 implies the statements of Theorem 1.5 on X_{438}, and Lemma 3.2 is used in Section 9.

We use the coordinates x, y, z, w of Theorem 1.5. The coordinates x_0, x_1, x_2, x_3 of $\mathbb{P}(219, 146, 61, 11)$ used in [Tot24] are w, z, y, x, respectively.

Lemma 3.1. *The surface X_{438} is a klt hypersurface with*

$$\mathcal{O}_{X_{438}}(K_{X_{438}}) \cong \mathcal{O}_{X_{438}}(1), \qquad K_{X_{438}}^2 = \frac{1}{48983}.$$

Proof. In the coordinates x, y, z, w, the equation in the paragraph after [Tot24, Theorem 2.1] in the case of dimension 2 is $w^2 + z^3 + xy^7 + x^{21}yz$. Hence X_{438} is the hypersurface of [Tot24, Theorem 2.1] in the case of dimension 2, so X_{438} is a klt hypersurface with

$$\mathcal{O}_{X_{438}}(K_{X_{438}}) \cong \mathcal{O}_{X_{438}}(1), \qquad K_{X_{438}}^2 = \frac{438}{11 \cdot 61 \cdot 146 \cdot 219} = \frac{1}{48983}.$$

□

Lemma 3.2. *We have* $\operatorname{mld}(X_{438}) \leq 5/61$.

Proof. Let $X := X_{438}$. The chart $(y = 1)$ of $\mathbb{P}(11, 61, 146, 219)$ is $\mathbb{A}^3_{x,z,w}/\mu_{61}$ with weights $(11, 24, 36)$, and X is given by $w^2 + z^3 + x + x^{21}z = 0$. Since the derivative in x is 1 at the origin, (w, z) are local coordinates on the cover, with weights $(36, 24)$. Since $39 \cdot 36 \equiv 1$ and $39 \cdot 24 \equiv 21 \pmod{61}$, X is near $[0 : 1 : 0 : 0]$ a cyclic quotient singularity of type $\frac{1}{61}(1, 21)$ (Definition-Lemma 2.11(3) with $r = 61$ and $Q = 21$), and $61/21 = [3, 11, 2]$. Let a_1, a_2, a_3 be the log discrepancies of the exceptional curves of the minimal resolution, in the order of the chain. Intersecting the pullback of K_X with the three curves, we obtain $3a_1 = 1 + a_2$, $11a_2 = a_1 + a_3$, and $2a_3 = 1 + a_2$, so $a_2 = 5/61$. □

4. Vanishing of low plurigenera

Let X be a klt stable surface with $K_X^2 \leq 1/48983$. The goal of this section is to prove Proposition 4.14: if $X \ncong X_{438}$, then $P_m(X) = 0$ for any $1 \leq m \leq 17$.

We go up to degree 17 because Section 6 derives a contradiction with the vanishing of $P_m(X)$, where m is the denominator of the pseudo-effective threshold p, and m is at most 17 except for the values 22, 23, and 29, which are treated by a separate argument.

4.1. **Log canonical models of pluricanonical sections.** In this subsection, we attach to a non-zero element of $R(X)_m$ a projective lc surface pair with non-zero reduced boundary and small volume.

Lemma 4.1. *Let $n \geq 1$ and $r_1, \dots, r_n \geq 2$ be integers, and $\delta := -2 + \sum_{i=1}^{n}(1 - 1/r_i)$. Assume that $\delta > 0$. Then the following hold, where $\{r_1, \dots, r_n\}$ denotes a multiset.*

(1) $\delta \geq 1/42$.
(2) If $\delta < 1/15$, then $\{r_1, \dots, r_n\}$ is $\{2,3,7\}$, $\{2,3,8\}$, $\{2,4,5\}$, or $\{2,3,9\}$, and $\delta = 1/42$, $1/24$, $1/20$, or $1/18$, respectively.

Proof. Since $1/2 \leq 1 - 1/r_i < 1$ for any i, we have

$$-2 + \frac{n}{2} \leq \delta < -2 + n.$$

Thus $n \geq 3$, and $\delta \geq 1/2$ if $n \geq 5$. If $n = 4$, then $\delta = 2 - \sum_{i=1}^{4} 1/r_i$. Hence $\delta = 0$ if $r_1 = \cdots = r_4 = 2$, and otherwise $\delta \geq 2 - 3/2 - 1/3 = 1/6$. In particular, (1) and (2) hold if $n \geq 4$.

Assume that $n = 3$ and $r_1 \leq r_2 \leq r_3$. Then $\delta = 1 - 1/r_1 - 1/r_2 - 1/r_3$.

Case 1. $r_1 \geq 3$. If $r_2 = 3$, then $\delta = 1/3 - 1/r_3$, so $r_3 \geq 4$ and $\delta \geq 1/12$. If $r_2 \geq 4$, then $\delta \geq 1 - 1/3 - 1/4 - 1/4 = 1/6$. Thus $\delta \geq 1/12$.

Case 2. $r_1 = 2$. Then $\delta = 1/2 - 1/r_2 - 1/r_3$, and $r_2 \geq 3$ since $\delta > 0$. If $r_2 = 3$, then $\delta = 1/6 - 1/r_3$, so $r_3 \geq 7$, and

$$\delta = \frac{1}{42}, \frac{1}{24}, \frac{1}{18}, \frac{1}{15}, \frac{5}{66} \quad \text{for } r_3 = 7,\ 8,\ 9,\ 10,\ 11, \text{ respectively,}$$

while $\delta \geq 1/6 - 1/12 = 1/12$ if $r_3 \geq 12$. If $r_2 = 4$, then $\delta = 1/4 - 1/r_3$, so $r_3 \geq 5$, and $\delta = 1/20$ for $r_3 = 5$, $\delta = 1/12$ for $r_3 = 6$, and $\delta \geq 1/4 - 1/7 = 3/28$ if $r_3 \geq 7$. If $r_2 \geq 5$, then $\delta \geq 1/2 - 1/5 - 1/5 = 1/10$.

Since $1/42 < 1/24 < 1/20 < 1/18 < 1/15 < 5/66 < 1/12 < 1/10$, the values listed above imply (1) and (2). □

A non-zero element of $R(X)_m$ only determines a Weil divisor $D \geq 0$ with $D \sim mK_X$. The log canonical model of the reduced divisor on a log resolution supported on the exceptional locus and the strict transform of $\operatorname{Supp} D$ is a pair with non-zero reduced boundary and volume at most $(m+1)^2 K_X^2$. Thus the bound $1/462$ (Theorem 1.2(2)) implies a lower bound for K_X^2, and if $(m+1)^2 K_X^2 < 1/42$, then [LS26, Theorem 1.3] implies that the boundary is a smooth curve.

Construction-Proposition 4.2. Let X be a klt stable surface, $m \geq 1$ an integer, $s \in R(X)_m$ non-zero, and $D \geq 0$ the Weil divisor of zeros of s, so that $D \sim mK_X$. Let $f : W \to X$ be a log resolution of $(X, \operatorname{Supp} D)$, let B be the reduced divisor on W with support $\operatorname{Exc}(f) \cup f_*^{-1} \operatorname{Supp} D$, and let $g : W \to X'$ be the morphism to the log canonical model [BCHM10, Definition 3.6.7] of (W, B), so that $K_{X'} + C$ is ample for $C := g_* B$. Then the following hold.

(1) $C \neq 0$, C is connected, (X', C) is lc, and $0 < (K_{X'} + C)^2 \leq (m+1)^2 K_X^2$.
(2) If $(m+1)^2 K_X^2 < 1/42$, then the following hold.
  (a) C is an ample smooth rational curve.
  (b) (X', C) is plt near C; in particular, X' is klt near C.
  (c) There exists a rational number $0 < p < 1$ with $K_{X'} + pC \sim_{\mathbb{Q}} 0$, and $p = 12/13$ or $p \leq 10/11$.
  (d) mp is an integer.
  (e) The prime divisor E on W with $g_* E = C$ is f-exceptional.
  (f) The coefficients b_E and c_E of E in $f^* K_X - K_W$ and in $f^* D$ satisfy $g_* f^* K_X = K_{X'} + b_E C$ and $c_E = m(b_E - p)$.
  (g) Regarded as a rational m-canonical form on X' via f and g, s satisfies $\operatorname{div}_{X'}(s) = -mpC$.

(3) In (2),
   (a) write $p = \alpha/\beta$ with coprime positive integers α, β, and
   (b) let t be the least positive integer such that $t(\beta K_{X'} + \alpha C)$ is Cartier near C.

   Then there exists a non-zero element of $R(X)_{\beta t}$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that g exists and that (X', C) is lc, and we prove the inequalities of (1).

Since X is klt, D is $\mathbb{Q}$-Cartier [KM98, Proposition 4.11], and $a(P, X, 0) > 0$ for any f-exceptional prime divisor P on W. We have

$$K_W + B = f^*K_X + F + f_*^{-1}\operatorname{Supp} D, \quad \text{where } F := \sum_P a(P, X, 0)P \geq 0,$$

the sum running over the f-exceptional prime divisors P on W, and F is a $\mathbb{Q}$-divisor since K_X is $\mathbb{Q}$-Cartier. Since $\operatorname{vol}(f^*K_X) = K_X^2 > 0$ and $F + f_*^{-1}\operatorname{Supp} D \geq 0$, the divisor $K_W + B$ is big (Lemma 2.5). Since $D \geq 0$ is $\mathbb{Q}$-Cartier, we have $f^*D \geq 0$. Since D has integer coefficients, $f_*^{-1}\operatorname{Supp} D \leq f_*^{-1}D \leq f^*D$, so

$$f^*(K_X + D) + F - (K_W + B) = f^*D - f_*^{-1}\operatorname{Supp} D \geq 0.$$

Since $K_X + D \sim_{\mathbb{Q}} (m+1)K_X$ is ample, Lemma 2.5(1) and (2) imply

$$\operatorname{vol}(K_W + B) \leq \operatorname{vol}(f^*(K_X + D) + F) = \operatorname{vol}(K_X + D) = (m+1)^2 K_X^2.$$

The pair (W, B) is lc and $K_W + B$ is pseudo-effective. By Theorem 2.4, there exists a sequence of extremal contractions

$$(W, B) = (W_0, B_0) \to (W_1, B_1) \to \cdots \to (W_l, B_l)$$

of the $(K_W + B)$-MMP such that (W_l, B_l) is lc and $K_{W_l} + B_l$ is semi-ample. The steps of the $(K_W + B)$-MMP are divisorial contractions since $K_W + B$ is pseudo-effective and surfaces have no small contractions, so the composite $g_1 : W \to W_l$ is a birational morphism. Each step $W_j \to W_{j+1}$ is the contraction of a $(K_{W_j} + B_j)$-negative extremal ray, so $a(P, W_j, B_j) \leq a(P, W_{j+1}, B_{j+1})$ for any prime divisor P over W_{j+1} [KM98, Lemma 3.38]. Hence

$$K_W + B = g_1^*(K_{W_l} + B_l) + G$$

for some g_1-exceptional $\mathbb{Q}$-divisor $G \geq 0$, and $\operatorname{vol}(K_{W_l} + B_l) = \operatorname{vol}(K_W + B) > 0$ by Lemma 2.5(2), since an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor is $\mathbb{Q}$-Cartier. Thus g_1 is a semi-ample model of $K_W + B$ in the sense of [BCHM10, Definition 3.6.5], so the ample model of $K_W + B$, that is, the log canonical model of (W, B), exists and is unique [BCHM10, Lemma 3.6.6(1)(3)]. It is $g = g_2 \circ g_1$, where $g_2 : W_l \to X'$ is the birational morphism defined by the semi-ample and big divisor $K_{W_l} + B_l$, and $K_{W_l} + B_l = g_2^*(K_{X'} + C)$ with $K_{X'} + C$ ample. Since B is reduced, C is reduced, and (X', C) is lc since the log discrepancies of (X', C) are those of (W_l, B_l) [KM98, Lemma 2.30]. Moreover,

$$K_W + B = g^*(K_{X'} + C) + G,$$

so Lemma 2.5(2), applied to the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $K_{X'} + C$, and $\operatorname{vol}(K_W + B) \leq (m+1)^2 K_X^2$ imply

$$0 < \operatorname{vol}(K_W + B) = (K_{X'} + C)^2 \leq (m+1)^2 K_X^2.$$

Step 2. In this step, we show that $C \neq 0$ and that C is connected. This completes the proof of (1).

Write $C = \sum_i C_i$, where the C_i are distinct prime divisors (the sum is empty if $C = 0$), and let E_i be the prime divisor on W with $g_*E_i = C_i$. Let b_i be the coefficient of E_i in $f^*K_X - K_W$, and c_i the coefficient of E_i in f^*D. Since $f^*K_X - K_W$ is f-exceptional, $b_i = 0$ if E_i is not f-exceptional, and $b_i = 1 - a(E_i, X, 0) < 1$ otherwise. Since $D \geq 0$, $c_i \geq 0$.

Since $D \sim mK_X$, we may write $D = mK_X + \mathrm{div}_X(h)$ for a rational function h. Then $f^*D = mf^*K_X + \mathrm{div}_W(h)$. The $\mathbb{Q}$-divisor f^*D is supported on $\operatorname{Supp} B$, and the components of B which are not contracted by g are the E_i. Pushing forward by g, we obtain

$$\sum_i c_i C_i = mK_{X'} + m\sum_i b_i C_i + \mathrm{div}_{X'}(h). \tag{4.1}$$

If $C = 0$, then (4.1) implies $mK_{X'} + \mathrm{div}_{X'}(h) = 0$, so $K_{X'} = K_{X'} + C$ is ample and $\mathbb{Q}$-linearly trivial, which contradicts $(K_{X'} + C)^2 > 0$. Thus $C \neq 0$. Let $\gamma_i := c_i/m + 1 - b_i$. Then $\gamma_i > 0$, and (4.1) implies

$$m(K_{X'} + C) + \mathrm{div}_{X'}(h) = m\sum_i \gamma_i C_i.$$

Hence $\sum_i \gamma_i C_i \sim_{\mathbb{Q}} K_{X'} + C$ is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor with support C. For some integer $q > 0$, $q\sum_i \gamma_i C_i \geq 0$ is an ample Cartier divisor with support C, so C is connected [Har77, Corollary III.7.9].

Step 3. In this step, we assume that $(m+1)^2 K_X^2 < 1/42$, and we show that C is a smooth prime divisor, that (X', C) is plt near C, and that X' is klt near C.

By **Step 1**, $(K_{X'} + C)^2 < 1/42$. Since (X', C) is a projective lc surface pair such that C is reduced and $K_{X'} + C$ is ample, no zero-dimensional lc center of (X', C) is contained in C [LS26, Theorem 1.3].

Suppose that (X', C) is not plt near C. Since (X', C) is lc, there exists an exceptional divisor P over X' whose center on X' is a point $y \in C$ and such that $a(P, X', C) = 0$ [KM98, Definition 2.34 and Corollary 2.31(3)]. Then y is a zero-dimensional lc center of (X', C) contained in C, a contradiction. Thus (X', C) is plt, hence dlt [KM98, Definitions 2.34 and 2.37], near C, so C is normal and the irreducible components of C are disjoint [KM98, Proposition 5.51]. Since C is connected (**Step 2**), C is a smooth prime divisor. Hence (X', C) is numerically dlt near C [KM98, Notation 4.1], so any Weil divisor on a neighborhood of C is $\mathbb{Q}$-Cartier [KM98, Proposition 4.11]. In particular, C is $\mathbb{Q}$-Cartier near C, so $(X', 0)$ is plt, hence klt, near C [KM98, Corollary 2.35(1)].

Step 4. In this step, we keep the assumption of **Step 3**, and we show that C is an ample smooth rational curve and that $K_{X'} + pC \sim_{\mathbb{Q}} 0$ for a rational number p with $0 < p < 1$ such that $p = 12/13$ or $p \leq 10/11$.

By **Step 3**, we may write $C = C_1$, and we let $E := E_1$, $b := b_1$, $c := c_1$, and $\gamma := \gamma_1$. By **Step 2**, $K_{X'} + C \sim_{\mathbb{Q}} \gamma C$ with $\gamma > 0$, so C is ample. In particular, C is $\mathbb{Q}$-Cartier. Let $\lambda := (K_{X'} + C) \cdot C > 0$. Then

$$(K_{X'} + C)^2 = \gamma\lambda.$$

For any real number $u \geq 0$, we have $K_{X'} + uC \equiv (u - 1 + \gamma)C$. If $u < 1 - \gamma$, then $K_{X'} + uC$ is not pseudo-effective, since C is ample and $(K_{X'} + uC) \cdot C = (u - 1 + \gamma)C^2 < 0$. If $u \geq 1 - \gamma$, then $K_{X'} + uC$ is pseudo-effective. Hence $\mathrm{pet}(X'; C) = \max\{0, 1 - \gamma\}$. Since $K_{X'} + C$ is ample, Theorem 2.17 implies $\mathrm{pet}(X'; C) \leq 12/13$, so $\gamma \geq 1/13$. Therefore

$$\lambda = \frac{(K_{X'} + C)^2}{\gamma} \leq 13(K_{X'} + C)^2 < \frac{13}{42} < \frac{1}{2}.$$

Let $r_1, \ldots, r_n$ be the orders of the singular points of X' contained in C. Since (X', C) is plt near C, Definition-Lemma 2.12 implies

$$\lambda = 2p_a(C) - 2 + \sum_{i=1}^{n} \left(1 - \frac{1}{r_i}\right).$$

If $p_a(C) \geq 2$, then $\lambda \geq 2$. If $p_a(C) = 1$, then $\lambda = 0$ or $\lambda \geq 1/2$. Both contradict $0 < \lambda < 1/2$, so $p_a(C) = 0$ and $C \cong \mathbb{P}^1$. Since $\lambda > 0$, we have $n \geq 1$, and Lemma 4.1(1) implies $\lambda \geq 1/42$. Thus

$$\gamma = \frac{(K_{X'} + C)^2}{\lambda} \leq 42(K_{X'} + C)^2 < 1.$$

Let $p := 1-\gamma$. Then p is a rational number with $0 < p < 1$, $K_{X'} + pC \sim_{\mathbb{Q}} 0$, and $\operatorname{pet}(X'; C) = p$. By Theorem 2.17, $p = 12/13$ or $p \le 10/11$.

Step 5. In this step, we prove the remaining parts of (2).

Since $c/m + 1 - b = \gamma = 1 - p$, we have $c = m(b-p)$. If E is not f-exceptional, then $b = 0$ and $c = -mp < 0$, which contradicts $c \ge 0$. Thus E is f-exceptional. Since $c - mb = -mp$, (4.1) implies

$$\operatorname{div}_{X'}(h) + mK_{X'} + mpC = 0. \tag{4.2}$$

The coefficients of $\operatorname{div}_{X'}(h)$ and $mK_{X'}$ are integers, so the coefficient mp of C in (4.2) is an integer. The $\mathbb{Q}$-divisor $f^*K_X - K_W$ is supported on $\operatorname{Supp} B$, and E is the only component of B which is not contracted by g, so $g_*f^*K_X = K_{X'} + bC$, where $b = b_E$. Since $\operatorname{div}_X(s) = D = mK_X + \operatorname{div}_X(h)$, (4.2) implies $\operatorname{div}_{X'}(s) = -mpC$. This proves (2).

Step 6. In this step, we show that βt divides m and that $t(\beta K_{X'} + \alpha C)$ is torsion in the class group of X'.

Since $mp = m\alpha/\beta$ is an integer and α, β are coprime, β divides m, and

$$\frac{m}{\beta}(\beta K_{X'} + \alpha C) = mK_{X'} + mpC = -\operatorname{div}_{X'}(h).$$

By **Step 3**, $\beta K_{X'} + \alpha C$ is $\mathbb{Q}$-Cartier near C, so t exists. The integers j such that $j(\beta K_{X'} + \alpha C)$ is Cartier near C form a subgroup of $\mathbb{Z}$, which is generated by t and contains m/β. Hence t divides m/β, that is, βt divides m. Let

$$\zeta := t(\beta K_{X'} + \alpha C) \quad \text{and} \quad k := \frac{m}{\beta t}.$$

Then ζ is Cartier near C, and $k\zeta = -\operatorname{div}_{X'}(h) \sim 0$ by (4.2). Let N be the exact order of ζ in the class group of X', and let ϕ be a rational function with $\operatorname{div}_{X'}(\phi) = N\zeta$.

Step 7. In this step, we show that $N = 1$, that is, $t(\beta K_{X'} + \alpha C) \sim 0$.

Let

$$\tau : X'' := \operatorname{Spec}_{X'} \bigoplus_{i=0}^{N-1} \mathcal{O}_{X'}(-i\zeta) \to X'$$

be the cyclic cover defined by ϕ [KM98, Definition 2.52], where the multiplication

$$\mathcal{O}_{X'}(-i\zeta) \otimes \mathcal{O}_{X'}(-j\zeta) \to \mathcal{O}_{X'}(-(i+j-N)\zeta) \quad \text{for } i+j \ge N$$

is the product with ϕ^{-1}. Then τ is finite of degree N. Let V be a neighborhood of C on which ζ is Cartier. Over V, the sheaf $\mathcal{O}_{X'}(-\zeta)$ is invertible and $\mathcal{O}_{X'}(-N\zeta) \cong \mathcal{O}_{X'}$ since $N\zeta = \operatorname{div}_{X'}(\phi)$, so $\tau^{-1}V \to V$ is the unramified cyclic cover of [KM98, Definition 2.49], which is finite étale of degree N.

Let X_1 be the normalization of an irreducible component of X'', and let $\tau_1 : X_1 \to X'$ be the induced morphism. Since the $\mathcal{O}_{X'}$-module $\tau_*\mathcal{O}_{X''}$ is torsion-free, the component dominates X', so τ_1 is finite and surjective, and X_1 is a normal projective surface. Since τ is étale over V and $\tau^{-1}V$ is normal, hence the disjoint union of the irreducible components of $\tau^{-1}V$, τ_1 is étale over V. Since $C \cong \mathbb{P}^1$, the Hurwitz formula [Har77, Corollary IV.2.4] implies that any connected component of $\tau_1^{-1}C$ maps isomorphically onto C. Since C is ample and $\mathbb{Q}$-Cartier, $\tau_1^*C \ge 0$ is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X_1 [Har77, Exercise III.5.7(d)] with support $\tau_1^{-1}C$, so $\tau_1^{-1}C$ is connected by [Har77, Corollary III.7.9] applied to a Cartier multiple of τ_1^*C. Hence $\tau_1^{-1}C \to C$ is an isomorphism, so the finite morphism τ_1, which is étale over V, has degree one. Since X' is normal, τ_1 is an isomorphism.

Thus τ has a section, that is, a homomorphism of $\mathcal{O}_{X'}$-algebras from $\bigoplus_{i=0}^{N-1} \mathcal{O}_{X'}(-i\zeta)$ to $\mathcal{O}_{X'}$. On $\mathcal{O}_{X'}(-\zeta)$ it is the multiplication by a rational function ψ, and since τ is defined by ϕ, we have $\psi^N = \phi^{-1}$. Thus $\operatorname{div}_{X'}(\psi) = -\zeta$, so $\zeta \sim 0$. Since N is the exact order of ζ, $N = 1$.

Step 8. In this step, we construct a non-zero element of $R(X)_{\beta t}$ and conclude the proof of (3).

By **Step 7**, there exists a rational function h_0 with $\mathrm{div}_{X'}(h_0) + t(\beta K_{X'} + \alpha C) = 0$. By (4.2),

$$\mathrm{div}_{X'}(h) = -(mK_{X'} + mpC) = -k\zeta = \mathrm{div}_{X'}(h_0^k).$$

Since X' is normal and projective, h/h_0^k is a non-zero constant. Hence $\mathrm{div}_X(h_0^k) = \mathrm{div}_X(h)$, and

$$\beta t K_X + \mathrm{div}_X(h_0) = \frac{1}{k}\big(mK_X + \mathrm{div}_X(h)\big) = \frac{1}{k}D \geq 0.$$

Thus $D/k \geq 0$ is a divisor linearly equivalent to $\beta t K_X$, so it is the divisor of zeros of a non-zero element of $R(X)_{\beta t}$. This proves (3). □

Corollary 4.3. *Let X be a klt stable surface and $m \geq 1$ an integer with $P_m(X) > 0$. Then $K_X^2 \geq 1/(462(m+1)^2)$.*

Proof. Pick a non-zero $s \in R(X)_m$, and let (X', C) be the pair of Construction-Proposition 4.2 attached to s. By Construction-Proposition 4.2(1), (X', C) is a projective lc surface pair, $C \neq 0$ is reduced, $K_{X'} + C$ is ample, and $(K_{X'} + C)^2 \leq (m+1)^2 K_X^2$. By Theorem 1.2(2), $(K_{X'} + C)^2 \geq 1/462$. Hence

$$K_X^2 \geq \frac{1}{462(m+1)^2}.$$

□

4.2. **Degrees at most twelve.** In this subsection, we show that for a klt stable surface X with $K_X^2 \leq 1/48983$ and an integer $1 \leq m \leq 12$, a non-zero element of $R(X)_m$ exists only for $m = 11$, and that its log canonical model is the pair of volume $1/462$ of [LL23, Example 3.2].

Proposition 4.4. *Let X be a klt stable surface with $K_X^2 \leq 1/48983$, let $1 \leq m \leq 12$ be an integer, and let $s \in R(X)_m$ be non-zero. Let $g : W \to X'$ and C be given by Construction-Proposition 4.2 for s. Then the following hold.*

(1) $m = 11$.
(2) (X', C) is isomorphic to the pair of [LL23, Example 3.2]*, and $K_{X'} + \frac{10}{11}C \sim_{\mathbb{Q}} 0$.*
(3) The singular points of X' contained in C are cyclic quotient singularities of orders 2, 3, and 7.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that C is a smooth rational curve containing exactly three singular points of X', of orders 2, 3, and 7, and that $(K_{X'} + C) \cdot C = 1/42$.

Since $m \leq 12$ and $169 \cdot 42 = 7098 < 48983$, we have

$$(m+1)^2 K_X^2 \leq \frac{169}{48983} < \frac{1}{42}.$$

By Construction-Proposition 4.2(2), C is an ample smooth rational curve, (X', C) is plt near C, and there exists a rational number $0 < p < 1$ such that $K_{X'} + pC \sim_{\mathbb{Q}} 0$, $mp \in \mathbb{Z}$, and $p = 12/13$ or $p \leq 10/11$. If $p = 12/13$, then $12m/13 \in \mathbb{Z}$, so 13 divides m, which is impossible as $1 \leq m \leq 12$. Thus $p \leq 10/11$, that is,

$$1 - p \geq \frac{1}{11}.$$

Since $K_{X'} + pC \sim_{\mathbb{Q}} 0$, we have $K_{X'} + C \sim_{\mathbb{Q}} (1-p)C$, hence

$$(K_{X'} + C)^2 = (1-p)^2 C^2 \quad \text{and} \quad (K_{X'} + C) \cdot C = (1-p)C^2.$$

In particular, $(K_{X'} + C) \cdot C = (K_{X'} + C)^2/(1-p)$. By Construction-Proposition 4.2(1), $(K_{X'} + C)^2 \leq (m+1)^2 K_X^2 \leq 169/48983$. Since $K_{X'} + C$ is ample and $24 \cdot 1859 = 44616 < 48983$, we obtain

$$0 < (K_{X'} + C) \cdot C \leq 11 \cdot \frac{169}{48983} = \frac{1859}{48983} < \frac{1}{24}.$$

Since (X', C) is plt near C and C is a smooth rational curve, Definition-Lemma 2.12 implies

$$(K_{X'} + C) \cdot C = -2 + \sum_y \Big(1 - \frac{1}{r_y}\Big),$$

where y runs over the singular points of X' contained in C, and $r_y \geq 2$ is the order of the cyclic quotient singularity $X' \ni y$. The right side is positive, so X' has at least one singular point on C, and it is less than $1/24 < 1/15$. By Lemma 4.1(2), the multiset $\{r_y\}$ is $\{2,3,7\}$, $\{2,3,8\}$, $\{2,4,5\}$, or $\{2,3,9\}$, and $(K_{X'} + C) \cdot C = 1/42$, $1/24$, $1/20$, or $1/18$, respectively. Only the first value is less than $1/24$. Thus C contains exactly three singular points y_2, y_3, and y_7 of X', of orders 2, 3, and 7, and

$$(K_{X'} + C) \cdot C = \frac{1}{42}.$$

Step 2. In this step, we show that $42C^2 = 11$, $p = 10/11$, and $m = 11$.

For $i \in \{2,3,7\}$, let Q_{y_i} be the integer of Definition-Lemma 2.12, so that the germ $(X' \ni y_i, C)$ is analytically isomorphic to the germ of Definition-Lemma 2.11 with $r = i$ and $Q = Q_{y_i}$, and

$$C^2 - \frac{Q_{y_2}}{2} - \frac{Q_{y_3}}{3} - \frac{Q_{y_7}}{7} \in \mathbb{Z}.$$

Definition-Lemma 2.11 takes $0 \leq Q \leq r-1$ and $\gcd(Q, r) = 1$. Since $i \geq 2$, we have $1 \leq Q_{y_i} \leq i-1$ and $\gcd(Q_{y_i}, i) = 1$. Hence $Q_{y_2} = 1$, $Q_{y_3} \in \{1,2\}$, and $Q_{y_7} \in \{1, \dots, 6\}$. Let $n := 42C^2$. Then

$$n = 42\Big(C^2 - \frac{1}{2} - \frac{Q_{y_3}}{3} - \frac{Q_{y_7}}{7}\Big) + 21 + 14Q_{y_3} + 6Q_{y_7}$$

is an integer, and $n \equiv 21 + 14Q_{y_3} + 6Q_{y_7} \pmod{42}$. Thus n is odd, $n \equiv 2Q_{y_3} \not\equiv 0 \pmod 3$, and $n \equiv 6Q_{y_7} \not\equiv 0 \pmod 7$. Therefore $\gcd(n, 42) = 1$.

By **Step 1**,

$$\frac{1}{42} = (K_{X'} + C) \cdot C = (1-p)C^2 = \frac{(1-p)n}{42},$$

so $n = 1/(1-p)$. Since $1 - p \geq 1/11$, we have $n \leq 11$. Moreover,

$$(K_{X'} + C)^2 = (1-p)^2 C^2 = \frac{1-p}{42} = \frac{1}{42n} \leq \frac{169}{48983},$$

so $n \geq 48983/7098 > 6$, as $6 \cdot 7098 = 42588 < 48983$. Among the integers $7 \leq n \leq 11$, only $n = 11$ is coprime to 42. Thus $n = 11$, and

$$p = \frac{10}{11}, \quad C^2 = \frac{11}{42}, \quad (K_{X'} + C)^2 = \frac{1}{462}.$$

Since $mp = 10m/11 \in \mathbb{Z}$ and $\gcd(10, 11) = 1$, 11 divides m. As $1 \leq m \leq 12$, we have $m = 11$. This proves (1).

Step 3. In this step, we prove (2) and (3).

By Construction-Proposition 4.2(1) and **Steps 1 and 2**, (X', C) is a projective lc surface pair, $C \neq 0$ is a reduced divisor, $K_{X'} + C$ is ample, and $(K_{X'} + C)^2 = 1/462$. By Theorem 1.3(2), (X', C) is isomorphic to the pair of [LL23, Example 3.2], which is the pair constructed in [AL19c, Section 5]. Since $K_{X'} + pC \sim_{\mathbb{Q}} 0$ with $p = 10/11$, together with **Steps 1 and 2**, this proves (2) and (3). □

4.3. **Toric valuations.** In this subsection, we collect the toric statements used for sections of degrees 11 and 13, and again in Section 8. Over a cyclic quotient point of a plt pair whose boundary curve is one axis, any divisor of log discrepancy less than one is toric (Lemma 4.5). The curves of X contracted on the log canonical model of Construction-Proposition 4.2 have log discrepancy less than one, so they correspond to lattice vectors.

Lemma 4.5. *Notation as in Definition-Lemma 2.11, and let $0 \leq b, b' < 1$ be rational numbers. Then any prime divisor G over X with $\operatorname{center}_X G = x$ and $a(G, X, bC + b'C') < 1$ is the toric divisor of a primitive vector $v_{(j,k)}$ with $j, k \geq 1$.*

Proof. Let $\Delta := bC + b'C'$. An element $(c_1, c_2) + c_3(Q, 1)/r$ of N, with $c_1, c_2, c_3 \in \mathbb{Z}$, equals $v_{(j,k)}$ for $j = c_1 r + c_3 Q$ and $k = c_2 r + c_3$, and $j \equiv Qk \bmod r$. Thus any vector of N in the interior of the cone is $v_{(j,k)}$ for integers $j, k \geq 1$. By Definition-Lemma 2.11(1), the toric divisor of any primitive vector of N in the cone has positive log discrepancy with respect to (X, Δ): it is $((1-b)j + (1-b')k)/r > 0$ for an interior vector $v_{(j,k)}$, and $1-b$ or $1-b'$ for the strict transforms of C and C'.

Let $X_0 \to X$ be the toric resolution given by a subdivision of the cone into smooth cones [CLS11, Theorem 10.2.3]. By [KM98, Lemma 2.45], there exists a sequence of blow-ups

$$X_n \to X_{n-1} \to \cdots \to X_0$$

such that $X_{i+1} \to X_i$ is the blow-up of the center y_i of G on X_i, which is a closed point, for $0 \leq i \leq n-1$, and the center of G on X_n is a prime divisor. For $1 \leq i \leq n$, let $E_i \subset X_i$ be the exceptional curve of $X_i \to X_{i-1}$. Then $G = E_n$ if $n \geq 1$, and $y_i \in E_i$ for $1 \leq i \leq n-1$, because the center of G on X_i maps to y_{i-1}. Let Σ_i be the reduced divisor on X_i whose components are the strict transforms of the torus-invariant prime divisors of X_0 and $E_1, \ldots, E_i$. Then Σ_i is an snc divisor, as the reduced total transform of the snc divisor Σ_0 under blow-ups of points, and the boundary Δ_i defined by $K_{X_i} + \Delta_i = (X_i \to X)^*(K_X + \Delta)$ is $\Delta_i = \sum_F (1 - a(F, X, \Delta))F$, where F runs over the components of Σ_i. By [KM98, Lemma 2.29] applied to (X_i, Δ_i) and by [KM98, Lemma 2.30], where all components below are components of Σ_i,

$$a(E_{i+1}, X, \Delta) = \begin{cases} 2 & \text{if } y_i \text{ is contained in no component,} \\ 1 + a(F, X, \Delta) & \text{if } F \text{ is the only component containing } y_i, \\ a(F_1, X, \Delta) + a(F_2, X, \Delta) & \text{if } y_i \in F_1 \cap F_2 \text{ for components } F_1 \neq F_2. \end{cases} \tag{4.3}$$

Suppose first that y_i is a torus-fixed point of X_i for $0 \leq i \leq n-1$. By [CLS11, Proposition 3.3.15], the blow-up of the torus-fixed point corresponding to a smooth two-dimensional cone τ spanned by u and u' is the toric morphism given by the subdivision of τ by the ray of $u + u'$. Hence X_i is a smooth toric surface over X for $0 \leq i \leq n$, and G is a torus-invariant prime divisor of X_n whose center on X is x. Thus G is the toric divisor of a primitive vector of N in the interior of the cone, that is, of $v_{(j,k)}$ with $j, k \geq 1$.

Suppose now that y_{i_1} is not a torus-fixed point of X_{i_1}, where $0 \leq i_1 \leq n-1$ is the least such index. Then X_{i_1} is a smooth toric surface over X, and the components of Σ_{i_1} are the torus-invariant prime divisors of X_{i_1}. Since y_{i_1} is not torus-fixed, y_{i_1} is contained in at most one component F of Σ_{i_1}, and $a(F, X, \Delta) > 0$. By (4.3) for $i = i_1$, $a(E_{i_1+1}, X, \Delta) > 1$. We show by induction that $a(E_i, X, \Delta) > 1$ for $i_1 + 1 \leq i \leq n$, and that any component of Σ_i has positive log discrepancy. If both statements hold for some $i \leq n-1$, then $y_i \in E_i$, and y_i is contained in at most one other component F of Σ_i, because Σ_i is snc. Hence

$$a(E_{i+1}, X, \Delta) \in \{1 + a(E_i, X, \Delta), a(E_i, X, \Delta) + a(F, X, \Delta)\},$$

and both numbers are greater than 1. Thus $a(G, X, \Delta) = a(E_n, X, \Delta) > 1$, a contradiction. □

The germs of Definition-Lemma 2.12 are only analytically isomorphic to the germ of Definition-Lemma 2.11. The next result extends Lemma 4.5 to such germs.

Definition-Lemma 4.6. Notation as in Definition-Lemma 2.11. Let X' be a normal surface, and let $x' \in X'$ be a closed point such that X' is klt near x'. Let ϕ be an isomorphism from the analytic germ $X' \ni x'$ onto the analytic germ $X \ni x$. Let $0 \leq b, b' < 1$ be rational numbers, $B := bC + b'C'$, and let B' be a $\mathbb{Q}$-divisor on X' whose germ at x' is mapped to B by ϕ. Then the following hold.

(1) The isomorphism ϕ induces a bijection $G' \to \phi_* G'$ from the prime divisors over X' with center x' onto the prime divisors over X with center x, and $a(G', X', B') = a(\phi_* G', X, B)$ for any such G'. We say that G' is the *toric divisor* of a vector of N if $\phi_* G'$ is.
(2) If G' is a prime divisor over X' with center x' and $a(G', X', B') < 1$, then G' is the toric divisor of a primitive vector $v_{(j,k)}$ with $j, k \geq 1$.

(3) Let $g' : Y' \to X'$ be a proper birational morphism from a normal surface Y' which is an isomorphism over $X' \backslash \{x'\}$, and assume that any g'-exceptional prime divisor is the toric divisor of a primitive vector of N in the interior of the cone. Let $g : Y \to X$ be the toric morphism of Definition-Lemma 2.11(2) given by the vectors of the g'-exceptional prime divisors. Then ϕ lifts to an isomorphism from an analytic neighborhood of $g'^{-1}(x')$ onto an analytic neighborhood of $g^{-1}(x)$ which maps any g'-exceptional prime divisor G' onto $\phi_* G'$.

Proof. (1) The minimal resolution of a normal surface is unique [BHPV04, Theorem III.6.2], so ϕ lifts to an isomorphism between analytic neighborhoods of the exceptional curves of the minimal resolutions of $X' \ni x'$ and $X \ni x$. Such a lift extends to the blow-ups of a closed point and of the image of the closed point. Any prime divisor over X' with center x' and any prime divisor over X with center x appears after finitely many blow-ups of closed points over the minimal resolution [KM98, Lemma 2.45]. Thus the last exceptional curve of a sequence of blow-ups over X' corresponds to the last exceptional curve of the corresponding sequence of blow-ups over X. This is the bijection of (1). Log discrepancies are computed from intersection numbers on a resolution [KM98, Notation 4.1], and the lift of ϕ preserves intersection numbers, so $a(G', X', B') = a(\phi_* G', X, B)$.

(2) This follows from (1) and Lemma 4.5.

(3) Let $\widehat{X}' \to X'$ be a composition of the minimal resolution of $X' \ni x'$ with blow-ups of closed points over x', and let $\widehat{X} \to X$ be the corresponding composition over X. By [BHPV04, Corollary III.4.4], we may choose them such that $\widehat{X}' \to X'$ factors through Y' and $\widehat{X} \to X$ factors through Y (a resolution of Y' factors through the minimal resolution of $X' \ni x'$, and the factorization is a composition of blow-ups, likewise for Y, so we first choose them for Y', transport the blow-ups to X, then blow up further for Y, and transport back). By the definition of g and by (1) (the lift of ϕ does not depend on the sequence of blow-ups, since two lifts coincide on a common refinement), the lift of ϕ maps the curves contracted by $\widehat{X}' \to Y'$ onto the curves contracted by $\widehat{X} \to Y$. Since the fibers of $\widehat{X}' \to Y'$ and $\widehat{X} \to Y$ are connected, the lift of ϕ descends to a homeomorphism $\bar{\phi}$ from an analytic neighborhood of $g'^{-1}(x')$ onto an analytic neighborhood of $g^{-1}(x)$, and $\bar{\phi}$ is biholomorphic outside finitely many points. Since Y' and Y are normal, $\bar{\phi}$ is biholomorphic by the Riemann extension theorem [GR84, Chapter 7, Section 4.2]. □

Let f and g be the morphisms of Construction-Proposition 4.2. A rational pluricanonical form σ on X' without poles outside C belongs to $R(X)$ exactly when the orders of σ at the g-exceptional curves that are not f-exceptional are non-negative. Lemma 4.7 shows that each such curve is a component of the divisor of the given section and a toric divisor over a singular point of X' contained in C, with explicit coordinates, so the test is a lattice computation.

Lemma 4.7. *Notation as in Construction-Proposition 4.2, and assume that* $(m+1)^2 K_X^2 < 1/42$. *Let* b_E *be the coefficient of* E *in* $f^* K_X - K_W$. *Then the following hold.*

(1) Let G *be a* g*-exceptional prime divisor on* W *that is not* f*-exceptional. Then* G *is a component of* $f_*^{-1} D$, $0 < a(G, X', b_E C) < 1$, *and*

$$m(a(G, X', pC) - 1) = \operatorname{mult}_G f_*^{-1} D.$$

In particular, $m(a(G, X', pC) - 1)$ *is a positive integer.*

(2) Let G *be as in (1). Then the center of* G *on* X' *is a singular point* x' *of* X' *contained in* C. *If* r *is the order of* $X' \ni x'$ *and* $(X' \ni x', C)$ *is identified with the germ of Definition-Lemma 2.11 by Definition-Lemma 2.12, then* G *is the toric divisor of a primitive vector* $v_{(j,k)}$ *with* $j, k \geq 1$ *and* $(1 - b_E)j + k < r$.

(3) Let $m' \geq 1$ *be an integer. Identify rational* m'*-canonical forms on* X, W, *and* X' *via the birational morphisms* f *and* g, *and regard* $R(X)_{m'}$ *as a space of rational* m'*-canonical*

forms on X'. Let σ be a non-zero rational m'-canonical form such that the Weil divisor $\operatorname{div}_{X'}(\sigma) \sim m'K_{X'}$ is ≥ 0 outside C. Then $\sigma \in R(X)_{m'}$ if and only if

$$\operatorname{mult}_F g^* \operatorname{div}_{X'}(\sigma) + m'(a(F, X', 0) - 1) \geq 0$$

for any g-exceptional prime divisor F on W that is not f-exceptional.

Proof. The proof is long, so we divide it into several steps. By Construction-Proposition 4.2(2), C is an ample smooth rational curve, (X', C) is plt near C, $K_{X'} + pC \sim_{\mathbb{Q}} 0$, and E is f-exceptional. Since C is ample, C is $\mathbb{Q}$-Cartier, and hence so are $K_{X'}$ and $K_{X'} + tC$ for any $t \in \mathbb{Q}$. In **Steps 3–5**, G is a g-exceptional prime divisor on W that is not f-exceptional.

Step 1. In this step, we describe how the divisor of a rational pluricanonical form changes under a birational morphism.

Let V be a normal projective surface such that K_V is $\mathbb{Q}$-Cartier, let $h : W \to V$ be a birational morphism, let $m' \geq 1$ be an integer, and let σ be a non-zero rational m'-canonical form. We claim that

$$\begin{aligned} \operatorname{div}_W(\sigma) &= h^* \operatorname{div}_V(\sigma) + m'(K_W - h^*K_V), \\ K_W - h^*K_V &= \sum_F (a(F, V, 0) - 1)F, \end{aligned} \tag{4.4}$$

where F runs over the h-exceptional prime divisors. To prove (4.4), let ω be a non-zero rational differential 2-form, and let $K_V := \operatorname{div}_V(\omega)$ and $K_W := \operatorname{div}_W(\omega)$, so that $h_*K_W = K_V$. Then $\sigma = \phi\omega^{\otimes m'}$ for a non-zero rational function ϕ, and

$$\operatorname{div}_W(\sigma) = \operatorname{div}_W(\phi) + m'K_W = h^* \operatorname{div}_V(\phi) + m'K_W = h^* \operatorname{div}_V(\sigma) + m'(K_W - h^*K_V).$$

The second identity of (4.4) is the definition of log discrepancies. We use (4.4) for $h = f$, $V = X$, and for $h = g$, $V = X'$.

Step 2. In this step, we compare f^*K_X and $g^*(K_{X'} + C)$ on W.

Let

$$\Theta := K_W + B - f^*K_X \quad \text{and} \quad \Gamma := K_W + B - g^*(K_{X'} + C).$$

Since $\operatorname{Exc}(f) \subset \operatorname{Supp} B$, (4.4) implies that the coefficient of Θ at an f-exceptional prime divisor F is $a(F, X, 0) > 0$, because X is klt, and the coefficient of Θ at a component of $f_*^{-1}D$ is 1. Thus $\Theta \geq 0$ and $\operatorname{Supp} \Theta = \operatorname{Supp} B$.

By Theorem 2.4, g is the composition of the steps $W = W_0 \to W_1 \to \cdots \to W_n$ of a $(K_W + B)$-MMP and of the morphism $W_n \to X'$ defined by the semi-ample divisor $K_{W_n} + B_n$, where B_n is the pushforward of B, and $K_{W_n} + B_n$ is the pullback of $K_{X'} + C$. For a prime divisor F on W, the coefficient of Γ at F is $a(F, X', C) - a(F, W, B)$. It is 0 if F is not g-exceptional, since $g_*B = C$. Each step of the MMP does not decrease log discrepancies [KM98, Lemma 3.38], and $W_n \to X'$ preserves them [KM98, Lemma 2.30]. Thus $\Gamma \geq 0$ and Γ is g-exceptional. In particular, the intersection matrix of the components of $\operatorname{Supp} \Gamma$ is negative definite [KM98, Lemma 3.40].

We show that $\Gamma \leq \Theta$. Write $\Theta - \Gamma = \Lambda_1 - \Lambda_2$, where $\Lambda_1, \Lambda_2 \geq 0$ have no common component, and suppose that $\Lambda_2 \neq 0$. Then $\operatorname{Supp} \Lambda_2 \subset \operatorname{Supp} \Gamma$, so $\Lambda_2^2 < 0$ and $g^*(K_{X'} + C) \cdot \Lambda_2 = 0$. Since K_X is ample, $f^*K_X \cdot \Lambda_2 \geq 0$, and $\Lambda_1 \cdot \Lambda_2 \geq 0$. Since $g^*(K_{X'} + C) = f^*K_X + \Lambda_1 - \Lambda_2$,

$$0 = g^*(K_{X'} + C) \cdot \Lambda_2 = f^*K_X \cdot \Lambda_2 + \Lambda_1 \cdot \Lambda_2 - \Lambda_2^2 > 0,$$

a contradiction. Thus $\Gamma \leq \Theta$.

Step 3. In this step, we show that G is a component of $f_*^{-1}D$ and that $0 < p \leq b_E < 1$.

Suppose that G is not a component of B. Then the coefficient of Θ at G is 0, so the coefficient of Γ at G is 0 by **Step 2**, and $(\Theta - \Gamma) \cdot G \geq 0$. Since G is not f-exceptional and K_X is ample, $f^*K_X \cdot G = K_X \cdot f_*G > 0$. Hence

$$g^*(K_{X'} + C) \cdot G = f^*K_X \cdot G + (\Theta - \Gamma) \cdot G > 0,$$

which is impossible because G is g-exceptional. Thus G is a component of B. Since G is not f-exceptional, G is a component of $f_*^{-1}D$.

Let c_E be the coefficient of E in f^*D. Then $c_E \geq 0$ since $D \geq 0$. By Construction-Proposition 4.2(2)(f), $c_E = m(b_E - p)$, so $b_E = p + c_E/m \geq p > 0$. Moreover, $b_E = 1 - a(E, X, 0) < 1$ because X is klt.

Step 4. In this step, we show that $a(G, X', b_E C) < 1$ and $m(a(G, X', pC) - 1) = \operatorname{mult}_G f_*^{-1}D$.

By Construction-Proposition 4.2(2)(f), $g_* f^* K_X = K_{X'} + b_E C$. Thus

$$g^*(K_{X'} + b_E C) - f^* K_X = \sum_F d_F F,$$

where F runs over the g-exceptional prime divisors. For any g-exceptional prime divisor F', $(f^*K_X - g^*(K_{X'} + b_E C)) \cdot F' = f^*K_X \cdot F' \geq 0$. By [KM98, Lemma 3.41(1)], $d_F \geq 0$ for any F. On the connected component of $\operatorname{Exc}(g)$ containing G, we have $f^*K_X \cdot G > 0$, so not all d_F vanish there, and [KM98, Lemma 3.41(2)] applied to the curves of the connected component implies $d_G > 0$. Since G is not f-exceptional, the coefficient of $f^*K_X - K_W$ at G is 0, while the coefficient of $g^*(K_{X'} + b_E C) - K_W$ at G is $1 - a(G, X', b_E C)$. Hence

$$a(G, X', b_E C) = 1 - d_G < 1.$$

We compute the coefficient of $\operatorname{div}_W(s)$ at G in two ways. By (4.4) for f and since G is not f-exceptional, it is $\operatorname{mult}_G f^*D = \operatorname{mult}_G f_*^{-1}D$. By (4.4) for g and Construction-Proposition 4.2(2)(g), it is

$$-m \operatorname{mult}_G g^*(pC) + m(a(G, X', 0) - 1).$$

Taking the coefficients at G of $K_W - g^*K_{X'}$ and $K_W - g^*(K_{X'} + pC)$, we obtain

$$\operatorname{mult}_G g^*(pC) = a(G, X', 0) - a(G, X', pC).$$

Thus

$$m(a(G, X', pC) - 1) = \operatorname{mult}_G f_*^{-1}D.$$

Since G is a component of $f_*^{-1}D$, $m(a(G, X', pC) - 1)$ is a positive integer.

Step 5. In this step, we prove (2), and complete the proof of (1).

Let x' be the center of G on X'. If $x' \notin C$, then $a(G, X', pC) = a(G, X', b_E C) < 1$, which contradicts **Step 4**. Suppose that $x' \in C$ is a smooth point of X'. Since C is smooth, $(X' \ni x', C)$ is analytically isomorphic to the germ of Definition-Lemma 2.11 with $r = 1$, where C corresponds to the first axis. Since $a(G, X', b_E C) < 1$ by **Step 4**, b_E is rational, as a coefficient of the $\mathbb{Q}$-divisor $f^*K_X - K_W$, and $0 < b_E < 1$ by **Step 3**, Definition-Lemma 4.6(2) with $b = b_E$ and $b' = 0$ (X' is klt near C by Construction-Proposition 4.2(2)(b)) implies that G is the toric divisor of $v_{(j,k)}$ with $j, k \geq 1$. By Definition-Lemma 2.11(1) and Definition-Lemma 4.6(1),

$$a(G, X', b_E C) = (1 - b_E)j + k \geq 1,$$

which contradicts **Step 4**. Thus x' is a singular point of X' contained in C.

Since (X', C) is a projective lc pair which is plt near C (Construction-Proposition 4.2), $(X' \ni x', C)$ is identified with the germ of Definition-Lemma 2.11 by Definition-Lemma 2.12. By **Step 4** and Definition-Lemma 4.6(2) with $b = b_E$ and $b' = 0$, G is the toric divisor of a primitive vector $v_{(j,k)}$ with $j, k \geq 1$, and Definition-Lemma 2.11(1) and Definition-Lemma 4.6(1) imply

$$0 < \frac{(1 - b_E)j + k}{r} = a(G, X', b_E C) < 1.$$

This proves (2), and together with **Steps 3 and 4**, also (1).

Step 6. In this step, we prove (3).

By definition, $\sigma \in R(X)_{m'}$ if and only if

$$\operatorname{div}_X(\sigma) = f_* \operatorname{div}_W(\sigma) \geq 0,$$

that is, if and only if the coefficient of $\operatorname{div}_W(\sigma)$ at any prime divisor F on W that is not f-exceptional is ≥ 0. By (4.4) for g,

$$\operatorname{div}_W(\sigma) = g^* \operatorname{div}_{X'}(\sigma) + m'(K_W - g^* K_{X'})$$

as $\mathbb{Q}$-divisors. If F is neither f-exceptional nor g-exceptional, then g_*F is a prime divisor on X' different from C, because E is the only prime divisor on W whose image is C, and E is f-exceptional. Hence the coefficient of $\operatorname{div}_W(\sigma)$ at F equals the coefficient of $\operatorname{div}_{X'}(\sigma)$ at g_*F, which is ≥ 0 by hypothesis. If F is g-exceptional and not f-exceptional, then by (4.4) the coefficient of $\operatorname{div}_W(\sigma)$ at F is

$$\operatorname{mult}_F g^* \operatorname{div}_{X'}(\sigma) + m'(a(F, X', 0) - 1).$$

This proves (3). □

4.4. **Sections of degree eleven.** In this subsection, we prove Theorem 4.10: a klt stable surface X with $K_X^2 \leq 1/48983$ and $P_{11}(X) > 0$ is isomorphic to X_{438}. From a non-zero element of degree 11, the proof constructs on the pair of volume $1/462$ the elements of degrees 61 and 146 and a rational 219-canonical form, which exhibit X as a double cover of $\mathbb{P}(11, 61, 146)$.

Definition 4.8. For algebraically independent homogeneous elements s_a, s_b, s_c of positive degrees a, b, and c in a graded domain with $\gcd(a, b, c) = 1$, we write $\mathbb{P}(a, b, c) := \operatorname{Proj} \mathbb{C}[s_a, s_b, s_c]$, which need not be well-formed, and for a positive multiple m of abc, we write $\mathcal{O}_{\mathbb{P}(a,b,c)}(m)$ for the invertible sheaf associated with the graded module $\mathbb{C}[s_a, s_b, s_c](m)$ (invertible since $D_+(s_a^{m/a})$, $D_+(s_b^{m/b})$, and $D_+(s_c^{m/c})$ cover $\mathbb{P}(a, b, c)$ [Gro61, Corollary (2.5.8)]).

Lemma 4.9. *Let X be a normal projective surface, D an ample $\mathbb{Q}$-Cartier Weil divisor on X, and $R(X, D) := \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(mD))$. Let a, b, c be positive integers with $\gcd(a, b, c) = 1$, and let $s_a, s_b, s_c \in R(X, D)$ be algebraically independent homogeneous elements of degrees a, b, and c. Then the rational map $\phi : X \dashrightarrow \mathbb{P}(a, b, c)$ defined by s_a, s_b, s_c is dominant and generically finite, and the degree δ of ϕ satisfies $\delta \leq abcD^2$. If $\delta = abcD^2$, then ϕ is a finite morphism.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix the integer m and show that ϕ is dominant and generically finite.

Let $A := \mathbb{C}[s_a, s_b, s_c]$ and $e := abc$. For a positive integer n, let $A^{(n)} := \bigoplus_{k \geq 0} A_{kn}$. Any monomial in s_a, s_b, s_c of degree divisible by e is the product of a monomial in s_a^{bc}, s_b^{ac}, s_c^{ab}, which are of degree e, and a monomial $s_a^i s_b^j s_c^k$ with $0 \leq i < bc$, $0 \leq j < ac$, and $0 \leq k < ab$, whose degree is divisible by e and smaller than $3e$. Thus $A^{(e)}$ is generated by A_e and A_{2e}. In a product of elements of A_e and A_{2e} of degree divisible by $2e$, the number of factors in A_e is even, and grouping the factors in A_e in pairs shows that $A^{(2e)}$ is generated by A_{2e}. For any positive integers k and n, we have $A_{2ken} = (A_{2e})^{kn} \subset (A_{2ke})^n$, so $A^{(2ke)}$ is generated by A_{2ke}.

Let m_0 be the product of $2e$ and the Cartier index of D, and let m be a positive multiple of m_0. Then mD is Cartier and $A^{(m)}$ is generated by A_m. Since $A \subset R(X, D)$, we have $A_m \subset H^0(X, \mathcal{O}_X(mD))$, and the subring of $\bigoplus_{k \geq 0} H^0(X, \mathcal{O}_X(kmD))$ generated by A_m is $A^{(m)}$. Thus ϕ is the rational map $X \dashrightarrow \mathbb{P}(A_m^\vee)$ defined by the linear system $A_m \subset H^0(X, \mathcal{O}_X(mD))$, the closure of the image of ϕ is $\operatorname{Proj} A^{(m)} = \operatorname{Proj} A = \mathbb{P}(a, b, c)$ (passing to a Veronese subring does not change Proj [Gro61, Proposition (2.4.7)(i)]), and $\mathcal{O}_{\mathbb{P}(a,b,c)}(m)$ is the restriction of $\mathcal{O}_{\mathbb{P}(A_m^\vee)}(1)$ [Gro61, (2.5.16)] and [Har77, Proposition II.5.12(c)]. Replacing m with a multiple of m composes the rational map defined by A_m with a Veronese embedding, so ϕ does not depend on m. Since s_a, s_b, s_c are algebraically independent, $\dim A = 3$, so $\dim \mathbb{P}(a, b, c) = 2 = \dim X$. Therefore ϕ is dominant and generically finite.

Step 2. In this step, we prove $\delta \leq abcD^2$.

Let $\mathfrak{b} \subset \mathcal{O}_X$ be the base ideal of A_m, that is, the image of $A_m \otimes \mathcal{O}_X(-mD) \to \mathcal{O}_X$. Let $f : W \to X$ be a resolution of X such that $\mathfrak{b} \cdot \mathcal{O}_W = \mathcal{O}_W(-N)$ for a divisor $N \geq 0$ on W [KM98, Theorem 0.2], and write

$$f^*(mD) = M + N.$$

Then the image of f^*A_m in $H^0(W, \mathcal{O}_W(M))$ is base point free and defines a morphism $h : W \to \mathbb{P}(a,b,c)$ such that $h = \phi \circ f$ on a dense open subset of W and $\mathcal{O}_W(M) \cong h^*\mathcal{O}_{\mathbb{P}(a,b,c)}(m)$. Thus M and $f^*(mD)$ are nef, and $\deg h = \delta$ since f is birational.

We compute $\mathcal{O}_{\mathbb{P}(a,b,c)}(m)^2$. The Hilbert series of A is

$$\sum_{n \geq 0} (\dim A_n) t^n = \frac{1}{(1-t^a)(1-t^b)(1-t^c)}.$$

Since $\gcd(a,b,c) = 1$, any root of unity other than 1 is a root of at most two of $1-t^a$, $1-t^b$, and $1-t^c$. Thus the Hilbert series has a pole of order three at $t = 1$ with leading coefficient $\frac{1}{abc}(1-t)^{-3}$ and poles of order at most two elsewhere, and the partial fraction decomposition implies $\dim A_n = \frac{n^2}{2abc} + O(n)$. On the other hand, for $k \gg 0$,

$$\dim A_{km} = h^0(\mathbb{P}(a,b,c), \mathcal{O}_{\mathbb{P}(a,b,c)}(km)) = \frac{k^2}{2}(\mathcal{O}_{\mathbb{P}(a,b,c)}(m)^2) + O(k)$$

by [Har77, Exercise II.5.9(b)] for the first equality (applied to $A^{(m)}$, which is generated by A_m) and [Laz04, Corollary 1.1.25] and Serre vanishing [Har77, Theorem III.5.2(b)] for the second. Comparing the coefficients of k^2, we obtain $\mathcal{O}_{\mathbb{P}(a,b,c)}(m)^2 = \frac{m^2}{abc}$. By the projection formula,

$$M^2 = \deg h \cdot \mathcal{O}_{\mathbb{P}(a,b,c)}(m)^2 = \frac{\delta m^2}{abc}, \qquad (f^*(mD))^2 = m^2 D^2.$$

Since $f^*(mD) + M$ is nef and $N \geq 0$,

$$m^2 D^2 - \frac{\delta m^2}{abc} = (f^*(mD))^2 - M^2 = (f^*(mD) + M) \cdot N \geq 0.$$

Thus $\delta \leq abcD^2$.

Step 3. In this step, we assume $\delta = abcD^2$ and show that ϕ is a finite morphism.

By **Step 2**, $f^*(mD) \cdot N + M \cdot N = 0$. Since $f^*(mD)$ and M are nef and $N \geq 0$, we have $f^*(mD) \cdot N = M \cdot N = 0$. For any prime divisor C on W that is not contracted by f, the projection formula implies $f^*(mD) \cdot C = m(D \cdot f_*C) > 0$ since D is ample. Thus any irreducible component of N is f-exceptional. Moreover,

$$N^2 = f^*(mD) \cdot N - M \cdot N = 0.$$

Since the intersection matrix of the f-exceptional curves is negative definite [KM98, Lemma 3.40], $N = 0$, that is, $\mathfrak{b} \cdot \mathcal{O}_W = \mathcal{O}_W$. If $\mathfrak{b} \neq \mathcal{O}_X$ at a point $p \in X$, then for any point $q \in f^{-1}(p)$, the ideal $\mathfrak{b} \cdot \mathcal{O}_{W,q}$ is contained in the maximal ideal of $\mathcal{O}_{W,q}$, a contradiction. Thus A_m is base point free on X, so ϕ extends to a morphism $\phi : X \to \mathbb{P}(a,b,c)$ with $\mathcal{O}_X(mD) \cong \phi^*\mathcal{O}_{\mathbb{P}(a,b,c)}(m)$. If ϕ contracted a curve C, then

$$mD \cdot C = \phi^*\mathcal{O}_{\mathbb{P}(a,b,c)}(m) \cdot C = 0,$$

contradicting the ampleness of D. Thus ϕ is quasi-finite. Since ϕ is projective, ϕ is finite [Har77, Exercise III.11.2]. □

Theorem 4.10. *Let X be a klt stable surface with $K_X^2 \leq 1/48983$. Then the following hold.*

(1) If $P_{11}(X) > 0$, then $X \cong X_{438}$.
(2) If $X \ncong X_{438}$, then $P_m(X) = 0$ for any $1 \leq m \leq 12$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct a hypersurface Y of degree 42 in $\mathbb{P}(6,11,14,21)$ and a prime divisor C_Y on Y, and we show that (Y, C_Y) is a projective plt pair such that $K_Y + C_Y$ is ample and $(K_Y + C_Y)^2 = 1/462$.

Let s_1, s_2, s_3, s_4 be variables of weights 6, 11, 14, and 21, let

$$F := s_4^2 + s_3^3 + s_1^7 + s_1 s_2^2 s_3 \in \mathbb{C}[s_1, s_2, s_3, s_4],$$

which is weighted homogeneous of degree 42, let

$$R_Y := \mathbb{C}[s_1, s_2, s_3, s_4]/(F), \qquad Y := \operatorname{Proj} R_Y \subset \mathbb{P}(6,11,14,21).$$

Since $s_3^3 + s_1^7 + s_1 s_2^2 s_3$ is of degree 3 in s_3, $-(s_3^3 + s_1^7 + s_1 s_2^2 s_3)$ is not a square in $\mathbb{C}[s_1, s_2, s_3]$. Thus F, a monic quadratic polynomial in s_4, is irreducible. The partial derivatives of F are

$$\frac{\partial F}{\partial s_1} = 7s_1^6 + s_2^2 s_3, \quad \frac{\partial F}{\partial s_2} = 2s_1 s_2 s_3, \quad \frac{\partial F}{\partial s_3} = 3s_3^2 + s_1 s_2^2, \quad \frac{\partial F}{\partial s_4} = 2s_4,$$

and the Euler formula reads

$$\sum_{i=1}^{4} (\deg s_i) s_i \frac{\partial F}{\partial s_i} = 42F. \tag{4.5}$$

If the four partial derivatives vanish at a point of $\mathbb{A}^4$, then $s_4 = 0$ and $s_1 s_2 s_3 = 0$ there, and each of $s_1 = 0$, $s_2 = 0$, and $s_3 = 0$ implies $s_1 = s_3 = 0$ by the formulas for $\partial F/\partial s_1$ and $\partial F/\partial s_3$. Thus the singular locus of the affine hypersurface $(F = 0) \subset \mathbb{A}^4$ is the s_2-axis $(s_1 = s_3 = s_4 = 0)$, which is of codimension two in $(F = 0)$. Since R_Y is a hypersurface ring, it is Cohen–Macaulay, and hence R_Y is normal by the Serre criterion [BH98, Theorem 2.2.22]. In particular, Y is a normal projective surface.

The greatest common divisor of any three of $6, 11, 14, 21$ is 1, and the greatest common divisor of two of them is 1, $\gcd(6,14) = 2$, $\gcd(6,21) = 3$, or $\gcd(14,21) = 7$, which divides 42. Thus Y is well-formed [IF00, Definition 6.9 and Section 6.10]. Explicitly, the restrictions of F to the lines $(s_2 = s_4 = 0)$, $(s_2 = s_3 = 0)$, and $(s_1 = s_2 = 0)$ are the non-zero polynomials $s_3^3 + s_1^7$, $s_4^2 + s_1^7$, and $s_4^2 + s_3^3$, and $(0:1:0:0)$ is the only coordinate point of $\mathbb{P}(6,11,14,21)$ on Y. Thus the points of Y with non-trivial stabilizer form a finite set.

For $1 \le l \le 4$, let $V_l \subset \mathbb{A}^3$ be the affine surface defined by F with $s_l = 1$, and let $\pi_l : V_l \to Y$ be the quotient map by $\mu_{\deg s_l}$ onto the open subset $(s_l \neq 0)$ of Y. Since the points of Y with non-trivial stabilizer form a finite set, π_l is étale outside a finite set. By (4.5), a singular point of V_l is a singular point of $(F = 0) \subset \mathbb{A}^4$. Thus V_l is smooth for $l \neq 2$, and V_2 is smooth outside the origin.

By the theorem of Demazure [Dem88], in the form [Wat81, Introduction and Notation (1.1)], there exists an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H on Y such that $(R_Y)_n = H^0(Y, \mathcal{O}_Y(nH))t^n$ for any $n \ge 0$, where $t := s_1^2 s_2^{-1}$ is of degree one. Since Y is well-formed, no curve on Y consists of points with non-trivial stabilizer, so the integer q_V of [Wat81, (1.3)] is 1 for any prime divisor V on Y. Thus H is a divisor, and the $\mathbb{Q}$-divisor D' of [Wat81, Theorem (2.8)] is zero. The graded canonical module of R_Y is $R_Y(42 - 6 - 11 - 14 - 21) = R_Y(-10)$ [BH98, Example 3.6.10 and Corollary 3.6.14], so

$$K_Y \sim -10H$$

by [Wat81, Theorem (2.8), Corollary (2.9), and Remark (2.10)].

For a quotient ξ of two homogeneous elements of R_Y, of degree n, let

$$\operatorname{div}_H(\xi) := \operatorname{div}(\xi t^{-n}) + nH.$$

Then $\operatorname{div}_H(\xi) \ge 0$ if $\xi \in R_Y$, and

$$\operatorname{div}_H(\xi\xi') = \operatorname{div}_H(\xi) + \operatorname{div}_H(\xi').$$

Since s_l generates $\mathcal{O}_Y((\deg s_l)H)$ on $(s_l \neq 0)$ [Wat81, Lemma (2.1)], $(\deg s_l)\operatorname{div}_H(\xi)$ is the divisor of the function $\xi^{\deg s_l} s_l^{-n}$ on $(s_l \neq 0)$. Since π_l is étale outside a finite set, we obtain

$$\pi_l^* \operatorname{div}_H(\xi) = \operatorname{div}(\xi|_{V_l}) \quad \text{for any homogeneous } \xi \in R_Y, \tag{4.6}$$

where $\xi|_{V_l}$ is the function on V_l obtained by setting $s_l = 1$. Fix a rational 2-form ω_0 on Y with $\operatorname{div}(\omega_0) = -10H$. For a quotient ξ of two homogeneous elements of R_Y, of degree $-10m$, let

$$\Psi(\xi) := \xi t^{10m} \omega_0^{\otimes m},$$

which is a rational m-canonical form on Y. For any integer m, the map Ψ on quotients of degree $-10m$ is $\mathbb{C}$-linear and injective. Moreover, $\Psi(\xi\xi') = \Psi(\xi)\Psi(\xi')$, and

$$\operatorname{div}(\Psi(\xi)) = \operatorname{div}(\xi t^{10m}) - 10mH = \operatorname{div}_H(\xi). \tag{4.7}$$

Let $C_Y := \operatorname{div}_H(s_2)$. Then $C_Y \sim 11H$, so $K_Y + C_Y \sim H$ is ample.

We show that (Y, C_Y) is plt. By (4.6), $Y \setminus \operatorname{Supp} C_Y = \pi_2(V_2)$. Since

$$F(s_1, 1, s_3, s_4) = s_4^2 + s_1 s_3 + s_3^3 + s_1^7$$

has the non-degenerate quadratic part $s_4^2 + s_1 s_3$, the origin of V_2 is an ordinary double point, that is, the quotient of a smooth germ by μ_2 acting freely outside the fixed point. Thus V_2 is klt [KM98, Proposition 4.18], and $Y \setminus \operatorname{Supp} C_Y$ is klt [KM98, Proposition 5.20]. Now let $l \in \{1, 3, 4\}$. At a point of V_l with $s_2 = 0$, we have $\partial F/\partial s_2 = 0$ and

$$\Big(\frac{\partial F}{\partial s_1}, \frac{\partial F}{\partial s_3}, \frac{\partial F}{\partial s_4}\Big) = (7s_1^6, 3s_3^2, 2s_4).$$

If at such a point $\partial F/\partial s_i = 0$ for any $i \notin \{2, l\}$, then $\partial F/\partial s_l = 0$ by (4.5), so $s_1 = s_3 = s_4 = 0$, which contradicts $s_l = 1$. Thus $(s_2 = 0) \subset V_l$ is a smooth curve along which $s_2|_{V_l}$ vanishes to order one, and $\pi_l^* C_Y = (s_2 = 0)$ by (4.6). Since V_l is smooth and π_l is étale outside a finite set, $\pi_l^*(K_Y + C_Y) = K_{V_l} + (s_2 = 0)$, and $(V_l, (s_2 = 0))$ is plt. Thus (Y, C_Y) is plt on $(s_l \neq 0)$ by [KM98, Proposition 5.20(2)]. Since $Y \setminus \operatorname{Supp} C_Y$ and the open subsets $(s_l \neq 0)$, $l \in \{1, 3, 4\}$, cover Y, (Y, C_Y) is plt. Moreover, C_Y is reduced, and C_Y is a prime divisor since $\mathbb{C}[s_1, s_3, s_4]/(s_4^2 + s_3^3 + s_1^7)$ is a domain, by the argument for F.

Finally, the Hilbert series of R_Y is $(1 - T^{42})/\prod_i (1 - T^{\deg s_i})$. The pole at $T = 1$ has order three with leading coefficient $42/(6 \cdot 11 \cdot 14 \cdot 21)$, and the other poles have order at most two, since no three of the weights have a common factor. Hence

$$\dim(R_Y)_n = \frac{42}{6 \cdot 11 \cdot 14 \cdot 21} \cdot \frac{n^2}{2} + O(n).$$

Comparing with the asymptotic Riemann–Roch formula [Laz04, Corollary 1.1.25] for $h^0(Y, \mathcal{O}_Y(nH)) = \dim(R_Y)_n$, for n divisible by an integer a with aH Cartier, using Serre vanishing [Har77, Proposition III.5.3] for the ample divisor aH, we obtain

$$(K_Y + C_Y)^2 = H^2 = \frac{42}{6 \cdot 11 \cdot 14 \cdot 21} = \frac{1}{462}.$$

Step 2. In this step, we identify the log canonical model of a non-zero element $x \in R(X)_{11}$ with (Y, C_Y), describe the singular points on C_Y, and show that x is $\Psi(s_2^{-10})$ up to a non-zero constant.

Let $x \in R(X)_{11}$ be non-zero, and let $f : W \to X$, $g : W \to X'$, C, E, and p be given by Construction-Proposition 4.2 applied to the section x, with $m = 11$. Let b_E be the coefficient of E in $f^*K_X - K_W$. Since $(11+1)^2 K_X^2 \leq 144/48983 < 1/42$, Construction-Proposition 4.2(2) and Lemma 4.7 hold for the section x. By Proposition 4.4(2) applied to the section x, $K_{X'} + \frac{10}{11}C \sim_{\mathbb{Q}} 0$. Since C is ample and $K_{X'} + pC \sim_{\mathbb{Q}} 0$, $p = 10/11$.

By Proposition 4.4(2), (X', C) is isomorphic to the pair of [LL23, Example 3.2]. By **Step 1** and Theorem 1.3(2), so is (Y, C_Y). Thus there exists an isomorphism $h : X' \to Y$ with $h(C) = C_Y$. From now on, we identify (X', C) with (Y, C_Y) by h.

Let o_2, o_3, and o_7 be the points of Y on the lines $(s_2 = s_4 = 0)$, $(s_2 = s_3 = 0)$, and $(s_1 = s_2 = 0)$. Let $i_2 := 4$, $i_3 := 3$, and $i_7 := 1$. For any $r \in \{2, 3, 7\}$, the points of the chart V_l in Table 1 form one orbit of $\mu_{\deg s_l}$, so the line of o_r intersects Y in one point, and the stabilizer of any of the points of the chart is the group of the table. At a point of C, at least two of s_1, s_3, s_4 are non-zero, since $F = s_4^2 + s_3^3 + s_1^7$ there. If all three are non-zero, then the stabilizer of the point is trivial since $\gcd(6, 14, 21) = 1$, so Y is smooth at the point. By Proposition 4.4(3), the singular points of X' on C are o_2, o_3, and o_7, of orders 2, 3, and 7.

point	chart	points of the chart	stabilizer	weights mod r	Q_{o_r}	lattice N
o_2	V_1	$s_2 = s_4 = 0,\ s_3^3 = -1$	$\mu_2 \subset \mu_6$	$(1, 1)$	1	$\mathbb{Z}^2 + \mathbb{Z}(1, 1)/2$
o_3	V_1	$s_2 = s_3 = 0,\ s_4^2 = -1$	$\mu_3 \subset \mu_6$	$(2, 2)$	1	$\mathbb{Z}^2 + \mathbb{Z}(1, 1)/3$
o_7	V_3	$s_1 = s_2 = 0,\ s_4^2 = -1$	$\mu_7 \subset \mu_{14}$	$(4, 6)$	3	$\mathbb{Z}^2 + \mathbb{Z}(3, 1)/7$

TABLE 1. The singular points of X' on C, where the weights are those of μ_r on (s_2, s_{i_r}).

At a point of V_l over o_r in Table 1, the partial derivative of F by the variable of V_l other than s_2 and s_{i_r} is non-zero: it is $\partial F/\partial s_3 = 3s_3^2$ for $r = 2$, and $\partial F/\partial s_4 = 2s_4$ for $r = 3, 7$. Thus (s_2, s_{i_r}) are local coordinates of V_l there, and μ_r acts on them with weights $(11, \deg s_{i_r})$ modulo r. Multiplying by the inverse of the second weight modulo r, the weights become $(Q_{o_r}, 1)$. Thus, by Definition-Lemma 2.11(3), the germ $(X' \ni o_r, C)$ is the germ of Definition-Lemma 2.11 with $Q = Q_{o_r}$, in which (s_2, s_{i_r}) correspond to the curves of $(1, 0)$ and $(0, 1)$, and $C = (s_2 = 0)$.

Definition-Lemma 2.12 only provides an isomorphism of germs carrying C to the curve of $(1, 0)$, and Definition-Lemma 4.6(2) holds for any such isomorphism. Thus Lemma 4.7(2) holds for the isomorphisms of germs given by the local coordinates (s_2, s_{i_r}) of V_l. Let G be the toric divisor of a primitive vector $v_{(j,k)}$ at o_r for such an isomorphism. By the definition of the toric divisor of a vector of N, the regular functions s_2^r and $s_{i_r}^r$ on the germ have orders j and k along G. Since their divisors near o_r are rC and $r \operatorname{div}_H(s_{i_r})$ by (4.6), we obtain

$$\operatorname{mult}_G g^* C = \frac{j}{r}, \qquad \operatorname{mult}_G g^* \operatorname{div}_H(s_{i_r}) = \frac{k}{r}. \tag{4.8}$$

Moreover, $s_1 \neq 0$ at o_2 and o_3, and $s_3 \neq 0$ at o_2 and o_7, since otherwise $F = 0$ forces $s_1 = s_3 = s_4 = 0$. Thus $o_2, o_3 \notin \operatorname{Supp} \operatorname{div}_H(s_1)$ and $o_2, o_7 \notin \operatorname{Supp} \operatorname{div}_H(s_3)$ by (4.6).

Regard x as a rational 11-canonical form on X' via f and g. By Construction-Proposition 4.2(2)(g), $\operatorname{div}_{X'}(x) = -11pC = -10C$. By (4.7), $\operatorname{div}(\Psi(s_2^{-10})) = -10C$ as well. Thus the rational function $x/\Psi(s_2^{-10})$ has neither zeros nor poles on the normal projective surface X', so

$$x = c\Psi(s_2^{-10}) \quad \text{for some } c \in \mathbb{C}^*.$$

Step 3. In this step, we construct $y \in R(X)_{61}$ and $z \in R(X)_{146}$.

Let

$$y := \Psi(s_1 s_2^{-56}), \qquad z := \Psi(s_3 s_2^{-134}).$$

Since $6 - 56 \cdot 11 = -10 \cdot 61$ and $14 - 134 \cdot 11 = -10 \cdot 146$, y and z are rational 61- and 146-canonical forms on X', and hence, via f and g, rational pluricanonical forms on X. By (4.7),

$$\operatorname{div}_{X'}(y) = \operatorname{div}_H(s_1) - 56C, \qquad \operatorname{div}_{X'}(z) = \operatorname{div}_H(s_3) - 134C,$$

and the restrictions of both to $X' \backslash C$ are ≥ 0. For a non-zero rational m'-canonical form σ on X' and a g-exceptional prime divisor G on W that is not f-exceptional, let

$$\lambda_G(\sigma) := \operatorname{mult}_G g^* \operatorname{div}_{X'}(\sigma) + m'\big(a(G, X', 0) - 1\big). \tag{4.9}$$

By Lemma 4.7(3), it suffices to show that $\lambda_G(y) \geq 0$ and $\lambda_G(z) \geq 0$ for any such G.

Fix such a G. By Lemma 4.7(2), the center of G on X' is o_r for some $r \in \{2, 3, 7\}$, and G is the toric divisor of a primitive vector $v_{(j,k)}$ with $j, k \geq 1$ and $(1 - b_E)j + k < r$. Since X is

klt, $b_E < 1$, so $1 \le k \le r-1$. In particular, $k = 1$ if $r = 2$. By Definition-Lemma 2.11(1) and Definition-Lemma 4.6(1),

$$a\Big(G, X', \frac{10}{11}C\Big) = \frac{j/11 + k}{r}, \qquad a(G, X', 0) = \frac{j+k}{r}.$$

By Lemma 4.7(1) with $p = 10/11$,

$$u := 11\Big(a\Big(G, X', \frac{10}{11}C\Big) - 1\Big) = \frac{j + 11k - 11r}{r}$$

is a positive integer, that is,

$$j = 11(r-k) + ru, \qquad u \ge 1. \tag{4.10}$$

Let $\epsilon_1 := \operatorname{mult}_G g^* \operatorname{div}_H(s_1)$ and $\epsilon_3 := \operatorname{mult}_G g^* \operatorname{div}_H(s_3)$. By (4.8), (4.9), and (4.10),

$$\lambda_G(y) = \epsilon_1 - \frac{56j}{r} + 61\Big(\frac{j+k}{r} - 1\Big) = \epsilon_1 + \frac{5j + 61k}{r} - 61 = \epsilon_1 + 5u + \frac{6k}{r} - 6,$$

$$\lambda_G(z) = \epsilon_3 - \frac{134j}{r} + 146\Big(\frac{j+k}{r} - 1\Big) = \epsilon_3 + \frac{12j + 146k}{r} - 146 = \epsilon_3 + 12u + \frac{14k}{r} - 14.$$

By (4.8) and the paragraph before the last one of **Step 2**, $\epsilon_1 = k/7$ if $r = 7$ and $\epsilon_1 = 0$ otherwise, and $\epsilon_3 = k/3$ if $r = 3$ and $\epsilon_3 = 0$ otherwise. Substituting, we obtain

	$r = 2$	$r = 3$	$r = 7$
$\lambda_G(y)$	$5u - 3$	$5u + 2k - 6$	$5u + k - 6$
$\lambda_G(z)$	$12u - 7$	$12u + 5k - 14$	$12u + 2k - 14$

Since $u \ge 1$ and $k \ge 1$, all of them are non-negative. Thus $y \in R(X)_{61}$ and $z \in R(X)_{146}$.

Step 4. In this step, we show that x, y, and z are algebraically independent and conclude the proof.

Since F is monic in s_4, R_Y is a free $\mathbb{C}[s_1, s_2, s_3]$-module with basis $1, s_4$. Thus s_1, s_2, s_3 are algebraically independent in R_Y. Suppose that $P(x, y, z) = 0$ in $R(X)$ for a non-zero polynomial P. Since $R(X)$ is graded and x, y, z are homogeneous of degrees $11, 61, 146$, we may assume that P is a linear combination of monomials $x^{a_1}y^{a_2}z^{a_3}$ with $11a_1 + 61a_2 + 146a_3 = m'$ for a fixed integer m'. By **Step 2** and since Ψ is multiplicative,

$$x^{a_1}y^{a_2}z^{a_3} = c^{a_1}\Psi\big(s_1^{a_2}s_2^{-10a_1-56a_2-134a_3}s_3^{a_3}\big),$$

and distinct (a_1, a_2, a_3) yield distinct exponent vectors $(a_2, -10a_1 - 56a_2 - 134a_3, a_3)$. Since Ψ is $\mathbb{C}$-linear and injective in degree $-10m'$, $P(x, y, z) = 0$ implies that a non-zero Laurent polynomial in s_1, s_2, s_3 vanishes in the fraction field of R_Y, a contradiction. Thus x, y, and z are algebraically independent.

Replacing x with x/c, we may assume that $c = 1$. Let $w := \Psi(s_4 s_2^{-201})$, which is a rational 219-canonical form on X, since $21 - 201 \cdot 11 = -10 \cdot 219$. Since $s_4^2 = -s_3^3 - s_1^7 - s_1 s_2^2 s_3$ in R_Y and Ψ is $\mathbb{C}$-linear and multiplicative,

$$w^2 = -\Psi(s_3^3 s_2^{-402}) - \Psi(s_1^7 s_2^{-402}) - \Psi(s_1 s_3 s_2^{-400}) = -\Phi, \qquad \Phi := z^3 + xy^7 + x^{21}yz.$$

Let $\mathbb{P} := \mathbb{P}(11, 61, 146)$, let $A := \mathbb{C}[x, y, z]$, let $\tau := y^2/x^{11}$, of degree one, and let $v := w\tau^{-219}$, a rational function on X. Then $v^2 = -\Phi\tau^{-438}$. Let $\phi : X \dashrightarrow \mathbb{P}$ be the rational map defined by x, y, z. If v were the pullback of a rational function on $\mathbb{P}$, that is, of a quotient of two elements of A of equal degree, then $\theta := v\tau^{219} \in \operatorname{Frac}(A)$ and $\theta^2 = -\Phi$. Since A is a UFD, $\theta \in A$, which is impossible since Φ has degree 3 in z. Thus v is not such a pullback, and in particular ϕ is not birational.

Since K_X is an ample $\mathbb{Q}$-Cartier Weil divisor with $R(X, K_X) = R(X)$ and $\gcd(11, 61, 146) = 1$, Lemma 4.9 implies that the degree δ of ϕ satisfies $2 \le \delta \le 97966K_X^2 \le 2$. Thus $\delta = 97966K_X^2 = 2$, ϕ is a finite morphism, and the function field of X is $\mathbb{C}(\mathbb{P})(v)$ with $v^2 = -\Phi\tau^{-438} \in \mathbb{C}(\mathbb{P})$. Since $(0 : 0 : 0 : 1) \notin X_{438}$, the projection $\pi : X_{438} \to \mathbb{P}$ that forgets the coordinate of weight 219 is a morphism, and π is finite since $\pi^*\mathcal{O}_{\mathbb{P}}(m) = \mathcal{O}_{X_{438}}(m)$ is ample for any sufficiently

divisible positive integer m. By the equation of X_{438}, the function field of X_{438} is $\mathbb{C}(\mathbb{P})(v')$ with $v'^2 = -\Phi\tau^{-438}$. Since X and X_{438} are normal (Lemma 3.1), both are the normalization of $\mathbb{P}$ in the extension of $\mathbb{C}(\mathbb{P})$ generated by a square root of $-\Phi\tau^{-438}$. Thus $X \cong X_{438}$, and (1) holds.

If $P_m(X) > 0$ for some $1 \le m \le 12$, then $m = 11$ by Proposition 4.4(1) applied to a non-zero element of $R(X)_m$, and $X \cong X_{438}$ by (1). Thus (2) holds. □

4.5. **Sections of degrees thirteen to seventeen.** In this subsection, we prove that $P_m(X) = 0$ for any $13 \le m \le 17$ and any klt stable surface X with $K_X^2 \le \frac{1}{48983}$. In Proposition 4.13, from a non-zero section of degree 13 we construct a pair (X', C) as in the following lemma, which determines such pairs up to isomorphism.

Lemma 4.11. *Let q_3, q_4, q_5 be the coordinates of weights $3, 4, 5$ on $\mathbb{P}(3,4,5)$, let*

$$S := (q_3^3 q_4 + q_4^2 q_5 + q_5^2 q_3 = 0) \subset \mathbb{P}(3,4,5),$$

and let $y_3 := (1:0:0)$. Let X be a normal projective surface and $C \subset X$ an ample prime divisor. Assume that $(X, \frac{12}{13}C)$ is lc, $K_X + \frac{12}{13}C \sim_{\mathbb{Q}} 0$, and $C^2 = \frac{13}{20}$. Then the following hold.

1. *There exists a projective birational morphism $h : X \to \mathbb{P}(3,4,5)$ such that $h_*C = S$, h is an isomorphism over $\mathbb{P}(3,4,5)\backslash\{y_3\}$, and $\mathrm{Exc}(h)$ is a prime divisor G_X. Moreover, G_X is the unique prime divisor G over $\mathbb{P}(3,4,5)$ such that $\mathrm{center}_{\mathbb{P}(3,4,5)} G = y_3$, $a(G, \mathbb{P}(3,4,5), \frac{12}{13}S) = 1$, and $a(G, \mathbb{P}(3,4,5), \frac{25}{26}S) = \frac{5}{6}$.*
2. *Let X_1 be a normal projective surface and $C_1 \subset X_1$ an ample prime divisor such that $(X_1, \frac{12}{13}C_1)$ is lc, $K_{X_1} + \frac{12}{13}C_1 \sim_{\mathbb{Q}} 0$, and $C_1^2 = \frac{13}{20}$. Then there exists an isomorphism $X \to X_1$ which maps C onto C_1.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we describe the pair $(\mathbb{P}(3,4,5), S)$.

Let $F := q_3^3 q_4 + q_4^2 q_5 + q_5^2 q_3$, and suppose that

$$\frac{\partial F}{\partial q_3} = 3q_3^2 q_4 + q_5^2, \quad \frac{\partial F}{\partial q_4} = q_3^3 + 2q_4 q_5, \quad \frac{\partial F}{\partial q_5} = q_4^2 + 2q_3 q_5$$

vanish at a point $(q_3, q_4, q_5) \neq (0,0,0)$ of $\mathbb{A}^3$. If one of q_3, q_4, q_5 is zero, then the three equations imply that all of them are zero. Thus $q_3 q_4 q_5 \neq 0$, and substituting $q_5 = -q_4^2/(2q_3)$ into the first two equations, we get $q_3^4 = q_4^3 = -12q_3^4$, a contradiction. Hence S is quasi-smooth [IF00, Definition 6.3], so S is a smooth curve (a finite quotient of a smooth curve is smooth). Since $3, 4, 5$ are pairwise coprime, the singular points of $\mathbb{P}(3,4,5)$ are y_3, $y_4 := (0:1:0)$, and $y_5 := (0:0:1)$, which are cyclic quotient singularities of orders $3, 4, 5$, and they are contained in S because F contains no power of a single coordinate.

Near y_3, y_4, y_5, $\mathbb{P}(3,4,5)$ is the quotient of $\mathbb{A}^2$ by μ_3, μ_4, μ_5 acting freely outside the origin, and the preimage of S is smooth since S is quasi-smooth. Since the quotient maps are étale in codimension one, $(\mathbb{P}(3,4,5), S)$ is plt [KM98, Proposition 5.20(2)]. Moreover, $\omega_{\mathbb{P}(3,4,5)} \cong \mathcal{O}_{\mathbb{P}(3,4,5)}(-12)$ [CLS11, Theorem 8.2.3 and Exercise 4.1.5] and $S \in |\mathcal{O}_{\mathbb{P}(3,4,5)}(13)|$, so

$$K_{\mathbb{P}(3,4,5)} + \frac{12}{13}S \sim_{\mathbb{Q}} 0 \quad \text{and} \quad S^2 = \frac{13^2}{3 \cdot 4 \cdot 5} = \frac{169}{60}.$$

Since $(\mathbb{P}(3,4,5), S)$ is plt, $(\mathbb{P}(3,4,5), \frac{12}{13}S)$ is klt.

Step 2. In this step, we construct h.

The pair $(X, \frac{12}{13}C)$ is projective and lc, $K_X + \frac{12}{13}C \sim_{\mathbb{Q}} 0$, and $\frac{12}{13} \ge \frac{10}{11}$. By [LS23, Theorem 3.1(3)(b)] with $b = b_1 = \frac{12}{13}$, where the curve $(x^3y + y^2z + z^2x = 0)$ of that theorem is S for $(x, y, z) = (q_3, q_4, q_5)$, there exists a projective birational morphism $h : X \to \mathbb{P}(3,4,5)$ such that

$$K_X + \frac{12}{13}C = h^*\left(K_{\mathbb{P}(3,4,5)} + \frac{12}{13}S\right) \tag{4.11}$$

and $a(G, \mathbb{P}(3,4,5), \frac{12}{13}S) = 1$ for any h-exceptional prime divisor G. In particular, $h_*C = S$.

Step 3. In this step, we show that h extracts at most one prime divisor over each singular point of $\mathbb{P}(3,4,5)$, and we compute C^2.

Let G be an h-exceptional prime divisor and y the center of G on $\mathbb{P}(3,4,5)$. By **Step 2**, $a(G,\mathbb{P}(3,4,5),\frac{12}{13}S)=1$. If $y\notin S$, then y is a smooth point of $\mathbb{P}(3,4,5)$ by **Step 1**, so $a(G,\mathbb{P}(3,4,5),0)\geq 2$ [KM98, Corollary 2.31(3)], a contradiction. Thus $y\in S$, $\operatorname{mult}_G h^*S>0$, and

$$a\Big(G,\mathbb{P}(3,4,5),\frac{25}{26}S\Big)=1-\frac{1}{26}\operatorname{mult}_G h^*S<1.$$

Let r be the order of y, with $r=1$ if y is a smooth point, and identify $(\mathbb{P}(3,4,5)\ni y,S)$ with the germ of Definition-Lemma 2.11 by Definition-Lemma 2.12 if $r\geq 2$ and by local coordinates if $r=1$. By Definition-Lemma 4.6(1) and (2) with $(b,b')=(\frac{25}{26},0)$ and Definition-Lemma 2.11(1), G is the toric divisor of a primitive vector $v_{(j,k)}$ with $j,k\geq 1$ and $\frac{j}{26}+k<r$. Thus $k\leq r-1$ and $r\geq 2$, so $y\in\{y_3,y_4,y_5\}$, and h is an isomorphism over $\mathbb{P}(3,4,5)\backslash\{y_3,y_4,y_5\}$ by Zariski's main theorem [Har77, Theorem V.5.2], as $\mathbb{P}(3,4,5)$ is normal. By Definition-Lemma 4.6(3), over a neighborhood of $y\in\{y_3,y_4,y_5\}$ of order r, h is the toric morphism of Definition-Lemma 2.11(2) given by primitive vectors $v_{(j_1,k_1)},\dots,v_{(j_l,k_l)}$, ordered from $(1,0)$ to $(0,1)$, with $1\leq k_i\leq r-1$ and $j_i\geq 1$. For $i\geq 2$, the ray of $v_{(j_i,k_i)}$ is not adjacent to the ray of $(1,0)$, so the exceptional curve of $v_{(j_i,k_i)}$ is disjoint from C [CLS11, Theorem 3.2.6(d)], which is impossible as C is ample. Thus $l\leq 1$.

Let $y\in\{y_3,y_4,y_5\}$ be a point over which h is not an isomorphism, let r_y be the order of y, let $v_{(j_y,k_y)}$ be the vector over y, and let G_y be the corresponding exceptional curve. By Definition-Lemma 2.11(1) and Definition-Lemma 4.6(1), $a(G_y,\mathbb{P}(3,4,5),bS)=((1-b)j_y+k_y)/r_y$ for any rational $0\leq b<1$. With $b=\frac{12}{13}$, **Step 2** implies

$$1=\frac{1}{r_y}\Big(\frac{j_y}{13}+k_y\Big),\quad\text{so}\quad j_y=13(r_y-k_y). \tag{4.12}$$

Comparing $b=0$ and $b=\frac{1}{2}$, we get $\operatorname{mult}_{G_y}h^*S=\frac{j_y}{r_y}$. By Definition-Lemma 2.11(2) with $w_0=(r_y,0)$, $w_1=(j_y,k_y)$, and $w_2=(0,r_y)$, we have $\det_{r_y}(w_0,w_1)=k_y$, so $C\cdot G_y=\frac{1}{k_y}$. Since $h^*S=C+\sum_y\frac{j_y}{r_y}G_y$, (4.12) implies

$$C^2=h^*S\cdot C-\sum_y\frac{j_y}{r_y}G_y\cdot C=S^2-\sum_y\frac{j_y}{r_yk_y}=\frac{169}{60}-13\sum_y\Big(\frac{1}{k_y}-\frac{1}{r_y}\Big), \tag{4.13}$$

where y runs over the points of $\{y_3,y_4,y_5\}$ over which h is not an isomorphism.

Step 4. In this step, we prove (1).

Since $C^2=\frac{13}{20}$, (4.13) implies

$$\sum_y\Big(\frac{1}{k_y}-\frac{1}{r_y}\Big)=\frac{1}{13}\Big(\frac{169}{60}-\frac{13}{20}\Big)=\frac{1}{6}.$$

The possible values of $\frac{1}{k_y}-\frac{1}{r_y}$ are listed in Table 2, and for each $r_y\in\{3,4,5\}$ at most one term occurs. Since all terms are positive, the terms greater than $\frac{1}{6}$ do not occur. Without the term $\frac{1}{6}$ for $r_y=3$, the remaining terms are $\frac{1}{12}$ for $r_y=4$ and one of $\frac{2}{15},\frac{1}{20}$ for $r_y=5$, so the sum is one of 0, $\frac{1}{12}$, $\frac{2}{15}$, $\frac{1}{20}$, $\frac{1}{12}+\frac{2}{15}=\frac{13}{60}$, $\frac{1}{12}+\frac{1}{20}=\frac{2}{15}$, none of which is $\frac{1}{6}$, and the sum of $\frac{1}{6}$ and a positive term is greater than $\frac{1}{6}$. Thus h is an isomorphism over $\mathbb{P}(3,4,5)\backslash\{y_3\}$, $\operatorname{Exc}(h)$ is a prime divisor G_X over y_3, and by (4.12), G_X is the toric divisor of $v_{(13,2)}$. By Definition-Lemma 2.11(1) and Definition-Lemma 4.6(1),

$$a\Big(G_X,\mathbb{P}(3,4,5),\frac{12}{13}S\Big)=\frac{1}{3}\Big(\frac{13}{13}+2\Big)=1,\quad a\Big(G_X,\mathbb{P}(3,4,5),\frac{25}{26}S\Big)=\frac{1}{3}\Big(\frac{13}{26}+2\Big)=\frac{5}{6}.$$

Conversely, let G be a prime divisor over $\mathbb{P}(3,4,5)$ such that $\operatorname{center}_{\mathbb{P}(3,4,5)} G = y_3$ and

$$a\Big(G, \mathbb{P}(3,4,5), \frac{12}{13}S\Big) = 1, \qquad a\Big(G, \mathbb{P}(3,4,5), \frac{25}{26}S\Big) = \frac{5}{6}.$$

By Definition-Lemma 4.6(2) with $(b, b') = (\frac{25}{26}, 0)$ (we use the coefficient $\frac{25}{26}$ since $a(G, \mathbb{P}(3,4,5), \frac{12}{13}S) = 1$ is not less than 1), G is the toric divisor of a primitive vector $v_{(j,k)}$ with $j, k \geq 1$, and by Definition-Lemma 2.11(1) and Definition-Lemma 4.6(1), $\frac{1}{3}(\frac{j}{13} + k) = 1$ and $\frac{1}{3}(\frac{j}{26} + k) = \frac{5}{6}$. Thus $(j,k) = (13,2)$ and $G = G_X$. This proves (1).

Step 5. In this step, we prove (2).

By (1) for (X_1, C_1), there exists a projective birational morphism $h_1 : X_1 \to \mathbb{P}(3,4,5)$ such that $h_{1*}C_1 = S$ and $\operatorname{Exc}(h_1)$ is a prime divisor G_{X_1}, and $G_{X_1} = G_X$ as prime divisors over $\mathbb{P}(3,4,5)$ by the uniqueness of G_X in (1). Thus G_X is the only exceptional divisor of h and the only exceptional divisor of h_1, and Lemma 2.15(2), applied to h and h_1, yields an isomorphism $X \to X_1$ over $\mathbb{P}(3,4,5)$. It maps $C = h_*^{-1}S$ onto $C_1 = h_{1*}^{-1}S$. □

r_y	$k_y = 1$	$k_y = 2$	$k_y = 3$	$k_y = 4$
3	$\frac{2}{3}$	$\frac{1}{6}$		
4	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{12}$	
5	$\frac{4}{5}$	$\frac{3}{10}$	$\frac{2}{15}$	$\frac{1}{20}$

TABLE 2. The values of $\frac{1}{k_y} - \frac{1}{r_y}$ for $r_y \in \{3,4,5\}$ and $1 \leq k_y \leq r_y - 1$.

Construction-Proposition 4.12. Let $u_{13}, u_4, u_{10}, u_{15}$ be the coordinates of weights $13, 4, 10, 15$ on $\mathbb{P}(13,4,10,15)$, let

$$U_0 := (u_{15}^2 + u_{10}^3 + u_4^5 u_{10} + u_4 u_{13}^2 = 0) \subset \mathbb{P}(13,4,10,15),$$

and let $C_0 := U_0 \cap (u_{13} = 0)$. Then the following hold.

(1) U_0 is a well-formed quasi-smooth hypersurface and a normal projective surface, C_0 is an ample prime divisor on U_0, (U_0, C_0) is plt, $\omega_{U_0} \cong \mathcal{O}_{U_0}(-12)$, $13K_{U_0} + 12C_0 \sim 0$, and $C_0^2 = \frac{13}{20}$.

(2) The singular points of U_0 contained in C_0 are the point $y_2 \in (u_{13} = u_{15} = 0)$ with $u_4 u_{10} \neq 0$, the point $y_4 := (0:1:0:0)$, and the point $y_5 \in (u_{13} = u_4 = 0)$, of orders 2, 4, and 5, respectively.

(3) $y_2, y_4 \notin (u_4 = 0)$ and $y_2, y_5 \notin (u_{10} = 0)$.

(4) Near y_5, U_0 is the quotient of a germ of a smooth surface in the chart $(u_{10} = 1)$ with local coordinates u_{13}, u_4 by μ_5 acting with weights $(3,4)$.

(5) Near y_4, U_0 is the quotient of the germ at the origin of the smooth surface

$$(u_{15}^2 + u_{10}^3 + u_{10} + u_{13}^2 = 0) \subset \mathbb{A}^3$$

by μ_4 acting on u_{13}, u_{10}, u_{15} with weights $(1,2,3)$, the functions u_{13}, u_{15} are local coordinates at the origin, and

$$u_{10}(1 + u_{10}^2) = -(u_{13}^2 + u_{15}^2).$$

Proof. Let $F_0 := u_{15}^2 + G_0$, where $G_0 := u_{10}^3 + u_4^5 u_{10} + u_4 u_{13}^2$. If the partial derivatives $2u_{15}$, $3u_{10}^2 + u_4^5$, $5u_4^4 u_{10} + u_{13}^2$, and $2u_4 u_{13}$ of F_0 vanish at a point of $\mathbb{A}^4$, then $u_{15} = 0$ and $u_4 u_{13} = 0$. If $u_4 = 0$, then $u_{13} = 0$ and $u_{10} = 0$. If $u_{13} = 0$, then $u_4^4 u_{10} = 0$ and $3u_{10}^2 = -u_4^5$, so $u_4 = u_{10} = 0$. Thus U_0 is quasi-smooth. The polynomial F_0 is irreducible, since $G_0|_{u_4=0} = u_{10}^3$ is not a square. Any three of the weights $13, 4, 10, 15$ are coprime, and the one-dimensional singular strata of

$\mathbb{P}(13,4,10,15)$ are $(u_{13}=u_{15}=0)$ and $(u_{13}=u_4=0)$, of generic orders 2 and 5, which intersect U_0 in finitely many points. Thus U_0 is a well-formed quasi-smooth hypersurface of degree 30. By [IF00, Definition 6.3 and the paragraph after it], U_0 is a projective variety which is locally the quotient of a manifold by a finite cyclic group, so U_0 is a normal projective surface with cyclic quotient singularities, and by [IF00, Section 6.14],

$$\omega_{U_0} \cong \mathcal{O}_{U_0}(30-13-4-10-15) = \mathcal{O}_{U_0}(-12).$$

Since $C_0 \in |\mathcal{O}_{U_0}(13)|$, $13K_{U_0}+12C_0 \sim 0$. Since $\mathcal{O}_{U_0}(1)$ is ample and the degree n part of $\mathbb{C}[u_{13},u_4,u_{10},u_{15}]/(F_0)$ is $H^0(U_0,\mathcal{O}_{U_0}(n))$ [IF00, Lemma 7.1], comparing the Hilbert series $(1-t^{30})/((1-t^{13})(1-t^4)(1-t^{10})(1-t^{15}))$, which has a pole of order three at $t=1$ and poles of order at most two at the other roots of unity, with the asymptotic Riemann–Roch formula [Laz04, Corollary 1.1.25] and Serre vanishing [Har77, Proposition III.5.3] for the invertible sheaf $\mathcal{O}_{U_0}(780)$ ($780=\operatorname{lcm}(13,4,10,15)$) implies

$$\mathcal{O}_{U_0}(1)^2 = \frac{30}{13\cdot 4\cdot 10\cdot 15} = \frac{1}{260},$$

so C_0 is ample and $C_0^2 = \frac{169}{260} = \frac{13}{20}$. The curve

$$C_0 = (u_{15}^2 + u_{10}^3 + u_4^5 u_{10} = 0) \subset \mathbb{P}(4,10,15)$$

is quasi-smooth, since if the partial derivatives $2u_{15}$, $3u_{10}^2+u_4^5$, and $5u_4^4u_{10}$ vanish at a point of $\mathbb{A}^3$, then $u_{15}=0$ and $u_4u_{10}=0$, so $u_4=u_{10}=0$. It is prime since $u_{10}^3+u_4^5u_{10}$ is not a square. Thus on the local covers of U_0, both U_0 and the preimage of C_0 are smooth, and the quotient maps are étale in codimension one as U_0 is well-formed. Hence (U_0,C_0) is plt [KM98, Proposition 5.20(2)]. This proves (1).

The points of $U_0 \cap (u_{13}=0)$ with non-trivial stabilizer are contained in $(u_{13}=u_{15}=0)$ or $(u_{13}=u_4=0)$, since $(0:0:1:0),(0:0:0:1) \notin U_0$. On $(u_{13}=u_{15}=0)$, $F_0 = u_{10}(u_{10}^2+u_4^5)$, which defines y_4 and y_2, and on $(u_{13}=u_4=0)$, $F_0 = u_{15}^2+u_{10}^3$, which defines y_5. The stabilizers of y_2,y_4,y_5 are $\mu_{\gcd(4,10)}=\mu_2$, μ_4, and $\mu_{\gcd(10,15)}=\mu_5$. Near y_4, in the chart $(u_4=1)$, U_0 is the quotient of $(u_{15}^2+u_{10}^3+u_{10}+u_{13}^2=0)$ by μ_4 acting with weights $(13,10,15)\equiv(1,2,3)$, and since the partial derivative in u_{10} is 1 at the origin, u_{13},u_{15} are local coordinates of weights $(1,3)$, which become $(3,1)$ after replacing a generator ζ of μ_4 with ζ^3. Near y_2, in the chart $(u_4=1)$, y_2 corresponds to $(0,u_{10},0)$ with $u_{10}^2=-1$, the partial derivative in u_{10} is $3u_{10}^2+1=-2\neq 0$, and u_{13},u_{15} are local coordinates of weights $(1,1)$ modulo 2. Near y_5, in the chart $(u_{10}=1)$, y_5 corresponds to $(0,0,u_{15})$ with $u_{15}^2=-1$, the partial derivative in u_{15} is non-zero, and u_{13},u_4 are local coordinates of weights $(13,4)\equiv(3,4)$ modulo 5, which become $(2,1)$ after replacing ζ with ζ^4. In each case C_0 is the image of $(u_{13}=0)$, and (2)–(5) follow. □

Proposition 4.13. *Let X be a klt stable surface with $K_X^2 \le \frac{1}{48983}$. Then $P_{13}(X)=0$.*

Proof. The proof is long, so we divide it into several steps. Suppose that $P_{13}(X)>0$, and let s be a non-zero element of $R(X)_{13}$. Let W,f,B,g,X',C,D,E,p be the objects of Construction-Proposition 4.2 for $m=13$ and s. Since $42\cdot 196 = 8232 < 48983$,

$$(13+1)^2K_X^2 \le \frac{196}{48983} < \frac{1}{42}.$$

By Construction-Proposition 4.2(2), C is an ample smooth rational curve, (X',C) is plt near C, $K_{X'}+pC \sim_{\mathbb{Q}} 0$, $p=\frac{12}{13}$ or $p\le\frac{10}{11}$, and $13p\in\mathbb{Z}$. Let $\lambda := (K_{X'}+C)\cdot C$.

Step 1. In this step, we show that $p=\frac{12}{13}$, that X' has exactly three singular points contained in C, of orders $2,4,5$, and that $C^2=\frac{13}{20}$.

Since $K_{X'}+C \sim_{\mathbb{Q}} (1-p)C$, we have $\lambda=(1-p)C^2$, and by Construction-Proposition 4.2(1),

$$(1-p)\lambda = (K_{X'}+C)^2 \le 196K_X^2 \le \frac{196}{48983}. \tag{4.14}$$

By Definition-Lemma 2.12,

$$\lambda = -2 + \sum_y \left(1 - \frac{1}{r_y}\right),$$

where y runs over the singular points of X' contained in C and r_y is the order of $X' \ni y$, and $\lambda > 0$ since $K_{X'} + C$ is ample.

Suppose that $p \leq \frac{10}{11}$. By (4.14),

$$\lambda \leq \frac{11 \cdot 196}{48983} = \frac{2156}{48983} < \frac{1}{20}.$$

By Lemma 4.1(2), the multiset $\{r_y\}$ is $\{2,3,7\}$ or $\{2,3,8\}$, and $\lambda = \frac{1}{\ell}$, where $\ell := \operatorname{lcm}\{r_y\}$ is 42 or 24, respectively. By Definition-Lemma 2.12, $C^2 - \sum_y Q_y/r_y \in \mathbb{Z}$, so $\ell C^2 \in \mathbb{Z}$. However,

$$\ell C^2 = \frac{\ell\lambda}{1-p} = \frac{1}{1-p} = \frac{13}{13-13p},$$

and $2 \leq 13 - 13p \leq 12$ because $13p \in \mathbb{Z}$ and $0 < p \leq \frac{10}{11}$. Thus $\ell C^2 \notin \mathbb{Z}$, a contradiction. Therefore $p = \frac{12}{13}$.

Since $K_{X'} + tC \sim_{\mathbb{Q}} (t-p)C$ for any t and C is ample, $\operatorname{pet}(X'; C) = p = \frac{12}{13}$. By Theorem 2.17, any singular point of X' contained in C has order at most 5. By (4.14),

$$\lambda \leq \frac{13 \cdot 196}{48983} = \frac{2548}{48983} < \frac{1}{15}.$$

By Lemma 4.1(2), $\{r_y\} = \{2,4,5\}$ and $\lambda = \frac{1}{20}$. Thus

$$C^2 = \frac{\lambda}{1-p} = \frac{13}{20}.$$

Step 2. In this step, we identify (X', C) with (U_0, C_0).

Since (X', C) is lc (Construction-Proposition 4.2(1)) and C is $\mathbb{Q}$-Cartier, $(X', \frac{12}{13}C)$ is lc [KM98, Corollary 2.35(1)]. Since $p = \frac{12}{13}$, $K_{X'} + \frac{12}{13}C \sim_{\mathbb{Q}} 0$. Together with **Step 1**, X' and C satisfy the assumptions of Lemma 4.11, and so do U_0 and C_0 by Construction-Proposition 4.12(1). By Lemma 4.11(2), there exists an isomorphism $\psi : X' \to U_0$ which maps C onto C_0. From now on we identify (X', C) with (U_0, C_0) by ψ. In particular, the singular points of X' contained in C are y_2, y_4, y_5 of Construction-Proposition 4.12(2).

Step 3. In this step, we normalize s and describe the g-exceptional prime divisors on W which are not f-exceptional.

As in Lemma 4.7(3), we regard $R(X)_{m'}$ as a space of rational m'-canonical forms on X'. Let

$$A := \mathbb{C}[u_{13}, u_4, u_{10}, u_{15}]/(u_{15}^2 + u_{10}^3 + u_4^5 u_{10} + u_4 u_{13}^2),$$

and let A_n be its degree n part. Since U_0 is quasi-smooth (Construction-Proposition 4.12(1)), the singular locus of the hypersurface of $\mathbb{A}^4$ defined by A is the origin, so A is normal [BH98, Theorem 2.2.22].

Let $\tau := u_{13} u_4^{-3}$, of degree one. By the theorem of Demazure [Dem88], in the form [Wat81, Introduction and Notation (1.1)], there exists an ample $\mathbb{Q}$-divisor H on U_0 with $A_n = H^0(U_0, \mathcal{O}_{U_0}(nH))\tau^n$ for $n \geq 0$. Since U_0 is well-formed (Construction-Proposition 4.12(1)), no curve on U_0 consists of points with non-trivial stabilizer, so the integer q_V of [Wat81, (1.3)] is 1 for any prime divisor V on U_0. Thus H is a divisor, and the $\mathbb{Q}$-divisor $\sum_V \frac{q_V - 1}{q_V} V$ of [Wat81, Theorem (2.8)] is zero. The graded canonical module of A is $A(30 - 13 - 4 - 10 - 15) = A(-12)$ [BH98, Example 3.6.10 and Corollary 3.6.14], so $K_{U_0} \sim -12H$ [Wat81, Theorem (2.8), Corollary (2.9), and Remark (2.10)].

For a quotient ξ of two homogeneous elements of A, of degree n, let

$$\operatorname{div}(\xi) := \operatorname{div}(\xi \tau^{-n}) + nH.$$

The divisor $\operatorname{div}(\xi)$ is additive in ξ, and, for a rational 2-form ω_0 with $\operatorname{div}(\omega_0) = -12H$, the map $\xi \to \xi\tau^{12m'}\omega_0^{\otimes m'}$ is a bijection from the homogeneous elements of degree $-12m'$ of the field of fractions of A onto the rational m'-canonical forms on X', compatibly with products and divisors. For $l \in \{13, 4, 10, 15\}$, let $V_l \subset \mathbb{A}^3$ be the affine surface defined by $u_{15}^2 + u_{10}^3 + u_4^5 u_{10} + u_4 u_{13}^2$ with $u_l = 1$, and let $\pi_l : V_l \to U_0$ be the quotient map by μ_l onto the open subset $(u_l \neq 0)$ of U_0. Since u_l generates $\mathcal{O}_{U_0}(lH)$ on $(u_l \neq 0)$ [Wat81, Lemma (2.1)], $l \operatorname{div}(\xi)$ is the divisor of the rational function $\xi^l u_l^{-n}$ on $(u_l \neq 0)$. Thus

$$\pi_l^* \operatorname{div}(\xi) = \operatorname{div}(\xi|_{V_l}),$$

where $\xi|_{V_l}$ is the rational function on V_l obtained by setting $u_l = 1$.

By Construction-Proposition 4.2(2)(g), $\operatorname{div}_{X'}(s) = -13pC = -12C$. The element u_{13}^{-12} has degree $-156 = -12 \cdot 13$ and divisor $-12C_0$. Thus s/u_{13}^{-12} is a rational function without zeros and poles on the normal projective surface U_0, hence a non-zero constant. Possibly replacing s with a non-zero multiple, we may assume that $s = u_{13}^{-12}$.

Let G be a g-exceptional prime divisor on W which is not f-exceptional. By Lemma 4.7(1), $\mu_G := \operatorname{mult}_G f_*^{-1} D$ is a positive integer and

$$13\Big(a\Big(G, X', \frac{12}{13}C\Big) - 1\Big) = \mu_G. \tag{4.15}$$

By Lemma 4.7(2), $\operatorname{center}_{X'} G$ is a point $y \in \{y_2, y_4, y_5\}$ of order r, and for any identification of $(X' \ni y, C)$ with the germ of Definition-Lemma 2.11 with r and $Q = Q_y$, G is the toric divisor of a primitive vector $v_{(j,k)}$ with $j, k \geq 1$ and $(1 - b_E)j + k < r$, where b_E is the coefficient of E in $f^*K_X - K_W$. Since X is klt, $b_E < 1$, so $k \leq r - 1$. By Definition-Lemma 2.11(1) and Definition-Lemma 4.6(1),

$$a(G, X', bC + b'C') = \frac{(1-b)j + (1-b')k}{r} \quad \text{for any rational } 0 \leq b, b' < 1. \tag{4.16}$$

Since (j, k) is determined by $a(G, X', bC)$ for $b = 0$ and $b = \frac{12}{13}$, (j, k) does not depend on the identification. Near y_5 we use the identification given by $u = u_{13}$ and $v = u_4$, and near y_4 the one given by $u = u_{13}$ and $v = u_{15}$ (Definition-Lemma 2.11(3) and Construction-Proposition 4.12(4) and (5), where a change of generator of μ_r turns the weights of (u, v) into $(2, 1)$ at y_5 and $(3, 1)$ at y_4, so that $Q_{y_5} = 2$ and $Q_{y_4} = 3$). Comparing the values of (4.16) for $(b, b') = (0, 0)$, $(\frac{1}{2}, 0)$, and $(0, \frac{1}{2})$, and using (4.15), we get

$$\operatorname{mult}_G g^*C = \frac{j}{r}, \quad \operatorname{mult}_G g^*C' = \frac{k}{r}, \quad a(G, X', 0) = \frac{j+k}{r}, \quad j = 13(r-k) + r\mu_G. \tag{4.17}$$

Let $t \geq 1$ and $m' \geq 1$ be integers, let $\varphi \in \{u_4, u_{10}\}$, let $d \in \{4, 10\}$ be the degree of φ, and assume that $13t - 12m' = d$. Then $\sigma := \varphi u_{13}^{-t}$ is a rational m'-canonical form on X' with

$$\operatorname{div}_{X'}(\sigma) = \operatorname{div}(\varphi) - tC,$$

and $\operatorname{div}_{X'}(\sigma)|_{X' \setminus C} \geq 0$ since $\operatorname{div}(\varphi) \geq 0$. By Lemma 4.7(3) and (4.17), $\sigma \in R(X)_{m'}$ if and only if for any g-exceptional prime divisor G on W which is not f-exceptional,

$$\begin{aligned} &\operatorname{mult}_G g^* \operatorname{div}(\varphi) - \frac{tj}{r} + m'\Big(\frac{j+k}{r} - 1\Big) \\ &\quad = \operatorname{mult}_G g^* \operatorname{div}(\varphi) - d\Big(1 - \frac{k}{r}\Big) + (m' - t)\mu_G \geq 0. \end{aligned} \tag{4.18}$$

Step 4. In this step, we construct elements of $R(X)_{43}$ and $R(X)_{114}$.

Let

$$s_{43} := u_4 u_{13}^{-40} \quad \text{and} \quad s_{114} := u_{10} u_{13}^{-106},$$

so that $(m', t, d) = (43, 40, 4)$ and $(114, 106, 10)$ in **Step 3**, as

$$13 \cdot 40 - 12 \cdot 43 = 4, \qquad 13 \cdot 106 - 12 \cdot 114 = 10.$$

We compute $\mathrm{mult}_G\, g^*\,\mathrm{div}(\varphi)$ for G as in **Step 3**. If $\mathrm{center}_{X'} G = y$ and $y \notin \mathrm{div}(\varphi)$, then $\mathrm{mult}_G\, g^*\,\mathrm{div}(\varphi) = 0$. Thus by Construction-Proposition 4.12(3), it remains to consider $\varphi = u_4$ at y_5 and $\varphi = u_{10}$ at y_4. Near y_5, $\mathrm{div}(u_4) = C'$, so $\mathrm{mult}_G\, g^*\,\mathrm{div}(u_4) = \frac{k}{5}$ by (4.17). The functions u_{10}/u_{15}^2 and $(u_{13}/u_{15})^2$ are μ_4-invariant, hence rational functions on X' near y_4, and by Construction-Proposition 4.12(5),

$$\frac{u_{10}}{u_{15}^2} = -\frac{1 + (u_{13}/u_{15})^2}{1 + u_{10}^2}.$$

Since $\mathrm{div}((u_{13}/u_{15})^2) = 2C - 2C'$ near y_4 and $j \geq 13 + 4 > k$ by (4.17), we have

$$\mathrm{mult}_G\, g^*\,\mathrm{div}\left((u_{13}/u_{15})^2\right) = \frac{2(j-k)}{4} > 0, \quad \text{so} \quad \mathrm{mult}_G\, g^*\,\mathrm{div}\left(1 + (u_{13}/u_{15})^2\right) = 0.$$

Since $1 + u_{10}^2$ is a unit near y_4 and $\mathrm{div}(u_{15}) = C'$ near y_4, we get

$$\mathrm{mult}_G\, g^*\,\mathrm{div}(u_{10}) = \mathrm{mult}_G\, g^*\,\mathrm{div}\left(\frac{u_{10}}{u_{15}^2}\right) + 2\,\mathrm{mult}_G\, g^* C' = \frac{k}{2}.$$

At y_2 we have $r = 2$ and $k = 1$. Table 3 lists the left-hand side of (4.18), and all entries are non-negative for $\mu_G \geq 1$ and $1 \leq k \leq r - 1$. Thus $s_{43} \in R(X)_{43}$ and $s_{114} \in R(X)_{114}$.

Step 5. In this step, we conclude the proof.

Since $u_{15}^2 + u_{10}^3 + u_4^5 u_{10} + u_4 u_{13}^2$ is monic in u_{15}, A is finite over $\mathbb{C}[u_{13}, u_4, u_{10}]$, so u_{13}, u_4, u_{10} are algebraically independent. The exponent vectors of $s = u_{13}^{-12}$, s_{43}, and s_{114} in u_{13}, u_4, u_{10} form the matrix

$$\begin{pmatrix} -12 & -40 & -106 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

of determinant $-12 \neq 0$, so distinct monomials in s, s_{43}, s_{114} are distinct Laurent monomials in u_{13}, u_4, u_{10}. Thus s, s_{43}, s_{114} are algebraically independent. Let δ be the degree of the rational map $X \dashrightarrow \mathbb{P}(13, 43, 114)$ defined by s, s_{43}, s_{114}. Since $\gcd(13, 43, 114) = 1$, Lemma 4.9 implies

$$\delta \leq 13 \cdot 43 \cdot 114 \cdot K_X^2 = 63726 K_X^2. \tag{4.19}$$

We show that $\delta \geq 2$. Let $w := u_{15} u_{13}^{-159}$, of degree $-2052 = -12 \cdot 171$, so w is a rational 171-canonical form on X'. The equation of U_0 implies $w^2 = -\Phi$, where $\Phi := s_{114}^3 + s_{43}^5 s_{114} s + s_{43} s^{23}$. Let $A_0 := \mathbb{C}[s, s_{43}, s_{114}]$, let $\tau' := s^{10}/s_{43}^3$, a rational 1-canonical form, and let $v := w\tau'^{-171}$, a rational function on X. If $\delta = 1$, then v is the pullback of a rational function on $\mathbb{P}(13, 43, 114)$, that is, of a quotient of two elements of A_0 of equal degree, so $\theta := v\tau'^{171} \in \mathrm{Frac}(A_0)$ and $\theta^2 = -\Phi$. Since A_0 is a UFD, $\theta \in A_0$, which is impossible since Φ has degree 3 in s_{114}. Thus $\delta \geq 2$.

By (4.19), $K_X^2 \geq \frac{2}{63726} = \frac{1}{31863} > \frac{1}{48983}$, a contradiction. □

	y_2 $(r = 2,\ k = 1)$	y_4 $(r = 4)$	y_5 $(r = 5)$
s_{43} $(d = 4,\ m' - t = 3)$	$3\mu_G - 2$	$3\mu_G + k - 4$	$3\mu_G + k - 4$
s_{114} $(d = 10,\ m' - t = 8)$	$8\mu_G - 5$	$8\mu_G + 3k - 10$	$8\mu_G + 2k - 10$

TABLE 3. The left-hand side of (4.18) for s_{43} and s_{114} at the three singular points of X' contained in C.

Proposition 4.14. *Let X be a klt stable surface with $K_X^2 \leq \frac{1}{48983}$. Then $P_m(X) = 0$ for any $14 \leq m \leq 17$. Moreover, if $X \ncong X_{438}$, then $P_m(X) = 0$ for any $1 \leq m \leq 17$.*

Proof. The proof is long, so we divide it into several steps. Suppose that $P_m(X) > 0$ for some $14 \le m \le 17$, and let $s \in R(X)_m$ be non-zero. Let $W, f, B, g, X', C, D, E, p$ be the objects of Construction-Proposition 4.2 for m and s. Since $42 \cdot 324 = 13608 < 48983$,

$$(m+1)^2 K_X^2 \le \frac{324}{48983} < \frac{1}{42}.$$

By Construction-Proposition 4.2(2), C is an ample smooth rational curve, (X', C) is plt near C, $K_{X'} + pC \sim_{\mathbb{Q}} 0$, $p = \frac{12}{13}$ or $p \le \frac{10}{11}$, and $mp \in \mathbb{Z}$. Let $\lambda := (K_{X'} + C) \cdot C$.

Step 1. In this step, we show that (m, mp) is one of the rows of Table 4.

Since $13 \nmid 12m$, $p \ne \frac{12}{13}$, so $p \le \frac{10}{11}$. By Definition-Lemma 2.12,

$$\lambda = -2 + \sum_y \left(1 - \frac{1}{r_y}\right) > 0,$$

where y runs over the singular points of X' contained in C and r_y is the order of $X' \ni y$, so $\lambda \ge \frac{1}{42}$ by Lemma 4.1(1). Since $K_{X'} + C \sim_{\mathbb{Q}} (1-p)C$, Construction-Proposition 4.2(1) implies

$$(1-p)\lambda = (K_{X'} + C)^2 \le (m+1)^2 K_X^2 \le \frac{(m+1)^2}{48983}. \tag{4.20}$$

Hence

$$m\Big(1 - \frac{42(m+1)^2}{48983}\Big) \le mp \le \frac{10m}{11}.$$

For $m = 14, 15, 16, 17$, the left-hand side is $\frac{553462}{48983}$, $\frac{573465}{48983}$, $\frac{589520}{48983}$, and $\frac{601375}{48983}$, which belong to $(11, 12)$, $(11, 12)$, $(12, 13)$, and $(12, 13)$, respectively, and the right-hand side is $\frac{140}{11}$, $\frac{150}{11}$, $\frac{160}{11}$, and $\frac{170}{11}$, which belong to $(12, 13)$, $(13, 14)$, $(14, 15)$, and $(15, 16)$, respectively. Since $mp \in \mathbb{Z}$, (m, mp) is one of the eight rows of Table 4. Since $1 - p = \frac{m - mp}{m}$, (4.20) implies

$$\lambda \le \frac{m(m+1)^2}{48983(m - mp)},$$

which is the third column of Table 4.

Step 2. In this step, we show that $m = 15$, $p = \frac{4}{5}$, $\{r_y\} = \{2, 3, 7\}$, and $42C^2 = 5$.

In any row of Table 4, $\lambda < \frac{1}{15}$. By Lemma 4.1(2), the multiset $\{r_y\}$ is $\{2, 3, 7\}$, $\{2, 3, 8\}$, $\{2, 4, 5\}$, or $\{2, 3, 9\}$, and $\lambda = \frac{1}{\operatorname{lcm}\{r_y\}}$ in each case. By Definition-Lemma 2.12, $C^2 - \sum_y Q_y / r_y \in \mathbb{Z}$, so

$$\operatorname{lcm}\{r_y\} \cdot C^2 = \frac{\operatorname{lcm}\{r_y\} \cdot \lambda}{1-p} = \frac{m}{m - mp} \in \mathbb{Z}. \tag{4.21}$$

By the fourth column of Table 4, (4.21) excludes the rows $(15, 13)$, $(16, 13)$, $(17, 13)$, $(17, 14)$, and $(17, 15)$.

If $\{r_y\} = \{2, 3, 7\}$, let y_2, y_3, y_7 be the singular points of X' contained in C, of orders $2, 3, 7$. Then $Q_{y_2} = 1$, $\gcd(Q_{y_3}, 3) = \gcd(Q_{y_7}, 7) = 1$, and

$$42C^2 \equiv 21 + 14Q_{y_3} + 6Q_{y_7} \bmod 42, \tag{4.22}$$

so $42C^2$ is odd and divisible by neither 3 nor 7. If $\{r_y\} = \{2, 3, 8\}$, let y_2, y_3, y_8 be the singular points of X' contained in C, of orders $2, 3, 8$. Then $Q_{y_2} = 1$, Q_{y_8} is odd, $\gcd(Q_{y_3}, 3) = 1$, and

$$24C^2 \equiv 12 + 8Q_{y_3} + 3Q_{y_8} \bmod 24,$$

so $24C^2$ is odd and not divisible by 3.

In the row $(14, 12)$, $\lambda < \frac{1}{24}$, so $\{r_y\} = \{2, 3, 7\}$ and $42C^2 = 7$, which is impossible. In the row $(16, 14)$, $\lambda < \frac{1}{20}$, so $\{r_y\} = \{2, 3, 7\}$ or $\{2, 3, 8\}$, and $\operatorname{lcm}\{r_y\} \cdot C^2 = 8$ is even, which is impossible. In the remaining row $(15, 12)$, $\lambda < \frac{1}{24}$, so $\{r_y\} = \{2, 3, 7\}$, $p = \frac{4}{5}$, $\lambda = \frac{1}{42}$, and $42C^2 = 5$.

Step 3. In this step, we exclude the row $(15, 12)$.

Since $42C^2 = 5$, (4.22) modulo 3 and modulo 7 implies $2Q_{y_3} \equiv 2 \bmod 3$ and $6Q_{y_7} \equiv 5 \bmod 7$, so $Q_{y_3} = 1$ and $Q_{y_7} = 2$. Let $y \in \{y_2, y_3, y_7\}$, and let r be the order of y. By Definition-Lemma 2.12 and Definition-Lemma 2.11(3), near y, X' is the quotient of $(\mathbb{A}^2, 0)$ with coordinates u, v by μ_r acting with weights $(Q_y, 1)$, where C is the image of $(u = 0)$. The rational 5-canonical form $(du \wedge dv)^{\otimes 5} u^{-4}$ generates $\mathcal{O}_{\mathbb{A}^2}(5K_{\mathbb{A}^2} + 4(u = 0))$ near the origin, and a generator ζ of μ_r multiplies it by

$$\zeta^{5(Q_y+1)-4Q_y} = \zeta^{Q_y+5}.$$

Since $Q_y + 5 = 6, 6, 7$ for $r = 2, 3, 7$, the form is μ_r-invariant. Since the quotient map is étale in codimension one, the form descends to a generator of $\mathcal{O}_{X'}(5K_{X'} + 4C)$ near y (freeness of a coherent module can be checked on the analytic local ring). Since X' is smooth at the other points of C, $5K_{X'} + 4C$ is Cartier near C. With $p = \frac{4}{5}$, the integer t of Construction-Proposition 4.2(3) is 1. By Construction-Proposition 4.2(3), there exists a non-zero element of $R(X)_5$. This contradicts Proposition 4.4(1).

Thus $P_m(X) = 0$ for any $14 \le m \le 17$. If $X \ncong X_{438}$, then $P_m(X) = 0$ for any $1 \le m \le 12$ by Theorem 4.10(2), and $P_{13}(X) = 0$ by Proposition 4.13. Hence $P_m(X) = 0$ for any $1 \le m \le 17$. □

m	mp	upper bound for λ	$\frac{m}{m-mp}$
14	12	$\frac{1575}{48983} < \frac{1}{24}$	7
15	12	$\frac{1280}{48983} < \frac{1}{24}$	5
15	13	$\frac{1920}{48983} < \frac{1}{24}$	$\frac{15}{2}$
16	13	$\frac{4624}{3 \cdot 48983} < \frac{1}{24}$	$\frac{16}{3}$
16	14	$\frac{2312}{48983} < \frac{1}{20}$	8
17	13	$\frac{1377}{48983} < \frac{1}{24}$	$\frac{17}{4}$
17	14	$\frac{1836}{48983} < \frac{1}{24}$	$\frac{17}{3}$
17	15	$\frac{2754}{48983} < \frac{1}{15}$	$\frac{17}{2}$

TABLE 4. The possible values of (m, mp) in the proof of Proposition 4.14, with the bound on λ from (4.20) and the number $\operatorname{lcm}\{r_y\} \cdot C^2$ of (4.21).

5. An upper bound for plurigenera

In this section, we prove Proposition 5.1, which is used in Section 9.

Proposition 5.1. *Let X be a klt stable surface with $K_X^2 \le 1/48983$. Then $P_m(X) \le 1$ for any $1 \le m \le 156$.*

Proof. Let $\pi : \widetilde{X} \to X$ be the minimal resolution and $P := \pi^* K_X = K_{\widetilde{X}} + B$, where $B = \sum_E b_E E$ with E running over the π-exceptional prime divisors. Since klt surface germs are numerically dlt, any E is a smooth rational curve [KM98, Notation 4.1 and Theorem 4.7]. We have $E^2 < 0$ [KM98, Lemma 3.40], and $E^2 \ne -1$ because π is minimal. Thus $n_E := -E^2 \ge 2$, and by adjunction [Har77, Exercise V.1.3(a)],

$$K_{\widetilde{X}} \cdot E = -2 - E^2 = n_E - 2 \ge 0.$$

Hence $-B = K_{\widetilde{X}} - P$ is π-nef. Since $\pi_* B = 0$, the negativity lemma [KM98, Lemma 3.39] implies that $B \ge 0$. Since X is klt, $a(E, X, 0) = 1 - b_E > 0$, so $0 \le b_E < 1$. Since $P \cdot E = 0$ by

the projection formula,

$$0 = P \cdot E = n_E - 2 - n_E b_E + \sum_{E' \neq E} b_{E'}(E' \cdot E),$$

so $n_E b_E \geq n_E - 2$, that is, $b_E \geq 1 - 2/n_E$. Since K_X is ample, P is nef and $P^2 = K_X^2$ by the projection formula.

Suppose that $P_m(X) \geq 2$ for some $1 \leq m \leq 156$. The non-zero elements of $R(X)_m$ are the rational functions s on X with $\mathrm{div}_X(s) + mK_X \geq 0$, that is, with

$$\pi^*(\mathrm{div}_X(s) + mK_X) = \mathrm{div}_{\widetilde{X}}(s) + mP \geq 0,$$

since $\pi_*\pi^* D = D$ and $\pi^* D \geq 0$ for any $\mathbb{Q}$-Cartier divisor $D \geq 0$ on X. Since $\mathrm{div}_{\widetilde{X}}(s)$ has integer coefficients,

$$H^0(\widetilde{X}, \mathcal{O}_{\widetilde{X}}(\lfloor mP \rfloor)) = R(X)_m.$$

Write $\lfloor mP \rfloor \sim M + N$, where $N \geq 0$ is the fixed part of $|\lfloor mP \rfloor|$ and $M \geq 0$. Then $h^0(\widetilde{X}, \mathcal{O}_{\widetilde{X}}(M)) \geq 2$ and $|M|$ has no fixed component. Hence M is nef, $M^2 \geq 0$, and $M \cdot N \geq 0$. Since $M \neq 0$ and $M^2 \geq 0$, M is not supported on π-exceptional curves [KM98, Lemma 3.40]. Thus $\pi_* M \neq 0$ and $P \cdot M = K_X \cdot \pi_* M > 0$. Let $R := mP - \lfloor mP \rfloor = \sum_E \{mb_E\} E \geq 0$, where $\{x\} := x - \lfloor x \rfloor$. Since $mP \equiv M + N + R$ and P and M are nef,

$$P \cdot M \leq mP^2 = mK_X^2 \quad \text{and} \quad M^2 + M \cdot N = mP \cdot M - M \cdot R \leq mP \cdot M \leq m^2 K_X^2 \leq \frac{156^2}{48983} < 1.$$

Thus $M^2 = M \cdot N = 0$, and

$$\sum_E \{mb_E\}(E \cdot M) = M \cdot R = mP \cdot M. \tag{5.1}$$

The integer $K_{\widetilde{X}} \cdot M = P \cdot M - B \cdot M$ is at most $P \cdot M < 1$, so $c := -K_{\widetilde{X}} \cdot M \geq 0$ and $B \cdot M = c + P \cdot M > c \geq 0$. Let $\mathcal{H}$ be the set of the E with $E \cdot M > 0$, and let $d := \sum_{E \in \mathcal{H}} E \cdot M$. Since $B \cdot M > 0$, $\mathcal{H} \neq \emptyset$, and since $b_E < 1$, we have $c < B \cdot M < d$, so $d - c \geq 1$.

Let $E \in \mathcal{H}$. Since $M \cdot N = 0$ and M is nef, E is not a component of N, so $N \cdot E \geq 0$. Since $P \cdot E = 0$,

$$E \cdot M \leq (M + N) \cdot E = \lfloor mP \rfloor \cdot E = -R \cdot E \leq n_E \{mb_E\}.$$

Multiplying by $(E \cdot M)/n_E$, summing over $\mathcal{H}$, and using (5.1), we obtain

$$\sum_{E \in \mathcal{H}} \frac{(E \cdot M)^2}{n_E} \leq mP \cdot M \leq m^2 K_X^2.$$

Since $b_E \geq 1 - 2/n_E$ and $E \cdot M \leq (E \cdot M)^2$,

$$d - 2 \sum_{E \in \mathcal{H}} \frac{(E \cdot M)^2}{n_E} \leq B \cdot M = c + P \cdot M.$$

Hence

$$1 \leq d - c \leq P \cdot M + 2 \sum_{E \in \mathcal{H}} \frac{(E \cdot M)^2}{n_E} \leq m(2m+1) K_X^2 \leq \frac{156 \cdot 313}{48983} = \frac{48828}{48983} < 1,$$

a contradiction. □

6. Extractions with large pseudo-effective threshold

In this section, we prove Theorem 6.2. Theorem 6.2 is used in Sections 7 and 8 to show that the pseudo-effective threshold is less than 115/134 unless $X \cong X_{438}$.

Lemma 6.1. *Let V be a normal projective surface, D a prime divisor on V, and $\phi : V \to W$ a contraction onto a curve W with general fiber G. Assume that $0 < t < 1$, $(K_V + tD) \cdot G = 0$, and $D \cdot G > 0$. Then $t \leq 2/3$.*

Proof. Since $\operatorname{Sing} V$ is finite, G is a smooth connected curve contained in the smooth locus of V [Har77, Corollary III.10.7], and $G^2 = 0$. Thus D is Cartier near G, and $D \cdot G$ is a positive integer. By the adjunction formula [Har77, Proposition V.1.5], computed on a resolution of V that is an isomorphism near G,

$$0 = (K_V + tD) \cdot G = 2g(G) - 2 + t(D \cdot G).$$

Since $t(D \cdot G) > 0$, we have $g(G) = 0$ and $t = 2/(D \cdot G)$. Since $t < 1$, we have $D \cdot G \geq 3$, so $t \leq 2/3$. □

Theorem 6.2. *Let X be a klt stable surface with $K_X^2 \leq 1/48983$. Let $f : Z \to X$ be a projective birational morphism from a klt surface Z whose exceptional locus is a prime divisor E with rational normalization. Write $K_Z + bE = f^*K_X$ and assume that $0 < b < 1$. Let $p := \inf\{t \geq 0 \mid K_Z + tE$ is pseudo-effective$\}$. If $p \geq 6/7 + 1/938 = 115/134$, then $X \cong X_{438}$.*

Proof. The proof is long, so we divide it into several steps. Suppose that $X \ncong X_{438}$.

Step 1. In this step, we run the $(K_Z + pE)$-MMP and obtain a klt Calabi–Yau pair (T', pE') with $\rho(T') = 1$.

Let $P := f^*K_X = K_Z + bE$. Since K_X is ample, P is nef and big, $P \cdot E = 0$, and $P \cdot C = K_X \cdot f_*C > 0$ for any curve $C \neq E$ on Z. Since X is klt and $K_Z + bE = f^*K_X$, $a(G, Z, bE) = a(G, X, 0) > 0$ for any prime divisor G over Z [KM98, Lemma 2.30], so (Z, bE) is klt. Since the big cone is open and the pseudo-effective cone is closed [Laz04, Theorem 2.2.26], $K_Z + tE = P - (b - t)E$ is big for any $t < b$ close to b, and $K_Z + pE$ is pseudo-effective. Thus $115/134 \leq p < b < 1$, and (Z, pE) is klt.

Since Z is klt, Z is $\mathbb{Q}$-factorial [KM98, Proposition 4.11]. By Theorem 2.4, applied to (Z, pE), there exists a sequence of steps of the $(K_Z + pE)$-MMP

$$Z = Z_0 \to Z_1 \to \cdots \to Z_k$$

such that $K_{Z_k} + pE_k$ is semi-ample, where E_i denotes the pushforward of E to Z_i. Each Z_i is $\mathbb{Q}$-factorial [Fuj12, Theorem 1.1]. Since $K_Z + pE$ is pseudo-effective, each step $\phi_i : Z_i \to Z_{i+1}$ is birational, so ϕ_i contracts exactly one curve C_i, with $s_i := -C_i^2 > 0$ and $(K_{Z_i} + pE_i) \cdot C_i < 0$, since the curves contracted by an extremal contraction are numerically proportional, and they are negative definite by [KM98, Lemma 3.40] on a resolution. For any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor L on Z_i, the $\mathbb{R}$-divisor $\phi_i^*\phi_{i*}L - L$ is supported on C_i and $(\phi_i^*\phi_{i*}L - L) \cdot C_i = -L \cdot C_i$, so

$$\phi_i^*\phi_{i*}L = L + \frac{L \cdot C_i}{s_i} C_i. \tag{6.1}$$

We show by induction on i that $E_i \neq 0$, that $P_i := K_{Z_i} + bE_i$ is nef and big, and that $P_i \cdot C > 0$ for any curve $C \neq E_i$ on Z_i. For $i = 0$, this was shown above. Assume that it holds for some $i < k$. If $C_i = E_i$, then

$$(K_{Z_i} + pE_i) \cdot E_i = P_i \cdot E_i + (b - p)s_i > 0,$$

a contradiction. Thus $C_i \neq E_i$, $E_{i+1} \neq 0$, and $P_i \cdot C_i > 0$. Since $P_{i+1} = \phi_{i*}P_i$, for any curve C' on Z_{i+1} with strict transform C on Z_i, (6.1) and the projection formula imply

$$P_{i+1} \cdot C' = P_i \cdot C + \frac{(P_i \cdot C_i)(C_i \cdot C)}{s_i} \geq P_i \cdot C, \qquad P_{i+1}^2 = P_i^2 + \frac{(P_i \cdot C_i)^2}{s_i} > 0.$$

Thus P_{i+1} is nef and big, and $P_{i+1} \cdot C' > 0$ for $C' \neq E_{i+1}$. This finishes the induction.

Applying (6.1) to $L = K_{Z_i} + pE_i$, we obtain

$$K_{Z_i} + pE_i = \phi_i^*(K_{Z_{i+1}} + pE_{i+1}) + c_iC_i, \qquad c_i := -\frac{(K_{Z_i} + pE_i) \cdot C_i}{s_i} > 0.$$

Thus, for any prime divisor G over Z_{i+1},

$$a(G, Z_{i+1}, pE_{i+1}) = a(G, Z_i, pE_i) + c_i \operatorname{mult}_G \pi^*C_i \geq a(G, Z_i, pE_i),$$

where π is a birational morphism onto Z_i from a variety on which G is a divisor. Since (Z, pE) is klt, (Z_i, pE_i) is klt for any i by induction on i. Let $Y := Z_k$, $E_Y := E_k$, $P_Y := P_k = K_Y + bE_Y$, and let $g : Z \to Y$ be the composition of $\phi_0, \dots, \phi_{k-1}$. The pullback of C_i to Z is a $\mathbb{Q}$-divisor ≥ 0 with coefficient one along the strict transform of C_i, and any g-exceptional prime divisor on Z is the strict transform of some C_i. Thus

$$K_Z + pE = g^*(K_Y + pE_Y) + N, \tag{6.2}$$

where $N \geq 0$ is g-exceptional and the coefficient of N along any g-exceptional prime divisor is positive. Moreover, (Y, pE_Y) is klt, $E_Y \neq 0$, P_Y is nef and big, and $P_Y \cdot C > 0$ for any curve $C \neq E_Y$ on Y.

Let $M := K_Y + pE_Y$, which is semi-ample. If M is big, then $K_Z + pE$ is big by (6.2), so $K_Z + tE$ is big for some $0 < t < p$ [Laz04, Theorem 2.2.26], contradicting the definition of p. Thus M is not big.

Suppose that $M \not\equiv 0$. By [Laz04, Theorem 2.1.27], applied to semi-ample Cartier divisors of which M is a positive combination, there exists a contraction $Y \to V$ onto a curve V such that M is $\mathbb{R}$-linearly equivalent to the pullback of an ample $\mathbb{R}$-divisor on V, as M is neither big nor numerically trivial. Let F be a general fiber of $Y \to V$. Then $M \cdot F = 0$, and $P_Y \cdot F > 0$ since $F \neq E_Y$. Since $P_Y = M + (b-p)E_Y$, we obtain $E_Y \cdot F > 0$, so Lemma 6.1 implies $p \leq 2/3$, contradicting $p \geq 115/134$. Therefore $M \equiv 0$.

Since $M \equiv 0$, $P_Y \equiv (b-p)E_Y$, so $(b-p)^2 E_Y^2 = P_Y^2 > 0$. Since $Y = Z_k$ is $\mathbb{Q}$-factorial, K_Y and E_Y are $\mathbb{Q}$-Cartier, so their intersection numbers are rational, and

$$p = -\frac{K_Y \cdot E_Y}{E_Y^2} \in \mathbb{Q}.$$

Moreover, $E_Y \cdot C = (P_Y \cdot C)/(b-p) > 0$ for any curve $C \neq E_Y$ on Y and $E_Y^2 > 0$, so E_Y is ample by the Nakai–Moishezon criterion [KM98, Theorem 1.37], applied to a Cartier multiple of E_Y.

Since $2/3 < p < 1$, Theorem 2.18, applied to (Y, pE_Y), implies that there exists a birational morphism $h : Y \to T'$ to a projective surface T' with $\rho(T') = 1$ such that $E' := h_* E_Y \neq 0$, (T', pE') is klt,

$$K_Y + pE_Y = h^*(K_{T'} + pE'), \tag{6.3}$$

and $K_{T'} + pE' \sim_{\mathbb{R}} 0$. Since $p > 5/6$, Theorem 2.18 also implies that (T', E') is lc. Since E_Y is a prime divisor and $h_* E_Y \neq 0$, E' is a prime divisor, and $E \to E'$ is birational, as $h \circ g$ does not contract E. Since $K_X^2 \leq 1/48983 < 1/143$, Theorem 2.7 implies that the minimal resolution of X is rational. Thus any resolution of T' is a smooth projective rational surface, as T' is birational to X. Write $p = n/m$ with coprime positive integers $n < m$.

Step 2. In this step, we show that

$$mK_{T'} + nE' \sim 0. \tag{6.4}$$

Let $L := mK_{T'} + nE'$. Then $L = m(K_{T'} + pE') \equiv 0$. By Theorem 2.19, applied to (T', pE'), one of the cases (1)–(4) of Theorem 2.19 holds.

In case (2), $T' \cong \mathbb{P}(3,4,5)$, $E' = (x^3y + y^2z + z^2x = 0)$, $m = 13$, and $n = 12$. The class group of $\mathbb{P}(3,4,5)$ is $\mathbb{Z}$, with the torus-invariant divisors of degrees 3, 4, and 5 [CLS11, Exercise 4.1.5]. Since $K_{T'}$ is minus the sum of the three torus-invariant divisors [CLS11, Theorem 8.2.3], $\deg K_{T'} = -12$, and $\deg E' = 13$. Thus $\deg L = 13 \cdot (-12) + 12 \cdot 13 = 0$, and $L \sim 0$.

In the other cases, we first show that L is Cartier. Let y be a singular point of T' on E' such that $(T' \ni y, E')$ is analytically isomorphic to the germ of Definition-Lemma 2.11 with $r = r_y$ and $Q = Q_y$, E' corresponding to C. In the coordinates of Definition-Lemma 2.11, the dual lattice of N is

$$\{(\alpha, \beta) \in \mathbb{Z}^2 \mid r_y \text{ divides } \alpha Q_y + \beta\},$$

so the class group of the germ is isomorphic to $\mathbb{Z}/r_y$, a Weil divisor $\alpha C + \beta C'$ with $\alpha, \beta \in \mathbb{Z}$ being sent to $\alpha Q_y + \beta \bmod r_y$ [CLS11, Theorem 4.1.3]. Thus $[C] = Q_y$ and $[C'] = 1$. Since $K = -C - C'$ [CLS11, Theorem 8.2.3], $[E'] = Q_y$ and $[K_{T'}] = -(1 + Q_y)$. Since the Cartier divisors on the germ are exactly the Weil divisors of class zero [CLS11, Proposition 4.2.2], and the invertibility of the reflexive sheaf $\mathcal{O}_{T'}(L)$ can be checked on the analytic germ, we obtain

$$L \text{ is Cartier at } y \iff r_y \text{ divides } m + (m-n)Q_y. \tag{6.5}$$

Similarly, at a cyclic quotient point of type $\frac{1}{r}(1, q)$ of T' off E', $L = mK_{T'}$ is Cartier if and only if r divides $m(1+q)$, that is, if and only if $r/\gcd(r, 1+q)$ divides m. Replacing q by the inverse of q modulo r changes neither the singularity nor $r/\gcd(r, 1+q)$. At a smooth point of T', L is Cartier.

In cases (3) and (4), all singular points of T' are contained in E', (T', E') is plt, and the germs $(T' \ni y, E')$ are those of Definition-Lemma 2.12. In case (3), $(m, n) = (11, 10)$ and $(r_y, Q_y) = (2, 1)$, $(5, 4)$, and $(7, 3)$, so $m+(m-n)Q_y = 12$, 15, and 14. In case (4), $(m, n) = (17, 15)$ and $(r_y, Q_y) = (3, 2)$, $(5, 4)$, and $(7, 2)$, so $m + (m-n)Q_y = 21$, 25, and 21. In both cases, r_y divides $m + (m-n)Q_y$, so L is Cartier by (6.5).

In case (1), E' is not a smooth rational curve. Since $E \to E'$ is birational and the normalization of E is rational, E' is not smooth. By [Abe97, Lemma 1.1], E' has at worst nodes, and any singular point y of T' on E' is a cyclic quotient point at which the strict transform of E' on the minimal resolution intersects one end curve of the chain transversally. Thus E' has a node y_0, y_0 is a smooth point of T', and at any singular point y of T' on E' the curve E' is smooth. Since (T', E') is lc by **Step 1**, it is plt at y [KM98, Theorem 4.15(3)]. By Definition-Lemma 2.12, $(T' \ni y, E')$ is analytically isomorphic to the germ of Definition-Lemma 2.11 with $r = r_y$ and $Q = Q_y$, where r_y/Q_y is the Hirzebruch–Jung continued fraction of the chain read from E'. Since $\rho(T') = 1$ and $E' \not\equiv 0$, p is the unique real number t with $K_{T'} + tE' \equiv 0$. By [Abe97, p. 9], p is the value $\max(b)$ of the third column of [Abe97, Table] in the row with second column 0. Table 5 lists the data of [Abe97, Table, pp. 11–13] and [Abe97, Lemma 1.1] that we use. In cases 12 and 16 of [Abe97, Table], [Abe97, p. 13] lists the trivial 8-complement $(T', \frac{7}{8}E')$ and the trivial 9-complement $(T', \frac{8}{9}E')$ of $(T', 0)$, that is, $8K_{T'} + 7E' \sim 0$ and $9K_{T'} + 8E' \sim 0$, which is (6.4). In the other cases of [Abe97, Table], T' has exactly two singular points, the cyclic quotient point $y_1 \in E'$ and the cyclic quotient point $y_2 \notin E'$ of type $\frac{1}{r_2}(1, q_2)$. In each row of Table 5, r_1 divides $m + (m-n)Q_1$ and $r_2/\gcd(r_2, 1+q_2)$ divides m. Thus L is Cartier.

case	$p = n/m$	m	chain at y_1	(r_1, Q_1)	$m + (m-n)Q_1$	chain at y_2	(r_2, q_2)	$r_2/\gcd(r_2, 1+q_2)$
4	8/9	9	$[2, 2, 2]$	$(4, 3)$	12	$[3]$	$(3, 1)$	3
5	9/10	10	$[2, 2]$	$(3, 2)$	12	$[3, 2]$	$(5, 2)$	5
6	7/8	8	$[3, 2]$	$(5, 2)$	10	$[2, 3, 2]$	$(8, 5)$	4
7	13/15	15	$[3, 2, 2]$	$(7, 3)$	21	$[2, 3]$	$(5, 3)$	5
8	19/22	22	$[4, 2]$	$(7, 2)$	28	$[2, 2, 3, 2]$	$(11, 8)$	11
10	10/11	11	$[2]$	$(2, 1)$	12	$[2, 3, 2, 2]$	$(11, 7)$	11
11	15/17	17	$[2, 2]$	$(3, 2)$	21	$[2, 4, 2, 2]$	$(17, 10)$	17
12	7/8	8	trivial 8-complement: $8K_{T'} + 7E' \sim 0$					
13	20/23	23	$[2, 2, 2]$	$(4, 3)$	32	$[2, 5, 2, 2]$	$(23, 13)$	23
14	25/29	29	$[2, 2, 2, 2]$	$(5, 4)$	45	$[2, 6, 2, 2]$	$(29, 16)$	29
16	8/9	9	trivial 9-complement: $9K_{T'} + 8E' \sim 0$					

TABLE 5. The cases of [Abe97, Table, pp. 11–13] in Theorem 2.19(1), with the value $p = n/m$ in lowest terms, the Hirzebruch–Jung chains of the singular point y_1 of T' on E' (read from E') and of the singular point y_2 of T' off E' together with the corresponding Cartier conditions, and, in cases 12 and 16, the trivial complements of [Abe97, p. 13].

It remains to show that $L \sim 0$ when L is Cartier. Let $\mu : \widetilde{T} \to T'$ be a resolution. By **Step 1**, $\widetilde{T}$ is a smooth projective rational surface. Numerical and linear equivalence of divisors coincide on $\widetilde{T}$, since $\widetilde{T}$ maps to $\mathbb{P}^2$ or to a rational ruled surface by a composition of blow-ups of points [Har77, Corollary V.5.4, Theorem V.5.8, Example V.5.8.2, and Remark V.5.8.4], so the Picard group of $\widetilde{T}$ is free and maps isomorphically onto the group of numerical classes [Har77, Corollary II.6.17, Propositions V.2.3 and V.3.2]. Since μ^*L is a numerically trivial Cartier divisor, $\mu^*L \sim 0$. Since T' is normal, $\mu_*\mathcal{O}_{\widetilde{T}} = \mathcal{O}_{T'}$, so the projection formula implies

$$\mathcal{O}_{T'}(L) \cong \mu_*\mathcal{O}_{\widetilde{T}}(\mu^*L) \cong \mathcal{O}_{T'}.$$

Thus $L \sim 0$.

Step 3. In this step, we construct a non-zero element of $R(X)_m$ and exclude the cases with $m \le 17$.

Let $\psi := h \circ g : Z \to T'$. By (6.2) and (6.3),

$$K_Z + pE = \psi^*(K_{T'} + pE') + N.$$

By (6.4), there exists a rational function u on T' with $\operatorname{div}_{T'}(u) + mK_{T'} + nE' = 0$. Multiplying the equality $K_Z + pE = \psi^*(K_{T'} + pE') + N$ by m and using $\psi^*(mK_{T'} + nE') = -\operatorname{div}_Z(u)$, we obtain

$$\operatorname{div}_Z(u) + mK_Z + nE = mN.$$

Since $mK_Z + nE$ and $\psi^*(mK_{T'} + nE') = -\operatorname{div}_Z(u)$ are divisors with integer coefficients, so is $mN \ge 0$. Applying f_* and $f_*E = 0$, we obtain

$$\operatorname{div}_X(u) + mK_X = D_X, \qquad D_X := f_*(mN) \ge 0. \tag{6.6}$$

Thus u defines a non-zero section $s \in R(X)_m$ with divisor of zeros D_X, and $P_m(X) > 0$. Since $\operatorname{mult}_G N > 0$ for any g-exceptional prime divisor G, $G \ne E$, and f contracts only E, the support of D_X is the union of the curves $f(G)$, and distinct G have distinct images $f(G)$.

In case (2) of Theorem 2.19, $m = 13$. In cases (3) and (4), $m = 11$ and $m = 17$. In case (1), Table 5 shows that $m \le 17$ except in the three rows with $m = 22$, 23, and 29. Since $X \ncong X_{438}$, Proposition 4.14 implies $P_m(X) = 0$ for $1 \le m \le 17$. Thus we are in one of the rows with $m = 22$, 23, and 29, where

$$1 - p = \frac{3}{22}, \quad \frac{3}{23}, \quad \text{and} \quad \frac{4}{29},$$

respectively.

Step 4. In this step, we show that E_Y is a smooth rational curve.

Let $w : W \to Z$ be a log resolution of $(Z, E + \operatorname{Exc}(g))$, and let $f_W := f \circ w : W \to X$ and $g_W := g \circ w : W \to Y$. The f_W-exceptional prime divisors are the w-exceptional prime divisors and the strict transform of E. By **Step 3**, the components of the strict transform of $\operatorname{Supp} D_X$ on W are the strict transforms of the g-exceptional prime divisors. The g_W-exceptional prime divisors are the w-exceptional prime divisors and the strict transforms of the g-exceptional prime divisors. Thus f_W is a log resolution of $(X, \operatorname{Supp} D_X)$, and the reduced divisor B on W with support $\operatorname{Exc}(f_W) \cup (f_W)_*^{-1} \operatorname{Supp} D_X$ satisfies

$$B = (g_W)_*^{-1} E_Y + \operatorname{Exc}(g_W)_{\mathrm{red}}. \tag{6.7}$$

Since (Y, pE_Y) is klt and $E_Y \ge 0$ is $\mathbb{Q}$-Cartier, Y is klt [KM98, Corollary 2.35(1)] and $\operatorname{lct}(Y; E_Y) \ge p > 5/6$. Thus $\operatorname{lct}(Y; E_Y) = 1$ by Theorem 2.16, and (Y, E_Y) is lc, since the log discrepancies of (Y, tE_Y) are affine functions of t. By (6.7) and the definition of log discrepancies,

$$K_W + B = g_W^*(K_Y + E_Y) + \sum_G a(G, Y, E_Y)G, \tag{6.8}$$

where G runs over the g_W-exceptional prime divisors, and $a(G, Y, E_Y) \ge 0$. Since $M \equiv 0$ and E_Y is ample, $K_Y + E_Y \equiv (1 - p)E_Y$ is ample. For any sufficiently divisible positive integer ℓ, $g_{W*}\mathcal{O}_W(\ell \sum_G a(G, Y, E_Y)G) = \mathcal{O}_Y$, since a rational function on the normal surface Y without

poles in codimension one is regular [Har77, Proposition II.6.3A]. Thus the projection formula and (6.8) imply that $H^0(W, \ell(K_W + B)) \cong H^0(Y, \ell(K_Y + E_Y))$ for any such ℓ, so $g_W : W \to Y$ is the morphism to the log canonical model of (W, B) [BCHM10, Lemma 3.6.6(4)].

We apply Construction-Proposition 4.2 to the section $s \in R(X)_m$ of (6.6): in the notation of Construction-Proposition 4.2, the section of degree m is s with divisor of zeros D_X, the log resolution is $f_W : W \to X$ with B as in (6.7), the log canonical model is Y, and $C = g_{W*}B = E_Y$, by (6.7) and (6.8). Since $m \leq 29$ and $900 \cdot 42 = 37800 < 48983$,

$$(m+1)^2 K_X^2 \leq \frac{900}{48983} < \frac{1}{42}.$$

Thus Construction-Proposition 4.2(2) implies that E_Y is a smooth rational curve.

Step 5. In this step, we show that E_Y has a node, which contradicts **Step 4**.

By **Step 2**, E' has a node y_0, and y_0 is a smooth point of T'. We show that h is an isomorphism over a neighborhood of y_0. Suppose not. If $h^{-1}(y_0)$ is finite, then h is an isomorphism over a neighborhood of y_0 by Zariski's main theorem [Har77, Theorem V.5.2], as T' is normal. Thus $h^{-1}(y_0)$ contains a curve Γ. Since $h_*E_Y \neq 0$, $\Gamma \neq E_Y$, so (6.3) implies

$$a(\Gamma, T', pE') = a(\Gamma, Y, pE_Y) = 1$$

[KM98, Lemma 2.30]. Let v be the coefficient of Γ in h^*E'. Since T' is smooth near y_0, $K_{T'}$ and E' are Cartier near y_0, so v is a positive integer, as $h(\Gamma) = y_0 \in E'$, and $a(\Gamma, T', E') = 1 + \mathrm{mult}_\Gamma(K_Y - h^*(K_{T'} + E'))$ is an integer. Since (T', E') is lc, $a(\Gamma, T', E') \geq 0$. Since log discrepancies are affine in the boundary,

$$1 = a(\Gamma, T', pE') = a(\Gamma, T', E') + (1-p)v.$$

Since $(1-p)v > 0$, the non-negative integer $a(\Gamma, T', E')$ is 0, so $v = 1/(1-p)$, which is $22/3$, $23/3$, or $29/4$ by **Step 3**. This contradicts the integrality of v. Thus h is an isomorphism over a neighborhood of y_0.

Since h is an isomorphism over a neighborhood of y_0 and $h_*E_Y = E'$, E_Y has a node at $h^{-1}(y_0)$. This contradicts **Step 4**. Therefore $X \cong X_{438}$. □

7. Extraction of a special divisorial valuation

In this section, we fix the data of the case $\mathrm{mld}(X) < 5/46$ and prove Construction-Theorem 7.3. Sections 7 and 8 prove Theorem 8.1.

Corollary 7.1. *Let (V, Δ) be a projective lc surface pair such that V is $\mathbb{Q}$-factorial and $K_V + \Delta$ is nef. Then $K_V + \Delta$ is semi-ample, and the ample model $h : V \to V'$ of $K_V + \Delta$ [BCHM10, Definition 3.6.5] is a morphism with connected fibers such that $K_V + \Delta \sim_{\mathbb{R}} h^*A$ for an ample $\mathbb{R}$-divisor A on V'. A curve C on V is contracted by h if and only if $(K_V + \Delta) \cdot C = 0$. If moreover $K_V + \Delta$ is big, then h is birational, the intersection matrix of the h-exceptional curves is negative definite, $K_{V'} + h_*\Delta$ is ample, and $K_V + \Delta = h^*(K_{V'} + h_*\Delta)$.*

Proof. Let $D := K_V + \Delta$. Since D is nef, no D-negative extremal ray exists, so the MMP of Theorem 2.4, applied to (V, Δ), has no step, and D is semi-ample. Thus the identity of V is a semi-ample model of D, so the ample model $h : V \to V'$ of D exists, h is a morphism with connected fibers, and $D \sim_{\mathbb{R}} h^*A$ for an ample $\mathbb{R}$-divisor A on V' [BCHM10, Lemma 3.6.6(1)(3)]. Since A is ample, a curve C on V is contracted by h if and only if $D \cdot C = 0$.

Assume that D is big. Then $\dim V' = 2$, so h is birational. Let $\mu : W \to V$ be a resolution. If $R \neq 0$ is an $\mathbb{R}$-linear combination of h-exceptional curves, then $\mu^*R \neq 0$ is supported on curves contracted by $h \circ \mu$, so $R^2 = (\mu^*R)^2 < 0$ [KM98, Lemma 3.40]. Thus the intersection matrix of the h-exceptional curves is negative definite. Since $h_*D \sim_{\mathbb{R}} A$, the $\mathbb{R}$-divisor $K_{V'} + h_*\Delta = h_*D$ is ample. The $\mathbb{R}$-divisor $h^*h_*D - D$ is h-exceptional and has zero intersection number with any h-exceptional curve, so it is zero. □

Set-up 7.2. Let X, E, f, Z, a, b, q, p, and n be as follows.

(1) X is a klt stable surface with $K_X^2 \le 1/48983$ and $a := \mathrm{mld}(X) < 5/46$.
(2) E is a special divisorial valuation over a point of X with $a(E, X, 0) = a$.
(3) $f : Z \to X$ is the extraction of E, $b := 1 - a$, so that $K_Z + bE = f^*K_X$, and $q := -E^2$, where E also denotes the exceptional curve of f.
(4) $p := \inf\{t \ge 0 \mid K_Z + tE \text{ is pseudo-effective}\}$ and $n := \inf\{t \ge 0 \mid K_Z + tE \text{ is nef}\}$.

Such E exists: $a = \mathrm{mld}(X \ni x) < 1$ for some point $x \in X$, so $X \ni x$ is not Du Val, and by Theorem 2.14(1), some special divisorial valuation over $X \ni x$ computes $\mathrm{mld}(X \ni x)$. The surface Z is klt by Lemma 2.15(1) and (2), the normalization of E is rational since the strict transform of E on the minimal resolution of X is a smooth rational curve [KM98, Theorem 4.7], and $0 < b < 1$. Thus, if $X \ncong X_{438}$, then $p < 115/134$ by Theorem 6.2.

Construction-Theorem 7.3. Notation and conditions as in Set-up 7.2. Then $K_Z + nE$ is semi-ample and big, $7/8 < n < b$, and the ample model $g : Z \to T$ of $K_Z + nE$ is birational, is not an isomorphism, and does not contract E. Let $E_T := g_*E$, $F := g^*E_T - E$, $\delta := E \cdot F$, and $\lambda := (K_T + E_T) \cdot E_T$, the *reduced adjunction number*. Then the following hold.

(1) The following hold.
  (a) T is klt.
  (b) (T, E_T) is lc.
  (c) $K_T + nE_T$ is ample.
  (d) The $\mathbb{Q}$-divisor $F \ge 0$ is g-exceptional with $\operatorname{Supp} F = \mathrm{Exc}(g)$.
  (e) Any g-exceptional curve intersects E.
  (f) $\lambda = -aq + (1 - n)\delta$.
  (g) $g^*(K_T + tE_T) = K_Z + tE + (t - n)F$ for any $t \in \mathbb{R}$.
  (h) $K_T + pE_T$ is pseudo-effective.
(2) If $\lambda \ge 0$, then $p > 20/23$.
(3) If $\lambda < 0$, let $q' := q - \delta$ and $b' := b + \delta(b - n)/q'$. Then the following hold.
  (a) $0 < \delta < aq/(1 - n) < q$.
  (b) $b < b' < 1$.
  (c) There exist a klt stable surface X' and birational morphisms $h : T \to X'$ and $f' : X \to X'$ with $h \circ g = f' \circ f$ such that h contracts exactly E_T.
  (d) $K_T + b'E_T = h^*K_{X'}$.

Proof. The proof is long, so we divide it into several steps. For any $t \in \mathbb{R}$, let $D_t := K_Z + tE$. We have $f^*K_X \cdot E = 0$. Since $K_Z + bE = f^*K_X$ and X is klt, (Z, bE) is klt [KM98, Lemma 2.30]. Thus Z is $\mathbb{Q}$-factorial [KM98, Proposition 4.11], and E and K_Z are $\mathbb{Q}$-Cartier.

Step 1. In this step, we prove $7/8 < n < b$, construct g, and prove the statements on T, (T, E_T), $K_T + nE_T$, and the g-exceptional curves in (1).

By Theorem 2.14(2) and (3), $q \ge 7/30$, $b > n$, and $K_X^2 \ge q(b - n)^2$. Since $a < 5/46$, we have $b > 41/46$. Suppose that $n \le 7/8$. Then $b - n > 41/46 - 7/8 > 0$, and

$$K_X^2 \ge q(b-n)^2 > \frac{7}{30}\left(\frac{41}{46} - \frac{7}{8}\right)^2 = \frac{21}{338560} > \frac{1}{48983},$$

where the last inequality holds because $21 \cdot 48983 = 1028643 > 338560$. This contradicts $K_X^2 \le 1/48983$. Thus $n > 7/8$.

Since $0 < n < b$ and (Z, bE) is klt, (Z, nE) is klt, and D_n is nef by the definition of n. Since Z is $\mathbb{Q}$-factorial, Corollary 7.1, applied to (Z, nE), implies that D_n is semi-ample, that the ample model $g : Z \to T$ of D_n is a morphism with connected fibers such that $D_n \sim_{\mathbb{R}} g^*A$ for an ample $\mathbb{R}$-divisor A on T, and that a curve C on Z is contracted by g if and only if $D_n \cdot C = 0$. Moreover,

$$D_n \cdot E = f^*K_X \cdot E - (b - n)E^2 = q(b - n) > 0.$$

If $\dim T = 0$, then $D_n \equiv 0$, which contradicts $D_n \cdot E > 0$. Suppose that $\dim T = 1$, and let G be a general fiber of g. Since $G \ne E$, f_*G is a curve, and since K_X is ample,

$$f^*K_X \cdot G = K_X \cdot f_*G > 0.$$

Since $f^*K_X = D_n + (b-n)E$ and $D_n \cdot G = 0$, we have $(b-n)(E\cdot G) = f^*K_X \cdot G > 0$, so $E\cdot G > 0$. Since $0 < n < 1$, Lemma 6.1 implies $n \le 2/3$, which contradicts $n > 7/8$. Thus $\dim T = 2$, g is birational, and $D_n^2 = A^2 > 0$, so D_n is big.

If g were an isomorphism, then D_n would be ample, and since the ample cone is open, $D_{n-\epsilon}$ would be ample for some real number $0 < \epsilon < n$, which contradicts the definition of n. Thus g is not an isomorphism. Since $D_n \cdot E > 0$, g does not contract E.

Since T is normal and g is a birational morphism that is not an isomorphism, a g-exceptional curve exists. Let C be a g-exceptional curve. Then $C \ne E$, so $f(C)$ is a curve, and since $D_n \cdot C = 0$,

$$(b-n)(E\cdot C) = f^*K_X \cdot C = K_X \cdot f_*C > 0,$$

so $E\cdot C > 0$. In particular, any g-exceptional curve intersects E. Moreover, $D_n \cdot C = 0$ implies $n = -(K_Z \cdot C)/(E\cdot C) \in \mathbb{Q}$.

Since D_n is big, Corollary 7.1 implies that the intersection matrix of the g-exceptional curves is negative definite, that $K_T + nE_T$ is ample, and that

$$g^*(K_T + nE_T) = D_n.$$

Since $n \in \mathbb{Q}$ and an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor is $\mathbb{Q}$-Cartier, $K_T + nE_T$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. Since (Z, nE) is klt, (T, nE_T) is klt [KM98, Lemma 2.30]. Thus T is $\mathbb{Q}$-factorial [KM98, Proposition 4.11], so E_T and K_T are $\mathbb{Q}$-Cartier and $C\cdot C' \ge 0$ for any two distinct curves C and C' on T. Moreover, T is klt [KM98, Corollary 2.35(1)]. Since (T, nE_T) is lc and $n > 5/6$, Theorem 2.16, applied to T and E_T, implies $\operatorname{lct}(T; E_T) = 1$. Since the log discrepancies of (T, tE_T) are affine functions of t, (T, E_T) is lc.

Step 2. In this step, we prove the statements on F, λ, $K_T + pE_T$, and $g^*(K_T + tE_T)$ in (1), and we compute the intersection numbers used in **Steps 3 and 4**.

Let $C_1, \dots, C_m$ be the g-exceptional curves, so that $m \ge 1$. Let $M := ((C_i \cdot C_j))_{1\le i,j\le m}$, and let $u := (u_1, \dots, u_m)^\top$, where $u_i := E\cdot C_i > 0$. The matrix M is negative definite by **Step 1**, and $C_i \cdot C_j \ge 0$ for $i \ne j$.

We show that $((-M)^{-1})_{ij} \ge 0$ for any i, j. Fix $1 \le j \le m$ and let $R := \sum_i ((-M)^{-1})_{ij} C_i$. Then $R\cdot C_k = -1$ if $k = j$ and $R \cdot C_k = 0$ if $k \ne j$. Let $\mu : W \to Z$ be a resolution. The curves contracted by $g\circ\mu$ are the strict transforms of $C_1, \dots, C_m$ and the μ-exceptional curves, and their intersection matrix is negative definite [KM98, Lemma 3.40]. We have $-\mu^*R \cdot \Gamma = 0$ for any μ-exceptional curve Γ and $-\mu^*R \cdot \mu_*^{-1}C_k = -R\cdot C_k \ge 0$ for any k. By [KM98, Lemma 3.41(1)], $\mu^*R \ge 0$. In particular, $((-M)^{-1})_{ij} \ge 0$ for any i.

Since $g_*(g^*E_T) = E_T = g_*E$, the $\mathbb{Q}$-divisor $F = g^*E_T - E$ is g-exceptional. Write $F = \sum_i r_i C_i$ and $r := (r_1, \dots, r_m)^\top$. Since $g^*E_T \cdot C_k = 0$ for any k, we have $\sum_i r_i (C_i \cdot C_k) = -u_k$, that is,

$$r = (-M)^{-1}u.$$

Since $-M$ is positive definite, $(-M)^{-1}$ is positive definite, so $((-M)^{-1})_{ii} > 0$ for any i. Since $((-M)^{-1})_{ij} \ge 0$ for any i, j and $u_i > 0$ for any i, we get

$$r_i \ge ((-M)^{-1})_{ii} u_i > 0$$

for any i. Thus $F \ge 0$ and $\operatorname{Supp} F = \operatorname{Exc}(g)$. Moreover,

$$\delta = E\cdot F = \sum_i u_i r_i = u^\top (-M)^{-1} u > 0,$$

because $(-M)^{-1}$ is positive definite and $u \ne 0$. Since $g^*E_T \cdot F = 0$, we have

$$E_T^2 = g^*E_T \cdot (E+F) = (E+F)\cdot E = -q + \delta. \tag{7.1}$$

For any $t \in \mathbb{R}$, since $g^*(K_T + nE_T) = D_n$ and $g^*E_T = E + F$,

$$g^*(K_T + tE_T) = D_n + (t-n)(E+F) = D_t + (t-n)F.$$

For any $t \in \mathbb{R}$, let $H_t := K_T + tE_T$. Since $g^*H_n = D_n$ and F is g-exceptional, $D_n \cdot F = 0$. Together with (7.1), we obtain

$$\begin{aligned} H_n \cdot E_T &= D_n \cdot (E+F) = q(b-n), \\ H_b \cdot E_T &= H_n \cdot E_T + (b-n)E_T^2 = \delta(b-n), \\ \lambda = H_1 \cdot E_T &= H_n \cdot E_T + (1-n)E_T^2 = -aq + (1-n)\delta, \\ H_n^2 &= D_n^2 = K_X^2 - q(b-n)^2. \end{aligned} \tag{7.2}$$

Here we use $a = 1-b$ in the third line and $D_n^2 = (f^*K_X)^2 - 2(b-n)(f^*K_X \cdot E) + (b-n)^2E^2$ in the fourth line.

Since $D_t = D_p + (t-p)E$ for $t \geq p$ and the pseudo-effective cone is closed, D_p is pseudo-effective. Thus $H_p = g_*D_p$ is pseudo-effective, because g_* preserves numerical equivalence by the projection formula and $g_*D \geq 0$ for any $\mathbb{R}$-divisor $D \geq 0$.

Step 3. In this step, we prove (2). Assume that $\lambda \geq 0$.

For any curve C on T other than E_T, since H_n is ample and $E_T \cdot C \geq 0$,

$$H_1 \cdot C = H_n \cdot C + (1-n)(E_T \cdot C) > 0.$$

Since $H_1 \cdot E_T = \lambda \geq 0$, H_1 is nef. Since $H_1 = H_n + (1-n)E_T$, H_n is ample, and $(1-n)E_T \geq 0$, H_1 is big, so $H_1^2 > 0$.

Suppose that $\lambda > 0$. Then $H_1 \cdot C > 0$ for any curve C on T and $H_1^2 > 0$. Since T is $\mathbb{Q}$-factorial, some multiple of H_1 is Cartier, and the Nakai–Moishezon criterion [KM98, Theorem 1.37], applied to it, implies that $H_1 = K_T + E_T$ is ample. Since (T, E_T) is a projective lc surface pair and $E_T \neq 0$ is a Weil divisor, Theorem 1.2(2) implies

$$H_1^2 \geq \frac{1}{462}.$$

Suppose that $\lambda = 0$. By Corollary 7.1, applied to (T, E_T), the ample model $h_0 : T \to Y$ of H_1 is birational, and a curve C on T is contracted by h_0 if and only if $H_1 \cdot C = 0$, that is, if and only if $C = E_T$. Thus h_0 contracts exactly E_T, K_Y is ample, and $K_T + E_T = h_0^*K_Y$. Since $K_T + E_T = h_0^*K_Y$ and (T, E_T) is lc, Y is lc, and $a(E_T, Y, 0) = a(E_T, T, E_T) = 0$ [KM98, Lemma 2.30], so Y is not klt. By Theorem 1.3(1),

$$H_1^2 = K_Y^2 \geq \frac{1}{825}.$$

Thus $H_1^2 \geq 1/825$. For a nef and big $\mathbb{Q}$-divisor H on T, we have $\mathrm{vol}(H) = H^2$: this follows from Lemma 2.5(2), applied to a resolution of T, and Lemma 2.5(3), applied to the pullback of H, which is nef and big, so its Zariski decomposition has zero negative part. Suppose that $p \leq 20/23$, and let $p_0 := 20/23$. Then $H_{p_0} = H_p + (p_0 - p)E_T$ is pseudo-effective. Let $\alpha := (n-p_0)/(1-p_0)$. Since $n \in \mathbb{Q}$ and $p_0 < 7/8 < n < 1$, α is a rational number with $0 < \alpha < 1$, and $H_n = \alpha H_1 + (1-\alpha)H_{p_0}$. By Lemma 2.5(1),

$$H_n^2 = \mathrm{vol}(H_n) \geq \mathrm{vol}(\alpha H_1) = \alpha^2 H_1^2 \geq \frac{(n-p_0)^2}{825(1-p_0)^2}.$$

By (7.2), the Cauchy–Schwarz inequality, $b > 41/46$, and $1/q \leq 30/7$,

$$K_X^2 = q(b-n)^2 + H_n^2 \geq \frac{(b-p_0)^2}{1/q + 825(1-p_0)^2} > \frac{(41/46 - 20/23)^2}{30/7 + 825(1-20/23)^2} = \frac{7}{271380} > \frac{1}{48983},$$

because $7 \cdot 48983 = 342881 > 271380$. This contradicts $K_X^2 \leq 1/48983$, so $p > 20/23$. This proves (2).

Step 4. In this step, we prove (3). Assume that $\lambda < 0$.

By (7.2), $(1-n)\delta < aq$. Since $a = 1-b < 1-n$ and $\delta > 0$ by **Step 2**,

$$0 < \delta < \frac{aq}{1-n} < q.$$

Thus $q' = q - \delta > 0$, and $E_T^2 = -q'$ by (7.1). Since $\delta > 0$ and $b - n > 0$, we have $b' > b$. Moreover,

$$-(1-b')q' = -a(q-\delta) + \delta(b-n) = -aq + (1-n)\delta = \lambda < 0,$$

so $b' < 1$.

By (7.2),

$$H_{b'} \cdot E_T = H_b \cdot E_T + (b'-b)E_T^2 = \delta(b-n) - \frac{\delta(b-n)}{q'}q' = 0.$$

For any curve C on T other than E_T, since $H_{b'} = H_n + (b'-n)E_T$, $b' > n$, and H_n is ample, $H_{b'} \cdot C \geq H_n \cdot C > 0$. Thus $H_{b'}$ is nef, and $H_{b'}$ is big because H_n is ample and $(b'-n)E_T \geq 0$. Since T is klt, (T, E_T) is lc, and $0 < b' < 1$, the log discrepancies $a(\cdot, T, b'E_T) = (1-b')a(\cdot, T, 0) + b'a(\cdot, T, E_T)$ are positive, so $(T, b'E_T)$ is klt. By Corollary 7.1, applied to $(T, b'E_T)$, the ample model $h : T \to X'$ of $H_{b'}$ is birational, and a curve C on T is contracted by h if and only if $H_{b'} \cdot C = 0$, that is, if and only if $C = E_T$. Thus h contracts exactly E_T, $K_{X'}$ is ample, and

$$K_T + b'E_T = h^* K_{X'}.$$

Since $(T, b'E_T)$ is klt, X' is klt [KM98, Lemma 2.30]. Thus X' is a klt stable surface.

Since X is normal and f is birational, $f_*\mathcal{O}_Z = \mathcal{O}_X$. The only positive-dimensional fiber of f is E, and $h(g(E)) = h(E_T)$ is a point, so $h \circ g$ contracts each fiber of f. By the rigidity lemma [Deb01, Lemma 1.15(b)], there exists a morphism $f' : X \to X'$ such that $h \circ g = f' \circ f$. Since X is projective and $h \circ g$ and f are birational, f' is a projective birational morphism.

This proves (3). □

8. Negative reduced adjunction

The goal of this section is to prove Theorem 8.1, which is proved at the end of Subsection 8.4.

Theorem 8.1. *Let X be a klt stable surface with $K_X^2 \leq 1/48983$ and $\mathrm{mld}(X) < 5/46$. Assume that X is a minimizer. Then $X \cong X_{438}$.*

In the notation of Set-up 7.2 and Construction-Theorem 7.3, the case $\lambda \geq 0$ follows from Theorem 6.2. For negative λ, we compare X with a surface Y over the surface X' of Construction-Theorem 7.3(3)(c), and Lemma 8.7 states the conditions on Y which each of the three configurations after Definition-Proposition 8.2 has to provide.

8.1. Structure of the contraction.

In this subsection, we describe the curves contracted by f' on the minimal resolution of X, and we prove the lemmas which reduce Theorem 8.1 to constructing a surface Y over X'.

Definition-Proposition 8.2. Notation and conditions as in Set-up 7.2 and Construction-Theorem 7.3, and assume that $\lambda < 0$. Let $x_0 := f(E)$, $x_0' := h(E_T)$, and let $C_1, \dots, C_m$ be the g-exceptional prime divisors. Let $\pi : \widetilde{X} \to X$ be the minimal resolution, $\widetilde{E} \subset \widetilde{X}$ the strict transform of E, and $D_i \subset \widetilde{X}$ the strict transform of $f(C_i)$ for $1 \leq i \leq m$. Then the following hold.

(1) The exceptional curves of f' are $f(C_1), \dots, f(C_m)$, $x_0 \in f(C_i)$ for any i, and $m \geq 1$.
(2) π factors through f, and the induced morphism $\pi_Z : \widetilde{X} \to Z$ is the minimal resolution of Z. The curves $D_1, \dots, D_m$ are pairwise disjoint smooth rational curves with $D_i^2 = -1$.
(3) The curves on $\widetilde{X}$ mapped to x_0' form a simple normal crossing divisor with dual graph $\mathcal{D}$. The graph $\mathcal{D}$ is a tree with at most one vertex of degree at least three, and such a vertex has degree three. Any D_i is a vertex of degree two in $\mathcal{D}$, and no path in $\mathcal{D}$ from $\widetilde{E}$ to D_i contains D_j for $j \neq i$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

By Construction-Theorem 7.3(1)(g) with $t = n$, we have $K_Z + nE = g^*(K_T + nE_T)$, so $(K_Z + nE) \cdot C_i = 0$ for any i. Since $C_i \neq E$ and f contracts only E, $f(C_i)$ is a curve. By Construction-Theorem 7.3(1)(e), $E \cap C_i \neq \emptyset$, so $E \cdot C_i > 0$, and

$$K_Z \cdot C_i = -n(E \cdot C_i) < 0. \tag{8.1}$$

Thus $x_0 \in f(C_i)$, $g(C_i) \in E_T$, and $(f' \circ f)(C_i) = (h \circ g)(C_i) = x_0'$. Conversely, let Σ be a curve on X contracted by f', and Σ_Z the strict transform of Σ on Z. Then $\Sigma_Z \neq E$ and $(h \circ g)(\Sigma_Z)$ is a point. If $g(\Sigma_Z)$ were a curve, then $g(\Sigma_Z) = E_T$ because h contracts exactly E_T, and $\Sigma_Z = E$ because g does not contract E, a contradiction. Thus Σ_Z is g-exceptional, that is, $\Sigma = f(C_i)$ for some i. Therefore the exceptional curves of f' are $f(C_1), \dots, f(C_m)$, and all of them are mapped to x_0'. Since g is not an isomorphism, $m \geq 1$.

Step 2. In this step, we prove (2).

By Definition 2.13, $\widetilde{E}$ is a π-exceptional curve. Let V be a smooth projective surface with birational morphisms $v_1 : V \to \widetilde{X}$ and $v_2 : V \to Z$ such that $\pi \circ v_1 = f \circ v_2$. Let Σ be a curve on V contracted by v_1. Then $f(v_2(\Sigma))$ is a point, so $v_2(\Sigma)$ is a point or $v_2(\Sigma) = E$. If $v_2(\Sigma) = E$, then Σ is the strict transform of E on V, so $v_1(\Sigma) = \widetilde{E}$ is a curve, a contradiction. Thus v_2 contracts any curve contracted by v_1. Since $\widetilde{X}$ is normal, the fibers of v_1 are connected [Har77, Corollary III.11.4], so v_2 factors through v_1 by the rigidity lemma [Deb01, Lemma 1.15(b)]. We obtain a birational morphism $\pi_Z : \widetilde{X} \to Z$ with $\pi = f \circ \pi_Z$, and the following commutative diagram:

$$\begin{array}{ccccc} \widetilde{X} & \xrightarrow{\pi_Z} & Z & \xrightarrow{g} & T \\ & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle h} \\ & & X & \xrightarrow{f'} & X'. \end{array}$$

The π_Z-exceptional curves are the π-exceptional curves other than $\widetilde{E}$. Since π is the minimal resolution, none of them is a (-1)-curve. Thus π_Z is the minimal resolution of Z.

Since D_i is not π_Z-exceptional and $f(\pi_Z(D_i)) = f(C_i)$, we have $\pi_Z(D_i) = C_i$. Write

$$\pi_Z^* K_Z = K_{\widetilde{X}} + \sum_G \beta_G G,$$

where G runs over the π_Z-exceptional curves. Then $\beta_G \geq 0$ for any G [KM98, Corollary 4.3], and $G \cdot D_i \geq 0$ for any G. By the projection formula and (8.1),

$$K_{\widetilde{X}} \cdot D_i = K_Z \cdot C_i - \sum_G \beta_G (G \cdot D_i) \leq K_Z \cdot C_i < 0.$$

Since D_i is contracted by $f' \circ \pi$, we have $D_i^2 < 0$ [KM98, Lemma 3.40]. Both $K_{\widetilde{X}} \cdot D_i$ and D_i^2 are integers, so by the adjunction formula [Har77, Exercise V.1.3(a)],

$$2p_a(D_i) - 2 = D_i^2 + K_{\widetilde{X}} \cdot D_i \leq -2.$$

Since $p_a(D_i) \geq 0$, we have $p_a(D_i) = 0$ and $D_i^2 = K_{\widetilde{X}} \cdot D_i = -1$. Thus $D_i \cong \mathbb{P}^1$ [Har77, Exercise IV.1.8(b)]. For $i \neq j$, the matrix

$$\begin{pmatrix} -1 & D_i \cdot D_j \\ D_i \cdot D_j & -1 \end{pmatrix}$$

is negative definite [KM98, Lemma 3.40], so $1 - (D_i \cdot D_j)^2 > 0$. Since $D_i \cdot D_j$ is a non-negative integer, $D_i \cdot D_j = 0$, that is, $D_i \cap D_j = \emptyset$.

Step 3. In this step, we prove that any curve G on $\widetilde{X}$ mapped to x_0' satisfies

$$0 < a(G, X', 0) \leq 1. \tag{8.2}$$

By Construction-Theorem 7.3(1)(g) with $t = b'$ and Construction-Theorem 7.3(3)(d),

$$(h \circ g)^* K_{X'} = g^*(K_T + b'E_T) = K_Z + b'E + (b' - n)F.$$

Since $h \circ g = f' \circ f$ and $f^*K_X = K_Z + bE$, we have $f^*(f'^*K_{X'} - K_X) = (b'-b)E + (b'-n)F$. Pushing forward by f, we obtain

$$f'^*K_{X'} = K_X + (b'-n)f_*F, \qquad f_*F = \sum_{i=1}^{m} (\mathrm{mult}_{C_i} F) f(C_i).$$

Since $b' > b > n$ and $F \geq 0$, we have $(b'-n)f_*F \geq 0$. Pulling back by π, we obtain $(f' \circ \pi)^*K_{X'} = \pi^*K_X + (b'-n)\pi^*f_*F$ with $\pi^*f_*F \geq 0$. The curves on $\widetilde{X}$ mapped to x_0' are the π-exceptional curves mapped to x_0' and the curves $D_1, \ldots, D_m$. Comparing coefficients, for a π-exceptional curve G mapped to x_0' we have

$$a(G, X', 0) = a(G, X, 0) - (b'-n)\,\mathrm{mult}_G\, \pi^*f_*F \leq a(G, X, 0) \leq 1,$$

where the last inequality holds because the discrepancies of the minimal resolution are non-positive [KM98, Corollary 4.3]. For D_i we have

$$a(D_i, X', 0) = 1 - (b'-n)\,\mathrm{mult}_{C_i} F \leq 1.$$

Since X' is klt, $a(G, X', 0) > 0$ for any such G. This proves (8.2).

Step 4. In this step, we prove the first two sentences of (3) and the first clause of the third.

While there exists a (-1)-curve on $\widetilde{X}$ mapped to x_0', we contract it [Har77, Theorem V.5.7], and we repeat the procedure on the new surface. Since the number of curves mapped to x_0' decreases, we obtain a sequence

$$\widetilde{X} = W_k \to W_{k-1} \to \cdots \to W_0$$

in which $W_j \to W_{j-1}$ is the blow-up of a smooth point y_j of W_{j-1}, with exceptional curve G_j' [Har77, Theorem V.5.7]. Since $f' \circ \pi$ contracts the exceptional curve of $W_j \to W_{j-1}$ and $W_j \to W_{j-1}$ has connected fibers, the morphism $W_j \to X'$ factors through W_{j-1} [Deb01, Lemma 1.15(b)], and y_j is mapped to x_0'. The morphism $W_0 \to X'$ is a resolution, and by construction no curve on W_0 mapped to x_0' is a (-1)-curve. Thus $W_0 \to X'$ is the minimal resolution of X' near x_0'.

If X' were smooth at x_0', then $W_0 \to X'$ would be an isomorphism near x_0', and $k \geq 1$ because $\widetilde{E}$ is mapped to x_0'. Then $y_1 = x_0'$, and $a(G_1', X', 0) = 2$ [KM98, Lemma 2.29], contradicting (8.2). Thus $X' \ni x_0'$ is a klt surface germ which is not smooth, and by [KM98, Theorem 4.7], the curves on W_0 mapped to x_0' are smooth rational curves forming a simple normal crossing divisor whose dual graph is a chain, or a tree with exactly one vertex of degree at least three, and that vertex has degree three. Since no curve on W_0 mapped to x_0' is a (-1)-curve, the self-intersection of any curve on W_0 mapped to x_0' is at most -2.

We show by induction on j that the curves on W_j mapped to x_0' are smooth rational curves forming a simple normal crossing divisor whose dual graph is obtained by subdividing edges of the dual graph of W_0 over x_0'. The case $j = 0$ is proved above. Let $j \geq 1$ and write

$$K_{W_{j-1}} + B_{j-1} = (\text{the pullback of } K_{X'}) \quad \text{over a neighborhood of } x_0',$$
$$B_{j-1} = \sum_G (1 - a(G, X', 0))G,$$

where G runs over the curves on W_{j-1} mapped to x_0'. By [KM98, Lemmas 2.29 and 2.30],

$$a(G_j', X', 0) = a(G_j', W_{j-1}, B_{j-1}) = 2 - \sum_G (1 - a(G, X', 0))\,\mathrm{mult}_{y_j} G.$$

If y_j is contained in at most one curve G mapped to x_0', then, since G is smooth, $a(G_j', X', 0) \geq 1 + a(G, X', 0) > 1$ or $a(G_j', X', 0) = 2$. The strict transform of G_j' on $\widetilde{X}$ is a curve mapped to x_0', so we obtain a contradiction to (8.2). By the induction hypothesis, y_j is the intersection point of exactly two curves G and H mapped to x_0', which intersect transversally at y_j. On W_j, the strict transforms of G and H are disjoint near G_j', and G_j' intersects each of them transversally

at one point and intersects no other curve mapped to x_0'. Thus the dual graph on W_j is the dual graph on W_{j-1} with the edge between G and H subdivided by the vertex G_j', and the induction is complete.

Subdividing an edge creates a vertex of degree two and does not change the degree of any other vertex. Thus $\mathcal{D}$ is a tree with at most one vertex of degree at least three, that vertex has degree three, and any vertex which is not the strict transform of a curve on W_0 has degree two. The self-intersection of the strict transform of a curve does not increase under the blow-up of a point [Har77, Proposition V.3.6], so the strict transforms on $\widetilde{X}$ of the curves on W_0 have self-intersection at most -2. Since $D_i^2 = -1$ by (2), D_i is not the strict transform of a curve on W_0, so D_i has degree two.

Step 5. In this step, we prove that no path in $\mathcal{D}$ from $\widetilde{E}$ to D_i contains D_j for $j \neq i$.

Fix i. By Construction-Theorem 7.3(1)(e), there exists a point $z \in E \cap C_i$. The fiber $\pi_Z^{-1}(z)$ is connected [Har77, Corollary III.11.4], and any curve in $\pi_Z^{-1}(z)$ is π_Z-exceptional and mapped to $(h \circ g)(z) = x_0'$. Since $\pi_Z(\widetilde{E}) = E$ and $\pi_Z(D_i) = C_i$ both contain z, the curves $\widetilde{E}$ and D_i both intersect $\pi_Z^{-1}(z)$. If $\pi_Z^{-1}(z)$ is a point, then $\widetilde{E}$ and D_i intersect. Otherwise $\pi_Z^{-1}(z)$ is a connected union of curves, and $\widetilde{E}$ and D_i intersect curves in $\pi_Z^{-1}(z)$. In both cases, $\mathcal{D}$ contains a path from $\widetilde{E}$ to D_i whose vertices other than $\widetilde{E}$ and D_i are π_Z-exceptional. Since $\mathcal{D}$ is a tree, the path from $\widetilde{E}$ to D_i is unique, and since D_j is not π_Z-exceptional for $j \neq i$, the path does not contain D_j. This proves (3). □

Definition 8.3. Let Z be a normal surface, E and C distinct prime divisors on Z, and $M \geq 2$ an integer. We say that a point $z \in E \cap C$ is a *cyclic quotient crossing* of E and C of order M if the analytic germ $(Z \ni z, E + C)$ is isomorphic to the quotient of $(\mathbb{C}^2 \ni 0, (\xi_1\xi_2 = 0))$ by a cyclic group of order M acting diagonally and freely outside the origin, with E and C the images of the coordinate axes.

For any i, the path in $\mathcal{D}$ from $\widetilde{E}$ to D_i leads to one of the following three configurations (see Lemma 8.9 and Proposition 8.10(2)): C_i intersects E at a smooth point of Z (Subsection 8.2), C_i intersects E at a cyclic quotient crossing (Subsection 8.3), or the path has an interior vertex of degree three (Subsection 8.4).

Lemma 8.4. *Let $\Gamma = \sum_{i=1}^k G_i$ be a reduced divisor on a smooth surface whose components are smooth rational curves with $G_i^2 \leq -2$ intersecting transversally, such that the dual graph of Γ is a tree and the intersection matrix of Γ is negative definite. Let $A := (-G_i \cdot G_j)_{1 \leq i,j \leq k}$, let $\alpha \in \mathbb{Q}^k$ be the solution of $A\alpha = (2 - d_i)_{1 \leq i \leq k}$, where d_i is the degree of G_i in the dual graph, and let $L_i := (A^{-1})_{ii}$. For a non-empty subset $I \subset \{1, \ldots, k\}$, the determinant of $\sum_{i \in I} G_i$ is $\det(-G_i \cdot G_j)_{i,j \in I}$. Then the following hold.*

(1) $(A^{-1})_{ij} > 0$ for all $1 \leq i, j \leq k$.
(2) If Γ is a chain with end curve G_1, $r := \det A$, and r' is the determinant of the chain $G_2 + \cdots + G_k$, with $r' := 1$ if $k = 1$, then $L_1 = r'/r$ and $\alpha_1 = (r' + 1)/r$.
(3) If Γ is a chain, G_1 is not an end curve, and r_1, r_2 are the determinants of the chains of $\Gamma - G_1$, then $\alpha_1 = L_1(1/r_1 + 1/r_2)$.
(4) If G_k is the only vertex of degree at least three in the dual graph of Γ, G_k has degree three, G_1 is an end curve, and r_1, r_2 are the determinants of the chains of $\Gamma - G_k$ that do not contain G_1, then $\alpha_1 = L_1 + (A^{-1})_{1k}(1/r_1 + 1/r_2 - 1)$.
(5) If Γ is a chain with end curve G_1, $G_j \cdot G_{j+1} = 1$ for $1 \leq j \leq k-1$, and ρ_j is the determinant of $G_{j+1} + \cdots + G_k$ for $1 \leq j < k$, with $\rho_k := 1$, then $(A^{-1})_{1j} = \rho_j / \det A$ for any j.
(6) If Γ is a connected component of the exceptional locus of the minimal resolution of a klt surface Z, then $\alpha_i = a(G_i, Z, 0)$ for any i.

Proof. Since the intersection matrix of Γ is negative definite, A is positive definite, and $\det(-G_i \cdot G_j)_{i,j\in I} > 0$ for any non-empty subset $I \subset \{1,\dots,k\}$. Since A is symmetric, so is A^{-1}.

(1) Fix j and let $D := -\sum_{i=1}^{k}(A^{-1})_{ij}G_i$. Then $D \cdot G_l = \sum_i (A^{-1})_{ij}A_{il} = (A^{-1}A)_{jl}$, which is 1 if $l = j$ and 0 otherwise. Since the dual graph of Γ is connected and $D \neq 0$, all coefficients of D are negative [KM98, Lemma 3.41], that is, $(A^{-1})_{ij} > 0$ for any i.

We use the following two facts. First, let $\Delta = \sum_{i=1}^{\ell} H_i$ be a chain with $H_i \cdot H_{i+1} = 1$ for $1 \le i \le \ell - 1$ and $B := (-H_i \cdot H_j)_{1\le i,j\le \ell}$. The minor of B obtained by deleting the last row and the first column is triangular with diagonal entries -1, so the cofactor at the end curve is

$$(B^{-1})_{1\ell} = \frac{(-1)^{\ell+1}(-1)^{\ell-1}}{\det B} = \frac{1}{\det B}. \tag{8.3}$$

Second, let G_c be a component of Γ, and let $\Delta = \sum_{i=1}^{\ell} H_i$ be a connected component of $\Gamma - G_c$ which is a chain with $H_i \cdot H_{i+1} = 1$ for $1 \le i \le \ell - 1$ such that H_1 is the only component of Δ intersecting G_c. Let $y \in \mathbb{Q}^k$ be a vector such that $(Ay)_l = 0$ for any G_l in Δ, and let y_Δ be the vector of entries of y at $H_1,\dots,H_\ell$. Since the dual graph is a tree, $(Ay)_l = (By_\Delta)_l - y_c$ if $G_l = H_1$, and $(Ay)_l = (By_\Delta)_l$ for any other G_l in Δ. Thus $y_\Delta = y_c B^{-1}e_1$, and by (8.3), the entry of y at H_ℓ is

$$y_c(B^{-1})_{\ell 1} = \frac{y_c}{\det B}. \tag{8.4}$$

Let $y := A^{-1}e_1$. Then $y_i = (A^{-1})_{1i}$, $y_1 = L_1$, $(Ay)_l = 0$ for $l \neq 1$, and

$$\alpha_1 = \sum_{i=1}^{k}(A^{-1})_{1i}(2 - d_i) = \sum_{i=1}^{k} y_i(2 - d_i).$$

(2) Renumbering $G_2,\dots,G_k$, we may assume that $G_i \cdot G_{i+1} = 1$ for $1 \le i \le k - 1$. The minor of A obtained by deleting the first row and the first column is the matrix of $G_2 + \cdots + G_k$, so $L_1 = r'/r$. If $k = 1$, then $d_1 = 0$ and $\alpha_1 = 2L_1 = 2/r = (r' + 1)/r$. If $k \ge 2$, then $2 - d_i = 1$ for $i \in \{1, k\}$ and $2 - d_i = 0$ otherwise, so by (8.3),

$$\alpha_1 = (A^{-1})_{11} + (A^{-1})_{1k} = \frac{r'}{r} + \frac{1}{r} = \frac{r' + 1}{r}.$$

(3) Since G_1 is not an end curve, $d_1 = 2$, and $\Gamma - G_1$ consists of two chains Δ_1 and Δ_2 with determinants r_1 and r_2, each intersecting G_1 at one end curve. The end curves of Γ are the curves H_ℓ of Δ_1 and Δ_2 in the numbering of (8.4) with $c = 1$ (so $H_\ell = H_1$ if Δ_ν has one component), and $2 - d_i = 0$ for any other G_i. By (8.4) with $c = 1$, the entries of y at the end curves of Γ are L_1/r_1 and L_1/r_2. Thus $\alpha_1 = L_1(1/r_1 + 1/r_2)$.

(4) Since G_k is the only vertex of degree at least three and has degree three, $\Gamma - G_k$ consists of three chains, and each of them intersects G_k at one end curve. Let Δ_1 and Δ_2 be the chains not containing G_1, with determinants r_1 and r_2. The vertices of degree one are G_1 and the curves H_ℓ of Δ_1 and Δ_2 in the numbering of (8.4) with $c = k$ (so $H_\ell = H_1$ if Δ_ν has one component), and $2 - d_k = -1$, while $2 - d_i = 0$ for any other G_i. By (8.4) with $c = k$, the entries of y at the curves H_ℓ of Δ_1 and Δ_2 are y_k/r_1 and y_k/r_2, where $y_k = (A^{-1})_{1k}$. Thus

$$\alpha_1 = L_1 + \frac{(A^{-1})_{1k}}{r_1} + \frac{(A^{-1})_{1k}}{r_2} - (A^{-1})_{1k} = L_1 + (A^{-1})_{1k}\Big(\frac{1}{r_1} + \frac{1}{r_2} - 1\Big).$$

(5) Deleting the j-th row and the first column of A, we get a block lower triangular matrix whose diagonal blocks are a lower triangular matrix of size $j - 1$ with diagonal entries -1 and the matrix of $G_{j+1} + \cdots + G_k$. Thus the corresponding minor is $(-1)^{j-1}\rho_j$, and Cramer's rule implies $(A^{-1})_{1j} = \rho_j / \det A$.

(6) Let $\sigma : \widetilde{Z} \to Z$ be the minimal resolution, and write $\sigma^* K_Z = K_{\widetilde{Z}} + \sum_i (1 - a(G_i, Z, 0))G_i$ near Γ. Intersecting with G_j and using $K_{\widetilde{Z}} \cdot G_j = -2 - G_j^2$, we see that the vector $(a(G_i, Z, 0))_i$ solves the equation defining α. Since A is invertible, $\alpha_i = a(G_i, Z, 0)$ for any i. □

Lemma 8.5 is used in Proposition 8.10, for the bound $t > 63/73$, and in Subsection 8.4.

Lemma 8.5. *Notation and conditions as in Set-up 7.2 and Construction-Theorem 7.3. Then* $1 - n < 5/44$.

Proof. By Theorem 2.14(3), $q(b-n)^2 \le K_X^2 \le 1/48983$. Since $b = 1-a$, we have $1-n = a+(b-n)$.

If $X \ni f(E)$ is not of type E-II, then $q > 1$ by Theorem 2.14(2), so $b - n < 1/\sqrt{48983}$. Since $5/44 - 5/46 = 5/1012$ and $1012^2 = 1024144 < 1224575 = 25 \cdot 48983$, we have

$$1 - n < \frac{5}{46} + \frac{1}{\sqrt{48983}} < \frac{5}{44}.$$

Assume that $X \ni f(E)$ is of type $[e_0; (r_1, q_1), (r_2, q_2), (r_3, q_3)]$ and of type E-II. Then $e_0 = 2$ and $r_2 = 3$, so by Lemma 2.10, $(r_1, r_2, r_3) = (2, 3, r_3)$ with $r_3 \in \{3, 4, 5\}$. Moreover, $q_1 = 1$, $q_2 \in \{1, 2\}$, and q_3 is an integer coprime to r_3 with $1 \le q_3 < r_3$. By Definition 2.13, the strict transform of E on the minimal resolution of $X \ni f(E)$ is the fork, so by Lemma 2.10, $a = \sigma/\eta$, where

$$\sigma = \frac{1}{2} + \frac{1}{3} + \frac{1}{r_3} - 1 = \frac{6 - r_3}{6r_3} \quad \text{and} \quad \eta = 2 - \frac{1}{2} - \frac{q_2}{3} - \frac{q_3}{r_3}.$$

Thus

$$\frac{1}{a} = \frac{\eta}{\sigma} = \frac{9r_3 - 2q_2r_3 - 6q_3}{6 - r_3}.$$

A direct calculation implies that $1/a \in \{1, 3, 5\}$ if $r_3 = 3$, $1/a \in \{1, 5, 7, 11\}$ if $r_3 = 4$, and $1/a \in \{1, 7, 11, 13, 17, 19, 23, 29\}$ if $r_3 = 5$. Since $a < 5/46$, we have $1/a > 46/5 > 9$, so $1/a \ge 11$, that is, $a \le 1/11$. By Theorem 2.14(2), $q \ge 7/30$, so $(b-n)^2 \le 30/(7 \cdot 48983)$. Since $7 \cdot 48983 = 342881 > 300000$, we have $b - n < 1/100$, and

$$1 - n < \frac{1}{11} + \frac{1}{100} = \frac{111}{1100} < \frac{125}{1100} = \frac{5}{44}.$$

□

Lemma 8.6. *Let X and Y be projective klt surfaces with K_X ample. Let $f : V \to X$ and $g : V \to Y$ be birational morphisms from a smooth projective surface V. Assume that $f^*K_X - g^*K_Y \ge 0$ and $f^*K_X \ne g^*K_Y$. If K_Y is big, then $\mathrm{vol}(K_Y) < K_X^2$, and there exists a klt stable surface Y' with $K_{Y'}^2 = \mathrm{vol}(K_Y)$.*

Proof. Since Y is klt, Y is $\mathbb{Q}$-factorial [KM98, Proposition 4.11], so $D := g^*K_Y$ is a $\mathbb{Q}$-divisor on V. Let $P := f^*K_X$ and $G := P - D$. Then $G \ge 0$, $G \ne 0$, P is nef, and $P^2 = K_X^2 > 0$.

Assume that K_Y is big. By Lemma 2.5(2), $\mathrm{vol}(D) = \mathrm{vol}(K_Y) > 0$, so D is big. Let $D = P_1 + N_1$ be the Zariski decomposition [Laz04, Theorem 2.3.19], where P_1 is nef and $N_1 \ge 0$. By Lemma 2.5(3), $\mathrm{vol}(K_Y) = P_1^2$. Let $F_1 := P - P_1 = G + N_1$. Then $F_1 \ge 0$ and $F_1 \ne 0$. Since P and P_1 are nef,

$$P^2 - P_1^2 = P \cdot F_1 + P_1 \cdot F_1 \ge 0.$$

Suppose that $P^2 = P_1^2$. Then $P \cdot F_1 = P_1 \cdot F_1 = 0$, so $F_1^2 = P \cdot F_1 - P_1 \cdot F_1 = 0$. Since $P^2 > 0$, the intersection form is negative definite on the classes ξ with $P \cdot \xi = 0$ by the Hodge index theorem [Har77, Theorem V.1.9 and Remark V.1.9.1], so $F_1 \equiv 0$. This is not possible, because $F_1 \cdot H_V > 0$ for an ample divisor H_V on V. Thus

$$0 < \mathrm{vol}(K_Y) = P_1^2 < P^2 = K_X^2.$$

Since Y is klt and K_Y is big, Theorem 2.4 applied to $(Y, 0)$ provides a sequence $Y = Y_0 \to Y_1 \to \cdots \to Y_l$ of steps of the K_Y-MMP such that K_{Y_l} is semi-ample. Each step $c_i : Y_{i-1} \to Y_i$ is the divisorial contraction of a $K_{Y_{i-1}}$-negative extremal ray (no step is of fiber type since K_Y is pseudo-effective), so Y_i is klt [KM98, Corollary 3.43(1)], hence $\mathbb{Q}$-factorial [KM98, Proposition 4.11]. If Σ_i is the curve contracted by c_i, then $K_{Y_{i-1}} = c_i^*K_{Y_i} + \epsilon_i\Sigma_i$ for some rational number ϵ_i, as K_{Y_i} is $\mathbb{Q}$-Cartier, and $\epsilon_i\Sigma_i^2 = K_{Y_{i-1}} \cdot \Sigma_i < 0$ with $\Sigma_i^2 < 0$, so $\epsilon_i > 0$. By Lemma 2.5(2), $\mathrm{vol}(K_{Y_{i-1}}) = \mathrm{vol}(K_{Y_i})$. Since K_{Y_l} is semi-ample and big, it defines a birational

morphism $c : Y_l \to Y'$ with $K_{Y_l} = c^*K_{Y'}$ and $K_{Y'}$ ample. Since c is crepant, Y' is klt [KM98, Lemma 2.30], and $K^2_{Y'} = \mathrm{vol}(K_{Y_l}) = \mathrm{vol}(K_Y)$ by Lemma 2.5(2). Thus Y' is a klt stable surface with $K^2_{Y'} = \mathrm{vol}(K_Y)$. □

Lemma 8.7. *Notation and conditions as in Set-up 7.2 and Construction-Theorem 7.3. Assume that $X \ncong X_{438}$. Let Y be a projective klt surface, and assume the following.*

(1) There exist birational morphisms $\phi_X : V \to X$ and $\phi_Y : V \to Y$ from a smooth projective surface V with $\phi_X^*K_X - \phi_Y^*K_Y \geq 0$ and $\phi_X^*K_X \neq \phi_Y^*K_Y$.

(2) There exist a rational number t with $115/134 < t < n$, a normal projective surface V', and birational morphisms $w : V' \to Z$ and $u : V' \to Y$ with $u^*K_Y - (g \circ w)^*(K_T + tE_T) \geq 0$.

Then X is not a minimizer.

Proof. By Set-up 7.2, $p < 115/134 < t$.

Since T is klt, T is $\mathbb{Q}$-factorial [KM98, Proposition 4.11]. By Construction-Theorem 7.3(1)(c) and (h), $K_T + nE_T$ is ample and $K_T + pE_T$ is pseudo-effective. Since $p < t < n$,

$$K_T + tE_T = \frac{n-t}{n-p}(K_T + pE_T) + \frac{t-p}{n-p}(K_T + nE_T)$$

is the sum of a pseudo-effective divisor and an ample divisor, so $K_T + tE_T$ is big [Laz04, Theorem 2.2.26]. By [Laz04, Proposition 2.2.43], (2), and Lemma 2.5(1),

$$\mathrm{vol}(u^*K_Y) \geq \mathrm{vol}((g \circ w)^*(K_T + tE_T)) = \mathrm{vol}(K_T + tE_T) > 0.$$

Thus $\mathrm{vol}(K_Y) = \mathrm{vol}(u^*K_Y) > 0$ [Laz04, Proposition 2.2.43], that is, K_Y is big.

By (1) and Lemma 8.6, there exists a klt stable surface Y' with $K^2_{Y'} = \mathrm{vol}(K_Y) < K^2_X$, so X is not a minimizer (Definition 2.3). □

8.2. **Contact at a smooth point.** In this subsection, we show that if $X \ncong X_{438}$ and a g-exceptional prime divisor intersects E at a smooth point of Z, then there exists a klt stable surface of smaller volume. We construct Y by extracting over X' the curve of the minimal resolution of Z that intersects the strict transform of such a divisor, together with the other g-exceptional prime divisors, and the inequality $\lambda < 0$ bounds $1 - n$ from above, so that the number t of Lemma 8.7(2) satisfies $t \geq 7/8 > 115/134$.

Proposition 8.8. *Notation and conditions as in Set-up 7.2, Construction-Theorem 7.3, and Definition-Proposition 8.2. Assume that $X \ncong X_{438}$ and that C_i intersects E at a smooth point of Z for some $1 \leq i \leq m$. Then X is not a minimizer.*

Proof. The proof is long, so we divide it into several steps. Renumbering $C_1, \dots, C_m$, we may assume that $i = 1$. By Definition-Proposition 8.2(2), $\pi = f \circ \pi_Z$, and π_Z is the minimal resolution of Z.

Step 1. In this step, we show that exactly one connected component Γ of $\mathrm{Exc}(\pi_Z)$ intersects D_1, that Γ is disjoint from $\widetilde{E} \cup D_2 \cup \cdots \cup D_m$, that $C_1 \cap C_j = \emptyset$ for $2 \leq j \leq m$, and that $E \cdot C_1 = 1$.

Any g-exceptional curve intersects E (Construction-Theorem 7.3(1)(e)), so g maps $C_1, \dots, C_m$ to points of E_T, and h contracts E_T to x'_0. Thus $\widetilde{E}, D_1, \dots, D_m$ are vertices of $\mathcal{D}$. A curve on $\widetilde{X}$ that is mapped to x'_0 and is not π_Z-exceptional is the strict transform of a curve on Z contracted by $h \circ g$, that is, of E or of some C_j. Hence any vertex of $\mathcal{D}$ other than $\widetilde{E}, D_1, \dots, D_m$ is π_Z-exceptional. By Definition-Proposition 8.2(3), the curves on $\widetilde{X}$ mapped to x'_0 form a simple normal crossing divisor, and $\mathcal{D}$ is a tree. Therefore $G \cdot G' \in \{0, 1\}$ for any two distinct vertices G, G' of $\mathcal{D}$.

Let $P \in C_1 \cap E$ be a smooth point of Z. Since π_Z is the minimal resolution of Z, π_Z is an isomorphism over a neighborhood of P, so $\pi_Z^{-1}(P) \in \widetilde{E} \cap D_1$. Thus $\widetilde{E} \cdot D_1 = 1$. By Definition-Proposition 8.2(3), D_1 is a vertex of degree two in $\mathcal{D}$. Hence there exists exactly one vertex $G_1 \neq \widetilde{E}$ of $\mathcal{D}$ with $G_1 \cdot D_1 = 1$, and $G \cdot D_1 = 0$ for any vertex $G \notin \{\widetilde{E}, D_1, G_1\}$ of $\mathcal{D}$. Since $D_1, \dots, D_m$ are pairwise disjoint, G_1 is π_Z-exceptional. Moreover, any π_Z-exceptional curve

intersecting D_1 is mapped to a point of C_1, hence to x_0', so it is a vertex of $\mathcal{D}$ adjacent to D_1, and it equals G_1.

Let Γ be the connected component of $\mathrm{Exc}(\pi_Z)$ containing G_1, and let $z_1 := \pi_Z(\Gamma)$. Since Z is normal, $\pi_Z^{-1}(z_1)$ is connected [Har77, Corollary III.11.4], so $\pi_Z^{-1}(z_1) = \Gamma$. By the previous paragraph, Γ is the only connected component of $\mathrm{Exc}(\pi_Z)$ intersecting D_1, and G_1 is the only component of Γ intersecting D_1.

Suppose that a component G' of Γ intersects $\widetilde{E}$. Since Γ is connected, there exists a path in $\mathcal{D}$ from G_1 to G' whose vertices are components of Γ. Adding the vertices $\widetilde{E}$ and D_1, we obtain a cycle in the tree $\mathcal{D}$, which is not possible. Thus $\Gamma \cap \widetilde{E} = \emptyset$. Suppose that a component G' of Γ intersects D_j for some $2 \le j \le m$. Then a path in Γ from G_1 to G' extends to the path $\widetilde{E}, D_1, G_1, \dots, G', D_j$ in $\mathcal{D}$ from $\widetilde{E}$ to D_j containing D_1, which contradicts Definition-Proposition 8.2(3). Thus $\Gamma \cap D_j = \emptyset$ for $2 \le j \le m$.

Since the strict transform of a curve on Z through z_1 intersects $\pi_Z^{-1}(z_1) = \Gamma$, we have

$$z_1 \notin E \cup C_2 \cup \cdots \cup C_m.$$

If z is a singular point of Z on C_1, then $\pi_Z^{-1}(z)$ is a connected union of π_Z-exceptional curves intersecting D_1, so $\pi_Z^{-1}(z) = \Gamma$ and $z = z_1$. Thus z_1 is the only singular point of Z on C_1. Let $z \in C_1 \cap C_j$ for some $2 \le j \le m$. If z is a smooth point of Z, then $\pi_Z^{-1}(z) \in D_1 \cap D_j$, which is not possible. If z is a singular point of Z, then $z = z_1 \in C_j$, which is not possible either. Hence $C_1 \cap C_j = \emptyset$ for $2 \le j \le m$. Finally, since $z_1 \notin E$, the $\mathbb{Q}$-divisor $\pi_Z^* E - \widetilde{E}$ is supported on π_Z-exceptional curves not in Γ, which are disjoint from D_1. Thus

$$E \cdot C_1 = \pi_Z^* E \cdot D_1 = \widetilde{E} \cdot D_1 = 1.$$

Step 2. In this step, we check the hypotheses of Lemma 8.4 for Γ, and we compute C_1^2, $\mathrm{mult}_{C_1} F$, and $a(G_1, Z, 0)$.

Write $\Gamma = \sum_{i=1}^k G_i$, where G_1 is the curve of **Step 1**. The curves $G_1, \dots, G_k$ are π-exceptional curves on the minimal resolution of the klt surface X, hence smooth rational curves forming a simple normal crossing divisor [KM98, Theorem 4.7]. They satisfy $G_i^2 < 0$ [KM98, Lemma 3.40], and $G_i^2 \ne -1$ since π_Z is minimal, so $G_i^2 \le -2$. The intersection matrix of Γ is negative definite by [KM98, Lemma 3.40] applied to π_Z, and the dual graph of Γ is a tree, since it is a connected subgraph of the tree $\mathcal{D}$ (Definition-Proposition 8.2(3)). Let $A := (-G_i \cdot G_j)_{1 \le i,j \le k}$, let d_i be the degree of G_i in the dual graph of Γ, let $\alpha \in \mathbb{Q}^k$ be the solution of $A\alpha = (2 - d_i)_{1 \le i \le k}$, and let $L_1 := (A^{-1})_{11}$. Then $(A^{-1})_{ij} > 0$ for all i, j (Lemma 8.4(1)), and in particular $L_1 > 0$.

Since z_1 is the only singular point of Z on C_1, we may write $\pi_Z^* C_1 = D_1 + \sum_{i=1}^k y_i G_i$. Since $D_1 \cdot G_j = 1$ if $j = 1$ and $D_1 \cdot G_j = 0$ if $j \ge 2$, the equalities $\pi_Z^* C_1 \cdot G_j = 0$ for $1 \le j \le k$ imply that $(y_1, \dots, y_k)$ is the first column of A^{-1}. Thus

$$\pi_Z^* C_1 = D_1 + \sum_{i=1}^k (A^{-1})_{i1} G_i \quad \text{and} \quad C_1^2 = \pi_Z^* C_1 \cdot D_1 = D_1^2 + (A^{-1})_{11} = L_1 - 1.$$

Let $s := -C_1^2 = 1 - L_1$. Since $g \circ \pi_Z$ is a resolution of T and $\pi_Z^* C_1 \ne 0$ is supported on $(g \circ \pi_Z)$-exceptional curves, $C_1^2 = (\pi_Z^* C_1)^2 < 0$ [KM98, Lemma 3.40]. Together with $L_1 > 0$, we have $0 < s < 1$.

By Construction-Theorem 7.3(1)(d), $F \ge 0$ and $\mathrm{Supp}\, F = \mathrm{Exc}(g) = C_1 \cup \cdots \cup C_m$. Since $C_1 \cap C_j = \emptyset$ for $2 \le j \le m$, we have

$$-s\, \mathrm{mult}_{C_1} F = F \cdot C_1 = g^* E_T \cdot C_1 - E \cdot C_1 = -1.$$

Thus $\mathrm{mult}_{C_1} F = 1/s$, and

$$F = \frac{1}{s} C_1 + F^\circ,$$

where $F^\circ \ge 0$ is supported on $C_2 \cup \cdots \cup C_m$. Since $E \cdot F^\circ \ge 0$, we have $\delta = E \cdot F \ge \frac{1}{s} E \cdot C_1 = \frac{1}{s}$.

The surface Z is klt by Lemma 2.15(1) and (2). By Lemma 8.4(6), $a(G_i, Z, 0) = \alpha_i$ for $1 \le i \le k$, so $\pi_Z^* K_Z = K_{\widetilde{X}} + \sum_{i=1}^k (1-\alpha_i) G_i + R$, where R is supported on π_Z-exceptional curves not in Γ. Since C_1 is g-exceptional and $K_Z + nE = g^*(K_T + nE_T)$ (Construction-Theorem 7.3(1)(g)), $K_{\widetilde{X}} \cdot D_1 = -2 - D_1^2 = -1$, $R \cdot D_1 = 0$, and $E \cdot C_1 = 1$, we have

$$0 = (K_Z + nE) \cdot C_1 = K_{\widetilde{X}} \cdot D_1 + (1-\alpha_1) G_1 \cdot D_1 + n = n - \alpha_1.$$

Thus $a(G_1, Z, 0) = \alpha_1 = n$. In particular, $n \in \mathbb{Q}$.

Step 3. In this step, we prove that either $s = 1 - n$, or there exist integers $r > e \ge 2$ such that $1 - n = (e-1)/r$ and $s = e/r$.

By Construction-Theorem 7.3(1)(g) with $t = 1$, $g^*(K_T + E_T) = K_Z + E + (1-n)F$. Thus

$$(g \circ \pi_Z)^*(K_T + E_T) = \pi_Z^* K_Z + \pi_Z^* E + \frac{1-n}{s} \pi_Z^* C_1 + (1-n) \pi_Z^* F^\circ.$$

Since $\pi_Z^* K_Z - K_{\widetilde{X}}$ is π_Z-exceptional and D_1 is not a component of $\pi_Z^* E$ or of $\pi_Z^* F^\circ$, we have

$$\operatorname{mult}_{D_1} \left((g \circ \pi_Z)^*(K_T + E_T) - K_{\widetilde{X}} \right) = \frac{1-n}{s}.$$

Since D_1 is exceptional over T and (T, E_T) is lc (Construction-Theorem 7.3(1)(b)), we have $(1-n)/s \le 1$ [KM98, Definition 2.34], that is,

$$1 - n \le s. \tag{8.5}$$

We recall that the determinant of a chain $H_1 + \cdots + H_l$ of smooth rational curves with $H_j^2 \le -2$ is at least 2. Let Δ_j be the determinant of $H_1 + \cdots + H_j$, and let $\Delta_0 := 1$. Then $\Delta_1 \ge 2$, and for $j \ge 2$,

$$\Delta_j = -H_j^2 \Delta_{j-1} - \Delta_{j-2} \ge 2\Delta_{j-1} - \Delta_{j-2}.$$

Thus $\Delta_j - \Delta_{j-1} \ge \Delta_1 - \Delta_0 \ge 1$ for any $j \ge 1$, and $\Delta_l \ge l + 1 \ge 2$.

By Definition-Proposition 8.2(3), $\mathcal{D}$ has at most one vertex of degree at least three, and such a vertex has degree three. Since the degree of G_i in the dual graph of Γ is at most the degree of G_i in $\mathcal{D}$, either Γ is a chain, or the dual graph of Γ has exactly one vertex of degree at least three, and that vertex has degree three. There are three cases.

Case 1. Γ is a chain with end curve G_1. Let $r := \det A$, and let r' be the determinant of the chain $G_2 + \cdots + G_k$, with $r' := 1$ if $k = 1$. By Lemma 8.4(2), $L_1 = r'/r$ and $n = \alpha_1 = (r'+1)/r$. Let $e := r - r'$. Then e and r are integers,

$$s = 1 - L_1 = \frac{e}{r}, \quad \text{and} \quad 1 - n = \frac{e-1}{r}.$$

Since $n < b < 1$, we have $e \ge 2$, and since $s < 1$, we have $e < r$.

Case 2. Γ is a chain and G_1 is not an end curve. Let r_1, r_2 be the determinants of the chains of $\Gamma - G_1$. Then $r_1, r_2 \ge 2$, so Lemma 8.4(3) implies

$$n = \alpha_1 = L_1 \Big(\frac{1}{r_1} + \frac{1}{r_2} \Big) \le L_1 = 1 - s.$$

Together with (8.5), we have $s = 1 - n$.

Case 3. Γ is not a chain. Since $G_1 \cdot D_1 = 1$ and D_1 is not a component of Γ, the degree of G_1 in $\mathcal{D}$ is larger than the degree of G_1 in the dual graph of Γ, so G_1 is not the vertex of degree three of Γ. If G_1 had degree two in the dual graph of Γ, then G_1 and the vertex of degree three of Γ would be two vertices of degree at least three in $\mathcal{D}$, which is not possible. Thus G_1 is an end curve of Γ. Renumbering $G_2, \ldots, G_k$, which changes neither L_1 nor α_1, we may assume that G_k is the vertex of degree three of Γ. Let r_1, r_2 be the determinants of the chains of $\Gamma - G_k$ that do not contain G_1. Then $r_1, r_2 \ge 2$ and $(A^{-1})_{1k} > 0$ (Lemma 8.4(1)), so Lemma 8.4(4) implies

$$n = \alpha_1 = L_1 + (A^{-1})_{1k} \Big(\frac{1}{r_1} + \frac{1}{r_2} - 1 \Big) \le L_1 = 1 - s.$$

Together with (8.5), we have $s = 1 - n$.

Since $e/r \le 2(e-1)/r$ for $e \ge 2$, in all cases we have

$$1 - n \le s \le 2(1-n). \tag{8.6}$$

Step 4. In this step, we prove that $1 - n < 1/9$, and we define rational numbers t and b_1 such that $7/8 \le t < n$ and $t < b_1 < b$.

By Construction-Theorem 7.3(1)(f), $\lambda = -aq + (1-n)\delta$, and $\lambda < 0$ by assumption. Since $\delta \ge 1/s$ (**Step 2**), (8.6) implies

$$aq > (1-n)\delta \ge \frac{1-n}{s} \ge \frac{1}{2}. \tag{8.7}$$

Since $a < 5/46$, we have $q > 1/(2a) > 23/5$. By Theorem 2.14(3),

$$(b-n)^2 \le \frac{K_X^2}{q} < \frac{5}{23 \cdot 48983} < \frac{1}{414^2},$$

where the last inequality holds since $5 \cdot 414^2 = 856980 < 1126609 = 23 \cdot 48983$. Since $b - n > 0$, we have $b - n < 1/414$, and

$$1 - n = a + (b-n) < \frac{5}{46} + \frac{1}{414} = \frac{1}{9}.$$

Let

$$t := 1 - \frac{1-n}{1-s}.$$

Then $t \in \mathbb{Q}$ since $n, s \in \mathbb{Q}$. If $s = 1 - n$, then $n > 8/9$ and $t = 1 - (1-n)/n > 1 - 1/8 = 7/8$. Otherwise, by **Step 3**, there exist integers $r > e \ge 2$ such that $1 - n = (e-1)/r$ and $s = e/r$. Then $1 - n < 1/9$ implies $r > 9(e-1)$, so $r \ge 9(e-1) + 1$ and $r - e \ge 8(e-1)$. Thus

$$t = 1 - \frac{e-1}{r-e} \ge 1 - \frac{1}{8} = \frac{7}{8}.$$

Moreover, $n - t = (1-n)s/(1-s) > 0$. Since $7 \cdot 134 = 938 > 920 = 8 \cdot 115$, we obtain

$$\frac{115}{134} < \frac{7}{8} \le t < n. \tag{8.8}$$

Since $q > 1$, we may let

$$b_1 := 1 - \frac{aq}{q-1}.$$

Then

$$b - b_1 = \frac{a}{q-1} > 0, \qquad 1 - b_1 = \frac{aq}{q-1} > 0,$$
$$b_1 - t = \frac{1-n}{1-s} - \frac{aq}{q-1} = \frac{q(b-n) + aqs - (1-n)}{(q-1)(1-s)} > 0,$$

where the last inequality holds since $b - n > 0$ and $aqs > 1 - n$ by (8.7).

Step 5. In this step, we construct a surface V' over Z, the surface Y over X', and a morphism $u : V' \to Y$ over X'.

Since $a(G_1, Z, 0) = n < 1$ (**Step 2**), by Lemma 2.15(1), there exists a projective birational morphism $w : V' \to Z$ from a $\mathbb{Q}$-factorial klt projective surface V' whose only exceptional divisor is G_1. Let $\mu : \widetilde{X} \dashrightarrow V'$ be the induced birational map over Z. If μ is not defined at a point $P \in \widetilde{X}$, then by [Har77, Theorem V.5.2], the total transform of P contains a curve on V' mapped to the point $\pi_Z(P)$ of Z, that is, the strict transform of G_1, while the center of G_1 on $\widetilde{X}$ is the curve G_1. Thus μ is a morphism, $\pi_Z = w \circ \mu$, and μ contracts exactly the π_Z-exceptional curves other than G_1. On V', we keep the letters E, G_1, $C_1, \dots, C_m$, and F° for the strict transforms. No μ-exceptional curve intersects D_1 (**Step 1**), and the fibers of μ are connected

[Har77, Corollary III.11.4], so μ is an isomorphism over a neighborhood of C_1. Thus V' is smooth near C_1, and **Step 1** implies

$$C_1^2 = -1, \quad K_{V'} \cdot C_1 = -1, \quad E \cdot C_1 = C_1 \cdot G_1 = 1.$$

Since $\pi_Z = w \circ \mu$, we have $\operatorname{mult}_{G_1} w^*C_1 = \operatorname{mult}_{G_1} \pi_Z^* C_1 = (A^{-1})_{11} = L_1$. Since $z_1 \notin E \cup C_2 \cup \cdots \cup C_m$ and $a(G_1, Z, 0) = n$, we have

$$w^*K_Z = K_{V'} + (1-n)G_1, \quad w^*E = E, \quad w^*C_1 = C_1 + L_1G_1, \quad w^*F^\circ = F^\circ. \tag{8.9}$$

In particular, $E^2 = w^*E \cdot E = -q$ on V', and since $K_Z \cdot E = (f^*K_X - bE) \cdot E = bq$ and $G_1 \cdot E = 0$, we have $K_{V'} \cdot E = w^*K_Z \cdot E = bq$.

By Construction-Theorem 7.3(1)(g) and (3)(d),

$$(h \circ g)^* K_{X'} = g^*(K_T + b'E_T) = K_Z + b'E + (b' - n)F.$$

Since $b' > b > n$ and $\operatorname{Supp} F = \operatorname{Exc}(g)$, we have $a(C_j, X', 0) < 1$ for $2 \le j \le m$. Pulling back by w and using (8.9), we have

$$(h \circ g \circ w)^* K_{X'} = K_{V'} + b'E + \Big(1 - n + \frac{(b'-n)L_1}{s}\Big)G_1 + \frac{b'-n}{s}C_1 + (b'-n)F^\circ,$$

so

$$a(G_1, X', 0) = n - \frac{(b'-n)L_1}{s} < 1.$$

The curves $G_1, C_2, \ldots, C_m$ are mapped to x_0', so they are exceptional over X'. Since X' is a klt stable surface, by Lemma 2.15(1), there exists a projective birational morphism $h_Y : Y \to X'$ from a $\mathbb{Q}$-factorial klt projective surface Y whose exceptional divisors are exactly $G_1, C_2, \ldots, C_m$.

Let $u : V' \dashrightarrow Y$ be the induced birational map over X'. If u is not defined at a point $P \in V'$, then by [Har77, Theorem V.5.2], the total transform of P contains a curve on Y mapped to the point $(h \circ g \circ w)(P)$ of X', that is, one of $G_1, C_2, \ldots, C_m$, while the center of each of $G_1, C_2, \ldots, C_m$ on V' is a curve. Thus u is a morphism and $h_Y \circ u = h \circ g \circ w$. The curves on V' contracted by $h \circ g \circ w$ are E, G_1, and $C_1, \ldots, C_m$, and $G_1, C_2, \ldots, C_m$ are divisors on Y. Thus u contracts exactly E and C_1.

Step 6. In this step, we compute u^*K_Y and verify conditions (1) and (2) of Lemma 8.7.

Since u contracts exactly E and C_1, we may write $u^*K_Y = K_{V'} + c_1E + c_2C_1$. Since $u^*K_Y \cdot E = u^*K_Y \cdot C_1 = 0$, **Step 5** implies

$$bq - qc_1 + c_2 = 0 \quad \text{and} \quad -1 + c_1 - c_2 = 0.$$

Since $q \ne 1$, the unique solution is $c_1 = (bq-1)/(q-1) = b_1$ and $c_2 = b_1 - 1$, where we recall that $b_1 = 1 - aq/(q-1)$ (**Step 4**). Thus

$$u^*K_Y = K_{V'} + b_1E + (b_1 - 1)C_1. \tag{8.10}$$

By (8.9), $(f \circ w)^*K_X = w^*(K_Z + bE) = K_{V'} + bE + (1-n)G_1$. Thus

$$(f \circ w)^*K_X - u^*K_Y = (b - b_1)E + (1 - b_1)C_1 + (1-n)G_1, \tag{8.11}$$

and all three coefficients are positive by **Step 4** and $n < 1$.

For condition (1), let $V := \widetilde{X}$, $\phi_X := \pi = f \circ w \circ \mu$, and $\phi_Y := u \circ \mu$. By (8.11),

$$\phi_X^*K_X - \phi_Y^*K_Y = \mu^*\big((b - b_1)E + (1 - b_1)C_1 + (1-n)G_1\big) \ge 0.$$

Since $\operatorname{mult}_{D_1}(\phi_X^*K_X - \phi_Y^*K_Y) = 1 - b_1 > 0$, we have $\phi_X^*K_X \ne \phi_Y^*K_Y$.

For condition (2), recall that $t = 1-(1-n)/(1-s)$ (**Step 4**). By Construction-Theorem 7.3(1)(g) and (8.9),

$$\begin{aligned}(g \circ w)^*(K_T + tE_T) &= w^*\big(K_Z + tE + (t-n)F\big) \\ &= K_{V'} + (1-n)G_1 + tE + \frac{t-n}{s}(C_1 + L_1G_1) + (t-n)F^\circ.\end{aligned}$$

Since $t-n=-(1-n)s/(1-s)$ and $L_1=1-s$, we have $(t-n)L_1/s=-(1-n)$ and $(t-n)/s=-(1-n)/(1-s)=t-1$. Thus

$$(g\circ w)^*(K_T+tE_T)=K_{V'}+tE+(t-1)C_1+(t-n)F^\circ.$$

By (8.10),

$$u^*K_Y-(g\circ w)^*(K_T+tE_T)=(b_1-t)(E+C_1)+(n-t)F^\circ\geq 0,$$

since $b_1>t$ and $n>t$ (**Step 4**). By (8.8), t is a rational number with $115/134<7/8\leq t<n$. Thus Y, V', w, u, and t satisfy condition (2) of Lemma 8.7. Since $X\ncong X_{438}$, Lemma 8.7 implies that X is not a minimizer. □

8.3. **Contact at a cyclic quotient crossing.** In this subsection, we construct a klt stable surface of smaller volume when $X\ncong X_{438}$ and a g-exceptional prime divisor C intersects E at a cyclic quotient crossing (Definition 8.3), and we show that for a minimizer other than X_{438} only fork configurations remain. Lemma 8.9 recognizes a cyclic quotient crossing from the dual graph on the minimal resolution.

Lemma 8.9. *Let Z be a klt surface, $z\in Z$ a singular point, and $\sigma:\widetilde{Z}\to Z$ the minimal resolution of $Z\ni z$, with exceptional curves $G_1,\dots,G_k$. Let E and C be distinct prime divisors on Z through z, with strict transforms $\widetilde{E}$ and $\widetilde{C}$ on $\widetilde{Z}$. Assume that, near $\sigma^{-1}(z)$, the reduced divisor $\widetilde{E}+\sum_{i=1}^k G_i+\widetilde{C}$ has simple normal crossings and dual graph a chain whose vertices, in order along the chain, are $\widetilde{E},G_1,\dots,G_k,\widetilde{C}$. Let $M:=\det(-G_i\cdot G_j)_{1\leq i,j\leq k}$. Then $M\geq 2$, and z is a cyclic quotient crossing of E and C of order M.*

Proof. Since Z is klt, $G_1,\dots,G_k$ are smooth rational curves [KM98, Theorem 4.7]. By [Kol13b, Theorem 3.32 and Definition 3.33], applied to the Henselization of Z at z and the boundary $E+C$, the germ $Z\ni z$ is isomorphic to $\mathbb{A}^2/\frac{1}{M}(1,d)$ for an integer d with $\gcd(d,M)=1$, and $E+C$ is contained in the image of the coordinate axes. Since E and C are distinct and each has one branch at z, as $\widetilde{E}$ and $\widetilde{C}$ each intersect $\sigma^{-1}(z)$ at one point, E and C are the images of the coordinate axes (in [Kol13b, 3.34] with $n=k$, $C_0=\widetilde{E}$, and $C_{k+1}=\widetilde{C}$, the functions u and v have divisors ME and MC on Z). The Henselization is the direct limit of the local rings of étale neighborhoods [Kol13b, 4.38], so an isomorphism of Henselizations is defined on a common étale neighborhood and induces an isomorphism of analytic germs. The cyclic group acts diagonally and freely outside the origin since $\gcd(d,M)=1$. Since z is a singular point, $M\geq 2$. □

Proposition 8.10. *Notation and conditions as in Set-up 7.2, Construction-Theorem 7.3, and Definition-Proposition 8.2.*

(1) If $X\ncong X_{438}$ and X is a minimizer, then there exist no i and no singular point $z\in E\cap C_i$ of Z such that, near $\pi_Z^{-1}(z)$, the divisor $\widetilde{E}+\sum_{G'\subset\pi_Z^{-1}(z)}G'+D_i$ has simple normal crossings and dual graph a chain with end vertices $\widetilde{E}$ and D_i.

(2) If $X\ncong X_{438}$ and X is a minimizer, then $\mathcal{D}$ has a vertex $G\neq\widetilde{E}$ of degree three, G is the only vertex of degree at least three, and for any i, G is an interior vertex of the path in $\mathcal{D}$ from $\widetilde{E}$ to D_i.

Proof. The proof is long, so we divide it into several steps. In **Steps 1–5**, we prove (1): suppose that $X\ncong X_{438}$, that X is a minimizer, and that there exist i and a singular point $z\in E\cap C_i$ of Z as in (1).

Step 1. In this step, we describe the curves on $\widetilde{X}$ near the strict transform of C and compute $E\cdot C$, C^2, and $K_Z\cdot C$.

By Lemma 2.15(1) and (2), Z is a $\mathbb{Q}$-factorial klt surface. Write $C=C_i$. By Definition-Proposition 8.2(2), $\pi=f\circ\pi_Z$, where π_Z is the minimal resolution of Z, and D_i is the strict transform of C on $\widetilde{X}$. The vertices of $\mathcal{D}$ are $\widetilde{E}$, $D_1,\dots,D_m$, and the π_Z-exceptional curves over points of $E\cup C_1\cup\cdots\cup C_m$, since $(h\circ g)^{-1}(x_0')=E\cup C_1\cup\cdots\cup C_m$ by Construction-Theorem 7.3(1) and (3). By Definition-Proposition 8.2(3), the curves mapped to x_0' form a simple normal crossing

divisor, $\mathcal{D}$ is a tree, D_i is a vertex of degree two, and no path in $\mathcal{D}$ from $\widetilde{E}$ to D_i contains D_j for $j \neq i$.

By Lemma 8.9, z is a cyclic quotient crossing of E and C of order M, where M is the determinant of the chain $\pi_Z^{-1}(z)$. By Definition-Lemma 2.11(3), applied to the generator of the cyclic group that multiplies the coordinate vanishing on the preimage of C by $e^{2\pi i/M}$, there exists an integer Q with $1 \leq Q \leq M-1$ and $\gcd(Q, M) = 1$ such that the analytic germ $(Z \ni z, E+C)$ is isomorphic to the germ of Definition-Lemma 2.11 with $r = M$, with E and C corresponding to the curves of $(1, 0)$ and $(0, 1)$. Let $\Lambda_1, \dots, \Lambda_\ell$ be the curves of $\pi_Z^{-1}(z)$, numbered along the chain so that Λ_1 intersects D_i and Λ_ℓ intersects $\widetilde{E}$, each at one point transversally, and the other curves Λ_l are disjoint from D_i and $\widetilde{E}$. Let $L_\Lambda := (A_\Lambda^{-1})_{11}$ and $\alpha_\Lambda := a(\Lambda_1, Z, 0)$, where

$$A_\Lambda := (-\Lambda_l \cdot \Lambda_{l'})_{1 \leq l, l' \leq \ell}.$$

The vertex D_i of $\mathcal{D}$ has exactly two neighbors. One of them is Λ_1. The other is not $\widetilde{E}$ or a curve Λ_l, since otherwise $\mathcal{D}$ would contain a cycle through Λ_1, and it is not D_j for $j \neq i$, since $D_1, \dots, D_m$ are pairwise disjoint. Hence it is a π_Z-exceptional curve over a point $z' \in C \backslash \{z\}$. Let $\Gamma = \sum_{l=1}^k G_l := \pi_Z^{-1}(z')$, with G_1 the curve intersecting D_i. The curves of Γ are disjoint from $\widetilde{E}$, since otherwise $\widetilde{E}, \Lambda_\ell, \dots, \Lambda_1, D_i, G_1$ and a path in Γ would form a cycle in $\mathcal{D}$. In particular, $z' \notin E$. The curves of Γ are disjoint from D_j for any $j \neq i$, since otherwise the path in $\mathcal{D}$ from $\widetilde{E}$ to D_j would pass through D_i.

The curves Γ and $\Lambda := \sum_{l=1}^\ell \Lambda_l$ satisfy the conditions of Lemma 8.4 by [KM98, Theorem 4.7 and Lemma 3.40] and the minimality of π_Z. Let $A := (-G_l \cdot G_{l'})_{1 \leq l, l' \leq k}$, let $\alpha \in \mathbb{Q}^k$ be the solution of $A\alpha = (2 - \deg_\Gamma G_l)_{1 \leq l \leq k}$, where $\deg_\Gamma G_l$ is the degree of G_l in the dual graph of Γ, and let $L_l := (A^{-1})_{ll}$. By Lemma 8.4(6), applied to Γ and to Λ, $\alpha_l = a(G_l, Z, 0)$ for any l, and α_Λ is the first entry of the vector of Lemma 8.4 for Λ. Let r'_Λ be the determinant of $\Lambda - \Lambda_1$, with $r'_\Lambda := 1$ if $\ell = 1$. By Lemma 8.4(2), $L_\Lambda = r'_\Lambda / M$ and $\alpha_\Lambda = (r'_\Lambda + 1)/M$, so

$$\alpha_\Lambda = L_\Lambda + \frac{1}{M}. \tag{8.12}$$

Since D_i intersects exactly Λ_1 and G_1 among the π_Z-exceptional curves, each transversally at one point, and $D_i \cap \widetilde{E} = \emptyset$,

$$\pi_Z^* C = D_i + \sum_{l=1}^{\ell} (A_\Lambda^{-1})_{l1} \Lambda_l + \sum_{l=1}^{k} (A^{-1})_{l1} G_l.$$

Since the curves of Γ are disjoint from $\widetilde{E}$ and Λ_ℓ is the only curve of Λ intersecting $\widetilde{E}$, the coefficients of Λ_1 and G_1 in $\pi_Z^* E$ are $(A_\Lambda^{-1})_{1\ell}$ and 0, and $(A_\Lambda^{-1})_{1\ell} = 1/M$ by Lemma 8.4(5). Thus

$$E \cdot C = \pi_Z^* E \cdot D_i = \frac{1}{M}, \qquad C^2 = \pi_Z^* C \cdot D_i = -1 + L_\Lambda + L_1.$$

Let $s := -C^2$. Since $\pi_Z^* C$ is a non-zero combination of curves contracted by $\widetilde{X} \to T$, $s > 0$ [KM98, Lemma 3.40]. By Construction-Theorem 7.3(1), $K_Z + nE = g^*(K_T + nE_T)$, so $(K_Z + nE) \cdot C = 0$ and $K_Z \cdot C = -n/M$. On the other hand, since D_i is a smooth rational curve with $D_i^2 = -1$, $K_{\widetilde{X}} \cdot D_i = -1$, so

$$-\frac{n}{M} = K_Z \cdot C = \pi_Z^* K_Z \cdot D_i = -1 + (1 - \alpha_\Lambda) + (1 - \alpha_1).$$

Together with $s = 1 - L_\Lambda - L_1$ and (8.12), we obtain

$$n = M(\alpha_\Lambda + \alpha_1 - 1) \quad \text{and} \quad s = \frac{1-n}{M} + \alpha_1 - L_1. \tag{8.13}$$

In particular, n and s are rational numbers.

Step 2. In this step, we bound s from below and show that s takes one of two forms.

By Construction-Theorem 7.3(1), $F = \sum_{l=1}^{m} r_l C_l$ with $r_l > 0$ for any l. Since $E \cdot C = 1/M$ and $(E+F)\cdot C = g^*E_T \cdot C = 0$,

$$sr_i = \frac{1}{M} + \sum_{l\neq i} r_l (C_l \cdot C) \geq \frac{1}{M}.$$

Hence

$$R := F - \frac{1}{Ms} C \geq 0.$$

By Construction-Theorem 7.3(1), $g^*(K_T + E_T) = K_Z + E + (1-n)F$, and (T, E_T) is lc, so $(1-n)r_i \leq 1$. Since $r_i \geq 1/(Ms)$,

$$s \geq \frac{1-n}{M}. \tag{8.14}$$

Since E is not a component of R, $\delta = E \cdot F \geq E \cdot C/(Ms) = 1/(M^2 s)$. By Construction-Theorem 7.3(3), $\delta < aq/(1-n)$, so

$$M^2 aqs > 1 - n. \tag{8.15}$$

The determinant of a non-empty chain of curves of self-intersection at most -2 is at least 2: if Δ_l is the determinant of the chain of the first l curves, counted from one end, and $\Delta_0 := 1$, then $\Delta_1 \geq 2$, and for $l \geq 2$, the expansion along the row of the l-th curve shows $\Delta_l \geq 2\Delta_{l-1} - \Delta_{l-2}$ whenever $\Delta_{l-1} \geq 0$, so $\Delta_l - \Delta_{l-1} \geq \Delta_1 - \Delta_0 \geq 1$ and $\Delta_l \geq l+1$ by induction on l. We show that one of the following holds.

(I) $s = (1-n)/M$.

(II) Γ is a chain with end curve G_1, $r_\Gamma := \det A \geq 2$, $s = (1-n)/M + 1/r_\Gamma$, and $Mr_\Gamma s$ is an integer with $Mr_\Gamma s > M$.

Suppose first that Γ is a chain with end curve G_1. Let r'_Γ be the determinant of $G_2 + \cdots + G_k$, with $r'_\Gamma := 1$ if $k = 1$. By Lemma 8.4(2), $L_1 = r'_\Gamma / r_\Gamma$ and $\alpha_1 = (r'_\Gamma + 1)/r_\Gamma$, so (8.13) implies $s = (1-n)/M + 1/r_\Gamma$, and $r_\Gamma \geq 2$. Moreover, $s = 1 - L_\Lambda - L_1 = 1 - r'_\Lambda/M - r'_\Gamma/r_\Gamma$, so

$$Mr_\Gamma s = Mr_\Gamma - r_\Gamma r'_\Lambda - Mr'_\Gamma \in \mathbb{Z}, \qquad Mr_\Gamma s = r_\Gamma(1-n) + M > M.$$

Thus (II) holds.

Suppose that Γ is a chain and G_1 is not an end curve. Then $\Gamma - G_1$ consists of two chains. Let $\rho_1, \rho_2 \geq 2$ be their determinants. By Lemma 8.4(3), $\alpha_1 = L_1(1/\rho_1 + 1/\rho_2) \leq L_1$.

Suppose that Γ is not a chain. By [KM98, Theorem 4.7], Γ has exactly one vertex of degree three. The degree of G_1 in $\mathcal{D}$ equals $\deg_\Gamma G_1 + 1$, since among $\widetilde{E}, D_1, \ldots, D_m$ only D_i intersects Γ. If G_1 were not an end curve of Γ, then either G_1 is not the vertex of degree three and $\mathcal{D}$ would have two vertices of degree at least three, or G_1 is the vertex of degree three and $\mathcal{D}$ would have a vertex of degree four, which contradicts Definition-Proposition 8.2(3). Hence G_1 is an end curve of Γ, and we number the other curves so that the vertex of degree three is G_k. Let $\rho_1, \rho_2 \geq 2$ be the determinants of the chains of $\Gamma - G_k$ that do not contain G_1. By Lemma 8.4(4) and (1),

$$\alpha_1 = L_1 + (A^{-1})_{1k}\Big(\frac{1}{\rho_1} + \frac{1}{\rho_2} - 1\Big) \leq L_1.$$

In the last two cases, $\alpha_1 \leq L_1$, so (8.13) implies $s \leq (1-n)/M$, and (8.14) implies (I).

Step 3. In this step, we choose a primitive vector $v_{(j,1)}$ and the number t, and we show that $63/73 < t < n$.

Let Q be as in **Step 1**, and let j be the largest integer such that

$$j \equiv Q \bmod M \quad \text{and} \quad j(1-n) < M - 1. \tag{8.16}$$

Since $0 < 1 - n < 1$ and $1 \leq Q \leq M-1$, Q satisfies (8.16), and the integers satisfying (8.16) are bounded from above. Thus j exists, $j \geq Q \geq 1$, and by maximality

$$(j + M)(1-n) \geq M - 1. \tag{8.17}$$

Let N be the lattice of Definition-Lemma 2.11 for $r = M$. Since $(0,1) = M \cdot (Q,1)/M - Q(1,0)$, the vectors $(1,0)$ and $(Q,1)/M$ form a basis of N. Since

$$v_{(j,1)} = \frac{j-Q}{M}(1,0) + \frac{(Q,1)}{M},$$

$v_{(j,1)}$ is a primitive vector of N in the interior of the cone. Let B be the toric divisor of $v_{(j,1)}$ (Definition-Lemma 4.6(1)), a prime divisor over Z with center z.

Let t be the rational number with

$$1 - t = \frac{Ms(M-1) + 1 - n}{1 + Msj}.$$

Since $j(1-n) < M-1$, $(1-n)(1+Msj) < Ms(M-1)+1-n$, so $1-t > 1-n$ and $t < n$. Let

$$\kappa := \frac{M^2 s}{1 - n + Ms(M-1)}.$$

In case (I) of **Step 2**, $\kappa = 1$. In case (II), substituting $s = (1-n)/M + 1/r_\Gamma$,

$$\kappa = \frac{r_\Gamma(1-n) + M}{r_\Gamma(1-n) + M - 1} = \frac{Mr_\Gamma s}{Mr_\Gamma s - 1} \leq \frac{3}{2},$$

since $Mr_\Gamma s$ is an integer with $Mr_\Gamma s > M \geq 2$. Multiplying (8.17) by Ms, we obtain

$$(1-n)(1+Msj) \geq 1 - n + Ms(M-1) - M^2 s(1-n) = (1 - n + Ms(M-1))(1 - \kappa(1-n)).$$

By Lemma 8.5, $1 - n < 5/44$, so $1 - \kappa(1-n) \geq 1 - 3(1-n)/2 > 0$. Hence

$$1 - t = \frac{(1-n)(Ms(M-1)+1-n)}{(1-n)(1+Msj)} \leq \frac{1-n}{1-\kappa(1-n)} \leq \frac{1-n}{1-3(1-n)/2} < \frac{5/44}{1-15/88} = \frac{10}{73}.$$

Thus

$$\frac{63}{73} < t < n \quad \text{and} \quad t > \frac{115}{134}, \tag{8.18}$$

where the last inequality follows from $63 \cdot 134 = 8442 > 8395 = 115 \cdot 73$.

Step 4. In this step, we construct V', Y, w, and u, and we verify condition (1) of Lemma 8.7.

The numbers b and b' are rational, since K_X and $K_{X'}$ are $\mathbb{Q}$-Cartier. Since $K_Z + bE = f^* K_X$, by [KM98, Lemma 2.30], Definition-Lemma 2.11(1), and Definition-Lemma 4.6(1) applied to $(Z \ni z, bE)$,

$$a(B, X, 0) = a(B, Z, bE) = \frac{1 + aj}{M}.$$

Since $a < a + (b-n) = 1 - n$, (8.16) implies $aj < M - 1$, so $a(B, X, 0) < 1$. Moreover, B is exceptional over X and $a(E, X, 0) = a < 1$. By Lemma 2.15(1) applied over X to E and B, there exists a projective birational morphism $V' \to X$ from a $\mathbb{Q}$-factorial klt surface V' whose exceptional divisors are exactly E and B.

By Construction-Theorem 7.3(1) and (3),

$$(f' \circ f)^* K_{X'} = (h \circ g)^* K_{X'} = K_Z + b'E + (b' - n)F,$$

where $n < b < b' < 1$. Thus, for any l, $a(C_l, X', 0) = 1 - (b' - n) r_l < 1$. Since $(b' - n)F \geq 0$, by [KM98, Lemma 2.30], Definition-Lemma 2.11(1), and Definition-Lemma 4.6(1) applied to $(Z \ni z, b'E)$,

$$a(B, X', 0) = a(B, Z, b'E + (b'-n)F) \leq a(B, Z, b'E) = \frac{1 + (1-b')j}{M} < \frac{1 + aj}{M} < 1.$$

By Definition-Proposition 8.2(1), $f(C_1), \ldots, f(C_m)$ are contracted by f' to x'_0, and B has center x'_0 on X'. By Lemma 2.15(1) applied over X' to B and the prime divisors C_l with $l \neq i$, there exist a projective $\mathbb{Q}$-factorial klt surface Y and a birational morphism $h_Y : Y \to X'$ whose exceptional divisors are exactly B and the strict transforms of C_l with $l \neq i$.

Let V be a smooth projective surface with birational morphisms $\psi : V \to V'$, $\psi_Z : V \to Z$, and $\psi_Y : V \to Y$ over X'. Since V' is normal, $\psi_*\mathcal{O}_V = \mathcal{O}_{V'}$, and any fiber of ψ is connected. Any curve P contracted by ψ is a prime divisor over X, exceptional over X, which is different from E and B. Since E is the only exceptional divisor of f, P is contracted by ψ_Z. Since P is exceptional over X' and the exceptional divisors of h_Y are divisors on V', P is contracted by ψ_Y. By [Deb01, Lemma 1.15(b)], there exist birational morphisms $w : V' \to Z$ and $u : V' \to Y$ with $\psi_Z = w \circ \psi$ and $\psi_Y = u \circ \psi$. The prime divisors on V' exceptional over X are E and B, so B is the only exceptional divisor of w, and $w(B) = z$. The prime divisors on V' exceptional over X' are E, B, and the strict transforms of $C_1, \dots, C_m$, and among them exactly B and the strict transforms of C_l with $l \neq i$ are divisors on Y. Let $E_{V'}$ and $C_{V'}$ be the strict transforms of E and C on V'. Thus the exceptional curves of u are exactly $E_{V'}$ and $C_{V'}$.

Since Z is normal and B is the only exceptional curve of w, w is an isomorphism over $Z\backslash\{z\}$, and Definition-Lemma 4.6(3) implies that, over an analytic neighborhood of z, w is isomorphic to the toric morphism given by the subdivision of the cone by the ray of $v_{(j,1)}$. We apply Definition-Lemma 2.11(2) with $l = 1$ and the vectors $(M, 0)$, $(j, 1)$, and $(0, M)$, where

$$\det_M((M,0),(j,1)) = 1, \qquad \det_M((j,1),(0,M)) = j, \qquad \det_M((M,0),(0,M)) = M.$$

Thus

$$B^2 = -\frac{M}{j}, \qquad E_{V'} \cdot B = 1, \qquad C_{V'} \cdot B = \frac{1}{j}.$$

Since $w^*E \cdot B = w^*C \cdot B = 0$, and since $a(B, Z, 0) = (1 + j)/M$ by Definition-Lemma 2.11(1) and Definition-Lemma 4.6(1),

$$w^*E = E_{V'} + \frac{j}{M}B, \qquad w^*C = C_{V'} + \frac{1}{M}B, \qquad w^*K_Z = K_{V'} + \frac{M-1-j}{M}B. \tag{8.19}$$

Since $E^2 = -q$, $C^2 = -s$, and $E \cdot C = 1/M$, (8.19) implies

$$E_{V'}^2 = -q - \frac{j}{M}, \qquad C_{V'}^2 = -s - \frac{1}{Mj}, \qquad E_{V'} \cdot C_{V'} = \frac{1}{M} - \frac{j}{M^2} \cdot \frac{M}{j} = 0.$$

Write $u^*K_Y = K_{V'} + c_E E_{V'} + c_C C_{V'}$. Since $(K_Z + bE) \cdot E = f^*K_X \cdot E = 0$, $K_Z \cdot E = bq$. Together with $K_Z \cdot C = -n/M$ and (8.19),

$$K_{V'} \cdot E_{V'} = bq - \frac{M-1-j}{M}, \qquad K_{V'} \cdot C_{V'} = -\frac{n}{M} - \frac{M-1-j}{Mj}.$$

The equations $u^*K_Y \cdot E_{V'} = u^*K_Y \cdot C_{V'} = 0$ imply

$$c_E = \frac{Mbq - (M-1-j)}{Mq+j} = 1 - \frac{Maq + M - 1}{Mq+j}, \qquad c_C = \frac{j(1-n) - (M-1)}{1 + Msj} < 0, \tag{8.20}$$

where $c_C < 0$ by (8.16). On the other hand, (8.19) implies

$$(f \circ w)^*K_X = w^*(K_Z + bE) = K_{V'} + bE_{V'} + \frac{M - 1 - aj}{M}B,$$

and $b - c_E = (M - 1 - aj)/(Mq + j)$. Since $aj < M - 1$ and $c_C < 0$, the three coefficients in

$$(f \circ w)^*K_X - u^*K_Y = \frac{M-1-aj}{Mq+j}E_{V'} - c_C C_{V'} + \frac{M-1-aj}{M}B \tag{8.21}$$

are positive, so $(f\circ w)^*K_X - u^*K_Y \geq 0$ and $(f \circ w)^*K_X \neq u^*K_Y$. Let $\phi_X := f \circ \psi_Z = f \circ w \circ \psi$ and $\phi_Y := \psi_Y = u \circ \psi$. Then $\phi_X^*K_X - \phi_Y^*K_Y$ is the pullback of (8.21) by ψ, hence $\phi_X^*K_X - \phi_Y^*K_Y \geq 0$ and $\phi_X^*K_X \neq \phi_Y^*K_Y$. Thus Y satisfies condition (1) of Lemma 8.7.

Step 5. In this step, we verify condition (2) of Lemma 8.7 for the number t of **Step 3** and conclude the proof of (1).

By the definition of t,

$$t - n = (1 - n) - (1 - t) = \frac{Ms(j(1-n) - (M-1))}{1 + Msj}, \quad \text{so} \quad \frac{t-n}{Ms} = c_C.$$

Moreover,

$$\Big(M-1-(1-t)j\Big)(1+Msj) = (M-1)(1+Msj) - j\big(Ms(M-1)+1-n\big)$$
$$= M-1-j(1-n) = -(1+Msj)c_C,$$

so

$$\frac{M-1-j}{M} + \frac{tj}{M} + \frac{t-n}{M^2 s} = \frac{M-1-(1-t)j+c_C}{M} = 0.$$

By Construction-Theorem 7.3(1), $g^*(K_T+tE_T) = K_Z + tE + (t-n)F$, and $F = \frac{1}{Ms}C + R$ by the definition of R. By (8.19), the identity $(t-n)/(Ms) = c_C$, and the last display,

$$(g\circ w)^*(K_T+tE_T) = K_{V'} + tE_{V'} + c_C C_{V'} - (n-t)w^*R,$$

so

$$u^*K_Y - (g\circ w)^*(K_T+tE_T) = (c_E - t)E_{V'} + (n-t)w^*R. \tag{8.22}$$

Since $R \geq 0$ and $n > t$, it suffices to show $c_E > t$. By (8.20) and the definition of t, $c_E - t = \Phi/((Mq+j)(1+Msj))$, where

$$\Phi := (Mq+j)\big(Ms(M-1)+1-n\big) - (Maq+M-1)(1+Msj).$$

A direct calculation shows

$$\Phi = Mq(b-n) + (M-1)(M^2qs-1) + j(1-n-M^2aqs).$$

By (8.15), $1-n-M^2aqs < 0$, and by (8.16), $j < (M-1)/(1-n)$. Hence

$$\Phi > Mq(b-n) + (M-1)(M^2qs-1) + \frac{M-1}{1-n}(1-n-M^2aqs)$$
$$= Mq(b-n) + \frac{(M-1)M^2qs(b-n)}{1-n} > 0,$$

where we use $1-n-a = b-n > 0$. Thus $c_E > t$, and $u^*K_Y - (g\circ w)^*(K_T+tE_T) \geq 0$ by (8.22). Since V' is a normal projective surface and $115/134 < t < n$ by (8.18), Y satisfies condition (2) of Lemma 8.7. By **Step 4** and Lemma 8.7, X is not a minimizer, a contradiction. This proves (1).

Step 6. In this step, we prove (2).

Assume that $X \ncong X_{438}$ and X is a minimizer.

Fix $1 \leq i \leq m$, and let P be the path in $\mathcal{D}$ from $\widetilde{E}$ to D_i. We show that the interior of P contains a vertex of degree three.

Suppose that P is a single edge, and let p be a point of $\widetilde{E} \cap D_i$. Since the curves mapped to x_0' form a simple normal crossing divisor, no π_Z-exceptional curve passes through p, so $\pi_Z(p)$ is a smooth point of Z in $E \cap C_i$, which contradicts Proposition 8.8.

Suppose that P has at least three vertices and no interior vertex of P has degree three in $\mathcal{D}$. The interior vertices of P are different from $\widetilde{E}$ and from D_j for any j, so they are π_Z-exceptional curves. Since they form a connected chain, they are mapped to one point $z \in Z$, which is a singular point of Z in $E \cap C_i$. Any interior vertex v of P has degree two in $\mathcal{D}$, so v intersects no curve mapped to x_0' other than the neighbors of v on P. Since $\pi_Z^{-1}(z)$ is connected and consists of curves mapped to x_0', $\pi_Z^{-1}(z)$ is the union of the interior vertices of P. Thus, near $\pi_Z^{-1}(z)$, the reduced divisor consisting of $\widetilde{E}$, the curves of $\pi_Z^{-1}(z)$, and D_i has simple normal crossings, and the dual graph is the chain P with ends $\widetilde{E}$ and D_i. This contradicts (1).

Hence, for any i, the path in $\mathcal{D}$ from $\widetilde{E}$ to D_i has an interior vertex of degree three. By Definition-Proposition 8.2(3), $\mathcal{D}$ has exactly one vertex G of degree at least three, G has degree three, and G is an interior vertex of the path from $\widetilde{E}$ to D_i for any i. In particular, $G \neq \widetilde{E}$. □

8.4. **The fork configurations.** In this subsection, we prove Theorem 8.1: by Propositions 8.8 and 8.10, it remains to exclude the case where the graph $\mathcal{D}$ of Definition-Proposition 8.2(3) has a vertex G of degree three, which is an interior vertex of the path from $\widetilde{E}$ to D_i for any i. If the curves of $\pi^{-1}(x_0)$ form a chain, then we construct a surface Y satisfying the conditions of Lemma 8.7, apart from one numerical case, which the inequality $K_X^2 \geq q(b-n)^2$ of Theorem 2.14(3) excludes.

Proof of Theorem 8.1. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix the notation and show that G is a curve over x_0.

Suppose that $X \ncong X_{438}$. Let E and f be as in Set-up 7.2. We use the notation of Set-up 7.2 and Construction-Theorem 7.3. If $\lambda \geq 0$, then $p > 20/23 > 115/134$ by Construction-Theorem 7.3(2), so $X \cong X_{438}$ by Theorem 6.2, a contradiction. Thus $\lambda < 0$. We use the notation of Construction-Theorem 7.3(3) and Definition-Proposition 8.2, so that $x_0 = f(E)$ and $\pi = f \circ \pi_Z$, where $\pi_Z : \widetilde{X} \to Z$ is the minimal resolution of Z (Definition-Proposition 8.2(2)). Thus the π_Z-exceptional curves are the π-exceptional curves other than $\widetilde{E}$.

Since $X \ncong X_{438}$ and X is a minimizer, Proposition 8.10(2) implies that $\mathcal{D}$ has a vertex $G \neq \widetilde{E}$ of degree three, that G is the only vertex of degree at least three, and that for any $1 \leq i \leq m$, G is an interior vertex of the path in $\mathcal{D}$ from $\widetilde{E}$ to D_i. In the rest of the proof, two curves are *adjacent* if they are joined by an edge of $\mathcal{D}$.

By Definition-Proposition 8.2(3), $\mathcal{D}$ is a tree. Since G is the only vertex of degree at least three, any other vertex has degree at most two. By Definition-Proposition 8.2(1), a curve on $\widetilde{X}$ mapped to x_0' is either one of $D_1, \dots, D_m$ or a π-exceptional curve over a point of $f'^{-1}(x_0')$, and π-exceptional curves over distinct points of X are disjoint. Since $f'(x_0) = h(g(E)) = x_0'$, the curves of $\pi^{-1}(x_0)$ are vertices of $\mathcal{D}$. Since X is normal, $\pi^{-1}(x_0)$ is connected [Har77, Corollary III.11.4], and the curves of $\pi^{-1}(x_0)$ form a chain or a tree with exactly one vertex of degree three [KM98, Theorem 4.7]. The π_Z-exceptional curves are smooth rational curves of self-intersection at most -2 intersecting transversally, and any connected component of $\mathrm{Exc}(\pi_Z)$ has dual graph a tree and negative definite intersection matrix [KM98, Theorem 4.7 and Lemma 3.40]. Thus the hypotheses of Lemma 8.4 hold for any connected component Γ' of $\mathrm{Exc}(\pi_Z)$, and we let $A_{\Gamma'}$ be the matrix $(-G' \cdot G'')$, where G' and G'' run through the curves of Γ'.

Since $G \neq \widetilde{E}$ and any D_i has degree two, G is a π_Z-exceptional curve. Let Γ be the connected component of $\mathrm{Exc}(\pi_Z)$ containing G. We show the following.

(a) Any vertex of the path from $\widetilde{E}$ to G is a curve of $\pi^{-1}(x_0)$. In particular, G is a π_Z-exceptional curve over x_0.
(b) Any vertex adjacent to a curve of $\pi^{-1}(x_0)$ is either a curve of $\pi^{-1}(x_0)$ or one of $D_1, \dots, D_m$.
(c) For any $1 \leq i \leq m$, D_i is adjacent to exactly one curve H_i of $\pi^{-1}(x_0)$, and H_i is a curve of Γ. The other vertex adjacent to D_i is an end curve of a connected component of $\mathrm{Exc}(\pi_Z)$ which is a chain and is disjoint from $\pi^{-1}(x_0)$.

For any j, the path from $\widetilde{E}$ to D_j passes through G. Since $\mathcal{D}$ is a tree, D_j is not a vertex of the path from $\widetilde{E}$ to G: otherwise the path from $\widetilde{E}$ to D_j would be the part of the path from $\widetilde{E}$ to G ending at D_j, which does not contain G. Hence the vertices of the path from $\widetilde{E}$ to G are π-exceptional curves. Since π-exceptional curves over distinct points of X are disjoint, any two adjacent π-exceptional curves are mapped by π to one point of X, and hence the vertices of the path from $\widetilde{E}$ to G are curves of $\pi^{-1}(x_0)$. Thus (a) holds. Since the π-exceptional curves over points other than x_0 are disjoint from the curves of $\pi^{-1}(x_0)$, (b) holds.

Fix $1 \leq i \leq m$. By Definition-Proposition 8.2(3), the path from $\widetilde{E}$ to D_i contains no D_j with $j \neq i$, and it passes through G. Thus the vertices of the path strictly between G and D_i are π-exceptional curves over x_0 other than $\widetilde{E}$, and they are connected to G through π_Z-exceptional curves, so they are curves of Γ. Let H_i be the vertex of the path adjacent to D_i, where $H_i = G$ is allowed. Since the curves of $\pi^{-1}(x_0)$ form a connected subgraph of the tree, H_i is the only curve

of $\pi^{-1}(x_0)$ adjacent to D_i. Since D_i has degree two, there exists exactly one other vertex W adjacent to D_i. The curve W is not a curve of $\pi^{-1}(x_0)$, and $W \neq D_j$ for any j since $D_1, \dots, D_m$ are pairwise disjoint. Hence W is a π-exceptional curve over a point $x_1 \neq x_0$, and the curves of $\pi^{-1}(x_1)$ form a connected component of $\mathrm{Exc}(\pi_Z)$ disjoint from $\pi^{-1}(x_0)$. Since G is not a curve of $\pi^{-1}(x_1)$, any curve of $\pi^{-1}(x_1)$ has degree at most two in $\mathcal{D}$, so the curves of $\pi^{-1}(x_1)$ form a chain. Since D_i is adjacent to W, W is adjacent to at most one curve of $\pi^{-1}(x_1)$, so W is an end curve. Thus (c) holds.

Let G_E be the vertex of the path from $\widetilde{E}$ to G adjacent to $\widetilde{E}$, where $G_E = G$ if $\widetilde{E}$ is adjacent to G. By (a), G_E is a curve of Γ, and since Γ is connected and $\mathcal{D}$ is a tree, G_E is the only curve of Γ adjacent to $\widetilde{E}$.

Step 2. In this step, we compute C_i^2, $E \cdot C_i$, and $K_Z \cdot C_i$ for a fixed i.

Fix $1 \leq i \leq m$, and let $H := H_i$. Let S be the vertex adjacent to D_i other than H, Γ_0 the connected component of $\mathrm{Exc}(\pi_Z)$ containing S, R_0 the determinant of Γ_0, and R_0' the determinant of $\Gamma_0 - S$, with $R_0' := 1$ if $\Gamma_0 = S$. Let $L_0 := R_0'/R_0$. By (c), Γ_0 is a chain with end curve S, and Γ_0 is disjoint from Γ and $\widetilde{E}$. By Lemma 8.4(2) and (6),

$$(A_{\Gamma_0}^{-1})_{SS} = L_0, \quad a(S, Z, 0) = \frac{R_0' + 1}{R_0}.$$

We have $D_i \cdot H = D_i \cdot S = 1$ and $D_i \cdot G' = 0$ for any other π_Z-exceptional curve G', and $\pi_Z^* C_i \cdot G' = 0$ for any π_Z-exceptional curve G'. Since the intersection matrix of $\mathrm{Exc}(\pi_Z)$ is negative definite,

$$\pi_Z^* C_i = D_i + \sum_{G' \subset \Gamma} (A_\Gamma^{-1})_{G'H} G' + \sum_{G' \subset \Gamma_0} (A_{\Gamma_0}^{-1})_{G'S} G'.$$

Since C_i is g-exceptional, $\pi_Z^* C_i$ is supported on the exceptional curves of the resolution $g \circ \pi_Z : \widetilde{X} \to T$, so $C_i^2 = (\pi_Z^* C_i)^2 < 0$ [KM98, Lemma 3.40]. By (c), $\widetilde{E}$ is not adjacent to D_i, and G_E is the only curve of $\Gamma \cup \Gamma_0$ adjacent to $\widetilde{E}$. Since $K_{\widetilde{X}} \cdot D_i = -1$ and $K_Z + nE = g^*(K_T + nE_T)$ (Construction-Theorem 7.3(1)), we obtain

$$C_i^2 = \pi_Z^* C_i \cdot D_i = -1 + (A_\Gamma^{-1})_{HH} + L_0 < 0, \tag{8.23}$$

$$E \cdot C_i = \widetilde{E} \cdot \pi_Z^* C_i = (A_\Gamma^{-1})_{G_E H}, \tag{8.24}$$

$$0 = (K_Z + nE) \cdot C_i = 1 - a(H, Z, 0) - \frac{R_0' + 1}{R_0} + n(A_\Gamma^{-1})_{G_E H}. \tag{8.25}$$

Here the last equality follows from

$$K_Z \cdot C_i = \pi_Z^* K_Z \cdot D_i = -1 + (1 - a(H, Z, 0)) + (1 - a(S, Z, 0)).$$

Step 3. In this step, we show that the curves of $\pi^{-1}(x_0)$ form a chain.

Suppose not. By [KM98, Theorem 4.7], the dual graph of $\pi^{-1}(x_0)$ has a vertex of degree three, the fork of $X \ni x_0$ (Definition 2.9(2)). The fork has degree at least three in $\mathcal{D}$, so the fork is G. Let $X \ni x_0$ be of type $[e_0; (r_1, q_1), (r_2, q_2), (r_3, q_3)]$. By Lemma 2.10, $r_2 \in \{2, 3\}$, and $e_0 \geq 2$, so $X \ni x_0$ is of type D-I, D-II, E-I, or E-II. Since $\widetilde{E} \neq G$, Definition 2.13(2) implies that $X \ni x_0$ is of type D-II. Thus $e_0 = 2$ and $(r_2, q_2) = (2, 1)$, and since $2 \leq r_1 \leq r_2$, $(r_1, q_1) = (2, 1)$. Hence the arms of (r_1, q_1) and (r_2, q_2) are (-2)-curves G_1 and G_2. By Definition 2.13(3), $\widetilde{E}$ is a curve of the arm of (r_3, q_3), and the curves strictly between G and $\widetilde{E}$ are (-2)-curves. Let $\ell \geq 0$ be the number of curves strictly between G and $\widetilde{E}$, and $N := \ell + 3$. Then Γ consists of G, G_1, G_2, and the curves strictly between G and $\widetilde{E}$, any curve of Γ is a (-2)-curve, the dual graph of Γ is the Dynkin diagram D_N if $\ell \geq 1$, and Γ is the chain $G_1 + G + G_2$ if $\ell = 0$.

Fix $1 \leq i \leq m$, and let H, S, Γ_0, R_0, R_0', and L_0 be as in **Step 2**. The vertices of $\mathcal{D}$ adjacent to G are G_1, G_2, and a curve of the arm of (r_3, q_3), since G has degree three. Thus the vertex after G on the path from $\widetilde{E}$ to D_i is G_1 or G_2, say G_1. Since G is the only curve of $\pi^{-1}(x_0)$ adjacent to G_1, the vertex after G_1 is D_i by (b) and Definition-Proposition 8.2(3), that is, $H = G_1$. Since

any curve of Γ is a (-2)-curve, $A_\Gamma(1,\dots,1)^\top$ is the vector $(2-d_{G'})_{G'\subset\Gamma}$, where $d_{G'}$ is the degree of G' in the dual graph of Γ. By Lemma 8.4(6), $a(G',Z,0)=1$ for any curve G' of Γ.

Suppose that $\ell\geq 1$. Then A_Γ is the Cartan matrix of type D_N, and $(A_\Gamma^{-1})_{G_1G_1}=N/4$ [Bou02, Chapter VI, Plate IV]. By (8.23),

$$C_i^2=-1+\frac{N}{4}+L_0\geq L_0>0,$$

a contradiction.

Suppose that $\ell=0$. Then $G_E=G$, and Γ is the chain G_1+G+G_2 of determinant 4. By Lemma 8.4(2), $(A_\Gamma^{-1})_{G_1G_1}=3/4$, and by Lemma 8.4(5), $(A_\Gamma^{-1})_{G_1G}=2/4=1/2$. By (8.23) and (8.25),

$$C_i^2=-\frac{1}{4}+\frac{R_0'}{R_0}<0,\quad n=\frac{2(R_0'+1)}{R_0}.$$

Thus $R_0>4R_0'$, so $R_0\geq 4R_0'+1$ since R_0 is an integer. Since $n>7/8$ (Construction-Theorem 7.3), $R_0<16(R_0'+1)/7$. Hence $7(4R_0'+1)<16(R_0'+1)$, that is, $12R_0'<9$, which contradicts $R_0'\geq 1$. Therefore the curves of $\pi^{-1}(x_0)$ form a chain.

Step 4. In this step, we show that $m=1$, that D_1 is adjacent to G, and that G is not an end curve of the chain $\pi^{-1}(x_0)$.

By **Step 3**, Γ consists of the curves of $\pi^{-1}(x_0)$ on the side of $\widetilde{E}$ containing G, and Γ is a chain with end curve G_E. We first show that no D_j is adjacent to the end curve of Γ other than G_E, or to $G=G_E$ in case $\Gamma=G$. Suppose the contrary, and let $z:=\pi_Z(\Gamma)$, a singular point of Z in $E\cap C_j$. The curve $\widetilde{E}$ is adjacent only to G_E among the curves of Γ, and D_j is adjacent only to H_j among them by (c). By Definition-Proposition 8.2(3), the divisor $\widetilde{E}+\sum_{G'\subset\Gamma}G'+D_j$ has simple normal crossings near $\pi_Z^{-1}(z)$, with dual graph the chain whose end vertices are $\widetilde{E}$ and D_j. Here $\Gamma=G$ is allowed, in which case G intersects $\widetilde{E}$ and D_j at distinct points. This contradicts Proposition 8.10(1).

Suppose that G is an end curve of the chain $\pi^{-1}(x_0)$. Since $\widetilde{E}\neq G$, G is adjacent to exactly one curve of $\pi^{-1}(x_0)$, so G is the end curve of Γ other than G_E, or $\Gamma=G$. Since G has degree three, G is adjacent to some D_j by (b), a contradiction.

Thus G is not an end curve of $\pi^{-1}(x_0)$. Then G is adjacent to exactly two curves of $\pi^{-1}(x_0)$, and by (b), the third vertex adjacent to G is some D_j. Possibly renumbering $C_1,\dots,C_m$, we may assume that D_1 is adjacent to G. Suppose that $m\geq 2$. By Definition-Proposition 8.2(3), the path from $\widetilde{E}$ to D_2 passes through G and does not contain D_1. Thus the vertex after G on the path is the curve of Γ adjacent to G on the side away from $\widetilde{E}$, and the path then runs along Γ until H_2. Any curve of Γ other than G and the end curves of Γ is adjacent to two curves of Γ and has degree at most two. Hence H_2 is the end curve of Γ other than G_E, a contradiction. Therefore $m=1$. We write $C:=C_1$ and $D:=D_1$, so that $H_1=G$.

Step 5. In this step, we show that the arm of G away from $\widetilde{E}$ is one (-2)-curve, that $1-a(S,Z,0)=1/R_0$ with R_0 odd, and we compute L, $E\cdot C$, and C^2.

Let r_E be the determinant of the chain of curves strictly between $\widetilde{E}$ and G, and r_E' the determinant of the chain obtained by removing from it the curve adjacent to G. If no such curve exists, then $r_E:=1$ and $r_E':=0$, and if exactly one exists, then $r_E':=1$. Thus $0\leq r_E'<r_E$, since removing an end curve from a non-empty chain of curves of self-intersection at most -2 lowers the determinant (by induction on the number of curves, expanding the determinant along the row of the removed curve). By **Step 4**, the curves of Γ on the side of G away from $\widetilde{E}$ form a non-empty chain, the arm of G away from $\widetilde{E}$. Let r_G be the determinant of the arm and r_G' the determinant of the arm minus the curve adjacent to G, with $r_G':=1$ if the arm is one curve. Let G'' be the end curve of Γ other than G_E, $e_G:=-G^2\geq 2$, r_Γ the determinant of Γ, and $L:=(A_\Gamma^{-1})_{GG}$. Expanding the determinant along the row of G,

$$r_\Gamma=e_Gr_Er_G-r_E'r_G-r_Er_G'.$$

The cofactor of the diagonal entry at G is $r_E r_G$, and by Lemma 8.4(5) applied from G_E and from G'', $(A_\Gamma^{-1})_{G_E G} = r_G/r_\Gamma$ and $(A_\Gamma^{-1})_{G'' G} = r_E/r_\Gamma$. Thus

$$L = \frac{r_E r_G}{r_\Gamma} = \frac{1}{e_G - r'_E/r_E - r'_G/r_G}, \quad (A_\Gamma^{-1})_{G_E G} = \frac{L}{r_E}, \quad (A_\Gamma^{-1})_{G'' G} = \frac{L}{r_G}. \tag{8.26}$$

The vector $(2 - d_{G'})_{G' \subset \Gamma}$ has entry 1 at G_E and at G'' and 0 elsewhere, so by Lemma 8.4(6), $a(G, Z, 0) = L(1/r_E + 1/r_G)$.

Let S, Γ_0, R_0, R'_0, and L_0 be the objects of **Step 2** for $i = 1$, and let $s := -C^2$. By (8.23), (8.24), and (8.25) with $H = G$,

$$s = 1 - L - L_0 > 0, \quad E \cdot C = \frac{L}{r_E}, \quad \frac{R_0 - R'_0 - 1}{R_0} = L\left(\frac{1-n}{r_E} + \frac{1}{r_G}\right). \tag{8.27}$$

Let $k := R_0 - R'_0 - 1$. Since $L > 0$ (Lemma 8.4(1)) and $n < b < 1$, k is a positive integer. Since $s > 0$, $L < 1 - L_0 = (k+1)/R_0$. By Lemma 8.5, $1 - n < 5/44$, and since $r_E \geq 1$ and $r_G \geq 2$,

$$\frac{1-n}{r_E} + \frac{1}{r_G} < \frac{5}{44} + \frac{1}{2} = \frac{27}{44}.$$

By (8.27), $k = R_0 L\left((1-n)/r_E + 1/r_G\right) < 27(k+1)/44$, so $17k < 27$ and $k = 1$. Thus

$$R'_0 = R_0 - 2, \quad 1 - a(S, Z, 0) = \frac{1}{R_0}, \quad L_0 = \frac{R_0 - 2}{R_0}.$$

Since $R'_0 \geq 1$, $R_0 \geq 3$. Expanding along the row of S, any common divisor of R_0 and R'_0 divides the determinant of Γ_0 minus S and the curve adjacent to S, and by induction along Γ_0, $\gcd(R_0, R'_0) = 1$. Hence R_0 is odd.

Since $R_0 L < k + 1 = 2$ and $R_0 L\left((1-n)/r_E + 1/r_G\right) = 1$, we have $(1-n)/r_E + 1/r_G > 1/2$. If $r_G \geq 3$, then $(1-n)/r_E + 1/r_G < 5/44 + 1/3 = 59/132 < 1/2$, a contradiction. Thus $r_G = 2$. A chain of $j \geq 2$ curves of self-intersection at most -2 has determinant at least $j + 1 \geq 3$ (by induction on j, expanding the determinant along the row of an end curve), so the arm of G away from $\widetilde{E}$ is one (-2)-curve, and $r'_G = 1$. By (8.26) and (8.27),

$$\frac{1}{R_0} = \frac{(1-n)/r_E + 1/2}{e_G - r'_E/r_E - 1/2}.$$

A direct calculation from the last equality and (8.26) shows

$$\begin{aligned} J := R_0(1-n) &= \left(e_G - \frac{R_0 + 1}{2}\right) r_E - r'_E, \\ L = \frac{2r_E}{R_0(r_E + 2(1-n))}, &\qquad E \cdot C = \frac{2}{R_0(r_E + 2(1-n))}. \end{aligned} \tag{8.28}$$

Since R_0 is odd, J is an integer, and $J > 0$ since $n < 1$. Moreover,

$$s = 1 - L - \frac{R_0 - 2}{R_0} = \frac{2}{R_0} - L = \frac{4(1-n)}{R_0(r_E + 2(1-n))} = 2(1-n)E \cdot C. \tag{8.29}$$

Since $m = 1$, $\operatorname{Supp} F = C$ by Construction-Theorem 7.3(1). Since $F \cdot C = (g^* E_T - E) \cdot C = -E \cdot C$ and $C^2 = -s$, (8.29) implies

$$F = \frac{E \cdot C}{s} C = \frac{1}{2(1-n)} C, \quad \delta = E \cdot F = \frac{(E \cdot C)^2}{s}, \quad qs > (E \cdot C)^2, \tag{8.30}$$

where the inequality follows from $\delta < q$ (Construction-Theorem 7.3(3)).

Step 6. In this step, we let

$$t := 1 - \frac{J}{R_0 - 2} = n - \frac{2(1-n)}{R_0 - 2} < n$$

and show that $t > 7/8$.

The second expression of t follows from

$$\frac{J}{R_0-2}=\frac{(1-n)R_0}{R_0-2}=(1-n)+\frac{2(1-n)}{R_0-2}.$$

We have $t>7/8$ if and only if $R_0>8J+2$. Since $1-n<5/44$, $R_0=J/(1-n)>44J/5$. If $J\geq 3$, then $44J/5\geq 8J+12/5>8J+2$. If $J=2$, then $R_0>88/5$, and since R_0 is odd, $R_0\geq 19>8J+2$. If $J=1$, then $R_0>44/5$, so $R_0\geq 9$, and since R_0 is odd, $R_0>8J+2$ unless $R_0=9$. For $(J,R_0)=(1,9)$, $t=6/7<115/134$, so Lemma 8.7 does not apply to t, and we exclude this case by the inequality $K_X^2\geq q(b-n)^2$ of Theorem 2.14(3).

Suppose that $(J,R_0)=(1,9)$. Then $1-n=1/9$, and by (8.28), $1=(e_G-5)r_E-r'_E$. Since $0\leq r'_E\leq r_E-1$, $1\leq(e_G-5)r_E\leq r_E$, so $e_G=6$ and $r'_E=r_E-1$. If $r_E\geq 2$, the Hirzebruch–Jung chain of $r_E/(r_E-1)$ is $[2,\dots,2]$ (Definition 2.9(1)), so the curves strictly between $\widetilde{E}$ and G are r_E-1 curves of self-intersection -2. Let r'_Γ be the determinant of $\Gamma-G_E$. A direct calculation by expansion along the row of G shows

$$r_\Gamma=12r_E-2(r_E-1)-r_E=9r_E+2\geq 11,\quad r'_\Gamma=r_\Gamma-9.$$

Let Γ_1 be the chain of the curves of $\pi^{-1}(x_0)$ on the side of $\widetilde{E}$ away from G, r_{Γ_1} the determinant of Γ_1, and r'_{Γ_1} the determinant of Γ_1 minus the curve adjacent to $\widetilde{E}$, with $r_{\Gamma_1}:=1$ and $r'_{\Gamma_1}:=0$ if Γ_1 is empty and $r'_{\Gamma_1}:=1$ if Γ_1 is one curve. Thus $0\leq r'_{\Gamma_1}\leq r_{\Gamma_1}-1$. Let $e_E:=-\widetilde{E}^2$. By Definition 2.13(1), $e_E\geq 3$. The curve $\widetilde{E}$ is adjacent to G_E, to one end curve of Γ_1 if Γ_1 is non-empty, and to no other vertex. By Lemma 8.4(2) and (6),

$$q=-\pi_Z^*E\cdot\widetilde{E}=e_E-\frac{r'_\Gamma}{r_\Gamma}-\frac{r'_{\Gamma_1}}{r_{\Gamma_1}}=e_E-1+\frac{9}{r_\Gamma}-\frac{r'_{\Gamma_1}}{r_{\Gamma_1}},$$

$$K_Z\cdot E=\pi_Z^*K_Z\cdot\widetilde{E}=e_E-2+\left(1-\frac{r'_\Gamma+1}{r_\Gamma}\right)+\left(1-\frac{r'_{\Gamma_1}+1}{r_{\Gamma_1}}\right)=q-\frac{1}{r_\Gamma}-\frac{1}{r_{\Gamma_1}}.$$

Since $(K_Z+bE)\cdot E=f^*K_X\cdot E=0$, $K_Z\cdot E=bq$, so $aq=q-K_Z\cdot E=1/r_\Gamma+1/r_{\Gamma_1}$. Since $a=1-b=1/9-(b-n)$,

$$q(b-n)=\frac{q}{9}-aq=\frac{\nu}{9r_{\Gamma_1}},\quad\text{where } \nu:=(e_E-1)r_{\Gamma_1}-r'_{\Gamma_1}-9\in\mathbb{Z}.$$

Since $b>n$, $\nu\geq 1$. Since $e_E\geq 3$ and $r'_{\Gamma_1}\leq r_{\Gamma_1}-1$, $\nu\geq r_{\Gamma_1}-8$, so $r_{\Gamma_1}\leq\nu+8\leq 9\nu$, and $\nu+9\leq 10\nu$. Since $q=(\nu+9)/r_{\Gamma_1}+9/r_\Gamma$ and $r_\Gamma\geq 11$,

$$r_{\Gamma_1}^2q=r_{\Gamma_1}(\nu+9)+\frac{9r_{\Gamma_1}^2}{r_\Gamma}\leq 90\nu^2+\frac{729}{11}\nu^2=\frac{1719}{11}\nu^2.$$

Therefore

$$q(b-n)^2=\frac{(q(b-n))^2}{q}=\frac{\nu^2}{81r_{\Gamma_1}^2q}\geq\frac{11}{81\cdot 1719}>\frac{1}{48983},$$

where the last inequality follows from $11\cdot 48983=538813>139239=81\cdot 1719$. This contradicts $q(b-n)^2\leq K_X^2\leq 1/48983$ (Theorem 2.14(3)). Hence $(J,R_0)\neq(1,9)$, and

$$\frac{115}{134}<\frac{7}{8}<t<n,\tag{8.31}$$

where $115/134<7/8$ since $8\cdot 115=920<938=7\cdot 134$.

Step 7. In this step, we extract S and conclude by Lemma 8.7.

Recall that S is the end curve of Γ_0 adjacent to D, and $1-a(S,Z,0)=1/R_0\in(0,1)$ by **Step 5**. By Lemma 2.15(1), there exists a projective birational morphism $w:V'\to Z$ from a $\mathbb{Q}$-factorial klt projective surface V' whose exceptional divisor is exactly the image $S_{V'}$ of S. The induced birational map $\widetilde{X}\dashrightarrow V'$ is a morphism $\pi_{V'}:\widetilde{X}\to V'$. In fact, on a smooth projective surface dominating $\widetilde{X}$ and V', any curve contracted to a point of $\widetilde{X}$ is mapped to a point of Z, hence to a point of V', since the only curve mapped onto $S_{V'}$ is the strict transform of S. Since

$\widetilde{X}$ is normal, the morphism to V' factors through $\widetilde{X}$ [Deb01, Lemma 1.15(b)]. Since $\pi_Z = w \circ \pi_{V'}$ and the only curve on $\widetilde{X}$ mapped onto $S_{V'}$ is S, $\pi_{V'}$ contracts the π_Z-exceptional curves other than S.

Let $E_{V'} := w_*^{-1}E$ and $C_{V'} := w_*^{-1}C$. Since $\pi_Z = w \circ \pi_{V'}$, the coefficients of $S_{V'}$ in w^*E and w^*C are the coefficients of S in π_Z^*E and π_Z^*C, which are 0 since $\widetilde{E}$ is disjoint from Γ_0, and L_0 by **Step 2**. Since $\pi_{V'}^*S_{V'} - S$ is supported on $\Gamma_0 - S$, and $\widetilde{E}$ and D are disjoint from $\Gamma_0 - S$, $C_{V'} \cdot S_{V'} = D \cdot \pi_{V'}^*S_{V'} = 1$ and $E_{V'} \cdot S_{V'} = \widetilde{E} \cdot \pi_{V'}^*S_{V'} = 0$. Since $w^*C \cdot S_{V'} = 0$, $S_{V'}^2 = -(C_{V'} \cdot S_{V'})/L_0 = -1/L_0$. By the definition of log discrepancies, $w^*K_Z = K_{V'} + S_{V'}/R_0$. Since $K_Z \cdot E = bq$ and $K_Z \cdot C = -nE \cdot C$, we obtain the following, where τ is any real number, and $s = -C^2$ and $E \cdot C$ are the numbers on Z of **Step 5**:

$$w^*E = E_{V'}, \quad w^*C = C_{V'} + L_0 S_{V'}, \tag{8.32}$$

$$S_{V'}^2 = -\frac{1}{L_0}, \quad C_{V'} \cdot S_{V'} = 1, \quad E_{V'} \cdot S_{V'} = 0, \tag{8.33}$$

$$E_{V'}^2 = -q, \quad E_{V'} \cdot C_{V'} = E \cdot C, \quad C_{V'}^2 = -s + L_0^2 S_{V'}^2 = -(s + L_0), \tag{8.34}$$

$$w^*K_Z = K_{V'} + \frac{1}{R_0}S_{V'}, \tag{8.35}$$

$$K_{V'} \cdot E_{V'} = bq, \quad K_{V'} \cdot C_{V'} = -nE \cdot C - \frac{1}{R_0}, \tag{8.36}$$

$$(g \circ w)^*(K_T + \tau E_T) = K_{V'} + \frac{1}{R_0}S_{V'} + \tau E_{V'} + \frac{\tau - n}{2(1-n)}(C_{V'} + L_0 S_{V'}). \tag{8.37}$$

Here (8.34) follows from (8.32) and (8.33), (8.36) follows from $w^*K_Z \cdot C_{V'} = K_Z \cdot C$ and (8.33), and (8.37) follows from $g^*(K_T + \tau E_T) = K_Z + \tau E + (\tau - n)F$ (Construction-Theorem 7.3(1)), (8.30), (8.32), and (8.35).

By Construction-Theorem 7.3(3), X' is a klt stable surface, $K_T + b'E_T = h^*K_{X'}$, and $b < b'$. By (8.37) with $\tau = b'$, the coefficient of $S_{V'}$ in $(h \circ g \circ w)^*K_{X'} - K_{V'}$ is

$$1 - a(S, X', 0) = \frac{1}{R_0} + \frac{(b' - n)L_0}{2(1-n)} > 0,$$

since $b' > b > n$. Hence $0 < a(S, X', 0) < 1$, where the first inequality holds since X' is klt. By Lemma 2.15(1), there exists a projective birational morphism $h_Y : Y \to X'$ from a $\mathbb{Q}$-factorial klt projective surface Y whose exceptional divisor is exactly the image of S. On a smooth common resolution of V' and Y over X', any curve contracted to a point of V' is mapped to a point of X', hence to a point of Y, since the only curve mapped onto the image of S on Y is the strict transform of S, which is mapped onto $S_{V'}$. Since V' is normal, the morphism to Y factors through V' [Deb01, Lemma 1.15(b)], and we obtain a birational morphism $u : V' \to Y$ with $h_Y \circ u = h \circ g \circ w$. The exceptional curves of $h \circ g \circ w$ are $E_{V'}$, $C_{V'}$, and $S_{V'}$, and $S_{V'}$ is not contracted by u, so u contracts exactly $E_{V'}$ and $C_{V'}$. Write

$$u^*K_Y = K_{V'} + c_E E_{V'} + c_C C_{V'}.$$

Intersecting with $E_{V'}$ and $C_{V'}$ and using (8.34) and (8.36),

$$0 = bq - c_E q + c_C E \cdot C, \quad 0 = -nE \cdot C - \frac{1}{R_0} + c_E E \cdot C - (s + L_0)c_C.$$

By (8.30), $s + L_0 - (E \cdot C)^2/q > L_0 > 0$, so

$$c_C = \frac{(b-n)E \cdot C - 1/R_0}{s + L_0 - (E \cdot C)^2/q}, \quad c_E = b + \frac{E \cdot C}{q}c_C.$$

By (8.28), $R_0(b-n)E\cdot C = 2(b-n)/(r_E+2(1-n)) < 1$, since $b-n<1-n$ and $r_E\geq 1$. Thus $c_C<0$ and $c_E<b$. Since $(f\circ w)^*K_X = w^*(K_Z+bE) = K_{V'}+S_{V'}/R_0+bE_{V'}$,

$$(f\circ w)^*K_X - u^*K_Y = (b-c_E)E_{V'} - c_C C_{V'} + \frac{1}{R_0}S_{V'} \tag{8.38}$$

has positive coefficients, so it is ≥ 0 and non-zero. Let V be a resolution of V', and let $\phi_X : V\to X$ and $\phi_Y : V\to Y$ be the compositions of $V\to V'$ with $f\circ w$ and u. By (8.38), $\phi_X^*K_X - \phi_Y^*K_Y\geq 0$ and $\phi_X^*K_X\neq\phi_Y^*K_Y$. Thus condition (1) of Lemma 8.7 holds.

Let t be the rational number of **Step 6**. Since $t-n = -2(1-n)/(R_0-2)$ and $L_0 = (R_0-2)/R_0$, the coefficient of $S_{V'}$ in (8.37) with $\tau = t$ is $1/R_0 - 1/R_0 = 0$, and

$$(g\circ w)^*(K_T+tE_T) = K_{V'}+tE_{V'}+c_tC_{V'},\quad c_t := \frac{t-n}{2(1-n)} = -\frac{1}{R_0-2} = -\frac{1}{R_0L_0}.$$

Let

$$P := u^*K_Y - (g\circ w)^*(K_T+tE_T) = (c_E-t)E_{V'} + (c_C-c_t)C_{V'}.$$

Since $C_{V'}$ is contracted by u and by $g\circ w$, $P\cdot C_{V'} = 0$, so by (8.34),

$$c_C - c_t = \frac{E\cdot C}{s+L_0}(c_E-t).$$

Since $E_{V'}$ is contracted by u,

$$P\cdot E_{V'} = -(g\circ w)^*(K_T+tE_T)\cdot E_{V'},$$

and by (8.34) and (8.36),

$$\left(q - \frac{(E\cdot C)^2}{s+L_0}\right)(c_E-t) = (b-t)q - \frac{E\cdot C}{R_0L_0}.$$

By (8.29), $n-t = 2(1-n)/(R_0-2) = s/(R_0L_0E\cdot C)$, so

$$(b-t)q - \frac{E\cdot C}{R_0L_0} = (b-n)q + \frac{qs-(E\cdot C)^2}{R_0L_0E\cdot C} > 0$$

by $b>n$ and (8.30). By (8.30), $q-(E\cdot C)^2/(s+L_0) > q-(E\cdot C)^2/s > 0$. Hence $c_E>t$ and $c_C>c_t$, and $P\geq 0$. By (8.31), $115/134 < t < n$, so condition (2) of Lemma 8.7 holds.

Since $X\ncong X_{438}$, Lemma 8.7 implies that X is not a minimizer, a contradiction. Therefore $X\cong X_{438}$. □

9. Surfaces with minimal log discrepancy at least 5/46

In this section, we prove Theorem 9.1.

Theorem 9.1. *Let X be a klt stable surface with* $\mathrm{mld}(X)\geq 5/46$. *Then* $K_X^2 > 1/48983$.

9.1. Local invariants and global identities.

In this subsection, we define the local invariants of a quotient singularity, write the plurigenera and the orbifold Bogomolov–Miyaoka–Yau inequality of a klt stable surface of small volume in terms of them, and prove the local cost inequality for quotient singularities that are not cyclic.

Definition 9.2. Let $X\ni x$ be a klt surface germ that is not smooth, and let $E_1,\dots,E_\ell$ be the exceptional curves of the minimal resolution of $X\ni x$. Let $\ell_x := \ell$, $A_x := (-E_i\cdot E_j)_{1\leq i,j\leq\ell}$, $k_x := (-E_i^2-2)_{1\leq i\leq\ell}$, $b_x := A_x^{-1}k_x$, and $\gamma_x := \ell_x - k_x^tb_x$. Let g_x be the order of the local fundamental group of $X\ni x$, and $\mu_x := \gamma_x + 1 - 1/g_x$. For an integer $m\geq 2$, let $\{mb_x\}$ be the vector of fractional parts of the entries of mb_x, and

$$c_x(m) := \frac{1}{2}\left(k_x^t\{mb_x\} - \{mb_x\}^tA_x\{mb_x\}\right).$$

Finally, let

$$(w_2,\dots,w_9) := (139754, 122195, 98574, 75044, 51760, 31572, 15936, 7272),$$

$$C_x := 57820\mu_x + \sum_{m=2}^{9} w_m c_x(m),$$

and

$$D_x := 57820\Big(\gamma_x + \frac{1}{2}\Big) + \sum_{m=2}^{9} w_m c_x(m) - 91(\ell_x + 1).$$

We call C_x the *cost* of $X \ni x$.

Lemma 9.3. *Let $X \ni x$ be a klt surface germ that is not smooth, $\pi : \widetilde{X} \to X$ the minimal resolution of $X \ni x$, and $E_1, \dots, E_\ell$ the exceptional curves of π. Let $k_{x,i}$ and $b_{x,i}$ be the i-th entries of k_x and b_x. Then $K_{\widetilde{X}} \cdot E_i = k_{x,i}$ and $a(E_i, X, 0) = 1 - b_{x,i}$ for any i, and*

$$\pi^* K_X = K_{\widetilde{X}} + \sum_{i=1}^{\ell} b_{x,i} E_i$$

near $\pi^{-1}(x)$. Moreover, $0 \le b_{x,i} < 1$ for any i, and $\operatorname{mld}(X \ni x) = \min_i(1 - b_{x,i})$.

Proof. The curves E_i are smooth rational curves forming a simple normal crossing divisor [KM98, Theorem 4.7], so $K_{\widetilde{X}} \cdot E_i = -E_i^2 - 2 = k_{x,i}$ by the adjunction formula [Har77, Proposition V.1.5]. Since $A_x b_x = k_x$, the $\mathbb{Q}$-divisor $K_{\widetilde{X}} + \sum_i b_{x,i} E_i$ has zero intersection number with any E_j. Since A_x is invertible [KM98, Lemma 3.40], this implies the formula for $\pi^* K_X$, so $a(E_i, X, 0) = 1 - b_{x,i}$. Since $X \ni x$ is klt, $b_{x,i} < 1$, and $b_{x,i} \ge 0$ by [KM98, Corollary 4.3]. The last equality follows from [KM98, Lemma 2.30, Corollary 2.31(3)]. □

Lemma 9.4. *Let X be a klt stable surface with $K_X^2 \le 1/48983$. Then the following hold, where x runs over the singular points of X.*

(1) $P_m(X) = 1 + \binom{m}{2} K_X^2 + \sum_x c_x(m)$ for any integer $m \ge 2$.
(2) $\sum_x \mu_x \le 12 - \frac{4}{3} K_X^2$.

Proof. Let $\pi : \widetilde{X} \to X$ be the minimal resolution, and write $\pi^* K_X = K_{\widetilde{X}} + B_{\widetilde{X}}$. Since $K_X^2 \le 1/48983 < 1/143$, Corollary 2.8 implies that $\chi(\mathcal{O}_X) = 1$ and $e(X) = 2 + \rho(X)$.

Fix a singular point x of X, and let $E_1, \dots, E_\ell$ be the exceptional curves over x. By Lemma 9.3, near $\pi^{-1}(x)$ we have $B_{\widetilde{X}} = \sum_{i=1}^{\ell} b_{x,i} E_i$. Hence the contribution of x to $B_{\widetilde{X}}^2$ is $-b_x^t A_x b_x = -k_x^t b_x$, so

$$K_{\widetilde{X}}^2 = (\pi^* K_X - B_{\widetilde{X}})^2 = K_X^2 - \sum_x k_x^t b_x.$$

(1) Let $m \ge 2$ be an integer. Since X is klt and $mK_X - K_X = (m-1)K_X$ is ample, $H^i(X, \mathcal{O}_X(mK_X)) = 0$ for $i > 0$ [KMM87, Theorem 1-2-5]. Thus $P_m(X) = \chi(X, \mathcal{O}_X(mK_X))$. By the Riemann–Roch formula of Blache [Bla95, Proposition 5.2], in the form [LL26, Theorem 6.12, Corollary 6.13],

$$\chi(X, \mathcal{O}_X(mK_X)) = \chi(\mathcal{O}_X) + \binom{m}{2} K_X^2 + \delta_m(X), \quad \delta_m(X) := \frac{1}{2}\big(K_{\widetilde{X}} + \{mB_{\widetilde{X}}\}\big) \cdot \{mB_{\widetilde{X}}\},$$

where $\delta_m(X)$ is defined in [LL26, Definition 6.11] and $\{mB_{\widetilde{X}}\}$ is the fractional part of $mB_{\widetilde{X}}$. Since $\{mB_{\widetilde{X}}\} = \sum_i \{mb_{x,i}\} E_i$ near $\pi^{-1}(x)$, $K_{\widetilde{X}} \cdot E_i = k_{x,i}$, and $E_i \cdot E_j = -(A_x)_{ij}$, the contribution of x to $\delta_m(X)$ is $\frac{1}{2}\big(k_x^t \{mb_x\} - \{mb_x\}^t A_x \{mb_x\}\big) = c_x(m)$. Together with $\chi(\mathcal{O}_X) = 1$, this implies (1).

(2) Let L be the number of π-exceptional curves. By Corollary 2.8 and the formula for $K_{\widetilde{X}}^2$ above,

$$\rho(X) + L = \rho(\widetilde{X}) = 10 - K_{\widetilde{X}}^2 = 10 - K_X^2 + \sum_x k_x^t b_x.$$

Since $L = \sum_x \ell_x$ and $\gamma_x = \ell_x - k_x^t b_x$, this implies $\rho(X) = 10 - K_X^2 - \sum_x \gamma_x$.

Since X is klt, X has quotient singularities [KM98, Proposition 4.18], and K_X is ample, hence nef. By the orbifold Bogomolov–Miyaoka–Yau inequality, due to Sakai [Sak80], Miyaoka [Miy84], Kobayashi–Nakamura–Sakai [KNS89], and Megyesi [Meg99], in the form [HK08, Theorem 3.1],

$$K_X^2 \le 3e_{\mathrm{orb}}(X), \quad e_{\mathrm{orb}}(X) := e(X) - \sum_x \Big(1 - \frac{1}{g_x}\Big).$$

Since $e(X) = 2 + \rho(X)$,

$$K_X^2 \le 3\Big(12 - K_X^2 - \sum_x \gamma_x - \sum_x \Big(1 - \frac{1}{g_x}\Big)\Big) = 36 - 3K_X^2 - 3\sum_x \mu_x.$$

Thus $3\sum_x \mu_x \le 36 - 4K_X^2$, which is (2). □

Lemma 9.5. *Let d, t, and u be integers with $d \ge 2$, $1 \le t < d$, $\gcd(d,t) = 1$, and $u \ge 1$. Let $X \ni x$ and $Y \ni y$ be klt surface germs of types $[2;(2,1),(2,1),(ud+t,(u-1)d+t)]$ and $[2;(2,1),(2,1),(d+t,t)]$, respectively. Then $\ell_x = \ell_y + u - 1$, $\gamma_x = \gamma_y + u - 1$, and $c_x(m) = c_y(m)$ for any integer $m \ge 2$.*

Proof. If $u = 1$, then $X \ni x$ and $Y \ni y$ are both of type $[2;(2,1),(2,1),(d+t,t)]$ and there is nothing to prove, so we may assume that $u \ge 2$. For an integer $j \ge 2$,

$$\frac{jd+t}{(j-1)d+t} = 2 - \frac{1}{\frac{(j-1)d+t}{(j-2)d+t}}.$$

By induction on u, the Hirzebruch–Jung chain of $(ud+t)/((u-1)d+t)$ is $[2,\dots,2,e_1,\dots,e_k]$ with $u-1$ entries 2, where $[e_1,\dots,e_k]$ is the Hirzebruch–Jung chain of $(d+t)/t$. Thus the minimal resolution of $X \ni x$ has the fork E_0 with $E_0^2 = -2$, the (-2)-curves F_1 and F_2 of the first two arms, and the third arm $G_1,\dots,G_{u-1},H_1,\dots,H_k$, read from E_0 outwards, with $G_i^2 = -2$ and $H_j^2 = -e_j$. The minimal resolution of $Y \ni y$ has the fork E_0', the curves F_1' and F_2', and the third arm $H_1',\dots,H_k'$, with $E_0'^2 = F_1'^2 = F_2'^2 = -2$ and $H_j'^2 = -e_j$, and H_1' intersects E_0'. In particular, $\ell_x = \ell_y + u - 1$.

For a curve E in the exceptional set of a germ, let $\deg(E)$ be the number of other exceptional curves meeting E. Since $(A_x\mathbf{1})_E = -E^2 - \deg(E)$ and $k_{x,E} = -E^2 - 2$, where $\mathbf{1} = (1,\dots,1)$, the vector $a_x := \mathbf{1} - b_x$ satisfies

$$(A_x a_x)_E = 2 - \deg(E)$$

for any exceptional curve E, and a_x is the unique vector a with $(A_x a)_E = 2 - \deg(E)$ for any E, since A_x is positive definite [KM98, Lemma 3.40]. Similarly, $a_y := \mathbf{1} - b_y$ is the unique vector a' with $(A_y a')_{E'} = 2 - \deg(E')$ for any exceptional curve E' over y. By Lemma 9.3, the entries of a_y are the log discrepancies $a(E,Y,0)$. For $Y \ni y$, let $\sigma := \frac{1}{2} + \frac{1}{2} + \frac{1}{d+t} - 1 = \frac{1}{d+t}$ and $\eta := 2 - \frac{1}{2} - \frac{1}{2} - \frac{t}{d+t} = \frac{d}{d+t}$. By Lemma 2.10,

$$a(E_0',Y,0) = \frac{\sigma}{\eta} = \frac{1}{d}, \quad a(H_1',Y,0) = \frac{t\cdot\frac{1}{d}+1}{d+t} = \frac{1}{d}.$$

Let a be the vector with entries $a_{E_0} = a_{G_1} = \cdots = a_{G_{u-1}} = \frac{1}{d}$, $a_{F_i} = a(F_i',Y,0)$, and $a_{H_j} = a(H_j',Y,0)$. We check $(A_x a)_E = 2 - \deg(E)$. At E_0, the left-hand side is $2a_{E_0} - a_{F_1} - a_{F_2} - a_{G_1}$, which equals the left-hand side of the equation of Y at E_0', since $a_{G_1} = \frac{1}{d} = a(H_1',Y,0)$. At G_i, it is $\frac{2}{d} - \frac{1}{d} - \frac{1}{d} = 0 = 2 - \deg(G_i)$. At H_1, the neighbor G_{u-1} has $a_{G_{u-1}} = \frac{1}{d} = a(E_0',Y,0)$, so the equation is the equation of Y at H_1'. At any other curve, the equation is the equation of Y at the corresponding curve. Thus $a = a_x$. Therefore the entries of b_x are the entries of b_y at the corresponding curves, together with the entry $1 - \frac{1}{d}$ at $G_1,\dots,G_{u-1}$.

Since $k_{x,G_i} = 0$ and the other entries of k_x and b_x are the entries of k_y and b_y at the corresponding curves, $k_x^t b_x = k_y^t b_y$, hence $\gamma_x = \gamma_y + u - 1$.

Let $m \geq 2$ be an integer, $f := \{mb_x\}$, $f' := \{mb_y\}$, and $\epsilon_m := \{m(1-\frac{1}{d})\}$, so that $f_{E_0} = f_{G_i} = f_{H_1} = \epsilon_m$ and $f'_{E'_0} = f'_{H'_1} = \epsilon_m$. Since $k_{x,G_i} = 0$ and the other entries of k_x and f are the entries of k_y and f' at the corresponding curves, $k_x^t f = k_y^t f'$. Moreover,

$$f^t A_x f = \sum_E (-E^2) f_E^2 - 2 \sum_{\{E,E'\}} f_E f_{E'},$$

where the second sum runs over the pairs of exceptional curves meeting each other, and similarly for $f'^t A_y f'$. Compared with $f'^t A_y f'$, the diagonal part of $f^t A_x f$ has the $u-1$ additional terms $2\epsilon_m^2$ at $G_1, \dots, G_{u-1}$, and the term $-2\epsilon_m^2$ of the pair $\{E'_0, H'_1\}$ is replaced by the u terms $-2\epsilon_m^2$ of the pairs $\{E_0, G_1\}, \{G_1, G_2\}, \dots, \{G_{u-1}, H_1\}$. Thus

$$f^t A_x f - f'^t A_y f' = 2(u-1)\epsilon_m^2 - 2(u-1)\epsilon_m^2 = 0,$$

and $c_x(m) = c_y(m)$. □

Proposition 9.6. *Let $X \ni x$ be a klt surface germ that is neither smooth nor a cyclic quotient singularity, with $\mathrm{mld}(X \ni x) \geq 5/46$. Then the following hold.*

1. *If $X \ni x$ is of type $[e_0; (2,1), (2,1), (r,q)]$ with $e_0 \geq 3$, then $e_0 \leq 6$, $r \leq 8/(e_0-2)$, and $(e_0-1)r - q \leq 9$.*
2. *If $X \ni x$ is of type $[2; (2,1), (2,1), (r, r-1)]$, then $b_x = 0$, $c_x(m) = 0$ for any integer $m \geq 2$, and $\gamma_x = \ell_x \geq 4$.*
3. *If $X \ni x$ is of type $[2; (2,1), (2,1), (r,q)]$ and $d := r - q \geq 2$, then $d \leq 9$, and $X \ni x$ is of type $[2; (2,1), (2,1), (ud+t, (u-1)d+t)]$ for some integers $u \geq 1$ and $1 \leq t < d$ with $\gcd(d,t) = 1$.*
4. *$D_x > 0$, and $C_x \geq D_x + 91(\ell_x + 1)$. In particular, $C_x \geq 91(\ell_x+1)$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we determine the possible types of $X \ni x$.

Since $X \ni x$ is neither smooth nor a cyclic quotient singularity, the dual graph of the minimal resolution of $X \ni x$ is a tree of smooth rational curves with exactly one vertex of degree three [KM98, Theorem 4.7]. By minimality, $E^2 \leq -2$ for any exceptional curve E. Thus $X \ni x$ is of type $[e_0; (r_1,q_1), (r_2,q_2), (r_3,q_3)]$ with $e_0 \geq 2$ (Definition 2.9). Let $\sigma := \sum_{i=1}^3 1/r_i - 1$ and $\eta := e_0 - \sum_{i=1}^3 q_i/r_i$. By Lemma 2.10, (r_1,r_2,r_3) is one of $(2,2,r)$ with $r \geq 2$, $(2,3,3)$, $(2,3,4)$, and $(2,3,5)$, and the fork E_0 has log discrepancy $a_0 := \sigma/\eta$. Since $a_0 \geq \mathrm{mld}(X \ni x) \geq 5/46$, we have

$$\frac{1}{a_0} = \frac{\eta}{\sigma} \leq \frac{46}{5}.$$

If $r_2 = 2$, then $(r_1,q_1) = (r_2,q_2) = (2,1)$ since $q_i < r_i$, and $X \ni x$ is of type D-I if $e_0 \geq 3$ and of type D-II if $e_0 = 2$. If $r_2 = 3$, then $(r_1,q_1) = (2,1)$, $r_3 \in \{3,4,5\}$, and $X \ni x$ is of type E-I or E-II.

Step 2. In this step, we treat the germs of type D-I and prove (1).

Write the type as $[e_0; (2,1), (2,1), (r,q)]$ with $e_0 \geq 3$, and let $d := (e_0-1)r - q$. Then $\sigma = \frac{1}{r}$ and $\eta = e_0 - 1 - \frac{q}{r} = \frac{d}{r}$, so $a_0 = \frac{1}{d}$ and d is a positive integer with $d \leq 46/5$, that is, $d \leq 9$. Since $q \leq r-1$, we have $(e_0-2)r + 1 \leq d \leq 9$. As $r \geq 2$, this implies $3 \leq e_0 \leq 6$ and $r \leq 8/(e_0-2)$, which proves (1). Listing the coprime integers $1 \leq q < r$ with $(e_0-1)r - q \leq 9$, we obtain nineteen triples (e_0,r,q). If $e_0 = 3$, then $r = 2,3,4,5,6,7,8$ and q is in $\{1\}$, $\{1,2\}$, $\{1,3\}$, $\{1,2,3,4\}$, $\{5\}$, $\{5,6\}$, $\{7\}$, respectively. If $e_0 = 4$, then $r = 2,3,4$ and q is in $\{1\}$, $\{1,2\}$, $\{3\}$, respectively. If $e_0 \in \{5,6\}$, then $(r,q) = (2,1)$. The nineteen triples are the rows of Table 6.

Step 3. In this step, we treat the germs of type D-II and prove (2) and (3).

Write the type as $[2; (2,1), (2,1), (r,q)]$ with $r \geq 2$, and let $d := r - q$. Then $\sigma = \frac{1}{r}$ and $\eta = 1 - \frac{q}{r} = \frac{d}{r}$, so $a_0 = \frac{1}{d}$ and $1 \leq d \leq 9$.

Suppose that $d = 1$. Since $\frac{r}{r-1} = 2 - 1/\frac{r-1}{r-2}$ for $r \geq 3$, the Hirzebruch–Jung chain of $r/(r-1)$ is $[2, \dots, 2]$ with $r-1$ entries. Thus all exceptional curves of $X \ni x$ are (-2)-curves, so $k_x = b_x = 0$, $\ell_x = r + 2 \geq 4$, $\gamma_x = \ell_x$, and $c_x(m) = 0$ for any integer $m \geq 2$, which proves (2). Hence

$$D_x = 57820\Big(\ell_x + \frac{1}{2}\Big) - 91(\ell_x + 1) = 57729\ell_x + 28819 \geq 57729 \cdot 4 + 28819 = 259735 > 0.$$

Suppose that $2 \leq d \leq 9$. Since $\gcd(r, d) = \gcd(r, q) = 1$ and $r > d$, we may write $r = ud + t$ with integers $u \geq 1$ and $1 \leq t < d$, and $\gcd(d, t) = 1$. Then $q = (u-1)d + t$, so $X \ni x$ is of type $[2; (2,1), (2,1), (ud + t, (u-1)d + t)]$, which proves (3). Let $Y \ni y$ be a germ of type $[2; (2,1), (2,1), (d + t, t)]$, which exists by the row of [HK08, Section 5, Table 1] with dual graph $\langle 2; 2, 1; 2, 1; d + t, t\rangle$, and is klt since it is a quotient singularity [KM98, Proposition 4.18]. By Lemma 9.5,

$$D_x = D_y + (57820 - 91)(u - 1) = D_y + 57729(u - 1) \geq D_y.$$

The pairs (d, t) with $2 \leq d \leq 9$, $1 \leq t < d$, and $\gcd(d, t) = 1$ are the twenty-seven rows of Table 7. Thus $D_x > 0$ if $D_y > 0$ for the germ $Y \ni y$ of each row of Table 7.

Step 4. In this step, we treat the germs of type E-I or E-II.

Write the type as $[e_0; (2,1), (3, q_2), (r_3, q_3)]$, where $r_3 \in \{3, 4, 5\}$, $1 \leq q_2 \leq 2$, $1 \leq q_3 < r_3$, $\gcd(r_3, q_3) = 1$, and $q_2 \leq q_3$ if $r_3 = 3$. Then $\sigma = \frac{1}{6}, \frac{1}{12}, \frac{1}{30}$ for $r_3 = 3, 4, 5$, respectively. Since $q_i \leq r_i - 1$,

$$\sum_{i=1}^{3} \frac{q_i}{r_i} \leq 3 - \sum_{i=1}^{3} \frac{1}{r_i} = 2 - \sigma.$$

Since $\eta = \sigma/a_0 \leq \frac{46}{5}\sigma$, we obtain

$$e_0 = \eta + \sum_{i=1}^{3} \frac{q_i}{r_i} \leq 2 + \frac{41}{5}\sigma.$$

Thus $e_0 \in \{2, 3\}$ if $r_3 = 3$, and $e_0 = 2$ if $r_3 \in \{4, 5\}$. Moreover, $1/a_0 = \eta/\sigma$ is a positive integer at most 9, and

$$\frac{1}{a_0} = \begin{cases} 6e_0 - 3 - 2(q_2 + q_3) & \text{if } r_3 = 3, \\ 18 - 4q_2 - 3q_3 & \text{if } r_3 = 4 \text{ and } e_0 = 2, \\ 45 - 10q_2 - 6q_3 & \text{if } r_3 = 5 \text{ and } e_0 = 2. \end{cases}$$

For $r_3 = 3$, the condition $1 \leq 1/a_0 \leq 9$ allows $(q_2, q_3) = (1,1), (1,2), (2,2)$ if $e_0 = 2$, and $(q_2, q_3) = (1,2), (2,2)$ if $e_0 = 3$. For $r_3 = 4$, it allows $(q_2, q_3) = (1,3), (2,1), (2,3)$. For $r_3 = 5$, it allows $(q_2, q_3) = (2,3), (2,4)$. Thus we obtain ten germs, the rows of Table 8.

Step 5. In this step, we explain how to check the rows of Tables 6, 7, and 8, and prove (4).

In each row, the exceptional curves are ordered as follows: the fork, the first arm, the second arm, and the third arm, each arm read from the fork outwards. The self-intersection numbers are determined by the type, and ℓ_x is the number of entries of Ib_x. In Table 7, the germ of a row is the germ $Y \ni y$ of **Step 3**, with y in place of x. Each row lists a positive integer I and a vector Ib_x with integer entries such that

$$A_x(Ib_x) = Ik_x,$$

which can be checked one curve at a time: the entry at a curve E is $-E^2$ times the entry of Ib_x at E minus the sum of the entries of Ib_x at the curves meeting E. Since A_x is invertible, the equation $A_x(Ib_x) = Ik_x$ determines b_x. For $2 \leq m \leq 9$, let V_m be the vector with entries in $\{0, \dots, I-1\}$ that is congruent to mIb_x modulo I entrywise. Then $\{mb_x\} = V_m/I$, $c_x(m) = \frac{1}{2I^2}\big(Ik_x^t V_m - V_m^t A_x V_m\big)$, and

$$2I^2 D_x = 57820\big((2\ell_x + 1)I^2 - 2Ik_x^t(Ib_x)\big) - 182I^2(\ell_x + 1) + \sum_{m=2}^{9} w_m\big(Ik_x^t V_m - V_m^t A_x V_m\big).$$

All values of D_x in Tables 6, 7, and 8 are positive. By **Steps 1–4**, $D_x > 0$ for any germ $X \ni x$ in the proposition.

Finally, $a_0 \le 1$ by Lemma 9.3, and $\sigma \le \frac{3}{2} - 1 = \frac{1}{2}$ since $r_i \ge 2$. By Lemma 2.10,

$$g_x = \frac{4\eta}{\sigma^2} = \frac{4}{a_0\sigma} \ge 8.$$

Thus $1 - \frac{1}{g_x} \ge \frac{1}{2}$, and

$$C_x - 91(\ell_x + 1) = D_x + 57820\Big(\frac{1}{2} - \frac{1}{g_x}\Big) \ge D_x > 0. \qquad \square$$

e_0	r	q	I	Ib_x	D_x
3	2	1	3	$(2,1,1,1)$	202097/3
3	3	1	5	$(4,2,2,3)$	40292
3	3	2	8	$(6,3,3,4,2)$	61227/4
3	4	1	7	$(6,3,3,5)$	28546
3	4	3	5	$(4,2,2,3,2,1)$	51947
3	5	1	9	$(8,4,4,7)$	209666/9
3	5	2	16	$(14,7,7,12,6)$	190261/8
3	5	3	7	$(6,3,3,5,4)$	36988
3	5	4	12	$(10,5,5,8,6,4,2)$	176045/6
3	6	5	7	$(6,3,3,5,4,3,2,1)$	45339
3	7	5	9	$(8,4,4,7,6,5)$	338836/9
3	7	6	16	$(14,7,7,12,10,8,6,4,2)$	303655/8
3	8	7	9	$(8,4,4,7,6,5,4,3,2,1)$	419939/9
4	2	1	5	$(4,2,2,2)$	28546
4	3	1	16	$(14,7,7,10)$	70723/8
4	3	2	7	$(6,3,3,4,2)$	20013
4	4	3	9	$(8,4,4,6,4,2)$	148570/9
5	2	1	7	$(6,3,3,3)$	11571
6	2	1	9	$(8,4,4,4)$	19400/9

TABLE 6. The germs of type $[e_0; (2,1), (2,1), (r,q)]$ with $e_0 \ge 3$ and $(e_0-1)r - q \le 9$, with $A_x(Ib_x) = Ik_x$ and the value D_x of $57820(\gamma_x + \frac{1}{2}) + \sum_{m=2}^{9} w_m c_x(m) - 91(\ell_x + 1)$.

9.2. **The local cost inequality.** In this subsection, we prove Theorem 9.10, the positivity of the cost for quotient singularities of minimal log discrepancy at least 5/46. Noncyclic germs, among them the germs of type D-II with a third arm of any length, are treated by Proposition 9.6. Among the cyclic quotient singularities with minimal log discrepancy at least 5/46 there are infinite families, in which a string of (-2)-curves has arbitrary length. Lemma 9.8 shows that on such a family $2rC_x$, where r is the order of the singularity, is piecewise a polynomial of degree at most two in the length, and Theorem 9.10 follows from Proposition 9.9 and Table 9.

Definition-Lemma 9.7. Let e, r_1, q_1, n, and t be integers such that $e \ge 3$, $t \ge 1$, $n \ge \max\{r_1, t\}$, $\gcd(n,t) = 1$, $n = t$ only if $n = t = 1$, and either $(r_1, q_1) = (1,0)$, or $1 \le q_1 < r_1$ and $\gcd(r_1, q_1) = 1$. Let $X \ni x$ be a cyclic quotient singularity whose minimal resolution has a chain of exceptional curves $F_k, \dots, F_1, G_0, G_1, \dots, G_{k'}$, with consecutive curves intersecting, such that $-G_0^2 = e$, $[-F_1^2, \dots, -F_k^2]$ is the Hirzebruch–Jung chain of r_1/q_1, and $[-G_1^2, \dots, -G_{k'}^2]$ is the Hirzebruch–Jung chain of $n/(n-t)$, where $k = 0$ if $r_1 = 1$ and $k' = 0$ if $n = 1$. We call $F_1, \dots, F_k$ the *first arm* and $G_1, \dots, G_{k'}$ the *second arm*. Let

$$\alpha := (e-1)r_1 - q_1, \quad \delta := \alpha - t, \quad r := \alpha n + r_1 t, \quad h := \lfloor n/t \rfloor - 1, \quad v := n - (h+1)t.$$

Then the following hold.

d	t	I	Ib_y	D_y
2	1	4	$(2,1,1,2)$	$116369/2$
3	1	3	$(2,1,1,2)$	$317191/3$
3	2	3	$(2,1,1,2,1)$	$375284/3$
4	1	8	$(6,3,3,6)$	$173953/4$
4	3	8	$(6,3,3,6,4,2)$	$292143/4$
5	1	5	$(4,2,2,4)$	74529
5	2	5	$(4,2,2,4,2)$	86275
5	3	5	$(4,2,2,4,3)$	98021
5	4	5	$(4,2,2,4,3,2,1)$	109676
6	1	12	$(10,5,5,10)$	$271705/6$
6	5	12	$(10,5,5,10,8,6,4,2)$	$522419/6$
7	1	7	$(6,3,3,6)$	60767
7	2	7	$(6,3,3,6,3)$	69300
7	3	7	$(6,3,3,6,4,2)$	77742
7	4	7	$(6,3,3,6,5)$	86275
7	5	7	$(6,3,3,6,5,4)$	94717
7	6	7	$(6,3,3,6,5,4,3,2,1)$	103068
8	1	16	$(14,7,7,14)$	$416977/8$
8	3	16	$(14,7,7,14,10)$	$532555/8$
8	5	16	$(14,7,7,14,12,6)$	$652093/8$
8	7	16	$(14,7,7,14,12,10,8,6,4,2)$	$765487/8$
9	1	9	$(8,4,4,8)$	$455401/9$
9	2	9	$(8,4,4,8,4)$	$538961/9$
9	4	9	$(8,4,4,8,6,4,2)$	$668131/9$
9	5	9	$(8,4,4,8,7)$	$729227/9$
9	7	9	$(8,4,4,8,7,6,5)$	$858397/9$
9	8	9	$(8,4,4,8,7,6,5,4,3,2,1)$	$939500/9$

TABLE 7. The germs $Y \ni y$ of type $[2;(2,1),(2,1),(d+t,t)]$ with $2 \le d \le 9$, $1 \le t < d$, and $\gcd(d,t) = 1$, with $A_y(Ib_y) = Ik_y$ and the value D_y of $57820(\gamma_y + \frac{1}{2}) + \sum_{m=2}^{9} w_m c_y(m) - 91(\ell_y + 1)$.

e_0	r_3	q_2	q_3	I	Ib_x	D_x
2	3	1	1	5	$(4,2,3,3)$	52038
2	3	1	2	9	$(6,3,5,4,2)$	$40802/9$
3	3	1	2	27	$(24,12,17,16,8)$	$394241/27$
2	3	2	2	1	$(0,0,0,0,0,0)$	375193
3	3	2	2	7	$(6,3,4,2,4,2)$	28455
2	4	1	3	5	$(4,2,3,3,2,1)$	63693
2	4	2	1	7	$(6,3,4,2,5)$	36988
2	4	2	3	1	$(0,0,0,0,0,0,0)$	432922
2	5	2	3	7	$(6,3,4,2,5,4)$	45430
2	5	2	4	1	$(0,0,0,0,0,0,0,0)$	490651

TABLE 8. The germs of type $[e_0;(2,1),(3,q_2),(r_3,q_3)]$ whose fork has log discrepancy at least $1/9$, with $A_x(Ib_x) = Ik_x$ and the value D_x of $57820(\gamma_x + \frac{1}{2}) + \sum_{m=2}^{9} w_m c_x(m) - 91(\ell_x + 1)$.

(1) The order of $X \ni x$ is r. For every exceptional curve G of $X \ni x$ there exists an integer y_G with $a(G,X,0) = y_G/r$. Moreover, $y_{G_0} = n + r_1$, $y_{F_1} = q_1 + (e-1)n + t$ if $k \ge 1$, $y_{G_1} = n + r_1 + \delta$ if $k' \ge 1$, and $-G^2 y_G = y_{G'} + y_{G''}$ for every exceptional curve G, where $y_{G'}$ and $y_{G''}$ are the values at the neighbors of G in the chain and the value r is used for each missing neighbor of an end curve.

(2) $\gcd(t,v)=1$, $v=0$ only if $t=1$, and $h\geq\max\{0,\lceil(r_1-v)/t\rceil-1\}$. If $k'\geq 1$, then $[-G_1^2,\dots,-G_{k'}^2]$ consists of h entries 2 followed by the Hirzebruch–Jung chain of $(t+v)/v$, which is empty if $v=0$, and $y_{G_i}=n+r_1+i\delta$ for $0\leq i\leq\min\{h+1,k'\}$.
(3) If $\operatorname{mld}(X\ni x)\geq 5/46$ and $a(G_0,X,0)=\operatorname{mld}(X\ni x)$, then $e\leq 18$, $r_1\leq 18$, $5(e-2)r_1\leq 92$, $t\leq\min\{\alpha,9\}$, and $(5\alpha-46)n\leq r_1(46-5t)$. In particular, $n\leq r_1(46-5t)/(5\alpha-46)$ if $\alpha\geq 10$.
(4) Conversely, for any cyclic quotient singularity $X\ni x$ with $\operatorname{mld}(X\ni x)\geq 5/46$ whose minimal resolution has an exceptional curve G with $G^2\leq -3$, there exist integers e, r_1, q_1, n, and t and a chain of exceptional curves as in the first two sentences with $a(G_0,X,0)=\operatorname{mld}(X\ni x)$.

Proof of Definition-Lemma 9.7. Let $\pi:\widetilde{X}\to X$ be the minimal resolution of $X\ni x$. For any exceptional curve G of $X\ni x$ let $a_G:=a(G,X,0)$, so that $K_{\widetilde{X}}=\pi^*K_X+\sum_G(a_G-1)G$. By Lemma 9.3, $a_G\leq 1$ for any G, and $\operatorname{mld}(X\ni x)=\min_G a_G$. Since G is a smooth rational curve, $K_{\widetilde{X}}\cdot G=-2-G^2$ by adjunction. Intersecting with G, we obtain

$$-2-G^2=(a_G-1)G^2+\sum_{G'}(a_{G'}-1),$$

where G' runs over the neighbors of G in the chain. The number of neighbors of G plus the number of missing neighbors of G equals 2. Thus

$$-G^2a_G=a_{G'}+a_{G''}, \tag{9.1}$$

where $a_{G'}$ and $a_{G''}$ are the values at the neighbors of G in the chain, and the value 1 is used for each missing neighbor of an end curve.

For integers $b_1,\dots,b_s\geq 2$, let $\det[b_1,\dots,b_s]$ be the determinant of the $s\times s$ matrix with diagonal entries $b_1,\dots,b_s$, entries -1 directly above and below the diagonal, and 0 elsewhere, and let $\det[]:=1$ for $s=0$. Expanding along the first row,

$$\det[b_1,\dots,b_s]=b_1\det[b_2,\dots,b_s]-\det[b_3,\dots,b_s]\quad\text{for } s\geq 2.$$

Let $\rho:=\det[b_1,\dots,b_s]$, $\sigma:=\det[b_2,\dots,b_s]$, and $\sigma':=\det[b_3,\dots,b_s]$ if $s\geq 2$, $\sigma':=0$ if $s=1$. By induction on s, $\sigma>\sigma'\geq 0$ and $\gcd(\sigma,\sigma')=1$. Since $\rho=b_1\sigma-\sigma'$, it follows that $\rho>\sigma\geq 1$, $\gcd(\rho,\sigma)=1$, and $\rho/\sigma=b_1-\sigma'/\sigma$. Thus $[b_1,\dots,b_s]$ is the Hirzebruch–Jung chain of ρ/σ. Since the Hirzebruch–Jung chain of a fraction is unique, the Hirzebruch–Jung chain $[b_1,\dots,b_s]$ of ρ/σ, for coprime integers $\rho>\sigma\geq 1$, satisfies $\det[b_1,\dots,b_s]=\rho$ and $\det[b_2,\dots,b_s]=\sigma$ (see also [BHPV04, Chapter III, Section 5]). For $(\rho,\sigma)=(1,0)$ the chain is empty, and we set $\det[b_2,\dots,b_s]:=0$, so that $\det[b_1,\dots,b_s]=\rho$ and $\det[b_2,\dots,b_s]=\sigma$ also hold for $(\rho,\sigma)=(1,0)$.

(1) Let $b_i:=-F_i^2$ for $1\leq i\leq k$ and $b'_j:=-G_j^2$ for $1\leq j\leq k'$. Then $\det[b_1,\dots,b_k]=r_1$, $\det[b_2,\dots,b_k]=q_1$, $\det[b'_1,\dots,b'_{k'}]=n$, and $\det[b'_2,\dots,b'_{k'}]=n-t$. Expanding the determinant of A_x along the row of G_0, we obtain

$$\begin{aligned}\det A_x=e\det[b_1,\dots,b_k]\det[b'_1,\dots,b'_{k'}]-\det[b_2,\dots,b_k]\det[b'_1,\dots,b'_{k'}]\\ -\det[b_1,\dots,b_k]\det[b'_2,\dots,b'_{k'}],\end{aligned}$$

that is,

$$\det A_x=er_1n-q_1n-r_1(n-t)=\alpha n+r_1t=r.$$

Thus the chain $[-F_k^2,\dots,-F_1^2,e,-G_1^2,\dots,-G_{k'}^2]$ is the Hirzebruch–Jung chain of r/q', where q' is the determinant of the chain without F_k (without G_0 if $k=0$). Hence $X\ni x$ is of type $\frac{1}{r}(1,q')$, and the order of $X\ni x$ is r.

For any exceptional curve G let $y_G:=ra_G$. Multiplying (9.1) by r, we obtain $-G^2y_G=y_{G'}+y_{G''}$, with the value r at each missing neighbor. We compute y_G along the first arm. Let $\mu_i:=\det[b_{i+1},\dots,b_k]$ for $0\leq i\leq k$, $\mu_{k+1}:=0$, $\lambda_0:=0$, and $\lambda_i:=\det[b_1,\dots,b_{i-1}]$ for $1\leq i\leq k+1$. Expanding along the first and the last row, respectively,

$$b_i\mu_i=\mu_{i-1}+\mu_{i+1},\qquad b_i\lambda_i=\lambda_{i-1}+\lambda_{i+1}\qquad\text{for } 1\leq i\leq k,$$

and $\mu_0 = r_1$, $\mu_1 = q_1$, $\lambda_1 = 1$, and $\lambda_{k+1} = r_1$, also when $k = 0$. Let $y'_0 := y_{G_0}$, $y'_i := y_{F_i}$ for $1 \le i \le k$, and $y'_{k+1} := r$. Then $b_i y'_i = y'_{i-1} + y'_{i+1}$ for $1 \le i \le k$. Let $d_i := r_1 y'_i - y'_0 \mu_i - r\lambda_i$ for $0 \le i \le k+1$. Then $b_i d_i = d_{i-1} + d_{i+1}$ for $1 \le i \le k$, and $d_0 = d_{k+1} = 0$. Since $\lambda_0 = 0$ and $\lambda_1 = 1$, induction on i shows $d_i = d_1 \lambda_i$. Thus $0 = d_{k+1} = d_1 r_1$, so $d_1 = 0$, that is,

$$r_1 y'_1 = q_1 y_{G_0} + r.$$

Replacing the first arm with the second arm, whose Hirzebruch–Jung chain is that of $n/(n-t)$, the computation of $r_1 y'_1$ above shows $n y''_1 = (n-t) y_{G_0} + r$, where $y''_1 := y_{G_1}$ if $k' \ge 1$ and $y''_1 := r$ if $k' = 0$. The equation $e y_{G_0} = y'_1 + y''_1$ at G_0, multiplied by $r_1 n$, becomes

$$e r_1 n y_{G_0} = n(q_1 y_{G_0} + r) + r_1\big((n-t) y_{G_0} + r\big),$$

that is, $(\alpha n + r_1 t) y_{G_0} = r(n + r_1)$. Since $r = \alpha n + r_1 t$, we obtain $y_{G_0} = n + r_1$. Hence, if $k \ge 1$,

$$y_{F_1} = \frac{q_1(n+r_1) + \alpha n + r_1 t}{r_1} = \frac{q_1 r_1 + (e-1) r_1 n + r_1 t}{r_1} = q_1 + (e-1)n + t,$$

and, if $k' \ge 1$,

$$y_{G_1} = \frac{(n-t)(n+r_1) + \alpha n + r_1 t}{n} = \frac{n(n + r_1 + \alpha - t)}{n} = n + r_1 + \delta.$$

Since y_{G_0}, y_{F_1}, and y_{G_1} are integers and $-G^2 y_G = y_{G'} + y_{G''}$ for any exceptional curve G, induction along each arm shows that y_G is an integer for any exceptional curve G. This proves (1).

(2) Since $v = n - (h+1)t$, $\gcd(t, v) = \gcd(t, n) = 1$. If $v = 0$, then t divides n, and $\gcd(n, t) = 1$ implies $t = 1$. Since $n \ge t$, $\lfloor n/t \rfloor \ge 1$, so $h \ge 0$. Since $n \ge r_1$, $(h+1)t \ge r_1 - v$, so $h \ge \lceil (r_1 - v)/t \rceil - 1$.

Assume that $k' \ge 1$. For $0 \le j \le h$ let $n_{(j)} := n - jt$. The Hirzebruch–Jung chain of a fraction ρ/σ with $\rho > \sigma \ge 1$ starts with $\lceil \rho/\sigma \rceil$ and continues with the Hirzebruch–Jung chain of $\sigma/(\lceil \rho/\sigma \rceil \sigma - \rho)$, and it ends when the denominator becomes 0. If $j \le h-1$, then $n_{(j)} \ge 2t$, so $\lceil n_{(j)}/(n_{(j)} - t) \rceil = 2$ and $2(n_{(j)} - t) - n_{(j)} = n_{(j+1)} - t$. Thus the first h entries of the Hirzebruch–Jung chain of $n/(n-t)$ are 2, and the remaining entries form the Hirzebruch–Jung chain of $n_{(h)}/(n_{(h)} - t) = (t+v)/v$, which is empty if $v = 0$. In particular, $k' \ge h$ and $-G_i^2 = 2$ for $1 \le i \le h$. By (1), $y_{G_0} = n + r_1$ and $y_{G_1} = n + r_1 + \delta$. For $2 \le i \le \min\{h+1, k'\}$, the equation at G_{i-1} is $2y_{G_{i-1}} = y_{G_{i-2}} + y_{G_i}$, so $y_{G_i} - y_{G_{i-1}} = y_{G_{i-1}} - y_{G_{i-2}}$. Thus $y_{G_i} = n + r_1 + i\delta$ for $0 \le i \le \min\{h+1, k'\}$.

(3) By (1), $a_{G_0} = (n + r_1)/r \ge 5/46$. The equation (9.1) at G_0 implies $e a_{G_0} \le 2$, so $e \le 92/5$, that is, $e \le 18$.

Since $q_1 < r_1$, $\alpha = (e-1)r_1 - q_1 > (e-2)r_1 \ge 1$. If $k' \ge 1$, then $a_{G_1} \ge a_{G_0}$, and (1) implies $\delta = y_{G_1} - y_{G_0} \ge 0$, that is, $t \le \alpha$. If $k' = 0$, then $t = n = 1 < \alpha$. Moreover,

$$t a_{G_0} \le 1 \iff t(n + r_1) \le \alpha n + r_1 t \iff t \le \alpha.$$

Thus $t \le 1/a_{G_0} \le 46/5$, that is, $t \le 9$.

Since $q_1 < r_1$, $\alpha n + r_1 t > \alpha n > (e-2) r_1 n$. Since $n \ge r_1$, $n + r_1 \le 2n$. Thus

$$\frac{5}{46} \le a_{G_0} = \frac{n + r_1}{\alpha n + r_1 t} < \frac{2n}{(e-2) r_1 n} = \frac{2}{(e-2) r_1},$$

so $5(e-2)r_1 < 92$. Since $e - 2 \ge 1$, $r_1 < 92/5$, that is, $r_1 \le 18$. Finally,

$$a_{G_0} \ge \frac{5}{46} \iff 46(n + r_1) \ge 5(\alpha n + r_1 t) \iff (5\alpha - 46) n \le r_1 (46 - 5t).$$

If $\alpha \ge 10$, then $5\alpha - 46 > 0$, and $n \le r_1(46 - 5t)/(5\alpha - 46)$.

(4) Let G be an exceptional curve with $a_G = \mathrm{mld}(X \ni x)$ and $G^2 = -2$. By (9.1), $2a_G = a_{G'} + a_{G''}$, where each of $a_{G'}$ and $a_{G''}$ is either the value at a neighbor of G, hence at least a_G, or $1 \ge a_G$. Thus $a_{G'} = a_{G''} = a_G$, and any neighbor of G also has log discrepancy $\mathrm{mld}(X \ni x)$. Starting from an exceptional curve of log discrepancy $\mathrm{mld}(X \ni x)$ and moving along the chain

towards an exceptional curve with self-intersection at most -3, we reach an exceptional curve G_0 with $-G_0^2 \geq 3$ and $a_{G_0} = \mathrm{mld}(X \ni x)$.

Let $e := -G_0^2$. The arms of the chain at G_0, read from G_0 outwards, are the Hirzebruch–Jung chains of ρ_1/σ_1 and ρ_2/σ_2, where ρ_j is the determinant of the j-th arm and σ_j is the determinant of the j-th arm without the curve meeting G_0, with $(\rho_j, \sigma_j) = (1, 0)$ for an empty arm. We order the arms so that $\rho_1 \leq \rho_2$, and we let $(r_1, q_1) := (\rho_1, \sigma_1)$, $n := \rho_2$, and $t := \rho_2 - \sigma_2$. Then $e \geq 3$, either $(r_1, q_1) = (1, 0)$ or $1 \leq q_1 < r_1$ and $\gcd(r_1, q_1) = 1$, $n \geq r_1$, and $1 \leq t \leq n$. Moreover, $\gcd(n, t) = \gcd(\rho_2, \sigma_2) = 1$, and $n = t$ if and only if $\sigma_2 = 0$, that is, if and only if the second arm is empty, in which case $n = t = 1$. Thus $X \ni x$ satisfies the conditions above with $a(G_0, X, 0) = \mathrm{mld}(X \ni x)$. □

The next lemma expresses $c_x(m)$ by ceilings of the integers y_G, and shows that, as h varies, $2rC_x$ is a polynomial of degree at most two in h on intervals of h.

Lemma 9.8. *Notation as in Definition-Lemma 9.7. Let $\ell_0 := \ell_x - h$, let $\mathcal{F}$ be the set of the curves $F_1, \dots, F_k$, G_0, and G_i with $h < i \leq k'$, and let $\mathcal{E}$ be the set of the pairs of consecutive curves of the chain other than $\{G_{i-1}, G_i\}$ with $1 \leq i \leq h$. Let $G_- := F_k$ if $k \geq 1$ and $G_- := G_0$ if $k = 0$, let $G_+ := G_{k'}$ if $k' \geq 1$ and $G_+ := G_0$ if $k' = 0$, let $k_G := -G^2 - 2$ for every exceptional curve G, and let $U := \sum_{G \in \mathcal{F}} k_G y_G$. For an integer $m \geq 2$, let $N_m(y) := \lceil my/r \rceil$ for $y \in \mathbb{Z}$, and $z_m := \lfloor m\delta/r \rfloor$. Then the following hold.*

(1) For every integer $m \geq 2$,

$$c_x(m) = \binom{m}{2}\frac{U}{r} - \sum_G k_G \binom{N_m(y_G)}{2} - \frac{1}{2}\sum_{\{G,G'\}} \big(N_m(y_G) - N_m(y_{G'})\big)^2 - \frac{1}{2}\big(m - N_m(y_{G_-})\big)^2 - \frac{1}{2}\big(m - N_m(y_{G_+})\big)^2,$$

where G runs over the exceptional curves of $X \ni x$, and $\{G, G'\}$ over the pairs of consecutive curves of the chain.

(2) $2rC_x = r(\theta_1 h + \theta_0) + 8106910U - 115640$, where

$$\theta_1 := 115640 + \sum_{m=2}^{9} w_m z_m(z_m + 1), \qquad \theta_0 := 115640\Big(\ell_0 + 1 - \sum_{G \in \mathcal{F}} k_G\Big) - \beta,$$

and $\beta := \sum_{m=2}^{9} w_m \beta_m$ with

$$\beta_m := \sum_{G \in \mathcal{F}} k_G N_m(y_G)\big(N_m(y_G) - 1\big) + \sum_{\{G,G'\} \in \mathcal{E}} \big(N_m(y_G) - N_m(y_{G'})\big)^2 + \big(m - N_m(y_{G_-})\big)^2 + \big(m - N_m(y_{G_+})\big)^2 + (2z_m + 1)\big(N_m(y_{G_h}) - N_m(y_{G_0})\big).$$

(3) Fix e, r_1, q_1, t, and v with $0 \leq v < t$, $\gcd(t, v) = 1$, and $v = 0$ only if $t = 1$, and let h vary over the integers with $h \geq \max\{0, \lceil (r_1 - v)/t \rceil - 1\}$, so that $n := (h+1)t + v$ satisfies the conditions of Definition-Lemma 9.7. Then k, $k' - h$, ℓ_0, and $\sum_{G \in \mathcal{F}} k_G$ do not depend on h, and r, U, y_{G_0}, y_{G_h}, y_{F_i} for $1 \leq i \leq k$, and $y_{G_{h+j}}$ for $1 \leq j \leq k' - h$ are affine functions of h. In particular, on any set of integers h on which, for $2 \leq m \leq 9$, the integers z_m, $N_m(y_{G_0})$, $N_m(y_{G_h})$, $N_m(y_{F_i})$ for $1 \leq i \leq k$, and $N_m(y_{G_{h+j}})$ for $1 \leq j \leq k' - h$ do not depend on h, $2rC_x$ is a polynomial in h of degree at most two.

Proof. For any exceptional curve G let $a_G := a(G, X, 0) = y_G/r$ (Definition-Lemma 9.7(1)), and let a_x be the vector $(a_G)_G$. By Definition-Lemma 9.7(2), $k_{G_i} = 0$ for $1 \leq i \leq h$, so

$$\sum_G k_G a_G = \frac{1}{r}\sum_{G \in \mathcal{F}} k_G y_G = \frac{U}{r}.$$

(1) Fix $m \geq 2$ and write $N_G := N_m(y_G)$, $N_\pm := N_{G_\pm}$, and $a_\pm := a_{G_\pm}$. By Lemma 9.3, $b_x = \mathbf{1} - a_x$, where $\mathbf{1}$ is the vector with all entries 1. Thus the entry of $\{mb_x\}$ at G is $\{-ma_G\} = N_G - ma_G$. Let $\mathbf{N}$ be the vector $(N_G)_G$, so that $\{mb_x\} = \mathbf{N} - ma_x$.

By Definition-Lemma 9.7(1), $-G^2 a_G = a_{G'} + a_{G''}$ for any exceptional curve G, where $a_{G'}$ and $a_{G''}$ are the values at the neighbors of G and the value 1 is used for each missing neighbor. Thus the entry of $A_x a_x$ at G is the number of missing neighbors of G. Thus

$$\mathbf{N}^t A_x a_x = N_- + N_+, \qquad a_x^t A_x a_x = a_- + a_+.$$

Moreover, since $\sum_G (2 - \deg G) N_G^2 = N_-^2 + N_+^2$, where $\deg G$ is the number of neighbors of G,

$$\mathbf{N}^t A_x \mathbf{N} = \sum_G (-G^2) N_G^2 - 2 \sum_{\{G,G'\}} N_G N_{G'} = \sum_G k_G N_G^2 + N_-^2 + N_+^2 + \sum_{\{G,G'\}} (N_G - N_{G'})^2.$$

Summing $-G^2 a_G = a_{G'} + a_{G''}$ over all exceptional curves G, we obtain

$$\sum_G (-G^2) a_G = \sum_G \deg(G) a_G + 2 = 2 \sum_G a_G - a_- - a_+ + 2,$$

that is,

$$\frac{U}{r} = \sum_G k_G a_G = 2 - a_- - a_+.$$

Substituting $\{mb_x\} = \mathbf{N} - ma_x$ into $2c_x(m) = k_x^t \{mb_x\} - \{mb_x\}^t A_x \{mb_x\}$, we obtain

$$\begin{aligned} 2c_x(m) &= k_x^t \mathbf{N} - m\frac{U}{r} - \mathbf{N}^t A_x \mathbf{N} + 2m \mathbf{N}^t A_x a_x - m^2 a_x^t A_x a_x \\ &= \sum_G k_G N_G - \sum_G k_G N_G^2 - \sum_{\{G,G'\}} (N_G - N_{G'})^2 - N_-^2 - N_+^2 \\ &\quad + 2m(N_- + N_+) - m\frac{U}{r} - m^2(a_- + a_+). \end{aligned}$$

Since $a_- + a_+ = 2 - U/r$, we have

$$-m\frac{U}{r} - m^2(a_- + a_+) = m(m-1)\frac{U}{r} - 2m^2,$$

$$-2m^2 + 2m(N_- + N_+) - N_-^2 - N_+^2 = -(m - N_-)^2 - (m - N_+)^2.$$

Thus

$$2c_x(m) = m(m-1)\frac{U}{r} - \sum_G k_G N_G (N_G - 1) - \sum_{\{G,G'\}} (N_G - N_{G'})^2 - (m - N_-)^2 - (m - N_+)^2,$$

which is (1).

(2) Fix $2 \leq m \leq 9$. For $1 \leq i \leq h$ let $\Delta_i := N_m(y_{G_i}) - N_m(y_{G_{i-1}})$. By Definition-Lemma 9.7(2), $y_{G_i} = y_{G_{i-1}} + \delta$. Let $u := m y_{G_{i-1}}/r$ and $u' := m\delta/r$, so that $\Delta_i = \lceil u + u' \rceil - \lceil u \rceil$ and $z_m = \lfloor u' \rfloor$. Since

$$\lceil u \rceil + \lfloor u' \rfloor \leq \lceil u + u' \rceil \leq \lceil u \rceil + \lfloor u' \rfloor + 1,$$

we obtain $\Delta_i \in \{z_m, z_m + 1\}$, so $\Delta_i^2 = (2z_m + 1)\Delta_i - z_m(z_m + 1)$. Since $\sum_{i=1}^h \Delta_i = N_m(y_{G_h}) - N_m(y_{G_0})$,

$$\sum_{i=1}^h \Delta_i^2 = (2z_m + 1)\big(N_m(y_{G_h}) - N_m(y_{G_0})\big) - h z_m(z_m + 1).$$

The pairs of consecutive curves are the pairs in $\mathcal{E}$ and the pairs $\{G_{i-1}, G_i\}$ with $1 \leq i \leq h$, and $k_G = 0$ for $G \notin \mathcal{F}$. Thus (1) implies

$$2r c_x(m) = m(m-1)U - r\beta_m + r h z_m(z_m + 1).$$

By Definition-Lemma 9.7(1), $X \ni x$ is a cyclic quotient singularity of order r. Thus $X \ni x$ is the quotient of $\mathbb{C}^2$ by a linear action of $\mathbb{Z}/r\mathbb{Z}$ that is free outside the origin [BHPV04, Chapter III,

Theorem 5.1 and Proposition 5.3], so $g_x = r$. Since $k_x^t b_x = \sum_G k_G(1-a_G) = \sum_{G\in\mathcal{F}} k_G - U/r$ and $\ell_x = h + \ell_0$,

$$\mu_x = \gamma_x + 1 - \frac{1}{r} = h + \ell_0 + 1 - \sum_{G\in\mathcal{F}} k_G + \frac{U}{r} - \frac{1}{r},$$

so

$$2r\cdot 57820\mu_x = 115640r\Big(h+\ell_0+1-\sum_{G\in\mathcal{F}} k_G\Big) + 115640U - 115640.$$

Moreover, $\sum_{m=2}^{9} w_m\binom{m}{2} = 3995635$, so

$$115640 + \sum_{m=2}^{9} w_m m(m-1) = 115640 + 7991270 = 8106910.$$

Summing, we obtain

$$\begin{aligned} 2rC_x &= 2r\cdot 57820\mu_x + \sum_{m=2}^{9} w_m\cdot 2rc_x(m)\\ &= rh\Big(115640 + \sum_{m=2}^{9} w_m z_m(z_m+1)\Big) + r\Big(115640\Big(\ell_0+1-\sum_{G\in\mathcal{F}} k_G\Big) - \beta\Big)\\ &\quad + 8106910U - 115640\\ &= r(\theta_1 h + \theta_0) + 8106910U - 115640. \end{aligned}$$

(3) Since $0 \le v < t$, $\lfloor n/t\rfloor - 1 = h$ and $n - (h+1)t = v$. Since $h \ge 0$, $n \ge t$. Since $h \ge \lceil (r_1 - v)/t\rceil - 1$, $n \ge r_1$. Moreover, $\gcd(n,t) = \gcd(v,t) = 1$, and if $n = t$, then $h = v = 0$, so $t = 1$ and $n = t = 1$. Thus n satisfies the conditions of Definition-Lemma 9.7.

The first arm $F_1,\dots,F_k$ is the Hirzebruch–Jung chain of r_1/q_1, which does not depend on h. By Definition-Lemma 9.7(2), the second arm consists of h curves $G_1,\dots,G_h$ with $-G_i^2 = 2$, followed by the curves $G_{h+1},\dots,G_{k'}$ forming the Hirzebruch–Jung chain of $(t+v)/v$, which does not depend on h. Thus k, $k'-h$, $\ell_0 = 1+k+(k'-h)$, $\sum_{G\in\mathcal{F}} k_G$, and the self-intersections of the curves in $\mathcal{F}$ do not depend on h. The integers α and δ do not depend on h, and $r = \alpha((h+1)t+v) + r_1 t$, $y_{G_0} = (h+1)t + v + r_1$, and $y_{G_h} = y_{G_0} + h\delta$ are affine functions of h. If $k \ge 1$, then $y_{F_1} = q_1 + (e-1)n + t$ is affine in h, and the recurrence $y_{F_{i+1}} = -F_i^2 y_{F_i} - y_{F_{i-1}}$ (with $F_0 := G_0$) has coefficients independent of h, so y_{F_i} is affine in h for $1 \le i \le k$. If $k' \ge h+1$, then $y_{G_{h+1}} = y_{G_0} + (h+1)\delta$ by Definition-Lemma 9.7(2), and the recurrence $y_{G_{h+j+1}} = -G_{h+j}^2 y_{G_{h+j}} - y_{G_{h+j-1}}$ has coefficients independent of h, so $y_{G_{h+j}}$ is affine in h for $1 \le j \le k'-h$. Hence $U = \sum_{G\in\mathcal{F}} k_G y_G$ is affine in h.

Let H be a set of integers h on which the integers listed in (3) do not depend on h. The curves G_- and G_+ belong to $\mathcal{F}\cup\{G_h\}$, and any pair in $\mathcal{E}$ consists of curves in $\mathcal{F}\cup\{G_h\}$. Thus β_m does not depend on $h \in H$ for $2 \le m \le 9$, so θ_1 and θ_0 do not depend on $h \in H$. Since r and U are affine in h, (2) implies that $2rC_x$ is a polynomial in h of degree at most two on H. □

Proposition 9.9. *Let $X \ni x$ be a cyclic quotient singularity of order $r \ge 20$ with $\mathrm{mld}(X \ni x) \ge 5/46$ that is not of type $\frac{1}{27}(1,2)$. Then $C_x \ge 500(r-1)$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove the bound $C_x \ge 500(r-1)$ when $X \ni x$ is a canonical cyclic quotient singularity.

Suppose that $X \ni x$ is a canonical cyclic quotient singularity of order $r \ge 20$, that is, $-G^2 = 2$ for any exceptional curve G of the minimal resolution. Then the Hirzebruch–Jung chain of $X \ni x$ is $[2,\dots,2]$ of length $\ell_x = r-1$, and $g_x = r$, since $X \ni x$ is the quotient of $\mathbb{C}^2$ by a linear action

of $\mathbb{Z}/r\mathbb{Z}$ that is free outside the origin [BHPV04, Chapter III, Theorem 5.1 and Proposition 5.3]. Thus $k_x = 0$, $b_x = 0$, $c_x(m) = 0$ for any $m \geq 2$, and $\gamma_x = \ell_x$. Hence

$$C_x = 57820\Big(\ell_x + 1 - \frac{1}{\ell_x + 1}\Big) \geq 57820\ell_x \geq 500(r-1).$$

Step 2. In this step, we introduce the integer T for the non-canonical cyclic quotient singularities.

Suppose that $X \ni x$ is a non-canonical cyclic quotient singularity of order $r \geq 20$ that is not of type $\frac{1}{27}(1,2)$. Then the minimal resolution has an exceptional curve G with $G^2 \leq -3$, and by Definition-Lemma 9.7(4), $X \ni x$ satisfies the conditions of Definition-Lemma 9.7 for some integers e, r_1, q_1, n, and t with $a(G_0, X, 0) = \mathrm{mld}(X \ni x)$. We use the notation of Definition-Lemma 9.7 and Lemma 9.8. By Definition-Lemma 9.7(1), the order of $X \ni x$ is $r = \alpha n + r_1 t$. By Definition-Lemma 9.7(3),

$$e \leq 18, \quad r_1 \leq 18, \quad 5(e-2)r_1 \leq 92, \quad t \leq \min\{\alpha, 9\}, \quad (5\alpha - 46)n \leq r_1(46 - 5t).$$

Let

$$T := 2r\big(C_x - 500(r-1)\big).$$

By Lemma 9.8(2),

$$T = r(\theta_1 h + \theta_0) + 8106910U - 115640 - 1000r(r-1).$$

It suffices to show that $T \geq 0$. The tables used below are in the subsection of Appendix A for Proposition 9.9.

Step 3. In this step, we prove that $T \geq 0$ when $\alpha \geq 10$.

Since $\alpha \geq 10$, $5\alpha - 46 > 0$, hence $n \leq r_1(46-5t)/(5\alpha-46)$. Together with the bounds of **Step 2**, there are only finitely many tuples (e, r_1, q_1, t, n). Under a header (e, r_1, q_1, t) of Table 11, an entry $n{:}T$ is the value of the integer T of **Step 2** at the cyclic quotient singularity with parameters (e, r_1, q_1, t, n). The value T is computed with integer arithmetic from Lemma 9.8(1) and Definition 9.2, or from the formula for T in **Step 2** with $h = \lfloor n/t \rfloor - 1$. Table 11 has an entry for any tuple satisfying the conditions of Definition-Lemma 9.7 and the bounds of **Step 2** with $\alpha \geq 10$ and $r \geq 20$, except the tuple $(e, r_1, q_1, t, n) = (14, 1, 0, 1, 2)$, and each entry of Table 11 is non-negative.

For $(e, r_1, q_1, t, n) = (14, 1, 0, 1, 2)$, we have $\alpha = 13$, $r = 27$, and the chain is $[14, 2]$, that is, $X \ni x$ is of type $\frac{1}{27}(1,2)$, which is excluded by hypothesis. Hence $T \geq 0$ when $\alpha \geq 10$.

Step 4. In this step, we prove that $T \geq 0$ when $\alpha \leq 9$.

Recall that $h = \lfloor n/t \rfloor - 1$ and $v = n - (h+1)t$, so $n = (h+1)t + v$ and $0 \leq v < t$. By Definition-Lemma 9.7(2), $\gcd(t, v) = 1$, $v = 0$ only if $t = 1$, and $h \geq \max\{0, \lceil (r_1 - v)/t \rceil - 1\}$. Since $\alpha \leq 9$, the inequality $(5\alpha - 46)n \leq r_1(46 - 5t)$ holds for any $n \geq 1$, and the bounds of **Step 2** leave finitely many headers (e, r_1, q_1, t, v). For a fixed header, we let h vary in Lemma 9.8(3), and we write $T(h)$ for the value of T at the cyclic quotient singularity with parameters $(e, r_1, q_1, t, (h+1)t + v)$. Since

$$r = \alpha t h + \alpha(t + v) + r_1 t,$$

the condition $r \geq 20$ is equivalent to $h \geq \lceil (20 - \alpha(t+v) - r_1 t)/(\alpha t) \rceil$.

Under a header (e, r_1, q_1, t, v) of Table 10, a row with columns $h_-, h_+, \tau_0, \tau_1, \tau_2$ means that

$$T(h_- + z) = \tau_0 + \tau_1 z + \tau_2 z(z-1) \quad \text{for any integer } 0 \leq z \leq h_+ - h_-,$$

where $h_+ = +\infty$ is allowed, and τ_1 is a dash if $h_+ = h_-$. The rows are obtained as follows. For $2 \leq m \leq 9$, the integers z_m and $N_m(y_G)$ of Lemma 9.8(3) change value only at the rational solutions h of $m\delta = jr$ and $my_G = jr$ with integers j, where r and y_G are affine functions of h by Lemma 9.8(3). On any interval of integers between two such solutions, T is a polynomial of degree at most two in h by Lemma 9.8(3), and adjacent intervals on which T is given by one polynomial in h are merged into one row. Each row can be checked with integer arithmetic. For rational h at least the lower bound of Lemma 9.8(3), there exist finitely many such solutions,

since $0 < y_G/r \le 1$ (for rational h, this follows from the case of integers, as y_G/r is monotone in h) implies $1 \le j \le m$ and $z_m = 0$ once $r > 9\delta$ (an equation that holds identically in h causes no change). Thus the last row under any header can have $h_+ = +\infty$.

Table 10 has rows under any header satisfying the conditions above, the rows under a header cover any integer

$$h \ge \max\left\{0, \left\lceil \frac{r_1 - v}{t} \right\rceil - 1, \left\lceil \frac{20 - \alpha(t+v) - r_1 t}{\alpha t} \right\rceil \right\},$$

and $\tau_0, \tau_1, \tau_2 \ge 0$ in any row. Since $z(z-1) \ge 0$ for any integer $z \ge 0$, $T(h) \ge 0$ for any such h. Hence $T \ge 0$ when $\alpha \le 9$.

By **Steps 1, 3, and 4**, $C_x \ge 500(r-1)$. □

Theorem 9.10. *Let $X \ni x$ be a klt surface germ that is not smooth, with* $\mathrm{mld}(X \ni x) \ge 5/46$. *Then $C_x > 0$.*

Proof. If $X \ni x$ is not a cyclic quotient singularity, then $C_x \ge 91(\ell_x+1) > 0$ by Proposition 9.6(4). Thus we may assume that $X \ni x$ is a cyclic quotient singularity.

Case 1. Suppose that all exceptional curves of $X \ni x$ are (-2)-curves. Then $k_x = 0$, so $b_x = 0$, $c_x(m) = 0$ for any $m \ge 2$, and $\gamma_x = \ell_x$. The chain $[2, \dots, 2]$ of length ℓ_x is the Hirzebruch–Jung chain of $(\ell_x+1)/\ell_x$, so $X \ni x$ is of type $\frac{1}{\ell_x+1}(1, \ell_x)$ and $g_x = \ell_x + 1$, since $X \ni x$ is the quotient of $\mathbb{C}^2$ by a linear action of $\mathbb{Z}/(\ell_x+1)\mathbb{Z}$ that is free outside the origin [BHPV04, Chapter III, Theorem 5.1 and Proposition 5.3]. Thus $\mu_x = \ell_x + 1 - 1/(\ell_x+1)$. Since $\ell_x + 1 \ge 2$,

$$\begin{aligned} C_x - 91(\ell_x+1) &= 57820\Big(\ell_x + 1 - \frac{1}{\ell_x+1}\Big) - 91(\ell_x+1) \\ &= 57729(\ell_x+1) - \frac{57820}{\ell_x+1} \ge 2 \cdot 57729 - \frac{57820}{2} > 0. \end{aligned}$$

Case 2. Suppose that the minimal resolution of $X \ni x$ has an exceptional curve G with $G^2 \le -3$. Let r be the order of $X \ni x$. If $r \ge 20$ and $X \ni x$ is not of type $\frac{1}{27}(1,2)$, then Proposition 9.9 implies $C_x \ge 500(r-1) > 0$. Otherwise, by Definition-Lemma 9.7(3) and (4), $X \ni x$ has parameters (e, r_1, q_1, t, n) with $a(G_0, X, 0) = \mathrm{mld}(X \ni x)$ satisfying the bounds of Definition-Lemma 9.7(3). Moreover, $r \le 19$, or $X \ni x$ is of type $\frac{1}{27}(1,2)$, whose chain is $[14, 2]$, so that $(e, r_1, q_1, t, n) = (14, 1, 0, 1, 2)$. Thus $T := 2r\big(C_x - 91(\ell_x+1)\big)$ is the entry of Table 9 for (e, r_1, q_1, t, n). The entries are non-negative, so $C_x \ge 91(\ell_x+1) > 0$. □

9.3. **The bound for the total cost and the periods.** In this subsection, we bound the total cost $\sum_x C_x$ of a klt stable surface X with $K_X^2 \le 1/48983$ and $\mathrm{mld}(X) \ge 5/46$, we show that the functions $m \to c_x(m)$ are periodic with controlled periods, and we deduce a positive lower bound for K_X^2.

Proposition 9.11. *Let X be a klt stable surface with $K_X^2 \le 1/48983$ and* $\mathrm{mld}(X) \ge 5/46$. *Then $\sum_x C_x < 151733$, where x runs over the singular points of X.*

Proof. By Proposition 4.4(1), $P_m(X) = 0$ for $2 \le m \le 9$. Thus Lemma 9.4(1) implies

$$\sum_x c_x(m) = -1 - \binom{m}{2} K_X^2 \quad \text{for any integer } 2 \le m \le 9.$$

We have

$$\sum_{m=2}^{9} w_m = 542107 \quad \text{and} \quad \sum_{m=2}^{9} \binom{m}{2} w_m = 3995635.$$

Since K_X is ample, $K_X^2 > 0$. Thus Definition 9.2 and Lemma 9.4(2) imply

$$\begin{aligned}\sum_x C_x &= 57820 \sum_x \mu_x + \sum_{m=2}^{9} w_m \sum_x c_x(m) \\ &\le 57820\Big(12 - \frac{4}{3}K_X^2\Big) - 542107 - 3995635K_X^2 \\ &= 151733 - \frac{12218185}{3}K_X^2 < 151733.\end{aligned}$$

□

Lemma 9.12. *Let $X \ni x$ be a klt surface germ that is neither smooth nor a cyclic quotient singularity, with $\mathrm{mld}(X \ni x) \ge 5/46$, and let d_x be the least positive integer such that $d_x b_x \in \mathbb{Z}^{\ell_x}$. Then the following hold.*

(1) If $X \ni x$ is of type $[e_0; (2,1), (2,1), (r,q)]$, then $d := (e_0 - 1)r - q$ satisfies $1 \le d \le 9$, and d_x divides $2d$.

(2) If $X \ni x$ is not of the type in (1), then $X \ni x$ is a germ of Table 8, and d_x divides the integer I of the row of $X \ni x$. Moreover, either $d_x \le 9$, or $X \ni x$ is of type $[3; (2,1), (3,1), (3,2)]$, $d_x = 27$, and $C_x \ge 408983/27$.

(3) In particular, either $d_x \le 18$, or $d_x = 27$ and $C_x > 500(d_x - 1)$.

Proof. Let $\pi : \widetilde{X} \to X$ be the minimal resolution. Since $X \ni x$ is a quotient singularity [KM98, Proposition 4.18], $X \ni x$ is of type $[e_0; (r_1, q_1), (r_2, q_2), (r_3, q_3)]$ (Definition 2.9(2)) by [KM98, Theorem 4.7]. Let E_0 be the fork. Let $\sigma := \sum_{i=1}^{3} 1/r_i - 1$ and $\eta := e_0 - \sum_{i=1}^{3} q_i/r_i$. By Lemma 2.10, $\sigma > 0$, $\eta > 0$, (r_1, r_2, r_3) is one of $(2,2,r)$ with $r \ge 2$, $(2,3,3)$, $(2,3,4)$, and $(2,3,5)$, the fork has log discrepancy $a_0 = \sigma/\eta$, and the curve on the i-th arm intersecting E_0 has log discrepancy $(q_i a_0 + 1)/r_i$.

Let $a_x := \mathbf{1} - b_x$, where $\mathbf{1}$ is the vector with all entries 1. By Lemma 9.3, the entries of a_x are the log discrepancies of the exceptional curves of π. Moreover, since $(A_x \mathbf{1})_j = -E_j^2 - \deg(j)$ and $A_x b_x = k_x$, we have $A_x a_x = (2 - \deg(j))_j$, where $\deg(j)$ is the number of exceptional curves other than E_j intersecting E_j. Thus d_x is the least positive integer such that $d_x a_x \in \mathbb{Z}^{\ell_x}$, and the set of integers M with $M a_x \in \mathbb{Z}^{\ell_x}$ is $d_x\mathbb{Z}$. Let $E_{i,1}, \dots, E_{i,s}$ be the curves of the i-th arm, with $E_{i,1}$ intersecting E_0 and $E_{i,j}$ intersecting $E_{i,j+1}$, let $a_{i,j}$ be the log discrepancy of $E_{i,j}$, and let $a_{i,0} := a_0$ and $a_{i,s+1} := 1$. The equation $A_x a_x = (2 - \deg(j))_j$ at $E_{i,j}$ reads

$$-E_{i,j}^2 a_{i,j} = a_{i,j-1} + a_{i,j+1} \quad \text{for } 1 \le j \le s.$$

Thus $a_{i,j} \in \mathbb{Z}a_0 + \mathbb{Z}a_{i,1}$ for $0 \le j \le s$.

(1) Here $\sigma = 1/r$ and $\eta = e_0 - 1 - q/r = d/r$. Since $\eta > 0$, $d \ge 1$. Moreover, $a_0 = \sigma/\eta = 1/d$. Since $a_0 \ge \mathrm{mld}(X \ni x) \ge 5/46$, $d \le 46/5$, hence $d \le 9$. Each of the first two arms consists of one curve, with log discrepancy $(a_0 + 1)/2 = (d+1)/(2d)$. On the third arm,

$$a_{3,1} = \frac{qa_0 + 1}{r} = \frac{q + d}{rd} = \frac{e_0 - 1}{d},$$

hence $a_{3,j} \in \frac{1}{d}\mathbb{Z}$ for any j. Therefore $2da_x \in \mathbb{Z}^{\ell_x}$, and d_x divides $2d$.

(2) By Lemma 2.10, (r_1, r_2, r_3) is one of $(2,3,3)$, $(2,3,4)$, and $(2,3,5)$, hence $X \ni x$ is of type $[e_0; (2,1), (3,q_2), (r_3, q_3)]$ with $r_3 \in \{3,4,5\}$. Here σ is $1/6$, $1/12$, or $1/30$, and $1/\sigma$ is a multiple of the denominators of $1/2$, $q_2/3$, and q_3/r_3, hence $1/a_0 = \eta/\sigma$ is a positive integer. Since $a_0 \ge \mathrm{mld}(X \ni x) \ge 5/46$, we have $1/a_0 \le 9$, that is, $a_0 \ge 1/9$. Thus $X \ni x$ is a germ of Table 8, and the column I of Table 8 is a common denominator of the entries of b_x. Thus $Ib_x \in \mathbb{Z}^{\ell_x}$, and d_x divides I. In any row of Table 8 other than the row of type $[3; (2,1), (3,1), (3,2)]$, $I \le 9$, hence $d_x \le 9$.

Suppose that $X \ni x$ is of type $[3; (2,1), (3,1), (3,2)]$. By the row of Table 8 with $(e_0, r_3, q_2, q_3) = (3,3,1,2)$, we have $27b_x = (24, 12, 17, 16, 8)$ and $D_x = 394241/27$. Thus

$d_x = 27$, since $\gcd(27, 24, 12, 17, 16, 8) = 1$. Since $\ell_x = 5$, Proposition 9.6(4) implies $C_x \geq D_x + 91(\ell_x + 1) = D_x + 546 = 408983/27$.

(3) In case (1), $d_x \leq 2d \leq 18$. In case (2), either $d_x \leq 9$, or $d_x = 27$ and $C_x \geq 408983/27 > 13000 = 500(d_x - 1)$. □

Proposition 9.13.

(1) Let $X \ni x$ be a klt surface germ that is not smooth, with $\mathrm{mld}(X \ni x) \geq 5/46$, and let d_x be the least positive integer such that $d_x b_x \in \mathbb{Z}^{\ell_x}$. Then $c_x(m + d_x) = c_x(m)$ for any integer $m \geq 2$, and $C_x \geq 500(d_x - 1)$ if $d_x \geq 20$.

(2) Let X be a klt stable surface with $K_X^2 \leq 1/48983$ and $\mathrm{mld}(X) \geq 5/46$. Then $P_a(X) > 0$ for some integer $2 \leq a \leq 426$. In particular, $K_X^2 \geq 1/84235998$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove the periodicity in (1) and the bound $C_x \geq 500(d_x - 1)$ if $d_x \geq 20$.

Since $d_x b_x \in \mathbb{Z}^{\ell_x}$, we have

$$\{(m + d_x)b_x\} = \{mb_x + d_x b_x\} = \{mb_x\},$$

hence $c_x(m + d_x) = c_x(m)$ for any $m \geq 2$ by Definition 9.2. Moreover, the set of integers M with $Mb_x \in \mathbb{Z}^{\ell_x}$ is $d_x\mathbb{Z}$. It remains to show that $C_x \geq 500(d_x - 1)$ if $d_x \geq 20$.

If $X \ni x$ is not a cyclic quotient singularity, then by Lemma 9.12(3), either $d_x \leq 18$, or $d_x = 27$ and $C_x > 500(d_x - 1)$.

If $X \ni x$ is a canonical cyclic quotient singularity, that is, $-G^2 = 2$ for any exceptional curve G of the minimal resolution, then $b_x = 0$ and $d_x = 1$.

Suppose that $X \ni x$ is a non-canonical cyclic quotient singularity of order r. Then the minimal resolution has an exceptional curve G with $G^2 \leq -3$. By Definition-Lemma 9.7(4) and (1), $a(G', X, 0) \in \frac{1}{r}\mathbb{Z}$ for any exceptional curve G' of the minimal resolution. Since $b_{x,j}$ is 1 minus the log discrepancy of the j-th exceptional curve, $rb_x \in \mathbb{Z}^{\ell_x}$, hence d_x divides r. If $X \ni x$ is of type $\frac{1}{27}(1, 2)$, then, reading the chain as $[14, 2]$,

$$A_x = \begin{pmatrix} 14 & -1 \\ -1 & 2 \end{pmatrix}, \qquad k_x = (12, 0)^t, \qquad b_x = A_x^{-1} k_x = \left(\frac{8}{9}, \frac{4}{9}\right)^t,$$

hence $d_x = 9$. Suppose that $d_x \geq 20$. Then $r \geq d_x \geq 20$, and $X \ni x$ is not of type $\frac{1}{27}(1, 2)$. By Proposition 9.9,

$$C_x \geq 500(r - 1) \geq 500(d_x - 1).$$

This proves (1).

Step 2. In this step, we prove (2).

For any singular point x of X, let d_x be the least positive integer such that $d_x b_x \in \mathbb{Z}^{\ell_x}$, as in (1). Let Ω be the set of complex numbers $\zeta \neq 1$ such that $\zeta^{d_x} = 1$ for some singular point x of X. An element $\zeta \in \Omega$ with $\zeta^{d_x} = 1$ for some $d_x \leq 19$ is a primitive j-th root of unity for some $2 \leq j \leq 19$. For each j, the number of primitive j-th roots of unity is the number of integers $1 \leq a \leq j$ coprime to j, and the sum of the numbers of such integers over $2 \leq j \leq 19$ is 119. Thus there are at most 119 such elements. A singular point x with $d_x \geq 20$ contributes at most $d_x - 1$ further elements, and $d_x - 1 \leq C_x/500$ by (1). By Theorem 9.10, $C_x > 0$ for any singular point x. By Proposition 9.11, $\sum_x C_x < 151733$. Therefore

$$|\Omega| \leq 119 + \sum_{d_x \geq 20} (d_x - 1) \leq 119 + \frac{1}{500} \sum_x C_x < 119 + \frac{151733}{500} < 423,$$

hence $|\Omega| \leq 422$.

For any singular point x, the function $m \to c_x(m)$ on the integers $m \geq 2$ is periodic with period d_x by (1). By the discrete Fourier expansion on $\mathbb{Z}/d_x\mathbb{Z}$, there exist complex numbers $\lambda_{x,\zeta}$,

$\zeta^{d_x} = 1$, such that $c_x(m) = \sum_{\zeta^{d_x}=1} \lambda_{x,\zeta}\zeta^m$ for any $m \geq 2$. By Lemma 9.4(1),

$$P_m(X) = f(m) + \sum_{\zeta\in\Omega} \lambda_\zeta \zeta^m \quad \text{for any integer } m \geq 2,$$

where $\lambda_\zeta := \sum_{x:\zeta^{d_x}=1} \lambda_{x,\zeta}$, and $f(m) := 1 + \binom{m}{2}K_X^2 + \sum_x \lambda_{x,1}$ is a polynomial in m of degree two with leading coefficient $K_X^2/2 > 0$.

Let $u := |\Omega| + 3$, and let

$$(z-1)^3 \prod_{\zeta\in\Omega}(z-\zeta) = z^u + \sum_{i=0}^{u-1} \kappa_i z^i$$

in $\mathbb{C}[z]$. Since the third difference of a polynomial of degree two is zero, and $\zeta^{m+1} - \zeta\cdot\zeta^m = 0$, the sequence $(P_m(X))_{m\geq 2}$ satisfies the linear recurrence

$$P_{m+u}(X) + \sum_{i=0}^{u-1} \kappa_i P_{m+i}(X) = 0 \quad \text{for any integer } m \geq 2.$$

Suppose that $P_a(X) = 0$ for any integer $2 \leq a \leq u+1$. Since $P_2(X), \dots, P_{u+1}(X)$ are u consecutive terms of the sequence, we have $P_m(X) = 0$ for any $m \geq 2$ by induction on m. However, $f(m) \to +\infty$ as $m \to +\infty$, while

$$\Big|\sum_{\zeta\in\Omega} \lambda_\zeta \zeta^m\Big| \leq \sum_{\zeta\in\Omega} |\lambda_\zeta| \quad \text{for any } m,$$

a contradiction. Hence $P_a(X) > 0$ for some integer

$$2 \leq a \leq u+1 = |\Omega| + 4 \leq 426.$$

By Corollary 4.3,

$$K_X^2 \geq \frac{1}{462(a+1)^2} \geq \frac{1}{462\cdot 427^2} = \frac{1}{84235998}.$$

□

9.4. **The finite list of singularity types.** In this subsection, we show that any singular point of a klt stable surface X with $K_X^2 \leq 1/48983$ and $\mathrm{mld}(X) \geq 5/46$ belongs to an explicit finite list of types of quotient singularities.

Theorem 9.14. *Let X be a klt stable surface with $K_X^2 \leq 1/48983$ and $\mathrm{mld}(X) \geq 5/46$, and let $x \in X$ be a singular point. Then $1 - b_{x,i} \geq 5/46$ for any entry $b_{x,i}$ of b_x, $C_x < 151733$, and $X \ni x$ is of one of the following types:*

(1) $\frac{1}{r}(1,q)$ with $2 \leq r \leq 304$, $1 \leq q < r$, and $\gcd(r,q) = 1$,
(2) $[e_0; (2,1), (2,1), (r,q)]$ with $3 \leq e_0 \leq 6$, $2 \leq r \leq \lfloor 8/(e_0-2)\rfloor$, $1 \leq q < r$, $\gcd(r,q) = 1$, and $(e_0-1)r - q \leq 9$,
(3) $[2; (2,1), (2,1), (ud+t, (u-1)d+t)]$ with $2 \leq d \leq 9$, $1 \leq t < d$, $\gcd(d,t) = 1$, and $1 \leq u \leq 3$, or
(4) $[e_0; (2,1), (3,q_2), (r,q_3)]$ with $r \in \{3,4,5\}$, $q_2 \in \{1,2\}$, $1 \leq q_3 < r$, $\gcd(r,q_3) = 1$, $q_2 \leq q_3$ if $r = 3$, and $e_0 \in \{2,3\}$ if $r = 3$, $e_0 = 2$ if $r \in \{4,5\}$.

Moreover, $g_x = r$ in case (1), and $g_x = 4\eta/\sigma^2$ in cases (2)–(4), where $\sigma := \sum_{i=1}^3 1/r_i - 1$ and $\eta := e_0 - \sum_{i=1}^3 q_i/r_i$ for the type $[e_0; (r_1,q_1), (r_2,q_2), (r_3,q_3)]$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove that $1 - b_{x,i} \geq 5/46$ for any i and that $C_x < 151733$.

Let $\pi : \widetilde{X} \to X$ be the minimal resolution of $X \ni x$, and let $E_1, \dots, E_{\ell_x}$ be the exceptional curves of π, indexed as in Definition 9.2. By Lemma 9.3,

$$\min_i(1 - b_{x,i}) = \mathrm{mld}(X \ni x) \geq \mathrm{mld}(X) \geq \frac{5}{46}.$$

Since $\mathrm{mld}(X) \geq 5/46$, Theorem 9.10 implies $C_{x'} > 0$ for any singular point $x' \in X$. Thus, by Proposition 9.11,

$$C_x \leq \sum_{x'} C_{x'} < 151733,$$

where x' runs over the singular points of X.

Step 2. In this step, we prove that $X \ni x$ is of one of the types (1)–(4), and we compute g_x.

Since X is klt, $X \ni x$ is a quotient singularity, and the dual graph of π is a chain or a tree with exactly one vertex of degree three [KM98, Proposition 4.18 and Theorem 4.7]. Thus $X \ni x$ is either of type $\frac{1}{r}(1,q)$ for some coprime integers $r > q \geq 1$, or of type $[e_0; (r_1,q_1), (r_2,q_2), (r_3,q_3)]$ (Definition 2.9). In the case of type $[e_0; (r_1,q_1), (r_2,q_2), (r_3,q_3)]$, Lemma 2.10 implies that (r_1,r_2,r_3) is one of $(2,2,r)$ with $r \geq 2$, $(2,3,3)$, $(2,3,4)$, and $(2,3,5)$, and that $g_x = 4\eta/\sigma^2$, where σ and η are defined in the statement of the theorem. Moreover, $e_0 \geq 2$, since the fork is a smooth rational curve [KM98, Theorem 4.7] and π is minimal.

Case 1. $X \ni x$ is of type $\frac{1}{r}(1,q)$. Then $1 \leq q < r$ and $\gcd(r,q) = 1$, so $r \geq 2$. If $r < 20$, then $r \leq 304$. If $X \ni x$ is of type $\frac{1}{27}(1,2)$, then $r = 27 \leq 304$. Otherwise, if $r \geq 20$, then Proposition 9.9 and **Step 1** imply

$$500(r-1) \leq C_x < 151733,$$

so $r - 1 \leq 303$, that is, $r \leq 304$. Thus $X \ni x$ is of type (1). Moreover, $X \ni x$ is the quotient of $\mathbb{C}^2$ by a linear action of $\mathbb{Z}/r\mathbb{Z}$ that is free outside the origin [BHPV04, Chapter III, Theorem 5.1 and Proposition 5.3], so $g_x = r$.

Case 2. $X \ni x$ is of type $[e_0; (2,1), (2,1), (r,q)]$ with $r \geq 2$, $1 \leq q < r$, and $\gcd(r,q) = 1$. By **Step 1** and Lemma 9.12(1), $(e_0 - 1)r - q \leq 9$. If $e_0 \geq 3$, then $X \ni x$ is of type (2) by Proposition 9.6(1). Suppose that $e_0 = 2$, and let $d := r - q$. If $d = 1$, then, by Proposition 9.6(2), $b_x = 0$, $c_x(m) = 0$ for any integer $m \geq 2$, and $\gamma_x = \ell_x \geq 4$, so $C_x \geq 57820 \cdot 4 = 231280 > 151733$, which contradicts **Step 1**. Therefore $2 \leq d \leq 9$, and, by Proposition 9.6(3), $X \ni x$ is of type $[2; (2,1), (2,1), (ud+t, (u-1)d+t)]$ for some integers $u \geq 1$ and $1 \leq t < d$ with $\gcd(d,t) = 1$. It remains to prove that $u \leq 3$.

Suppose that $u \geq 4$. Let $Y \ni y$ be a germ of type $[2; (2,1), (2,1), (d+t, t)]$. It exists by the row of [HK08, Section 5, Table 1] with dual graph $\langle 2; 2,1; 2,1; d+t, t\rangle$, and $Y \ni y$ is klt since it is a quotient singularity [KM98, Proposition 4.18]. By the row (d,t) of Table 7, $D_y > 0$. By Lemma 9.5 and Definition 9.2,

$$D_x = D_y + (57820 - 91)(u-1) = D_y + 57729(u-1).$$

By Proposition 9.6(4), $C_x \geq D_x + 91(\ell_x + 1) > D_x$, so $C_x > D_y + 57729(u-1) \geq 173187 > 151733$, which contradicts **Step 1**. Therefore $u \leq 3$, and $X \ni x$ is of type (3).

Case 3. $X \ni x$ is of type $[e_0; (2,1), (3,q_2), (r,q_3)]$ with $r \in \{3,4,5\}$. By **Step 1** and Lemma 9.12(2), $X \ni x$ is a germ of Table 8, hence of type (4).

Cases 1–3 cover all singular points of X, and $g_x = 4\eta/\sigma^2$ in Cases 2 and 3 by Lemma 2.10. This completes the proof. □

9.5. **The integer system.** In this subsection, we turn the conditions on a klt stable surface X with $K_X^2 \leq 1/48983$ and $\mathrm{mld}(X) \geq 5/46$ into linear constraints in the numbers n_j of singular points of each type of Theorem 9.14 and the plurigenera p_m, which are integer variables.

Definition 9.15. For a type j in Theorem 9.14, let b_j, γ_j, g_j, C_j, and $c_j(m)$ be the invariants of Definition 9.2 of a germ of type j (such germs exist by [HK08, Section 5, Table 1]), let $b_{j,i}$ be the entries of b_j, and let $s_j(m) := c_j(m) - \binom{m}{2} c_j(2)$. Let J be the set of the types j in Theorem 9.14 with $\min_i(1 - b_{j,i}) \geq 5/46$ and $C_j < 151733$, taken up to isomorphism of germs: the cyclic types $\frac{1}{r}(1,q)$ and $\frac{1}{r}(1,q')$ with $qq' \equiv 1 \bmod r$ are identified. The *integer system* Σ has the integer variables n_j for $j \in J$ and p_m for $18 \leq m \leq 45$, where $p_m := 0$ for $2 \leq m \leq 17$ by convention, and consists of the following constraints:

(1) $\sum_j n_j s_j(3) = 2$, and $\sum_j n_j(s_j(m) - 2s_j(m-1) + s_j(m-2)) - p_m + 2p_{m-1} - p_{m-2} = 1$ for $4 \le m \le 45$,
(2) $\sum_j n_j c_j(2) \le -1 - 1/84235998$ and $-\sum_j n_j c_j(2) \le 1 + 1/48983$,
(3) $\sum_j n_j(\gamma_j + 1 - 1/g_j - \frac{4}{3}c_j(2)) \le 40/3$,
(4) $\sum_j n_j C_j \le 151733$,
(5) $p_a \le p_{ka}$ for $18 \le a \le 45$ and $2 \le k \le \lfloor 45/a \rfloor$,
(6) $p_a + p_b \le 1$ for $18 \le a < b \le 45$ with $\gcd(a,b) \le 17$ and $\operatorname{lcm}(a,b) \le 156$,
(7) $p_a + p_b - p_{a+b} \le 1$ for $18 \le a \le b$ with $a + b \le 45$, and
(8) for $j \in J$,
$$0 \le n_j \le \lceil 151733/C_j \rceil - 1,$$
and $0 \le p_m \le 1$ for $18 \le m \le 45$.

Remark 9.16. The bounds in Definition 9.15(8) make sense because $C_j > 0$ for any $j \in J$ by Theorem 9.10, as $\min_i(1 - b_{j,i})$ is the minimal log discrepancy of a germ of type j (Lemma 9.3), and such germs exist by [HK08, Section 5, Table 1].

Lemma 9.17. *Let X be a normal projective variety, D a Weil divisor on X, and $R := \bigoplus_{m\ge 0} H^0(X, \mathcal{O}_X(mD))$. Let $a, b \ge 1$ be integers, $s \in R_a$ and $t \in R_b$ non-zero elements, $e := \gcd(a,b)$, and $L := \operatorname{lcm}(a,b)$. Assume that $R_e = 0$. Then $\dim R_L \ge 2$.*

Proof. We regard a non-zero element $f \in R_m$ as a rational function on X with $\operatorname{div}(f) + mD \ge 0$, and the multiplication of R as the product of rational functions. Since X is a variety, R is a domain, so $s^{b/e}$ and $t^{a/e}$ are non-zero elements of R_L.

Suppose that they are linearly dependent, so that $s^{b/e} = \lambda t^{a/e}$ for some $\lambda \in \mathbb{C}^*$. Let $E_s := \operatorname{div}(s) + aD$ and $E_t := \operatorname{div}(t) + bD$. Then $E_s \ge 0$, $E_t \ge 0$, and
$$\frac{b}{e}E_s = \frac{b}{e}\operatorname{div}(s) + \frac{ab}{e}D = \frac{a}{e}\operatorname{div}(t) + \frac{ab}{e}D = \frac{a}{e}E_t.$$
Since $\gcd(a/e, b/e) = 1$, any coefficient of E_s is divisible by a/e. Thus there exists a Weil divisor $E \ge 0$ on X with $E_s = \frac{a}{e}E$ and $E_t = \frac{b}{e}E$. Pick integers p and q with $p\frac{a}{e} + q\frac{b}{e} = 1$, and let $u := s^p t^q$, a non-zero rational function on X. Since $pa + qb = e$, we have
$$\operatorname{div}(u) + eD = p(E_s - aD) + q(E_t - bD) + eD = \Big(p\frac{a}{e} + q\frac{b}{e}\Big)E = E \ge 0,$$
so u is a non-zero element of R_e, which contradicts $R_e = 0$. Thus $s^{b/e}$ and $t^{a/e}$ are linearly independent, and $\dim R_L \ge 2$. □

Proposition 9.18. *Let X be a klt stable surface with $K_X^2 \le 1/48983$ and $\operatorname{mld}(X) \ge 5/46$. For $j \in J$ let n_j be the number of singular points $x \in X$ such that $X \ni x$ is of type j, and for $18 \le m \le 45$ let $p_m := P_m(X)$. Then $(n_j)_{j\in J}$ and $(p_m)_{18\le m\le 45}$ form an integer solution of the system Σ of Definition 9.15.*

Proof. We first attach to any singular point of X a type in J. Let $x \in X$ be a singular point. By Theorem 9.14, $X \ni x$ is of one of the types (1)–(4) of Theorem 9.14, $1 - b_{x,i} \ge 5/46$ for any entry $b_{x,i}$ of b_x, and $C_x < 151733$. The type of $X \ni x$ is read off from the weighted dual graph of the minimal resolution of $X \ni x$.

The invariants A_x, k_x, b_x, γ_x, $c_x(m)$ of Definition 9.2 depend only on the weighted dual graph by definition, and so does g_x by the formula for g_x in Theorem 9.14. Hence μ_x and C_x depend only on the weighted dual graph as well.

A chain has no fork, and reversing the Hirzebruch–Jung chain of $\frac{1}{r}(1,q)$ yields the chain of $\frac{1}{r}(1,q')$ with $qq' \equiv 1 \bmod r$ [BHPV04, Chapter III, Section 5]. An isomorphism between the weighted dual graphs of types with a fork fixes the fork and preserves the pairs (r_i, q_i) of the arms, so the ordering of the arms in Definition 2.9(2) implies that the types are equal. Hence two listed types have isomorphic weighted dual graphs exactly when they are equal or are identified in Definition 9.15. Thus any singular point x of X has exactly one type $j \in J$, and

then $c_x(m) = c_j(m)$ for any $m \geq 2$, $\gamma_x = \gamma_j$, $\mu_x = \gamma_j + 1 - 1/g_j$, and $C_x = C_j$. Summing over the singular points x of X, we obtain

$$\begin{aligned} &\sum_x c_x(m) = \sum_j n_j c_j(m) \ (m \geq 2), \quad \sum_x \mu_x = \sum_j n_j\Big(\gamma_j + 1 - \frac{1}{g_j}\Big), \\ &\text{and} \quad \sum_x C_x = \sum_j n_j C_j. \end{aligned} \tag{9.2}$$

The numbers n_j and p_m are non-negative integers. We check the constraints of Definition 9.15, proving (8) before (5)–(7).

Since $\mathrm{mld}(X_{438}) \leq 5/61 < 5/46 \leq \mathrm{mld}(X)$ (Lemma 3.2), $X \ncong X_{438}$. Hence $P_m(X) = 0$ for any $1 \leq m \leq 17$ by Proposition 4.14. Together with the convention $p_m = 0$ for $2 \leq m \leq 17$, we have $p_m = P_m(X)$ for any $2 \leq m \leq 45$. By Lemma 9.4(1) for $m = 2$, $P_2(X) = 0$, and (9.2),

$$K_X^2 = -1 - \sum_j n_j c_j(2). \tag{9.3}$$

Substituting (9.3) and (9.2) into Lemma 9.4(1), we obtain

$$\begin{aligned} p_m &= 1 + \binom{m}{2}\Big(-1 - \sum_j n_j c_j(2)\Big) + \sum_j n_j c_j(m) \\ &= 1 - \binom{m}{2} + \sum_j n_j s_j(m) \quad \text{for } 2 \leq m \leq 45. \end{aligned} \tag{9.4}$$

For $m = 3$, since $p_3 = 0$ and $\binom{3}{2} = 3$, (9.4) implies $\sum_j n_j s_j(3) = 2$. For $4 \leq m \leq 45$, since

$$\binom{m}{2} - 2\binom{m-1}{2} + \binom{m-2}{2} = 1,$$

the second difference of (9.4) is

$$p_m - 2p_{m-1} + p_{m-2} = -1 + \sum_j n_j\big(s_j(m) - 2s_j(m-1) + s_j(m-2)\big).$$

Thus (1) holds. By (9.3), $K_X^2 \geq 1/84235998$ (Proposition 9.13(2)) is equivalent to $\sum_j n_j c_j(2) \leq -1 - 1/84235998$, and $K_X^2 \leq 1/48983$ is equivalent to $-\sum_j n_j c_j(2) \leq 1 + 1/48983$. Thus (2) holds.

By Lemma 9.4(2), (9.2), and (9.3),

$$\sum_j n_j\Big(\gamma_j + 1 - \frac{1}{g_j}\Big) = \sum_x \mu_x \leq 12 - \frac{4}{3}K_X^2 = \frac{40}{3} + \frac{4}{3}\sum_j n_j c_j(2),$$

so $\sum_j n_j(\gamma_j + 1 - 1/g_j - \frac{4}{3}c_j(2)) \leq 40/3$. Thus (3) holds.

By Proposition 9.11 and (9.2),

$$\sum_j n_j C_j < 151733. \tag{9.5}$$

In particular, (4) holds.

Let $j \in J$. By Theorem 9.10 and the remark after Definition 9.15, $C_{j'} > 0$ for any $j' \in J$, so (9.5) implies $n_j C_j < 151733$. Since n_j is an integer and $C_j > 0$, we have $n_j \leq \lceil 151733/C_j \rceil - 1$. Moreover, $0 \leq p_m \leq 1$ for $18 \leq m \leq 45$ by Proposition 5.1, since $m \leq 156$. Thus (8) holds, and $p_m \in \{0, 1\}$ for $18 \leq m \leq 45$.

For (5)–(7), note that $R(X)$ is a domain, since X is a normal variety. Let $18 \leq a \leq 45$ and $2 \leq k \leq \lfloor 45/a \rfloor$. If $p_a = 0$, then $p_a \leq p_{ka}$ since $p_{ka} \geq 0$. If $p_a = 1$, then for a non-zero $f \in R(X)_a$, the element $f^k \in R(X)_{ka}$ is non-zero, so $p_{ka} \geq 1 = p_a$. Thus (5) holds.

Let $18 \le a < b \le 45$ with $\gcd(a,b) \le 17$ and $\operatorname{lcm}(a,b) \le 156$, and suppose that $p_a = p_b = 1$. Since $1 \le \gcd(a,b) \le 17$, $R(X)_{\gcd(a,b)} = 0$ by Proposition 4.14. By Lemma 9.17 with $D = K_X$ and non-zero elements of $R(X)_a$ and $R(X)_b$, we have $P_{\operatorname{lcm}(a,b)}(X) = \dim R(X)_{\operatorname{lcm}(a,b)} \ge 2$. Since $\operatorname{lcm}(a,b) \le 156$, this contradicts Proposition 5.1. Thus $p_a + p_b \le 1$, and (6) holds.

Let $18 \le a \le b$ with $a + b \le 45$. If $p_a = p_b = 1$, then for non-zero $f \in R(X)_a$ and $f' \in R(X)_b$, where we take $f' = f$ if $a = b$, the product $ff' \in R(X)_{a+b}$ is non-zero, so $p_{a+b} \ge 1$ and $p_a + p_b - p_{a+b} \le 1$. Otherwise $p_a + p_b \le 1$ since $p_a, p_b \in \{0,1\}$, and $p_a + p_b - p_{a+b} \le 1$ since $p_{a+b} \ge 0$. Thus (7) holds.

Therefore, $(n_j)_{j\in J}$ and $(p_m)_{18\le m\le 45}$ form an integer solution of Σ. □

9.6. **The infeasibility proof.** In this subsection, we prove that the integer system Σ of Definition 9.15 has no integer solution (Proposition 9.21), and we conclude the proof of Theorem 9.1.

We prove Proposition 9.21 by an exact branch-and-bound derivation in the VIPR format of [CGS16], given in an ancillary file of this paper, the *proof file*. A branch-and-bound derivation splits into cases by assumptions and discharges the assumptions afterwards, so the rules of Definition 9.19 include assumptions and their discharge. We prove the soundness of the rules in Lemma 9.20.

Definition 9.19. Let Σ be a finite system of linear equations and inequalities with rational coefficients in integer variables x_k, where k runs over a finite set. A *derivation* from Σ is a finite sequence $\ell_1, \dots, \ell_N$ of linear equations or inequalities with rational coefficients in the variables x_k, together with finite sets $A_1, \dots, A_N$ of assumptions, such that for any $1 \le i \le N$ one of the following holds:

(1) ℓ_i is a constraint of Σ and $A_i = \emptyset$,
(2) (lin) there exist a set I of indices $i' < i$ and rational numbers $\lambda_{i'}$ for $i' \in I$, with $\lambda_{i'} \ge 0$ if $\ell_{i'}$ is an inequality, such that the following holds. Write any inequality among the $\ell_{i'}$ with $i' \in I$ as $\sum_k a_{i',k} x_k \le \beta_{i'}$, multiplying it by -1 if it has the form $\ge$, write any equation among them as $\sum_k a_{i',k} x_k = \beta_{i'}$, and let $a_k := \sum_{i'\in I} \lambda_{i'} a_{i',k}$ and $\beta := \sum_{i'\in I} \lambda_{i'} \beta_{i'}$. Then A_i is the union of the $A_{i'}$ with $i' \in I$, and ℓ_i is $\sum_k a_k x_k \le \beta'$ for a rational number $\beta' \ge \beta$, or -1 times such an inequality, or $\sum_k a_k x_k = \beta$ if all $\ell_{i'}$ with $i' \in I$ are equations, or, if all a_k are zero and $\beta < 0$, any linear equation or inequality,
(3) (rnd) for a rational linear combination ℓ as in (lin) of some $\ell_{i'}$ with $i' < i$ whose left-hand coefficients are integers, ℓ_i is ℓ with the right-hand side rounded up to an integer if ℓ is an inequality of the form $\ge$, and rounded down to an integer if ℓ is of the form $\le$, possibly with the right-hand side of the rounded inequality weakened, and A_i is the union of the corresponding $A_{i'}$,
(4) (asm) ℓ_i is $\sum_k a_k x_k \le \beta$ or $\sum_k a_k x_k \ge \beta + 1$ for integers a_k and β, and $A_i = \{\ell_i\}$, or
(5) (uns) there exist $i_1, i_2 < i$, integers a_k and β, such that any rational vector $(x_k)_k$ satisfying ℓ_{i_1} satisfies ℓ_i, any rational vector $(x_k)_k$ satisfying ℓ_{i_2} satisfies ℓ_i, and

$$A_i = \Big(A_{i_1} \setminus \Big\{ \sum_k a_k x_k \le \beta \Big\}\Big) \cup \Big(A_{i_2} \setminus \Big\{ \sum_k a_k x_k \ge \beta + 1 \Big\}\Big).$$

We call ℓ_i a *line* of the derivation, and a (lin), (rnd), (asm), or (uns) line if it satisfies the corresponding condition.

Lemma 9.20. *Let $\ell_1, \dots, \ell_N$ be a derivation from Σ with assumptions $A_1, \dots, A_N$ as in Definition 9.19. Then for any $1 \le i \le N$ and any integer solution of Σ satisfying all assumptions in A_i, the constraint ℓ_i holds. In particular, if $A_N = \emptyset$ and ℓ_N is an inequality between rational numbers that does not hold, then Σ has no integer solution.*

Proof. Let $\xi = (\xi_k)_k$ be an integer solution of Σ. We prove by induction on i that if ξ satisfies all assumptions in A_i, then ξ satisfies ℓ_i. Suppose that ξ satisfies all assumptions in A_i, and that the claim holds for all $i' < i$. We consider the five rules of Definition 9.19.

If ℓ_i is a constraint of Σ, then ξ satisfies ℓ_i since ξ is a solution of Σ.

Assume that ℓ_i is obtained by (lin). In the notation of (lin) of Definition 9.19, for any $i' \in I$ we have $A_{i'} \subset A_i$, hence ξ satisfies $\ell_{i'}$ by induction. Since $\lambda_{i'} \geq 0$ if $\ell_{i'}$ is an inequality, we obtain $\sum_k a_k \xi_k \leq \beta$, with equality if all $\ell_{i'}$ with $i' \in I$ are equations. Thus ξ satisfies ℓ_i if ℓ_i is $\sum_k a_k x_k \leq \beta'$ with $\beta' \geq \beta$, -1 times such an inequality, or $\sum_k a_k x_k = \beta$. If all a_k are zero and $\beta < 0$, then $0 \leq \beta < 0$, which is impossible. Thus no integer solution of Σ satisfies all assumptions in A_i, and the claim holds for any ℓ_i.

Assume that ℓ_i is obtained by (rnd) from a combination ℓ, say ℓ is $\sum_k a_k x_k \geq \beta$ with $a_k \in \mathbb{Z}$. By the previous paragraph, ξ satisfies ℓ. Since a_k and ξ_k are integers, $\sum_k a_k \xi_k$ is an integer, hence

$$\sum_k a_k \xi_k \geq \lceil \beta \rceil.$$

If ℓ is $\sum_k a_k x_k \leq \beta$ with $a_k \in \mathbb{Z}$, then similarly $\sum_k a_k \xi_k \leq \lfloor \beta \rfloor$. In both cases, ξ also satisfies any weakening of the rounded inequality.

If ℓ_i is obtained by (asm), then $\ell_i \in A_i$, and ξ satisfies ℓ_i by assumption.

Assume that ℓ_i is obtained by (uns), with i_1, i_2, a_k, and β as in (uns) of Definition 9.19. Since a_k, β, and ξ_k are integers, $\sum_k a_k \xi_k$ is an integer, hence

$$\sum_k a_k \xi_k \leq \beta \quad \text{or} \quad \sum_k a_k \xi_k \geq \beta + 1.$$

In the first case, ξ satisfies all assumptions in A_{i_1}, since any assumption in A_{i_1} other than $\sum_k a_k x_k \leq \beta$ belongs to A_i. By induction ξ satisfies ℓ_{i_1}, hence ξ satisfies ℓ_i. In the second case, ξ satisfies all assumptions in A_{i_2}, hence ξ satisfies ℓ_{i_2} by induction, and ξ satisfies ℓ_i.

This proves the first statement. If $A_N = \emptyset$, then any integer solution of Σ satisfies ℓ_N. Since ℓ_N is an inequality between rational numbers that does not hold, Σ has no integer solution. □

Proposition 9.21. *The system Σ of Definition 9.15 has no integer solution.*

Proof. By Lemma 9.20, it suffices to show that there exists a derivation from the system Σ of Definition 9.15, whose variables are n_j and p_m, with $A_N = \emptyset$ and ℓ_N an inequality between rational numbers that does not hold. The derivation is the proof file `large_mld_proof.vipr`, an ancillary file of this paper, shipped compressed as `large_mld_proof.vipr.xz`. The proof file has SHA256 hash

`c3ea6f374e31789e5d559fd3929605e339536e7a93941559fc27b8d26fd50650`.

It is written in the VIPR file format, version 1.0, whose technical specification is in the repository https://github.com/scipopt/vipr at the commit given below. The format is described at a conceptual level in [CGS16, Sections 2–4]. It lists the 1396 integer variables n_j for $j \in J$ and p_m for $18 \leq m \leq 45$, the 2901 constraints of Σ, namely 109 constraints and 2792 bounds, and 136707 further lines. The last line is a false inequality between rational numbers with no undischarged assumption. No line of the proof file uses a feasible solution or an objective bound, and all numbers in the proof file are exact rational numbers.

The file `README.txt` describes how the proof file was produced.

We checked the proof file with the checker `viprchk` of [CGS16, Section 5], at commit `99eec197a49b2bd9ad33c1f4c7afabe93caa6511` of https://github.com/scipopt/vipr, with the file `code/viprchk.cpp` unmodified. The checker was built against GMP, with a text input and output adapter for GMP rationals consisting of two functions. The adapter only parses and prints integers and fractions, with `mpq_set_str` and `mpq_get_str`, rejects malformed input and zero denominators, and does not change the checking logic. The checker verified all further lines and the final contradiction, and printed `Successfully verified infeasibility` (return code 0).

For each (uns) line, `viprchk` checks that the line named as an assumption for the premise ℓ_{i_1} and the line named as an assumption for the premise ℓ_{i_2} are, after exchanging i_1 and i_2

if necessary, $\sum_k a_k x_k \le \beta$ and $\sum_k a_k x_k \ge \beta + 1$ for some rational numbers a_k and β (the function `canUnsplit` in `code/viprchk.cpp`). It also checks that each of the premises ℓ_{i_1} and ℓ_{i_2} dominates the derived constraint ℓ_i, that is, the premise and ℓ_i have equal left-hand sides and the right-hand side of the premise is at least as strong as that of ℓ_i, or the premise is a false relation with zero left-hand side (the function `Constraint::dominates` in `code/viprchk.cpp`). Such a premise satisfies the implication in (uns), and the domination check also covers the weakening of the right-hand side in (lin) and (rnd). A separate script in the ancillary directory `provenance/` confirmed that all (asm) lines have integral coefficients and right-hand sides, as (asm) requires, and that each line refers only to earlier lines.

The assumptions that `viprchk` keeps for a line are (asm) lines, and for an (uns) line it removes the named lines from the assumptions of ℓ_{i_1} and ℓ_{i_2}, respectively, and takes the union. Hence, if a named line is not an (asm) line, then ℓ_i is a (lin) line whose only premise is the corresponding ℓ_{i_1} or ℓ_{i_2}. If both named lines are (asm) lines, then the numbers a_k and β of (uns) are integers. By induction on i, the set A_i of Definition 9.19 is contained in the set of assumptions that `viprchk` keeps for ℓ_i, and `viprchk` keeps no assumption for the last line.

Finally, the constraint section of the proof file coincides with Σ. The script `regenerate_and_compare.py` regenerates the 1368 types of J by running through the parameter ranges of Theorem 9.14, taking the tuple with $q \le q'$ for a cyclic type, and applying the conditions $\min_i(1 - b_{j,i}) \ge 5/46$ and $C_j < 151733$ of Definition 9.15, recomputes the invariants b_j, γ_j, g_j, C_j, and $c_j(m)$ for $2 \le m \le 45$, generates all constraints of Definition 9.15, and compares them with the constraint section of the proof file. The constraints are compared as multisets of exact normalized rational coefficient vectors, where an inequality may be scaled by a positive rational number and an equation by a non-zero rational number, and a scaled constraint is a (lin) line whose only premise is a constraint of Σ. The multiset of constraints of Σ and the multiset of constraints in the proof file coincide, with 1368 types, 1396 variables, 109 constraints, and 2792 bounds.

Thus the proof file is a derivation from Σ in the sense of Definition 9.19, whose last line is an inequality between rational numbers that does not hold and has an empty set of assumptions. By Lemma 9.20, Σ has no integer solution. □

Proof of Theorem 9.1. Let X be a klt stable surface with $\mathrm{mld}(X) \ge 5/46$, and suppose that $K_X^2 \le 1/48983$. By Proposition 9.18, the numbers n_j of singular points of type $j \in J$ and the plurigenera $p_m = P_m(X)$ for $18 \le m \le 45$ form an integer solution of Σ. This contradicts Proposition 9.21. Thus $K_X^2 > 1/48983$. □

10. Proof of the main theorem

In this section, we prove Theorem 1.5.

Proof of Theorem 1.5. We first consider X_{438}. By Lemma 3.1, X_{438} is a klt hypersurface whose canonical sheaf is $\mathcal{O}_{X_{438}}(1)$, so $K_{X_{438}}$ is ample since $\mathcal{O}_{X_{438}}(1)$ is ample. Since X_{438} is a closed subvariety of $\mathbb{P}(11, 61, 146, 219)$, X_{438} is projective. Thus X_{438} is a klt stable surface (Definition 2.3). Moreover, by Lemma 3.1,

$$\mathcal{O}_{X_{438}}(K_{X_{438}}) \cong \mathcal{O}_{X_{438}}(1) \quad \text{and} \quad K_{X_{438}}^2 = \frac{1}{48983}.$$

Next we prove the inequality. By Lemma 2.6, there exists a minimizer $X_{\min}$, that is, a klt stable surface $X_{\min}$ such that $K_{X_{\min}}^2 \le K_Y^2$ for any klt stable surface Y, and

$$K_{X_{\min}}^2 \le \frac{1}{48983}.$$

By Theorem 9.1, $\mathrm{mld}(X_{\min}) < 5/46$. Since $X_{\min}$ is a minimizer, Theorem 8.1 implies that $X_{\min} \cong X_{438}$. Therefore, for any klt stable surface X,

$$K_X^2 \ge K_{X_{\min}}^2 = K_{X_{438}}^2 = \frac{1}{48983}.$$

Finally, let X be a klt stable surface such that $K_X^2 = 1/48983$. By the inequality, $K_Y^2 \geq 1/48983 = K_X^2$ for any klt stable surface Y, so X is a minimizer. Since $K_X^2 \leq 1/48983$, Theorem 9.1 implies that $\mathrm{mld}(X) < 5/46$, and Theorem 8.1 implies that $X \cong X_{438}$. Conversely, if $X \cong X_{438}$, then $K_X^2 = K_{X_{438}}^2 = 1/48983$. □

11. Applications to pluricanonical systems

Throughout this section, X is a normal stable surface and $v = K_X^2$. We use the notation $R(X)_m$ and $P_m(X)$ of Definition 2.3 also when X is not klt.

Set

$$\mu(X) = \min\{m \geq 1 : P_m(X) \geq 2\}, \qquad N_{\mathrm{opt}} = \sup_X \mu(X),$$

where the supremum is over the normal stable surfaces.

Theorem 11.1. *One has*

$$m \geq 5, \quad (m-1)^2 v > 16 \quad \Longrightarrow \quad P_m(X) \geq 2. \tag{11.1}$$

Consequently:

(1) $P_m(X) \geq 2$ *for every integer* $m \geq 887$.
(2) If X *is not klt,* $P_m(X) \geq 2$ *for every* $m \geq 116$.
(3) $292 \leq N_{\mathrm{opt}} \leq 887$, *and* $\mu(X_{438}) = 292$.

The proof proceeds by local positivity at two independently very general points on a resolution. We include the fiber estimate and the adjoint evaluation argument, since a canonical volume bound alone does not control the Cartier index.

11.1. **Resolution and a lower bound for fiber degree.** Choose a log resolution $f : Y \to X$, an isomorphism over the smooth locus, and put $E = \mathrm{Exc}(f)_{\mathrm{red}}$. We arrange that E has simple normal crossings. Choose canonical divisors compatibly. Log canonicity gives the Zariski decomposition

$$D = K_Y + E = P + A, \qquad P = f^*K_X, \quad A \geq 0 \text{ exceptional}, \quad P \cdot A = 0, \quad P^2 = v. \tag{11.2}$$

For every $n \geq 1$, pullback and pushforward of effective divisors give

$$H^0(Y, nD) = R(X)_n. \tag{11.3}$$

Indeed, for $u \in \mathbb{C}(X)$, $\mathrm{div}_Y(u) + nD = f^*(\mathrm{div}_X(u) + nK_X) + nA$; the pullback of an effective $\mathbb{Q}$-Cartier divisor is effective. The converse follows by pushforward. Also, since D is integral and $P \leq D$,

$$K_Y + \lceil tP \rceil \leq K_Y + tD \leq (t+1)D \qquad (t \in \mathbb{Z}_{>0}). \tag{11.4}$$

Lemma 11.2 (Fiber degree)**.** *More generally, let* $D = K_Y + E = P + A$ *be the Zariski decomposition of a big log canonical divisor on a smooth projective surface with reduced boundary having simple normal crossings. If* $h : Y \to C$ *is a surjective morphism to a smooth projective curve with connected fibers and* F *is a general fiber, then, writing* $s = P \cdot F$ *and* $v = P^2$, *one has*

$$1 \leq D \cdot F \leq s + \frac{2s^2}{v}, \qquad s \geq c(v) := \frac{2}{1 + \sqrt{1 + 8/v}}. \tag{11.5}$$

Proof. Write $D + 2F = Q + N$ for its Zariski decomposition, and put $\delta = D \cdot F$. The Hodge index theorem gives $P \cdot F > 0$; thus δ is a positive integer. We first show that N is vertical.

Set $g = g(C)$ and $R = K_{Y/C} + E$. The bundles $\mathcal{V}_k = h_*\mathcal{O}_Y(kR)$ are weakly positive by [Fuj17, Theorem 1.1], hence nef on the smooth curve C. Choose a line bundle H of degree at least $2g$ (take $H = \mathcal{O}_C$ when $g = 0$). Then $\mathcal{V}_k \otimes H$ is globally generated: for $c \in C$, Serre duality identifies its obstruction group with

$$\mathrm{Hom}(\mathcal{V}_k, \omega_C \otimes H^{-1}(c)) = 0.$$

The image of a proposed nonzero map would be a line bundle quotient of $\mathcal{V}_k$ of negative degree, contradicting nefness. For large k, the restriction $kR|_F = k(K_F + E|_F)$ is basepoint-free. Generic base change therefore shows that $|kR + h^*H|$ has no horizontal fixed component.

The pseudo-effective divisor $R_k = R + k^{-1}h^*H$ has a Zariski decomposition $P_k + N_k$. Components of its negative part are fixed in every effective multiple, so N_k is vertical. Adding $2gF$ does not change this negative part, because $P_k + 2gF$ is nef and still orthogonal to N_k. Numerically, $R_k + 2gF \equiv D + 2F + k^{-1}h^*H$. Continuity of Zariski decompositions inside the big cone gives $N \cdot F = 0$ and $Q \cdot F = \delta$ as $k \to \infty$; see [Laz04, Section 2.3] for these decomposition properties.

We next have $N \le A$. To see this, write $N - A = T - U$ with effective T, U having no common components. If $T \neq 0$, then $Q \cdot T = 0$ and $T^2 < 0$, whereas

$$0 \le (P + 2F) \cdot T = (Q + T - U) \cdot T < 0,$$

a contradiction. Thus $B = A - N \ge 0$, $P \cdot B = 0$, and $Q = P + 2F + B$. Since Q is nef,

$$Q^2 = Q \cdot P + 2Q \cdot F + Q \cdot B \ge v + 2s + 2\delta.$$

The Hodge index theorem gives the other bound $Q^2 \le (P \cdot Q)^2/v = (v + 2s)^2/v$. Comparing gives $\delta \le s + 2s^2/v$. Solving $2s^2 + vs - v \ge 0$ proves the assertion. □

11.2. **Two general points and independent sections.**

Lemma 11.3 (Two-point local positivity)**.** *In the setting of Lemma 11.2, there exist two independently very general distinct points $y_1, y_2 \in Y$ such that every integral curve C_0 meeting at least one of them satisfies*

$$\frac{P \cdot C_0}{\operatorname{mult}_{y_1} C_0 + \operatorname{mult}_{y_2} C_0} \ge e(v) := \min\left\{c(v), \frac{\sqrt{v}}{2}\right\}. \tag{11.6}$$

Proof. We may discard countably many proper closed subsets of Y and Y^2 arising from pointed families of curves, stratified by numerical class and multiplicities. Curves through the remaining points belong to families dominating the relevant point space.

First consider a curve through just one marked point, of multiplicity r. For $r \ge 2$, the moving-curve inequality $C_0^2 \ge r(r-1)$ [EL93, Corollary 1.2] and Hodge index give $(P \cdot C_0)/r \ge \sqrt{v/2} > c(v)$. For $r = 1$ and $C_0^2 > 0$, the bound is at least $\sqrt{v}$. If $r = 1$ and $C_0^2 = 0$, distinct general curves in the covering family are disjoint and give a fibration. It is a morphism: indeterminacy would make general members meet. After Stein factorization their class is the general-fiber class. Apply Lemma 11.2.

Now suppose the curve contains both points, with multiplicities $a \ge b \ge 1$. Fix the first point and vary the second point in a one-parameter subfamily in which the curve varies. Such a subfamily exists because the pointed family dominates Y^2; a fixed curve cannot contain a general second point. The deformation calculation in [EL93, Lemma 1.1 and proof of Corollary 1.2] gives

$$C_0^2 \ge a^2 + b(b-1). \tag{11.7}$$

Here is the additional fixed-point detail. The first-order normal section vanishes to order at least a at the fixed point, and to order at least $b-1$ at the moving point. Their contributions to its zero divisor on the normalization are at least a^2 and $b(b-1)$. The section is nonzero after choosing a generically nontrivial embedded deformation in characteristic zero. Its degree is C_0^2.

Finally,

$$4(a^2 + b^2 - b) - (a+b)^2 = 3(a-b)^2 + 4b(a-1) \ge 0.$$

Hodge index and (11.7) therefore give $(P \cdot C_0)/(a+b) \ge \sqrt{v}/2$. This proves (11.6). □

Proposition 11.4 (Adjoint evaluation)**.** *For every positive integer t satisfying*

$$t > \max\left\{\frac{4}{\sqrt{v}},\ 1 + \sqrt{1 + 8/v}\right\}, \tag{11.8}$$

one has $P_{t+1}(X) \ge 2$.

Proof. Take y_1, y_2 as above, outside E, and blow them up: $\pi : W \to Y$, with exceptional curves T_1, T_2. The divisor

$$L = \pi^*(tP) - 2T_1 - 2T_2$$

is nef by Lemma 11.3, since $te(v) > 2$. It is big because $L^2 = t^2v - 8 > 8 > 0$. Its fractional part is supported on the strict transform of E. Kawamata–Viehweg vanishing [KMM87, Theorem 1-2-3] gives $H^1(W, \mathcal{O}_W(K_W + \lceil L \rceil)) = 0$. For $M = K_Y + \lceil tP \rceil$, integrality of P near the two points gives

$$K_W + \lceil L \rceil = \pi^* M - T_1 - T_2.$$

The restriction sequence therefore yields a surjection

$$H^0(Y, M) \longrightarrow \mathcal{O}_Y(M) \otimes \mathbb{C}(y_1) \ \oplus \ \mathcal{O}_Y(M) \otimes \mathbb{C}(y_2).$$

The target has dimension two. Equations (11.4) and (11.3) inject this space of sections into $R(X)_{t+1}$. □

Proof of the upper bounds in Theorem 11.1. Put $t = m - 1 \geq 4$. If $t^2v > 16$, then $t(t-2)v \geq t^2v/2 > 8$, which is equivalent to $t > 1 + \sqrt{1 + 8/v}$. Thus (11.8) holds. By Theorems 1.5 and 1.3(1), $v \geq 1/48983$. At the limiting value the exact arithmetic is

$$885^2 = 783225 < 16 \cdot 48983 = 783728 < 784996 = 886^2.$$

Hence $m = 887$ works, and so does every larger integer. For non-klt X, Theorem 1.3(1) gives $v \geq 1/825$. Now $114^2 < 16 \cdot 825 = 13200 < 115^2$, so $m = 116$ works. □

11.3. **The first pluricanonical pencil on X_{438}.**

Proof of the lower bound in Theorem 11.1. We use the coordinates x, y, z, w of Theorem 1.5. Let A be the weighted homogeneous coordinate ring of this hypersurface. Its affine cone is smooth away from a subset of codimension at least two: this follows directly from the partial derivatives of $w^2 + z^3 + xy^7 + x^{21}yz$. A hypersurface ring is Cohen–Macaulay, hence satisfies S_2; regularity in codimension one therefore proves normality by Serre's criterion. Moreover $\operatorname{depth}_{A_+} A = 3$. The graded local-cohomology sequence consequently gives

$$A_m = H^0(\operatorname{Proj} A, \widetilde{A(m)}).$$

The weighted quotient is free in codimension one on the punctured cone. Normality extends the associated weight-m sheaf reflexively across the remaining codimension-two locus. Weighted adjunction identifies its weight-one sheaf with ω_X, so $\widetilde{A(m)} = \omega_X^{[m]}$. This proves that the full reflexive canonical ring is A. In degrees below 438 its pieces are therefore counted by weighted monomials without a defining relation.

The possible weights of monomials in y, z, w of weight less than 292 are

$$0, 61, 122, 146, 183, 207, 219, 244, 268, 280.$$

They have pairwise distinct residues modulo 11, namely $0, 6, 1, 3, 7, 9, 10, 2, 4, 5$. Thus at most one can be completed to a monomial of any given weight $m < 292$ by multiplying by a power of x. In degree 292 the only two monomials are z^2 and $x^{21}y$. There is no relation in that degree. Hence $P_m(X_{438}) \leq 1$ for $1 \leq m < 292$ and $P_{292}(X_{438}) = 2$, proving $\mu(X_{438}) = 292$ and $N_{\mathrm{opt}} \geq 292$. □

Remark 11.5. The nonvanishing estimate

$$n > 2 + \sqrt{1 + 16/v} \quad \Longrightarrow \quad P_n(X) > 0 \tag{11.9}$$

also follows from Proposition 11.4: for $t = n - 1$, its hypothesis implies both inequalities in (11.8).

11.4. **Effective birationality.** The volume bound also makes Hisamatsu's log-pluricanonical birationality estimate uniform.

Corollary 11.6. *For every normal stable surface X, the rational map defined by $|mK_X|$ is birational onto its image for every integer $m \geq 1882$.*

Proof. Use the log resolution and Zariski decomposition (11.2). By [His03, Theorem 1.1], the system $|mD|$ defines a birational map whenever

$$m \geq \frac{6\sqrt{2}}{\sqrt{P^2}} + 4.$$

Here $P^2 = v \geq 1/48983$ by Theorems 1.5 and 1.3(1). Since

$$72 \cdot 48983 = 3526776 < 3526884 = 1878^2,$$

we have $6\sqrt{2}/\sqrt{P^2} + 4 < 1882$. Finally, the equality of section spaces in (11.3) preserves ratios of sections, so the maps defined by $|mD|$ and $|mK_X|$ agree on their common function field. □

Remark 11.7. The proof of the estimate in [His03] is presented as an outline in Section 3. The word "unless" in its Theorem 1.1 is read as "if", in accordance with Proposition 3.5 and the concluding estimate on p. 100. Corollaries 11.6 and 11.8 use that estimate.

Corollary 11.8. *Let X be a connected non-normal stable surface. Then $|mK_X|$ defines a rational map birational onto its image for every integer $m \geq 275$. Consequently, for every stable surface, normal or not, the pluricanonical map is birational onto its image for every $m \geq 1882$.*

Proof. Here we allow X to be non-normal. Let $\nu : \overline{X} \to X$ be its normalization, with reduced conductor divisor $\overline{B}$, and write $\overline{X} = \coprod_i X_i$ and $B_i = \overline{B}|_{X_i}$. Then (X_i, B_i) is lc and

$$H_i := K_{X_i} + B_i = (\nu^* K_X)|_{X_i}$$

is ample. Every B_i is non-zero: otherwise X_i would give a normal connected component of X. Thus $v_i := H_i^2 \geq 1/462$ by Theorem 1.2(2).

Choose a log resolution $f_i : Y_i \to X_i$, and let E_i be the reduced sum of its exceptional divisor and the strict transform of B_i. As in (11.2), the Zariski decomposition is

$$D_i := K_{Y_i} + E_i = P_i + A_i, \qquad P_i = f_i^* H_i, \quad A_i \geq 0 \text{ exceptional}, \quad P_i^2 = v_i.$$

The proof of Proposition 11.4 applies to this reduced boundary as well: Lemmas 11.2 and 11.3 are stated for big log canonical divisors. Since

$$87^2 > 16 \cdot 462,$$

it gives a non-zero section s_i of $K_{Y_i} + \lceil 87P_i \rceil$. The divisor inequality

$$K_{Y_i} + \lceil 87P_i \rceil \leq K_{Y_i} + 87D_i = 88D_i - E_i$$

therefore regards s_i as a section of $88D_i$ vanishing along E_i.

By the estimate of [His03, Theorem 1.1], $|kD_i|$ is birational for every $k \geq 187$, since

$$\frac{6\sqrt{2}}{\sqrt{v_i}} + 4 \leq 6\sqrt{924} + 4 < 187 \qquad (72 \cdot 462 < 183^2).$$

For $m \geq 275$, multiplication by s_i embeds $H^0(Y_i, (m-88)D_i)$ into $H^0(Y_i, mD_i - E_i)$ and preserves ratios of sections. Pushing forward gives a birational subsystem of

$$H^0\big(X_i, \mathcal{O}_{X_i}(mH_i - B_i)\big).$$

Extend these sections by zero on the other components of $\overline{X}$. Their residues along the conductor are zero, so they descend to sections of $\omega_X^{[m]}$ by [LR14, Proposition 2.13]. The resulting subsystems are birational on each component and separate distinct components. Hence the full pluricanonical system is birational onto its image.

For an arbitrary stable surface, apply this result to its non-normal connected components and Corollary 11.6 to its normal components. □

11.5. **Stable surface pairs with reduced boundary.** The doubling construction extends the preceding birationality bounds to stable surface pairs.

Corollary 11.9. *Let (X, Δ) be a connected projective slc surface pair with reduced boundary and ample $K_X + \Delta$. Then $|m(K_X + \Delta)|$ defines a rational map birational onto its image for every integer $m \geq 1882$. If $\Delta \neq 0$ or X is non-normal, the bound improves to $m \geq 275$.*

Proof. If $\Delta = 0$, this is Corollary 11.8. Suppose $\Delta \neq 0$. By [LR, Lemma 5.1], doubling along Δ gives a connected non-normal stable surface X^{d}. Write $\pi : \overline{X} \to X$ for the normalization, $\overline{B}$ for its conductor divisor, and $\overline{\Delta}$ for the strict transform of Δ. The normalization of X^{d} is the disjoint union of two copies of $\overline{X}$, and the pullback of its canonical divisor to either copy is

$$K_{\overline{X}} + \overline{B} + \overline{\Delta} = \pi^*(K_X + \Delta).$$

We compare global sections on the normalizations, rather than restricting reflexive sheaves to a component of the double. A section of $\omega_{X^{\mathrm{d}}}^{[m]}$ pulls back to a pair (s_1, s_2) of sections of $\mathcal{O}_{\overline{X}}(m(K_{\overline{X}} + \overline{B} + \overline{\Delta}))$ satisfying the residue gluing conditions. Along $\overline{B}$, the gluing involution on each copy is the original involution for X. Thus s_1 satisfies precisely the residue condition for descent to $\mathcal{O}_X(m(K_X + \Delta))$; see [LR14, Proposition 2.13]. Projection to s_1 therefore defines a map

$$H^0(X^{\mathrm{d}}, \omega_{X^{\mathrm{d}}}^{[m]}) \longrightarrow H^0(X, \mathcal{O}_X(m(K_X + \Delta)))$$

for every m, whose image agrees with restriction on the common smooth locus away from the boundary.

For $m \geq 275$, Corollary 11.8 gives a birational pluricanonical map on X^{d}. The image of the displayed map induces its restriction to the first copy of X on a dense open subset of every irreducible component. This subsystem is birational onto its image, and hence so is the complete log-pluricanonical system. □

Appendix A. Tables

The tables below are part of the proofs of Theorem 9.10 and Proposition 9.9, with one subsection for each. Any row of the tables is an exact identity on an interval of integers or an exact value, which can be checked with integer arithmetic from the formulas of Lemma 9.8.

A.1. **Table for Theorem 9.10.** In Table 9, r is the order of the cyclic quotient singularity $X \ni x$ of Definition-Lemma 9.7 with parameters (e, r_1, q_1, t, n), and $T := 2r(C_x - 91(\ell_x + 1))$, with C_x as in Definition 9.2.

A.2. **Tables for Proposition 9.9.** In this subsection, we list the tables for **Steps 3 and 4** of the proof of Proposition 9.9. Table 10 is used in **Step 4**, where α is at most 9 and a header (e, r_1, q_1, t, v) fixes the parameters of Definition-Lemma 9.7 except h. Table 11 is used in **Step 3**, where α is at least 10.

We use the notation of Definition-Lemma 9.7 and Lemma 9.8 and the integer T of **Step 2** of the proof of Proposition 9.9. Each identity of Table 10 can be checked by exact integer arithmetic from the formulas of Lemma 9.8. Indeed, on the interval of a row, r, U, and the integers y_G for the curves G in $\mathcal{F}$ and for G_h are affine functions of h by Lemma 9.8(3), and ℓ_0 does not depend on h. The integers z_m and $N_m(y_G)$ of Lemma 9.8(3) change only at finitely many rational values of h (the quotients $m\delta/r$ and my_G/r are monotone and bounded in h), at which $m\delta$ or my_G is an integer multiple of r, and any such value of h is the solution of a linear equation in h (an equation that holds identically in h causes no change). Between two such values, and beyond the last one, Lemma 9.8(2) implies that T is a polynomial in h of degree at most two with integer coefficients, which is determined by three values. Thus a row reduces to finitely many identities between integers, also if the interval of the row is unbounded, and any value of T is computed with integer ceilings, addition, and multiplication from Lemma 9.8(1) and (2).

TABLE 9. Values for Case 2 of the proof of Theorem 9.10: one row for any tuple (e, r_1, q_1, t, n) with $r \leq 19$ satisfying the conditions of Definition-Lemma 9.7 and the bounds of Definition-Lemma 9.7(3), and one row for the tuple $(14, 1, 0, 1, 2)$.

(e, r_1, q_1, t, n)	r	T
$(3, 1, 0, 1, 1)$	3	0
$(3, 1, 0, 1, 2)$	5	0
$(3, 1, 0, 1, 3)$	7	0
$(3, 1, 0, 1, 4)$	9	44928
$(3, 1, 0, 1, 5)$	11	449880
$(3, 1, 0, 1, 6)$	13	1316664
$(3, 1, 0, 1, 7)$	15	2645280
$(3, 1, 0, 1, 8)$	17	4435728
$(3, 1, 0, 1, 9)$	19	6688008
$(3, 1, 0, 2, 3)$	8	1730232
$(3, 1, 0, 2, 5)$	12	4038664
$(3, 1, 0, 2, 7)$	16	7270760
$(3, 2, 1, 1, 2)$	8	354208
$(3, 2, 1, 1, 3)$	11	90864
$(3, 2, 1, 1, 4)$	14	349468
$(3, 2, 1, 1, 5)$	17	342908
$(3, 2, 1, 2, 3)$	13	441944
$(3, 2, 1, 2, 5)$	19	2269384
$(3, 2, 1, 3, 4)$	18	4034296
$(3, 3, 1, 1, 3)$	18	391212
$(3, 3, 2, 1, 3)$	15	930580
$(3, 3, 2, 1, 4)$	19	620768
$(3, 3, 2, 2, 3)$	18	391212
$(4, 1, 0, 1, 1)$	4	345464
$(4, 1, 0, 1, 2)$	7	0
$(4, 1, 0, 1, 3)$	10	348740
$(4, 1, 0, 1, 4)$	13	170524
$(4, 1, 0, 1, 5)$	16	332112
$(4, 1, 0, 1, 6)$	19	583040
$(4, 1, 0, 2, 3)$	11	138016
$(4, 1, 0, 2, 5)$	17	1502168
$(4, 1, 0, 3, 4)$	15	2761710
$(4, 1, 0, 3, 5)$	18	4034296
$(4, 2, 1, 1, 2)$	12	1730232
$(4, 2, 1, 1, 3)$	17	863094
$(4, 2, 1, 2, 3)$	19	482794
$(5, 1, 0, 1, 1)$	5	0
$(5, 1, 0, 1, 2)$	9	920206
$(5, 1, 0, 1, 3)$	13	288290
$(5, 1, 0, 1, 4)$	17	385246
$(5, 1, 0, 2, 3)$	14	352016
$(5, 1, 0, 3, 4)$	19	1233812
$(5, 2, 1, 1, 2)$	16	353776
$(6, 1, 0, 1, 1)$	6	342552
$(6, 1, 0, 1, 2)$	11	106052
$(6, 1, 0, 1, 3)$	16	1722952
$(6, 1, 0, 2, 3)$	17	541212
$(7, 1, 0, 1, 1)$	7	0
$(7, 1, 0, 1, 2)$	13	176638
$(7, 1, 0, 1, 3)$	19	800910
$(8, 1, 0, 1, 1)$	8	330896
$(8, 1, 0, 1, 2)$	15	903280
$(9, 1, 0, 1, 1)$	9	0
$(9, 1, 0, 1, 2)$	17	315042
$(10, 1, 0, 1, 1)$	10	328720
$(10, 1, 0, 1, 2)$	19	378804
$(11, 1, 0, 1, 1)$	11	0
$(12, 1, 0, 1, 1)$	12	256080
$(13, 1, 0, 1, 1)$	13	61760
$(14, 1, 0, 1, 1)$	14	301056
$(14, 1, 0, 1, 2)$	27	559180
$(15, 1, 0, 1, 1)$	15	216240
$(16, 1, 0, 1, 1)$	16	317264
$(17, 1, 0, 1, 1)$	17	157920
$(18, 1, 0, 1, 1)$	18	0

Each entry $n{:}T$ of Table 11 is an exact integer, computed with integer arithmetic from Definition-Lemma 9.7 and Lemma 9.8: the integers y_G follow from the recurrence of Definition-Lemma 9.7(1), the integers $N_m(y_G)$ and z_m are ceilings and integer parts of quotients of integers, and T is the value of $2rC_x$ from Lemma 9.8(2) minus $1000r(r-1)$. Here $2rC_x$ is a polynomial with integer coefficients in r, h, ℓ_0, and the integers k_G, y_G, $N_m(y_G)$, and z_m, by Lemma 9.8(2).

TABLE 10. Identities for **Step 4** of the proof of Proposition 9.9, with the notation of Definition-Lemma 9.7 and Lemma 9.8, where $\alpha \leq 9$: the 450 headers (e, r_1, q_1, t, v) are the tuples with the conditions of **Step 4**, the rows of a header cover all integers h with $h \geq \max\{0, \lceil (r_1 - v)/t \rceil - 1\}$ and $r \geq 20$, a row $h_-, h_+, \tau_0, \tau_1, \tau_2$ means that for any integer z with $0 \leq z \leq h_+ - h_-$, $T(h_- + z) = \tau_0 + \tau_1 z + \tau_2 z(z-1)$, where $T(h) = 2r(C_x - 500(r-1))$, and all coefficients in the table are non-negative.

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (3, 1, 0, 1, 0)$				
9	∞	9024162	3098134	227280
$(e, r_1, q_1, t, v) = (3, 1, 0, 2, 1)$				
3	∞	11068360	4916160	446560
$(e, r_1, q_1, t, v) = (3, 2, 1, 1, 0)$				
5	∞	474600	1057874	337920
$(e, r_1, q_1, t, v) = (3, 2, 1, 2, 1)$				
2	∞	4909620	4280626	657840
$(e, r_1, q_1, t, v) = (3, 2, 1, 3, 1)$				
1	∞	8549200	6727320	959760
$(e, r_1, q_1, t, v) = (3, 2, 1, 3, 2)$				
0	∞	5138364	5450916	959760
$(e, r_1, q_1, t, v) = (3, 3, 1, 1, 0)$				
3	3	401352	–	1618600
4	4	1039752	–	1101000
5	5	744624	–	625920
6	∞	1148664	1510440	553200
$(e, r_1, q_1, t, v) = (3, 3, 1, 2, 1)$				
0	0	535696	–	2152000
1	∞	646306	2223410	1056400
$(e, r_1, q_1, t, v) = (3, 3, 1, 3, 1)$				
0	∞	1094210	3935750	1509600
$(e, r_1, q_1, t, v) = (3, 3, 1, 3, 2)$				
0	∞	2083036	4947610	1509600
$(e, r_1, q_1, t, v) = (3, 3, 1, 4, 1)$				
0	∞	3803984	7351090	1912800
$(e, r_1, q_1, t, v) = (3, 3, 1, 4, 3)$				
0	∞	7078362	9296670	1912800
$(e, r_1, q_1, t, v) = (3, 3, 1, 5, 1)$				
0	∞	7246630	10434950	2266000
$(e, r_1, q_1, t, v) = (3, 3, 1, 5, 2)$				
0	∞	8998360	11355000	2266000
$(e, r_1, q_1, t, v) = (3, 3, 1, 5, 3)$				
0	∞	10926820	12270500	2266000
$(e, r_1, q_1, t, v) = (3, 3, 1, 5, 4)$				
0	∞	13051120	13190550	2266000
$(e, r_1, q_1, t, v) = (3, 3, 2, 1, 0)$				
4	4	186276	–	632224
5	5	473482	–	504736
6	∞	1319296	1738934	446560
$(e, r_1, q_1, t, v) = (3, 3, 2, 2, 1)$				
1	∞	1195244	2815892	861120
$(e, r_1, q_1, t, v) = (3, 3, 2, 3, 1)$				
0	∞	2346740	4692346	1243680
$(e, r_1, q_1, t, v) = (3, 3, 2, 3, 2)$				
0	∞	3645038	5525834	1243680
$(e, r_1, q_1, t, v) = (3, 3, 2, 4, 1)$				
0	∞	6293320	7987200	1594240
$(e, r_1, q_1, t, v) = (3, 3, 2, 4, 3)$				
0	∞	9953920	9607664	1594240

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (3, 4, 1, 1, 0)$				
3	3	1657296	–	2252040
4	4	1434652	–	1527400
5	5	1362184	–	1085392
6	6	1771284	–	862288
7	∞	3032320	2781996	760480
$(e, r_1, q_1, t, v) = (3, 4, 1, 2, 1)$				
1	1	1055968	–	1626576
2	∞	2517582	4307534	1422960
$(e, r_1, q_1, t, v) = (3, 4, 1, 3, 1)$				
0	∞	842520	3033310	1987440
$(e, r_1, q_1, t, v) = (3, 4, 1, 3, 2)$				
0	∞	1429078	4365914	1987440
$(e, r_1, q_1, t, v) = (3, 4, 1, 4, 1)$				
0	∞	2201722	6765206	2453920
$(e, r_1, q_1, t, v) = (3, 4, 1, 4, 3)$				
0	∞	5077380	9265018	2453920
$(e, r_1, q_1, t, v) = (3, 4, 1, 5, 1)$				
0	∞	4529004	9921030	2822400
$(e, r_1, q_1, t, v) = (3, 4, 1, 5, 2)$				
0	∞	6099300	11069100	2822400
$(e, r_1, q_1, t, v) = (3, 4, 1, 5, 3)$				
0	∞	7889200	12210800	2822400
$(e, r_1, q_1, t, v) = (3, 4, 1, 5, 4)$				
0	∞	9925094	13358870	2822400
$(e, r_1, q_1, t, v) = (3, 4, 1, 6, 1)$				
0	∞	6939406	12416614	3092880
$(e, r_1, q_1, t, v) = (3, 4, 1, 6, 5)$				
0	∞	15257648	16843106	3092880
$(e, r_1, q_1, t, v) = (3, 4, 1, 7, 1)$				
0	∞	9457504	14447454	3265360
$(e, r_1, q_1, t, v) = (3, 4, 1, 7, 2)$				
0	∞	11187834	15416086	3265360
$(e, r_1, q_1, t, v) = (3, 4, 1, 7, 3)$				
0	∞	13043800	16375800	3265360
$(e, r_1, q_1, t, v) = (3, 4, 1, 7, 4)$				
0	∞	15021580	17326596	3265360
$(e, r_1, q_1, t, v) = (3, 4, 1, 7, 5)$				
0	∞	17158120	18286310	3265360
$(e, r_1, q_1, t, v) = (3, 4, 1, 7, 6)$				
0	∞	19457242	19254942	3265360
$(e, r_1, q_1, t, v) = (3, 4, 3, 1, 0)$				
3	3	1355544	–	1618600
4	4	812058	–	785280
5	5	755484	–	625920
6	∞	1383534	1734450	553200
$(e, r_1, q_1, t, v) = (3, 4, 3, 2, 1)$				
1	∞	939982	2651430	1056400
$(e, r_1, q_1, t, v) = (3, 4, 3, 3, 1)$				
0	∞	1657296	4547780	1509600
$(e, r_1, q_1, t, v) = (3, 4, 3, 3, 2)$				
0	∞	2851224	5559640	1509600
$(e, r_1, q_1, t, v) = (3, 4, 3, 4, 1)$				
0	∞	4990540	8127130	1912800

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (3, 4, 3, 4, 3)$				
0	∞	8659494	10072710	1912800
$(e, r_1, q_1, t, v) = (3, 4, 3, 5, 1)$				
0	∞	8998360	11355000	2266000
$(e, r_1, q_1, t, v) = (3, 4, 3, 5, 2)$				
0	∞	10936830	12275050	2266000
$(e, r_1, q_1, t, v) = (3, 4, 3, 5, 3)$				
0	∞	13051120	13190550	2266000
$(e, r_1, q_1, t, v) = (3, 4, 3, 5, 4)$				
0	∞	15362160	14110600	2266000
$(e, r_1, q_1, t, v) = (3, 5, 1, 1, 0)$				
4	4	4266720	–	1945800
5	5	3709004	–	1377504
6	6	3739000	–	1090656
7	∞	4830420	3010940	959760
$(e, r_1, q_1, t, v) = (3, 5, 1, 2, 1)$				
1	1	2897730	–	2593008
2	∞	3789124	4406434	1757520
$(e, r_1, q_1, t, v) = (3, 5, 1, 3, 1)$				
1	∞	4619956	7204794	2393280
$(e, r_1, q_1, t, v) = (3, 5, 1, 3, 2)$				
0	∞	2497800	4023582	2393280
$(e, r_1, q_1, t, v) = (3, 5, 1, 4, 1)$				
0	∞	2608720	6434474	2867040
$(e, r_1, q_1, t, v) = (3, 5, 1, 4, 3)$				
0	∞	5245234	9360518	2867040
$(e, r_1, q_1, t, v) = (3, 5, 1, 5, 1)$				
0	∞	4183006	9566050	3178800
$(e, r_1, q_1, t, v) = (3, 5, 1, 5, 2)$				
0	∞	5635656	10862140	3178800
$(e, r_1, q_1, t, v) = (3, 5, 1, 5, 3)$				
0	∞	7334784	12150040	3178800
$(e, r_1, q_1, t, v) = (3, 5, 1, 5, 4)$				
0	∞	9314060	13446130	3178800
$(e, r_1, q_1, t, v) = (3, 5, 1, 6, 1)$				
0	∞	5788492	11704746	3328560
$(e, r_1, q_1, t, v) = (3, 5, 1, 6, 5)$				
0	∞	13781550	16531950	3328560
$(e, r_1, q_1, t, v) = (3, 5, 1, 7, 1)$				
0	∞	7452786	13101914	3316320
$(e, r_1, q_1, t, v) = (3, 5, 1, 7, 2)$				
0	∞	9002856	14095298	3316320
$(e, r_1, q_1, t, v) = (3, 5, 1, 7, 3)$				
0	∞	10678640	15077216	3316320
$(e, r_1, q_1, t, v) = (3, 5, 1, 7, 4)$				
0	∞	12475224	16047668	3316320
$(e, r_1, q_1, t, v) = (3, 5, 1, 7, 5)$				
0	∞	14439746	17029586	3316320
$(e, r_1, q_1, t, v) = (3, 5, 1, 7, 6)$				
0	∞	16577120	18022970	3316320
$(e, r_1, q_1, t, v) = (3, 5, 1, 8, 1)$				
0	∞	8044024	13210858	3142080
$(e, r_1, q_1, t, v) = (3, 5, 1, 8, 3)$				
0	∞	10780742	14793886	3142080

Table 10, continued.

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (3, 5, 1, 8, 5)$				
0	∞	14071644	16448194	3142080
$(e, r_1, q_1, t, v) = (3, 5, 1, 8, 7)$				
0	∞	17623690	18031222	3142080
$(e, r_1, q_1, t, v) = (3, 5, 1, 9, 1)$				
0	∞	6561550	11488554	2805840
$(e, r_1, q_1, t, v) = (3, 5, 1, 9, 2)$				
0	∞	8410824	12590136	2805840
$(e, r_1, q_1, t, v) = (3, 5, 1, 9, 4)$				
0	∞	11269720	14110200	2805840
$(e, r_1, q_1, t, v) = (3, 5, 1, 9, 5)$				
0	∞	12622644	14763204	2805840
$(e, r_1, q_1, t, v) = (3, 5, 1, 9, 7)$				
0	∞	16055452	16283268	2805840
$(e, r_1, q_1, t, v) = (3, 5, 1, 9, 8)$				
0	∞	18756180	17384850	2805840
$(e, r_1, q_1, t, v) = (3, 5, 2, 1, 0)$				
4	4	2530530	–	1737600
5	5	2338404	–	1232448
6	6	2666982	–	977472
7	∞	3946968	3002226	861120
$(e, r_1, q_1, t, v) = (3, 5, 2, 2, 1)$				
1	1	1836720	–	1826944
2	∞	3200740	4552500	1594240
$(e, r_1, q_1, t, v) = (3, 5, 2, 3, 1)$				
1	∞	4412124	7381766	2199360
$(e, r_1, q_1, t, v) = (3, 5, 2, 3, 2)$				
0	∞	1954700	4458022	2199360
$(e, r_1, q_1, t, v) = (3, 5, 2, 4, 1)$				
0	∞	2497800	6926872	2676480
$(e, r_1, q_1, t, v) = (3, 5, 2, 4, 3)$				
0	∞	5416680	9655800	2676480
$(e, r_1, q_1, t, v) = (3, 5, 2, 5, 1)$				
0	∞	4601058	10132330	3025600
$(e, r_1, q_1, t, v) = (3, 5, 2, 5, 2)$				
0	∞	6187654	11364410	3025600
$(e, r_1, q_1, t, v) = (3, 5, 2, 5, 3)$				
0	∞	8008836	12589210	3025600
$(e, r_1, q_1, t, v) = (3, 5, 2, 5, 4)$				
0	∞	10095544	13821290	3025600
$(e, r_1, q_1, t, v) = (3, 5, 2, 6, 1)$				
0	∞	6722316	12503228	3246720
$(e, r_1, q_1, t, v) = (3, 5, 2, 6, 5)$				
0	∞	15186616	17178076	3246720
$(e, r_1, q_1, t, v) = (3, 5, 2, 7, 1)$				
0	∞	8901630	14262990	3339840
$(e, r_1, q_1, t, v) = (3, 5, 2, 7, 2)$				
0	∞	10608136	15257998	3339840
$(e, r_1, q_1, t, v) = (3, 5, 2, 7, 3)$				
0	∞	12441680	16242814	3339840
$(e, r_1, q_1, t, v) = (3, 5, 2, 7, 4)$				
0	∞	14397894	17217438	3339840
$(e, r_1, q_1, t, v) = (3, 5, 2, 7, 5)$				
0	∞	16520094	18202254	3339840
$(e, r_1, q_1, t, v) = (3, 5, 2, 7, 6)$				
0	∞	18812648	19197262	3339840
$(e, r_1, q_1, t, v) = (3, 5, 2, 8, 1)$				
0	∞	10099152	14925664	3304960
$(e, r_1, q_1, t, v) = (3, 5, 2, 8, 3)$				
0	∞	13232200	16588800	3304960
$(e, r_1, q_1, t, v) = (3, 5, 2, 8, 5)$				
0	∞	16926256	18315296	3304960
$(e, r_1, q_1, t, v) = (3, 5, 2, 8, 7)$				
0	∞	20912040	19978432	3304960
$(e, r_1, q_1, t, v) = (3, 5, 3, 1, 0)$				
4	4	1048000	–	1527400
5	5	1243050	–	1085392
6	6	1887796	–	862288
7	∞	3369934	3003098	760480
$(e, r_1, q_1, t, v) = (3, 5, 3, 2, 1)$				
1	1	1115100	–	1626576
2	∞	2990918	4721738	1422960
$(e, r_1, q_1, t, v) = (3, 5, 3, 3, 1)$				
1	∞	4668584	7587496	1987440
$(e, r_1, q_1, t, v) = (3, 5, 3, 3, 2)$				
0	∞	1836720	4945220	1987440
$(e, r_1, q_1, t, v) = (3, 5, 3, 4, 1)$				
0	∞	2897730	7481614	2453920
$(e, r_1, q_1, t, v) = (3, 5, 3, 4, 3)$				
0	∞	6138148	9981426	2453920
$(e, r_1, q_1, t, v) = (3, 5, 3, 5, 1)$				
0	∞	5637276	10746540	2822400
$(e, r_1, q_1, t, v) = (3, 5, 3, 5, 2)$				
0	∞	7375404	11894610	2822400
$(e, r_1, q_1, t, v) = (3, 5, 3, 5, 3)$				
0	∞	9332226	13036310	2822400
$(e, r_1, q_1, t, v) = (3, 5, 3, 5, 4)$				
0	∞	11535952	14184380	2822400
$(e, r_1, q_1, t, v) = (3, 5, 3, 6, 1)$				
0	∞	8377622	13323226	3092880
$(e, r_1, q_1, t, v) = (3, 5, 3, 6, 5)$				
0	∞	17343508	17749718	3092880
$(e, r_1, q_1, t, v) = (3, 5, 3, 7, 1)$				
0	∞	11171272	15407168	3265360
$(e, r_1, q_1, t, v) = (3, 5, 3, 7, 2)$				
0	∞	13043800	16375800	3265360
$(e, r_1, q_1, t, v) = (3, 5, 3, 7, 3)$				
0	∞	15040690	17335514	3265360
$(e, r_1, q_1, t, v) = (3, 5, 3, 7, 4)$				
0	∞	17158120	18286310	3265360
$(e, r_1, q_1, t, v) = (3, 5, 3, 7, 5)$				
0	∞	19435584	19246024	3265360
$(e, r_1, q_1, t, v) = (3, 5, 3, 7, 6)$				
0	∞	21876904	20214656	3265360
$(e, r_1, q_1, t, v) = (3, 5, 4, 1, 0)$				
4	4	1687280	–	1315200
5	5	1297246	–	745104
6	∞	1713852	1732286	657840
$(e, r_1, q_1, t, v) = (3, 5, 4, 2, 1)$				
1	∞	1048000	2495608	1243680
$(e, r_1, q_1, t, v) = (3, 5, 4, 3, 1)$				
1	∞	5844826	7925214	1757520
$(e, r_1, q_1, t, v) = (3, 5, 4, 3, 2)$				
0	∞	2530530	5588406	1757520
$(e, r_1, q_1, t, v) = (3, 5, 4, 4, 1)$				
0	∞	4266720	8236340	2199360
$(e, r_1, q_1, t, v) = (3, 5, 4, 4, 3)$				
0	∞	7936668	10475036	2199360
$(e, r_1, q_1, t, v) = (3, 5, 4, 5, 1)$				
0	∞	7855820	11580730	2569200
$(e, r_1, q_1, t, v) = (3, 5, 4, 5, 2)$				
0	∞	9797476	12624790	2569200
$(e, r_1, q_1, t, v) = (3, 5, 4, 5, 3)$				
0	∞	11937934	13663390	2569200
$(e, r_1, q_1, t, v) = (3, 5, 4, 5, 4)$				
0	∞	14302674	14707450	2569200
$(e, r_1, q_1, t, v) = (3, 5, 4, 6, 1)$				
0	∞	11424520	14371200	2867040
$(e, r_1, q_1, t, v) = (3, 5, 4, 6, 5)$				
0	∞	21059976	18453336	2867040
$(e, r_1, q_1, t, v) = (3, 6, 5, 1, 0)$				
5	5	1862896	–	1085392
6	∞	2046450	1704514	760480
$(e, r_1, q_1, t, v) = (3, 6, 5, 2, 1)$				
2	∞	3535620	5138490	1422960
$(e, r_1, q_1, t, v) = (3, 6, 5, 3, 1)$				
1	∞	5557928	8170624	1987440
$(e, r_1, q_1, t, v) = (3, 6, 5, 3, 2)$				
1	∞	7866752	9503228	1987440
$(e, r_1, q_1, t, v) = (3, 6, 5, 4, 1)$				
1	∞	11912122	13110958	2453920
$(e, r_1, q_1, t, v) = (3, 6, 5, 4, 3)$				
0	∞	7316730	10702930	2453920
$(e, r_1, q_1, t, v) = (3, 6, 5, 5, 1)$				
0	∞	6879312	11578420	2822400
$(e, r_1, q_1, t, v) = (3, 6, 5, 5, 2)$				
0	∞	8786546	12726490	2822400
$(e, r_1, q_1, t, v) = (3, 6, 5, 5, 3)$				
0	∞	10911564	13868190	2822400
$(e, r_1, q_1, t, v) = (3, 6, 5, 5, 4)$				
0	∞	13284396	15016260	2822400
$(e, r_1, q_1, t, v) = (3, 6, 5, 6, 1)$				
0	∞	9964100	14237482	3092880
$(e, r_1, q_1, t, v) = (3, 6, 5, 6, 5)$				
0	∞	19582726	18663974	3092880
$(e, r_1, q_1, t, v) = (3, 6, 5, 7, 1)$				
0	∞	13043800	16375800	3265360
$(e, r_1, q_1, t, v) = (3, 6, 5, 7, 2)$				
0	∞	15059800	17344432	3265360
$(e, r_1, q_1, t, v) = (3, 6, 5, 7, 3)$				
0	∞	17198888	18304146	3265360
$(e, r_1, q_1, t, v) = (3, 6, 5, 7, 4)$				
0	∞	19457242	19254942	3265360
$(e, r_1, q_1, t, v) = (3, 6, 5, 7, 5)$				
0	∞	21876904	20214656	3265360
$(e, r_1, q_1, t, v) = (3, 6, 5, 7, 6)$				
0	∞	24461696	21183288	3265360
$(e, r_1, q_1, t, v) = (3, 7, 5, 1, 0)$				
6	6	4051600	–	1090656
7	∞	5599916	3467836	959760
$(e, r_1, q_1, t, v) = (3, 7, 5, 2, 1)$				
2	∞	4724104	5248226	1757520
$(e, r_1, q_1, t, v) = (3, 7, 5, 3, 1)$				
1	∞	6090504	8359482	2393280

Table 10, continued.

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (3, 7, 5, 3, 2)$				
1	∞	8379010	9964830	2393280
$(e, r_1, q_1, t, v) = (3, 7, 5, 4, 1)$				
1	∞	11619182	13564138	2867040
$(e, r_1, q_1, t, v) = (3, 7, 5, 4, 3)$				
0	∞	7136542	10756102	2867040
$(e, r_1, q_1, t, v) = (3, 7, 5, 5, 1)$				
1	∞	17189892	17488130	3178800
$(e, r_1, q_1, t, v) = (3, 7, 5, 5, 2)$				
0	∞	7830368	12426620	3178800
$(e, r_1, q_1, t, v) = (3, 7, 5, 5, 3)$				
0	∞	9846032	13714520	3178800
$(e, r_1, q_1, t, v) = (3, 7, 5, 5, 4)$				
0	∞	12143664	15010610	3178800
$(e, r_1, q_1, t, v) = (3, 7, 5, 6, 1)$				
0	∞	8168160	13366122	3328560
$(e, r_1, q_1, t, v) = (3, 7, 5, 6, 5)$				
0	∞	17355274	18193326	3328560
$(e, r_1, q_1, t, v) = (3, 7, 5, 7, 1)$				
0	∞	10198982	14788186	3316320
$(e, r_1, q_1, t, v) = (3, 7, 5, 7, 2)$				
0	∞	12000140	15781570	3316320
$(e, r_1, q_1, t, v) = (3, 7, 5, 7, 3)$				
0	∞	13924464	16763488	3316320
$(e, r_1, q_1, t, v) = (3, 7, 5, 7, 4)$				
0	∞	15967040	17733940	3316320
$(e, r_1, q_1, t, v) = (3, 7, 5, 7, 5)$				
0	∞	18180102	18715858	3316320
$(e, r_1, q_1, t, v) = (3, 7, 5, 7, 6)$				
0	∞	20568564	19709242	3316320
$(e, r_1, q_1, t, v) = (3, 7, 5, 8, 1)$				
0	∞	10898476	14850026	3142080
$(e, r_1, q_1, t, v) = (3, 7, 5, 8, 3)$				
0	∞	14047650	16433054	3142080
$(e, r_1, q_1, t, v) = (3, 7, 5, 8, 5)$				
0	∞	17766848	18087362	3142080
$(e, r_1, q_1, t, v) = (3, 7, 5, 8, 7)$				
0	∞	21731350	19670390	3142080
$(e, r_1, q_1, t, v) = (3, 7, 5, 9, 1)$				
0	∞	9145314	13008618	2805840
$(e, r_1, q_1, t, v) = (3, 7, 5, 9, 2)$				
0	∞	11269720	14110200	2805840
$(e, r_1, q_1, t, v) = (3, 7, 5, 9, 4)$				
0	∞	14527080	15630264	2805840
$(e, r_1, q_1, t, v) = (3, 7, 5, 9, 5)$				
0	∞	16055452	16283268	2805840
$(e, r_1, q_1, t, v) = (3, 7, 5, 9, 7)$				
0	∞	19886724	17803332	2805840
$(e, r_1, q_1, t, v) = (3, 7, 5, 9, 8)$				
0	∞	22862584	18904914	2805840
$(e, r_1, q_1, t, v) = (3, 7, 6, 1, 0)$				
6	6	2836008	–	977472
7	∞	4547886	3434118	861120
$(e, r_1, q_1, t, v) = (3, 7, 6, 2, 1)$				
2	∞	4051600	5352284	1594240
$(e, r_1, q_1, t, v) = (3, 7, 6, 3, 1)$				
1	∞	5858006	8485442	2199360
$(e, r_1, q_1, t, v) = (3, 7, 6, 3, 2)$				
1	∞	8228680	9960418	2199360
$(e, r_1, q_1, t, v) = (3, 7, 6, 4, 1)$				
1	∞	12009440	13623400	2676480
$(e, r_1, q_1, t, v) = (3, 7, 6, 4, 3)$				
0	∞	7342776	10999368	2676480
$(e, r_1, q_1, t, v) = (3, 7, 6, 5, 1)$				
1	∞	18232450	17702990	3025600
$(e, r_1, q_1, t, v) = (3, 7, 6, 5, 2)$				
0	∞	8476608	12883870	3025600
$(e, r_1, q_1, t, v) = (3, 7, 6, 5, 3)$				
0	∞	10605322	14108670	3025600
$(e, r_1, q_1, t, v) = (3, 7, 6, 5, 4)$				
0	∞	13001382	15340750	3025600
$(e, r_1, q_1, t, v) = (3, 7, 6, 6, 1)$				
0	∞	9255680	14134580	3246720
$(e, r_1, q_1, t, v) = (3, 7, 6, 6, 5)$				
0	∞	18894020	18809428	3246720
$(e, r_1, q_1, t, v) = (3, 7, 6, 7, 1)$				
0	∞	11859972	15942234	3339840
$(e, r_1, q_1, t, v) = (3, 7, 6, 7, 2)$				
0	∞	13816562	16937242	3339840
$(e, r_1, q_1, t, v) = (3, 7, 6, 7, 3)$				
0	∞	15897642	17922058	3339840
$(e, r_1, q_1, t, v) = (3, 7, 6, 7, 4)$				
0	∞	18098844	18896682	3339840
$(e, r_1, q_1, t, v) = (3, 7, 6, 7, 5)$				
0	∞	20468580	19881498	3339840
$(e, r_1, q_1, t, v) = (3, 7, 6, 7, 6)$				
0	∞	23011218	20876506	3339840
$(e, r_1, q_1, t, v) = (3, 7, 6, 8, 1)$				
0	∞	13232200	16588800	3304960
$(e, r_1, q_1, t, v) = (3, 7, 6, 8, 3)$				
0	∞	16783696	18251936	3304960
$(e, r_1, q_1, t, v) = (3, 7, 6, 8, 5)$				
0	∞	20912040	19978432	3304960
$(e, r_1, q_1, t, v) = (3, 7, 6, 8, 7)$				
0	∞	25316272	21641568	3304960
$(e, r_1, q_1, t, v) = (3, 8, 7, 1, 0)$				
7	∞	6362600	3734234	959760
$(e, r_1, q_1, t, v) = (3, 8, 7, 2, 1)$				
3	∞	11344938	9260062	1757520
$(e, r_1, q_1, t, v) = (3, 8, 7, 3, 1)$				
2	∞	16347112	13837236	2393280
$(e, r_1, q_1, t, v) = (3, 8, 7, 3, 2)$				
1	∞	9816432	10656024	2393280
$(e, r_1, q_1, t, v) = (3, 8, 7, 4, 1)$				
1	∞	13510070	14413730	2867040
$(e, r_1, q_1, t, v) = (3, 8, 7, 4, 3)$				
1	∞	20214884	17339774	2867040
$(e, r_1, q_1, t, v) = (3, 8, 7, 5, 1)$				
1	∞	19639440	18460120	3178800
$(e, r_1, q_1, t, v) = (3, 8, 7, 5, 2)$				
1	∞	22903664	19756210	3178800
$(e, r_1, q_1, t, v) = (3, 8, 7, 5, 3)$				
0	∞	11716936	14686510	3178800
$(e, r_1, q_1, t, v) = (3, 8, 7, 5, 4)$				
0	∞	14211696	15982600	3178800
$(e, r_1, q_1, t, v) = (3, 8, 7, 6, 1)$				
1	∞	24404170	21081630	3328560
$(e, r_1, q_1, t, v) = (3, 8, 7, 6, 5)$				
0	∞	19915602	19251714	3328560
$(e, r_1, q_1, t, v) = (3, 8, 7, 7, 1)$				
0	∞	12270032	15896972	3316320
$(e, r_1, q_1, t, v) = (3, 8, 7, 7, 2)$				
0	∞	14234684	16890356	3316320
$(e, r_1, q_1, t, v) = (3, 8, 7, 7, 3)$				
0	∞	16321228	17872274	3316320
$(e, r_1, q_1, t, v) = (3, 8, 7, 7, 4)$				
0	∞	18524750	18842726	3316320
$(e, r_1, q_1, t, v) = (3, 8, 7, 7, 5)$				
0	∞	20900032	19824644	3316320
$(e, r_1, q_1, t, v) = (3, 8, 7, 7, 6)$				
0	∞	23451988	20818028	3316320
$(e, r_1, q_1, t, v) = (3, 8, 7, 8, 1)$				
0	∞	13093940	15973210	3142080
$(e, r_1, q_1, t, v) = (3, 8, 7, 8, 3)$				
0	∞	16525242	17556238	3142080
$(e, r_1, q_1, t, v) = (3, 8, 7, 8, 5)$				
0	∞	20534488	19210546	3142080
$(e, r_1, q_1, t, v) = (3, 8, 7, 8, 7)$				
0	∞	24781118	20793574	3142080
$(e, r_1, q_1, t, v) = (3, 8, 7, 9, 1)$				
0	∞	11269720	14110200	2805840
$(e, r_1, q_1, t, v) = (3, 8, 7, 9, 2)$				
0	∞	13569642	15211782	2805840
$(e, r_1, q_1, t, v) = (3, 8, 7, 9, 4)$				
0	∞	17102134	16731846	2805840
$(e, r_1, q_1, t, v) = (3, 8, 7, 9, 5)$				
0	∞	18756180	17384850	2805840
$(e, r_1, q_1, t, v) = (3, 8, 7, 9, 7)$				
0	∞	22862584	18904914	2805840
$(e, r_1, q_1, t, v) = (3, 8, 7, 9, 8)$				
0	∞	26013960	20006496	2805840
$(e, r_1, q_1, t, v) = (4, 1, 0, 1, 0)$				
6	∞	1096748	1507342	337920
$(e, r_1, q_1, t, v) = (4, 1, 0, 2, 1)$				
2	∞	3766746	3839882	657840
$(e, r_1, q_1, t, v) = (4, 1, 0, 3, 1)$				
1	∞	6724584	6084204	959760
$(e, r_1, q_1, t, v) = (4, 1, 0, 3, 2)$				
1	∞	8549200	6727320	959760
$(e, r_1, q_1, t, v) = (4, 2, 1, 1, 0)$				
3	3	310784	–	1618600
4	4	457164	–	1101000
5	5	784936	–	785280
6	∞	965876	1287340	553200
$(e, r_1, q_1, t, v) = (4, 2, 1, 2, 1)$				
1	∞	444604	1797210	1056400
$(e, r_1, q_1, t, v) = (4, 2, 1, 3, 1)$				
0	∞	659900	3326450	1509600
$(e, r_1, q_1, t, v) = (4, 2, 1, 3, 2)$				
0	∞	1444534	4338310	1509600
$(e, r_1, q_1, t, v) = (4, 2, 1, 4, 1)$				
0	∞	2780826	6578690	1912800
$(e, r_1, q_1, t, v) = (4, 2, 1, 4, 3)$				
0	∞	5662448	8524270	1912800
$(e, r_1, q_1, t, v) = (4, 2, 1, 5, 1)$				
0	∞	5688920	9519450	2266000

Table 10, continued.

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (4, 2, 1, 5, 2)$				
0	∞	7254820	10439500	2266000
$(e, r_1, q_1, t, v) = (4, 2, 1, 5, 3)$				
0	∞	8998360	11355000	2266000
$(e, r_1, q_1, t, v) = (4, 2, 1, 5, 4)$				
0	∞	10936830	12275050	2266000
$(e, r_1, q_1, t, v) = (4, 3, 1, 1, 0)$				
2	2	1607292	–	6086304
3	3	4392570	–	3766464
4	4	3412944	–	1737600
5	5	2688174	–	1232448
6	6	2547852	–	977472
7	∞	3388026	2562414	861120
$(e, r_1, q_1, t, v) = (4, 3, 1, 2, 1)$				
0	0	3188320	–	7404928
1	1	1940716	–	2336896
2	∞	2489112	3736876	1594240
$(e, r_1, q_1, t, v) = (4, 3, 1, 3, 1)$				
0	0	1323200	–	2548416
1	∞	3178810	6254330	2199360
$(e, r_1, q_1, t, v) = (4, 3, 1, 3, 2)$				
0	∞	1470826	3330586	2199360
$(e, r_1, q_1, t, v) = (4, 3, 1, 4, 1)$				
0	∞	1542344	5551624	2676480
$(e, r_1, q_1, t, v) = (4, 3, 1, 4, 3)$				
0	∞	3760488	8280552	2676480
$(e, r_1, q_1, t, v) = (4, 3, 1, 5, 1)$				
0	∞	2940616	8573270	3025600
$(e, r_1, q_1, t, v) = (4, 3, 1, 5, 2)$				
0	∞	4209940	9805350	3025600
$(e, r_1, q_1, t, v) = (4, 3, 1, 5, 3)$				
0	∞	5715670	11030150	3025600
$(e, r_1, q_1, t, v) = (4, 3, 1, 5, 4)$				
0	∞	7485106	12262230	3025600
$(e, r_1, q_1, t, v) = (4, 3, 1, 6, 1)$				
0	∞	4525528	10824356	3246720
$(e, r_1, q_1, t, v) = (4, 3, 1, 6, 5)$				
0	∞	11784108	15499204	3246720
$(e, r_1, q_1, t, v) = (4, 3, 1, 7, 1)$				
0	∞	6281280	12528306	3339840
$(e, r_1, q_1, t, v) = (4, 3, 1, 7, 2)$				
0	∞	7729782	13523314	3339840
$(e, r_1, q_1, t, v) = (4, 3, 1, 7, 3)$				
0	∞	9307870	14508130	3339840
$(e, r_1, q_1, t, v) = (4, 3, 1, 7, 4)$				
0	∞	11011176	15482754	3339840
$(e, r_1, q_1, t, v) = (4, 3, 1, 7, 5)$				
0	∞	12877920	16467570	3339840
$(e, r_1, q_1, t, v) = (4, 3, 1, 7, 6)$				
0	∞	14912470	17462578	3339840
$(e, r_1, q_1, t, v) = (4, 3, 1, 8, 1)$				
0	∞	7289512	13199168	3304960
$(e, r_1, q_1, t, v) = (4, 3, 1, 8, 3)$				
0	∞	9988272	14862304	3304960
$(e, r_1, q_1, t, v) = (4, 3, 1, 8, 5)$				
0	∞	13232200	16588800	3304960
$(e, r_1, q_1, t, v) = (4, 3, 1, 8, 7)$				
0	∞	16783696	18251936	3304960

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (4, 3, 2, 1, 0)$				
2	2	3534736	–	5332516
3	3	2412384	–	2252040
4	4	1860352	–	1527400
5	5	1521640	–	1085392
6	6	1696368	–	862288
7	∞	2737576	2562168	760480
$(e, r_1, q_1, t, v) = (4, 3, 2, 2, 1)$				
0	0	1607292	–	4406080
1	1	1064562	–	1626576
2	∞	2114520	3895878	1422960
$(e, r_1, q_1, t, v) = (4, 3, 2, 3, 1)$				
0	∞	720202	2457826	1987440
$(e, r_1, q_1, t, v) = (4, 3, 2, 3, 2)$				
0	∞	1113840	3790430	1987440
$(e, r_1, q_1, t, v) = (4, 3, 2, 4, 1)$				
0	∞	1618796	6053894	2453920
$(e, r_1, q_1, t, v) = (4, 3, 2, 4, 3)$				
0	∞	4132242	8553706	2453920
$(e, r_1, q_1, t, v) = (4, 3, 2, 5, 1)$				
0	∞	3551766	9101890	2822400
$(e, r_1, q_1, t, v) = (4, 3, 2, 5, 2)$				
0	∞	4955504	10249960	2822400
$(e, r_1, q_1, t, v) = (4, 3, 2, 5, 3)$				
0	∞	6579756	11391660	2822400
$(e, r_1, q_1, t, v) = (4, 3, 2, 5, 4)$				
0	∞	8449092	12539730	2822400
$(e, r_1, q_1, t, v) = (4, 3, 2, 6, 1)$				
0	∞	5646176	11517646	3092880
$(e, r_1, q_1, t, v) = (4, 3, 2, 6, 5)$				
0	∞	13321870	15944138	3092880
$(e, r_1, q_1, t, v) = (4, 3, 2, 7, 1)$				
0	∞	7898674	13496658	3265360
$(e, r_1, q_1, t, v) = (4, 3, 2, 7, 2)$				
0	∞	9488080	14465290	3265360
$(e, r_1, q_1, t, v) = (4, 3, 2, 7, 3)$				
0	∞	11204396	15425004	3265360
$(e, r_1, q_1, t, v) = (4, 3, 2, 7, 4)$				
0	∞	13043800	16375800	3265360
$(e, r_1, q_1, t, v) = (4, 3, 2, 7, 5)$				
0	∞	15040690	17335514	3265360
$(e, r_1, q_1, t, v) = (4, 3, 2, 7, 6)$				
0	∞	17198888	18304146	3265360
$(e, r_1, q_1, t, v) = (4, 4, 3, 1, 0)$				
3	3	5783560	–	4228272
4	4	4733624	–	1945800
5	5	3912936	–	1377504
6	6	3711832	–	1090656
7	∞	4586696	2794384	959760
$(e, r_1, q_1, t, v) = (4, 4, 3, 2, 1)$				
1	1	2996372	–	2593008
2	∞	3490654	4009322	1757520
$(e, r_1, q_1, t, v) = (4, 4, 3, 3, 1)$				
0	0	2211024	–	3646512
1	∞	4087590	6663126	2393280
$(e, r_1, q_1, t, v) = (4, 4, 3, 3, 2)$				
0	∞	2325454	3481914	2393280
$(e, r_1, q_1, t, v) = (4, 4, 3, 4, 1)$				
0	∞	2225530	5784250	2867040

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (4, 4, 3, 4, 3)$				
0	∞	4530376	8710294	2867040
$(e, r_1, q_1, t, v) = (4, 4, 3, 5, 1)$				
0	∞	3485620	8843270	3178800
$(e, r_1, q_1, t, v) = (4, 4, 3, 5, 2)$				
0	∞	4790984	10139360	3178800
$(e, r_1, q_1, t, v) = (4, 4, 3, 5, 3)$				
0	∞	6343736	11427260	3178800
$(e, r_1, q_1, t, v) = (4, 4, 3, 5, 4)$				
0	∞	8175726	12723350	3178800
$(e, r_1, q_1, t, v) = (4, 4, 3, 6, 1)$				
0	∞	4867230	10945410	3328560
$(e, r_1, q_1, t, v) = (4, 4, 3, 6, 5)$				
0	∞	12310828	15772614	3328560
$(e, r_1, q_1, t, v) = (4, 4, 3, 7, 1)$				
0	∞	6370040	12342022	3316320
$(e, r_1, q_1, t, v) = (4, 4, 3, 7, 2)$				
0	∞	7806458	13335406	3316320
$(e, r_1, q_1, t, v) = (4, 4, 3, 7, 3)$				
0	∞	9369864	14317324	3316320
$(e, r_1, q_1, t, v) = (4, 4, 3, 7, 4)$				
0	∞	11055344	15287776	3316320
$(e, r_1, q_1, t, v) = (4, 4, 3, 7, 5)$				
0	∞	12907488	16269694	3316320
$(e, r_1, q_1, t, v) = (4, 4, 3, 7, 6)$				
0	∞	14931210	17263078	3316320
$(e, r_1, q_1, t, v) = (4, 4, 3, 8, 1)$				
0	∞	6922930	12486410	3142080
$(e, r_1, q_1, t, v) = (4, 4, 3, 8, 3)$				
0	∞	9477204	14069438	3142080
$(e, r_1, q_1, t, v) = (4, 4, 3, 8, 5)$				
0	∞	12577742	15723746	3142080
$(e, r_1, q_1, t, v) = (4, 4, 3, 8, 7)$				
0	∞	15947344	17306774	3142080
$(e, r_1, q_1, t, v) = (4, 4, 3, 9, 1)$				
0	∞	5585580	10835550	2805840
$(e, r_1, q_1, t, v) = (4, 4, 3, 9, 2)$				
0	∞	7309180	11937132	2805840
$(e, r_1, q_1, t, v) = (4, 4, 3, 9, 4)$				
0	∞	9992628	13457196	2805840
$(e, r_1, q_1, t, v) = (4, 4, 3, 9, 5)$				
0	∞	11269720	14110200	2805840
$(e, r_1, q_1, t, v) = (4, 4, 3, 9, 7)$				
0	∞	14527080	15630264	2805840
$(e, r_1, q_1, t, v) = (4, 4, 3, 9, 8)$				
0	∞	17102134	16731846	2805840
$(e, r_1, q_1, t, v) = (5, 1, 0, 1, 0)$				
4	4	539518	–	884800
5	5	344520	–	504736
6	∞	737218	1285818	446560
$(e, r_1, q_1, t, v) = (5, 1, 0, 2, 1)$				
1	∞	397080	1941660	861120
$(e, r_1, q_1, t, v) = (5, 1, 0, 3, 1)$				
1	∞	4331184	5916358	1243680
$(e, r_1, q_1, t, v) = (5, 1, 0, 3, 2)$				
0	∞	1777184	4262486	1243680
$(e, r_1, q_1, t, v) = (5, 1, 0, 4, 1)$				
0	∞	3456064	6366736	1594240

Table 10, continued.

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (5, 1, 0, 4, 3)$				
0	∞	6293320	7987200	1594240
$(e, r_1, q_1, t, v) = (5, 2, 1, 1, 0)$				
2	2	963778	–	5332516
3	3	3193780	–	3302656
4	4	2311086	–	1527400
5	5	1704856	–	1085392
6	6	1643938	–	862288
7	∞	2464044	2341066	760480
$(e, r_1, q_1, t, v) = (5, 2, 1, 2, 1)$				
0	0	2175360	–	6507312
1	1	1125594	–	2072784
2	∞	1761348	3481674	1422960
$(e, r_1, q_1, t, v) = (5, 2, 1, 3, 1)$				
0	0	675000	–	2292864
1	∞	2553520	5853400	1987440
$(e, r_1, q_1, t, v) = (5, 2, 1, 3, 2)$				
0	∞	874444	3211124	1987440
$(e, r_1, q_1, t, v) = (5, 2, 1, 4, 1)$				
0	∞	1131116	5337486	2453920
$(e, r_1, q_1, t, v) = (5, 2, 1, 4, 3)$				
0	∞	3279802	7837298	2453920
$(e, r_1, q_1, t, v) = (5, 2, 1, 5, 1)$				
0	∞	2683904	8276380	2822400
$(e, r_1, q_1, t, v) = (5, 2, 1, 5, 2)$				
0	∞	3919810	9424450	2822400
$(e, r_1, q_1, t, v) = (5, 2, 1, 5, 3)$				
0	∞	5377140	10566150	2822400
$(e, r_1, q_1, t, v) = (5, 2, 1, 5, 4)$				
0	∞	7078644	11714220	2822400
$(e, r_1, q_1, t, v) = (5, 2, 1, 6, 1)$				
0	∞	4472452	10611034	3092880
$(e, r_1, q_1, t, v) = (5, 2, 1, 6, 5)$				
0	∞	11500502	15037526	3092880
$(e, r_1, q_1, t, v) = (5, 2, 1, 7, 1)$				
0	∞	6465480	12536944	3265360
$(e, r_1, q_1, t, v) = (5, 2, 1, 7, 2)$				
0	∞	7912688	13505576	3265360
$(e, r_1, q_1, t, v) = (5, 2, 1, 7, 3)$				
0	∞	9488080	14465290	3265360
$(e, r_1, q_1, t, v) = (5, 2, 1, 7, 4)$				
0	∞	11187834	15416086	3265360
$(e, r_1, q_1, t, v) = (5, 2, 1, 7, 5)$				
0	∞	13043800	16375800	3265360
$(e, r_1, q_1, t, v) = (5, 2, 1, 7, 6)$				
0	∞	15059800	17344432	3265360
$(e, r_1, q_1, t, v) = (6, 1, 0, 1, 0)$				
3	3	689010	–	1618600
4	4	429100	–	1101000
5	5	413726	–	785280
6	6	821520	–	625920
7	∞	1884850	2169730	553200
$(e, r_1, q_1, t, v) = (6, 1, 0, 2, 1)$				
1	1	323956	–	1201840
2	∞	1693146	3481990	1056400
$(e, r_1, q_1, t, v) = (6, 1, 0, 3, 1)$				
0	∞	344356	2714420	1509600
$(e, r_1, q_1, t, v) = (6, 1, 0, 3, 2)$				
0	∞	923888	3726280	1509600
$(e, r_1, q_1, t, v) = (6, 1, 0, 4, 1)$				
0	∞	1908326	5802650	1912800
$(e, r_1, q_1, t, v) = (6, 1, 0, 4, 3)$				
0	∞	4395372	7748230	1912800
$(e, r_1, q_1, t, v) = (6, 1, 0, 5, 1)$				
0	∞	4309760	8599400	2266000
$(e, r_1, q_1, t, v) = (6, 1, 0, 5, 2)$				
0	∞	5688920	9519450	2266000
$(e, r_1, q_1, t, v) = (6, 1, 0, 5, 3)$				
0	∞	7246630	10434950	2266000
$(e, r_1, q_1, t, v) = (6, 1, 0, 5, 4)$				
0	∞	8998360	11355000	2266000
$(e, r_1, q_1, t, v) = (6, 2, 1, 1, 0)$				
1	1	1347320	–	15992370
2	2	350344	–	6838092
3	3	1009224	–	4228272
4	4	2648736	–	2877480
5	5	4246088	–	1945800
6	6	3559000	–	1090656
7	∞	3976968	2337488	959760
$(e, r_1, q_1, t, v) = (6, 2, 1, 2, 1)$				
0	0	455790	–	8818128
1	1	3198720	–	3729600
2	∞	2851210	3167530	1757520
$(e, r_1, q_1, t, v) = (6, 2, 1, 3, 1)$				
0	0	2302468	–	5351400
1	∞	3024346	5508438	2393280
$(e, r_1, q_1, t, v) = (6, 2, 1, 3, 2)$				
0	0	2029818	–	2785968
1	∞	4357044	7113786	2393280
$(e, r_1, q_1, t, v) = (6, 2, 1, 4, 1)$				
0	∞	1540198	4388666	2867040
$(e, r_1, q_1, t, v) = (6, 2, 1, 4, 3)$				
0	∞	3134140	7314710	2867040
$(e, r_1, q_1, t, v) = (6, 2, 1, 5, 1)$				
0	∞	2168104	7278790	3178800
$(e, r_1, q_1, t, v) = (6, 2, 1, 5, 2)$				
0	∞	3155112	8574880	3178800
$(e, r_1, q_1, t, v) = (6, 2, 1, 5, 3)$				
0	∞	4391328	9862780	3178800
$(e, r_1, q_1, t, v) = (6, 2, 1, 5, 4)$				
0	∞	5904962	11158870	3178800
$(e, r_1, q_1, t, v) = (6, 2, 1, 6, 1)$				
0	∞	3086170	9284034	3328560
$(e, r_1, q_1, t, v) = (6, 2, 1, 6, 5)$				
0	∞	9335712	14111238	3328560
$(e, r_1, q_1, t, v) = (6, 2, 1, 7, 1)$				
0	∞	4238220	10655750	3316320
$(e, r_1, q_1, t, v) = (6, 2, 1, 7, 2)$				
0	∞	5423550	11649134	3316320
$(e, r_1, q_1, t, v) = (6, 2, 1, 7, 3)$				
0	∞	6738416	12631052	3316320
$(e, r_1, q_1, t, v) = (6, 2, 1, 7, 4)$				
0	∞	8177904	13601504	3316320
$(e, r_1, q_1, t, v) = (6, 2, 1, 7, 5)$				
0	∞	9781508	14583422	3316320
$(e, r_1, q_1, t, v) = (6, 2, 1, 7, 6)$				
0	∞	11554142	15576806	3316320
$(e, r_1, q_1, t, v) = (6, 2, 1, 8, 1)$				
0	∞	4674622	10847242	3142080
$(e, r_1, q_1, t, v) = (6, 2, 1, 8, 3)$				
0	∞	6816440	12430270	3142080
$(e, r_1, q_1, t, v) = (6, 2, 1, 8, 5)$				
0	∞	9488682	14084578	3142080
$(e, r_1, q_1, t, v) = (6, 2, 1, 8, 7)$				
0	∞	12445828	15667606	3142080
$(e, r_1, q_1, t, v) = (6, 2, 1, 9, 1)$				
0	∞	3575728	9315486	2805840
$(e, r_1, q_1, t, v) = (6, 2, 1, 9, 2)$				
0	∞	5024196	10417068	2805840
$(e, r_1, q_1, t, v) = (6, 2, 1, 9, 4)$				
0	∞	7309180	11937132	2805840
$(e, r_1, q_1, t, v) = (6, 2, 1, 9, 5)$				
0	∞	8410824	12590136	2805840
$(e, r_1, q_1, t, v) = (6, 2, 1, 9, 7)$				
0	∞	11269720	14110200	2805840
$(e, r_1, q_1, t, v) = (6, 2, 1, 9, 8)$				
0	∞	13569642	15211782	2805840
$(e, r_1, q_1, t, v) = (7, 1, 0, 1, 0)$				
3	3	2175360	–	2836848
4	4	1413490	–	1315200
5	5	945692	–	936336
6	6	980070	–	745104
7	∞	1792000	2127610	657840
$(e, r_1, q_1, t, v) = (7, 1, 0, 2, 1)$				
0	0	1343680	–	5601696
1	1	535536	–	1800672
2	∞	1286432	3238256	1243680
$(e, r_1, q_1, t, v) = (7, 1, 0, 3, 1)$				
0	0	252774	–	2019312
1	∞	2189880	5452146	1757520
$(e, r_1, q_1, t, v) = (7, 1, 0, 3, 2)$				
0	∞	519928	3115338	1757520
$(e, r_1, q_1, t, v) = (7, 1, 0, 4, 1)$				
0	∞	974556	5130916	2199360
$(e, r_1, q_1, t, v) = (7, 1, 0, 4, 3)$				
0	∞	3065568	7369612	2199360
$(e, r_1, q_1, t, v) = (7, 1, 0, 5, 1)$				
0	∞	2694554	7938950	2569200
$(e, r_1, q_1, t, v) = (7, 1, 0, 5, 2)$				
0	∞	3896934	8983010	2569200
$(e, r_1, q_1, t, v) = (7, 1, 0, 5, 3)$				
0	∞	5301756	10021610	2569200
$(e, r_1, q_1, t, v) = (7, 1, 0, 5, 4)$				
0	∞	6927220	11065670	2569200
$(e, r_1, q_1, t, v) = (7, 1, 0, 6, 1)$				
0	∞	4683432	10289064	2867040
$(e, r_1, q_1, t, v) = (7, 1, 0, 6, 5)$				
0	∞	11424520	14371200	2867040
$(e, r_1, q_1, t, v) = (8, 1, 0, 1, 0)$				
2	2	94676	–	5332516
3	3	999950	–	3302656
4	4	2787400	–	2252040
5	5	1912378	–	1085392
6	6	1614540	–	862288
7	∞	2212270	2118690	760480

Table 10, continued.

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (8, 1, 0, 2, 1)$				
0	0	418546	–	6507312
1	1	1240156	–	2072784
2	∞	1459158	3064922	1422960
$(e, r_1, q_1, t, v) = (8, 1, 0, 3, 1)$				
0	0	708552	–	2962176
1	∞	2003944	5270272	1987440
$(e, r_1, q_1, t, v) = (8, 1, 0, 3, 2)$				
0	∞	712528	2627996	1987440
$(e, r_1, q_1, t, v) = (8, 1, 0, 4, 1)$				
0	∞	740866	4615982	2453920
$(e, r_1, q_1, t, v) = (8, 1, 0, 4, 3)$				
0	∞	2522244	7115794	2453920
$(e, r_1, q_1, t, v) = (8, 1, 0, 5, 1)$				
0	∞	1928148	7444500	2822400
$(e, r_1, q_1, t, v) = (8, 1, 0, 5, 2)$				
0	∞	2994948	8592570	2822400
$(e, r_1, q_1, t, v) = (8, 1, 0, 5, 3)$				
0	∞	4284082	9734270	2822400
$(e, r_1, q_1, t, v) = (8, 1, 0, 5, 4)$				
0	∞	5816480	10882340	2822400
$(e, r_1, q_1, t, v) = (8, 1, 0, 6, 1)$				
0	∞	3421510	9696778	3092880
$(e, r_1, q_1, t, v) = (8, 1, 0, 6, 5)$				
0	∞	9796820	14123270	3092880
$(e, r_1, q_1, t, v) = (8, 1, 0, 7, 1)$				
0	∞	5161744	11568312	3265360
$(e, r_1, q_1, t, v) = (8, 1, 0, 7, 2)$				
0	∞	6465480	12536944	3265360
$(e, r_1, q_1, t, v) = (8, 1, 0, 7, 3)$				
0	∞	7898674	13496658	3265360
$(e, r_1, q_1, t, v) = (8, 1, 0, 7, 4)$				
0	∞	9457504	14447454	3265360
$(e, r_1, q_1, t, v) = (8, 1, 0, 7, 5)$				
0	∞	11171272	15407168	3265360
$(e, r_1, q_1, t, v) = (8, 1, 0, 7, 6)$				
0	∞	13043800	16375800	3265360
$(e, r_1, q_1, t, v) = (9, 1, 0, 1, 0)$				
2	2	314290	–	6086304
3	3	578940	–	3766464
4	4	1859310	–	2565760
5	5	3198720	–	1737600
6	6	2597418	–	977472
7	∞	3005700	2130522	861120
$(e, r_1, q_1, t, v) = (9, 1, 0, 2, 1)$				
0	0	194612	–	7870336
1	1	2294824	–	3347200
2	∞	2043436	2937092	1594240
$(e, r_1, q_1, t, v) = (9, 1, 0, 3, 1)$				
0	0	1540770	–	4828800
1	∞	2292704	5150654	2199360
$(e, r_1, q_1, t, v) = (9, 1, 0, 3, 2)$				
0	0	1318320	–	2548416
1	∞	3545230	6625630	2199360
$(e, r_1, q_1, t, v) = (9, 1, 0, 4, 1)$				
0	∞	983512	4208056	2676480
$(e, r_1, q_1, t, v) = (9, 1, 0, 4, 3)$				
0	∞	2516760	6936984	2676480

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (9, 1, 0, 5, 1)$				
0	∞	1733974	7053810	3025600
$(e, r_1, q_1, t, v) = (9, 1, 0, 5, 2)$				
0	∞	2693946	8285890	3025600
$(e, r_1, q_1, t, v) = (9, 1, 0, 5, 3)$				
0	∞	3892144	9510690	3025600
$(e, r_1, q_1, t, v) = (9, 1, 0, 5, 4)$				
0	∞	5352228	10742770	3025600
$(e, r_1, q_1, t, v) = (9, 1, 0, 6, 1)$				
0	∞	2823716	9193004	3246720
$(e, r_1, q_1, t, v) = (9, 1, 0, 6, 5)$				
0	∞	8908256	13867852	3246720
$(e, r_1, q_1, t, v) = (9, 1, 0, 7, 1)$				
0	∞	4181082	10849062	3339840
$(e, r_1, q_1, t, v) = (9, 1, 0, 7, 2)$				
0	∞	5379500	11844070	3339840
$(e, r_1, q_1, t, v) = (9, 1, 0, 7, 3)$				
0	∞	6710052	12828886	3339840
$(e, r_1, q_1, t, v) = (9, 1, 0, 7, 4)$				
0	∞	8168370	13803510	3339840
$(e, r_1, q_1, t, v) = (9, 1, 0, 7, 5)$				
0	∞	9787578	14788326	3339840
$(e, r_1, q_1, t, v) = (9, 1, 0, 7, 6)$				
0	∞	11572044	15783334	3339840
$(e, r_1, q_1, t, v) = (9, 1, 0, 8, 1)$				
0	∞	5009200	11536032	3304960
$(e, r_1, q_1, t, v) = (9, 1, 0, 8, 3)$				
0	∞	7289512	13199168	3304960
$(e, r_1, q_1, t, v) = (9, 1, 0, 8, 5)$				
0	∞	10099152	14925664	3304960
$(e, r_1, q_1, t, v) = (9, 1, 0, 8, 7)$				
0	∞	13232200	16588800	3304960
$(e, r_1, q_1, t, v) = (10, 1, 0, 1, 0)$				
2	2	932736	–	7974684
3	3	660658	–	4228272
4	4	1533844	–	2877480
5	5	2468390	–	1945800
6	6	3253320	–	1377504
7	∞	3404890	2071090	959760
$(e, r_1, q_1, t, v) = (10, 1, 0, 2, 1)$				
0	0	432654	–	8818128
1	1	1678932	–	3729600
2	∞	2332610	2670734	1757520
$(e, r_1, q_1, t, v) = (10, 1, 0, 3, 1)$				
0	0	1044540	–	5351400
1	∞	2318232	4817244	2393280
$(e, r_1, q_1, t, v) = (10, 1, 0, 3, 2)$				
0	0	1783408	–	3646512
1	∞	3419440	6422592	2393280
$(e, r_1, q_1, t, v) = (10, 1, 0, 4, 1)$				
0	∞	1117326	3539074	2867040
$(e, r_1, q_1, t, v) = (10, 1, 0, 4, 3)$				
0	∞	2279916	6465118	2867040
$(e, r_1, q_1, t, v) = (10, 1, 0, 5, 1)$				
0	∞	1403576	6306800	3178800
$(e, r_1, q_1, t, v) = (10, 1, 0, 5, 2)$				
0	∞	2193456	7602890	3178800
$(e, r_1, q_1, t, v) = (10, 1, 0, 5, 3)$				
0	∞	3233454	8890790	3178800

h_-	h_+	τ_0	τ_1	τ_2
$(e, r_1, q_1, t, v) = (10, 1, 0, 5, 4)$				
0	∞	4549960	10186880	3178800
$(e, r_1, q_1, t, v) = (10, 1, 0, 6, 1)$				
0	∞	2058306	8225646	3328560
$(e, r_1, q_1, t, v) = (10, 1, 0, 6, 5)$				
0	∞	7559020	13052850	3328560
$(e, r_1, q_1, t, v) = (10, 1, 0, 7, 1)$				
0	∞	2997412	9546964	3316320
$(e, r_1, q_1, t, v) = (10, 1, 0, 7, 2)$				
0	∞	4019248	10540348	3316320
$(e, r_1, q_1, t, v) = (10, 1, 0, 7, 3)$				
0	∞	5171894	11522266	3316320
$(e, r_1, q_1, t, v) = (10, 1, 0, 7, 4)$				
0	∞	6450436	12492718	3316320
$(e, r_1, q_1, t, v) = (10, 1, 0, 7, 5)$				
0	∞	7891820	13474636	3316320
$(e, r_1, q_1, t, v) = (10, 1, 0, 7, 6)$				
0	∞	9500960	14468020	3316320
$(e, r_1, q_1, t, v) = (10, 1, 0, 8, 1)$				
0	∞	3332006	9724058	3142080
$(e, r_1, q_1, t, v) = (10, 1, 0, 8, 3)$				
0	∞	5191696	11307086	3142080
$(e, r_1, q_1, t, v) = (10, 1, 0, 8, 5)$				
0	∞	7573890	12961394	3142080
$(e, r_1, q_1, t, v) = (10, 1, 0, 8, 7)$				
0	∞	10248908	14544422	3142080
$(e, r_1, q_1, t, v) = (10, 1, 0, 9, 1)$				
0	∞	2302776	8213904	2805840
$(e, r_1, q_1, t, v) = (10, 1, 0, 9, 2)$				
0	∞	3575728	9315486	2805840
$(e, r_1, q_1, t, v) = (10, 1, 0, 9, 4)$				
0	∞	5585580	10835550	2805840
$(e, r_1, q_1, t, v) = (10, 1, 0, 9, 5)$				
0	∞	6561550	11488554	2805840
$(e, r_1, q_1, t, v) = (10, 1, 0, 9, 7)$				
0	∞	9145314	13008618	2805840
$(e, r_1, q_1, t, v) = (10, 1, 0, 9, 8)$				
0	∞	11269720	14110200	2805840

TABLE 11. Values for the proof of Proposition 9.9 for the cyclic quotient singularities not covered by Table 10: an entry $n{:}T$ under the header (e, r_1, q_1, t), where $\alpha = (e-1)r_1 - q_1 \geq 10$, is the value $T = 2r(C_x - 500(r-1))$, with C_x as in Definition 9.2, at the cyclic quotient singularity of Definition-Lemma 9.7 with parameters (e, r_1, q_1, t, n) and of order $r = \alpha n + r_1 t \geq 20$.

e	r_1	q_1	t	$n{:}T$
3	6	1	1	6:6879312 7:6580660 8:7536920 9:10795260 10:16355680 11:24218180 12:34382760 13:46849420 14:61618160 15:78688980 16:98061880 17:119736860 18:143713920 19:169993060 20:198574280 21:229457580 22:262642960 23:298130420 24:335919960 25:376011580 26:418405280 27:463101060
3	6	1	2	7:6059362 9:10601608 11:19264014 13:32046580 15:48949306 17:69972192 19:95115238 21:124378444 23:157761810
3	6	1	3	7:6483530 8:8385676 10:13892296 11:17624534 13:26755302 14:32317632 16:45072548 17:52464970 19:68844034 20:78066548
3	6	1	4	7:6649288 9:10360572 11:16051130 13:22842458 15:31757292 17:41628664
3	6	1	5	6:5167504 7:6520964 8:8131776 9:10040890 11:14310854 12:17028424 13:19993336 14:23266560
3	6	1	6	7:6095282
3	7	1	1	7:9255680 8:9979660 9:13372280 10:19433540 11:28163440 12:39561980 13:53629160 14:70364980 15:89769440
3	7	1	2	7:8168160 9:12619978 11:21733076 13:35507454
3	7	1	3	7:8186864 8:9976610 10:15412242 11:19208824
3	7	1	4	7:7890302 9:11256670
3	7	1	5	7:7236432
3	7	2	1	7:8476608 8:9505054 9:13020860 10:19024026 11:27514552 12:38492438 13:51957684 14:67910290 15:86350256 16:107277582 17:130692268 18:156594314 19:184983720 20:215860486
3	7	2	2	7:7830368 9:12684052 11:21936456 13:35587580 15:53637424 17:76085988
3	7	2	3	7:8187270 8:10200088 10:16009320 11:19946602 13:29565450 14:35427196
3	7	2	4	7:8227632 9:12075832 11:17993200 13:25009448
3	7	2	5	7:7911960 8:9559290 9:11508454
3	7	2	6	7:7236432
3	7	3	1	7:7342776 8:8609190 9:12177684 10:18048258 11:26220912 12:36695646 13:49472460 14:64551354 15:81932328 16:101615382 17:123600516 18:147887730 19:174477024 20:203368398 21:234561852 22:268057386 23:303855000 24:341954694 25:382356468 26:425060322 27:470066256 28:517374270 29:566984364 30:618896538 31:673110792
3	7	3	2	7:7136542 9:12255096 11:21493810 13:34852684 15:52331718 17:73930912 19:99650266 21:129489780 23:163449454 25:201529288 27:243729282
3	7	3	3	7:7830368 8:9999760 10:16037596 11:20037080 13:29699064 14:35528640 16:48814772 17:56474440 19:73384720 20:82874480 22:103408908 23:114728760
3	7	3	4	7:8206380 9:12399416 11:18584838 13:25857918 15:35267616 17:45620740 19:58254714
3	7	3	5	7:8227632 8:10062418 9:12196416 11:16903408 12:19845862 13:23034748 14:26532856 16:33827928
3	7	3	6	7:7890302 11:16283852
3	7	3	7	8:9026844
3	7	4	1	7:5858006 8:7296436 9:10847666 10:16511696 11:24288526 12:34178156 13:46180586 14:60295816 15:76523846 16:94864676 17:115318306 18:137884736 19:162563966 20:189355996 21:218260826 22:249278456 23:282408886 24:317652116 25:355008146 26:394476976 27:436058606 28:479753036 29:525560266 30:573480296 31:623513126 32:675658756 33:729917186 34:786288416 35:844772446 36:905369276 37:968078906 38:1032901336 39:1099836566 40:1168884596 41:1240045426 42:1313319056 43:1388705486 44:1466204716 45:1545816746 46:1627541576 47:1711379206 48:1797329636 49:1885392866 50:1975568896 51:2067857726 52:2162259356 53:2258773786 54:2357401016 55:2458141046 56:2560993876 57:2665959506 58:2773037936 59:2882229166 60:2993533196 61:3106950026 62:3222479656 63:3340122086 64:3459877316 65:3581745346 66:3705726176 67:3831819806 68:3960026236 69:4090345466 70:4222777496 71:4357322326
3	7	4	2	7:6090504 9:11338024 11:20411144 13:33309864 15:50034184 17:70584104 19:94959624 21:123160744 23:155187464 25:191039784 27:230717704 29:274221224 31:321550344 33:372705064 35:427685384 37:486491304 39:549122824 41:615579944 43:685862664 45:759970984 47:837904904 49:919664424 51:1005249544 53:1094660264 55:1187896584 57:1284958504 59:1385846024 61:1490559144 63:1599097864

Table 11, continued.

e	r_1	q_1	t	n:T
3	7	4	3	7:7119980 8:9379994 10:15502530 11:19486264 13:29023480 14:34730934 16:47682830 17:55114004 19:71480580 20:80635474 22:100416730 23:111295344 25:134491280 26:147093614 28:173704230 29:188030284 31:218055580 32:234105354 34:267545330 35:285318824 37:322173480 38:341670694 40:381940030 41:403160964 43:446844980 44:469789634 46:516888330 47:541556704 49:592070080 50:618462174 52:672390230 53:700506044
3	7	4	4	7:7830368 9:12232336 11:18533908 13:25895916 15:35288648 17:45610696 19:58094588 21:71376676 23:86951728 25:103193856 27:121860068 29:141062236 31:162819608 33:184981816 35:209830348 37:234952596 39:262892288 41:290974576 43:322005428 45:353047756
3	7	4	5	7:8187270 8:10143560 9:12392440 11:17341180 12:20385020 13:23672310 14:27261290 16:34762830 17:39146770 18:43765060 19:48694140 21:58748480 22:64472520 23:70421810 24:76690990 26:89298130 27:96362270 28:103642560 29:111251840 31:126411780 32:134816020 33:143427310 34:152376690 36:170089430
3	7	4	6	7:8186864 11:17113456 13:21046584 17:34856736 19:40583104 23:59276816 25:66796424
3	7	4	7	8:9800974 9:11496376 10:13309620 11:15235246 12:17329310 13:19597272 15:23698844 16:26358006 17:29122270 18:31986176 19:35031260
3	7	4	8	9:9980492
3	8	1	1	8:12270032 9:15699292 10:22147752 11:31615412
3	8	1	2	9:14365092
3	8	1	3	8:11380680
3	8	3	1	8:11716936 9:15497190 10:21946084 11:31063618 12:42849792 13:57304606 14:74428060 15:94220154 16:116680888 17:141810262
3	8	3	2	9:15021118 11:24857484 13:39355130 15:58514056
3	8	3	3	8:12270032 10:18375840 11:22509148 13:32586040
3	8	3	4	9:14041498
3	8	3	5	8:11238120
3	8	5	1	8:9816432 9:13706264 10:19898176 11:28392168 12:39188240 13:52286392 14:67686624 15:85388936 16:105393328 17:127699800 18:152308352 19:179218984 20:208431696 21:239946488 22:273763360 23:309882312 24:348303344 25:389026456 26:432051648 27:477378920 28:525008272 29:574939704 30:627173216 31:681708808 32:738546480 33:797686232 34:859128064 35:922871976 36:988917968
3	8	5	2	9:14096240 11:23933630 13:37891180 15:55968890 17:78166760 19:104484790 21:134922980 23:169481330 25:208159840 27:250958510 29:297877340 31:348916330
3	8	5	3	8:11827816 10:18419236 11:22697150 13:32912718 14:39020724 16:52860440 17:60798538 19:78262402 20:88030592 22:109118604 23:120716886 25:145429046 26:158857420
3	8	5	4	9:14696916 11:21399570 13:29176770 15:39103700 17:49960944 19:63112150 21:77049438 23:93424920
3	8	5	5	8:12270032 9:14640098 11:19806486 12:22985008 13:26409052 14:30143228 16:37897696 17:42440328 18:47218472
3	8	5	6	11:19258748 13:23165176
3	8	5	7	8:11380680 9:13098630
3	9	1	1	9:16309440
3	9	2	1	9:17841286 10:24636368
3	9	2	2	9:16309440
3	9	4	1	9:18267930 10:25170084 11:34918158 12:47512152 13:62952066 14:81237900 15:102369654
3	9	4	2	9:17659216 11:28082608 13:43413840
3	9	4	3	10:21045546 11:25370412
3	9	4	4	9:16309440
3	9	5	1	9:17679340 10:24517250 11:34023800 12:46198990 13:61042820 14:78555290 15:98736400 16:121586150 17:147104540 18:175291570 19:206147240
3	9	5	2	9:17520300 11:28082698 13:43306376 15:63191334 17:87737572
3	9	5	3	10:21475664 11:25947080 13:36696912 14:43175164
3	9	5	4	9:16993972 11:23744702
3	9	5	5	9:16309440

Table 11, continued.

e	r_1	q_1	t	n:T
3	9	7	1	9:15588072 10:22133286 11:30980580 12:42129954 13:55581408 14:71334942 15:89390556 16:109748250 17:132408024 18:157369878 19:184633812 20:214199826 21:246067920 22:280238094 23:316710348 24:355484682 25:396561096 26:439939590 27:485620164 28:533602818 29:583887552 30:636474366 31:691363260 32:748554234 33:808047288 34:869842422 35:933939636 36:1000338930 37:1069040304 38:1140043758 39:1213349292 40:1288956906 41:1366866600
3	9	7	2	9:16352164 11:26852158 13:41472312 15:60212626 17:83073100 19:110053734 21:141154528 23:176375482 25:215716596 27:259177870 29:306759304 31:358460898 33:414282652 35:474224566
3	9	7	3	10:21305172 11:25893480 13:36726560 14:43144960 16:57602188 17:65850680 19:83932056 20:94010640 22:115716164 23:127624840 25:152954512 26:166693280 28:195647100 29:211215960 31:243793928
3	9	7	4	9:17520300 11:24804114 13:33149362 15:43657452 17:55082744 19:68815110 21:83320446 23:100277088 25:117862468
3	9	7	5	9:17659216 11:23348928 12:26795482 13:30486648 14:34488856 16:42766648 17:47577312 18:52622578 19:57988896 21:68854768
3	9	7	6	11:22918560 13:27268356
3	9	7	7	9:16309440 10:18359758 11:20510040

e	r_1	q_1	t	n:T
3	9	8	1	9:14324464 10:20664894 11:29118124 12:39684154 13:52362984 14:67154614 15:84059044 16:103076274 17:124206304 18:147449134 19:172804764 20:200273194 21:229854424 22:261548454 23:295355284 24:331274914 25:369307344 26:409452574 27:451710604 28:496081434 29:542565064 30:591161494 31:641870724 32:694692754 33:749627584 34:806675214 35:865835644 36:927108874 37:990494904 38:1055993734 39:1123605364 40:1193329794 41:1265167024 42:1339117054 43:1415179884 44:1493355514 45:1573643944 46:1656045174 47:1740559204 48:1827186034 49:1915925664 50:2006778094 51:2099743324 52:2194821354 53:2292012184 54:2391315814 55:2492732244 56:2596261474 57:2701903504 58:2809658334 59:2919525964 60:3031506394 61:3145599624 62:3261805654 63:3380124484 64:3500556114 65:3623100544 66:3747757774 67:3874527804 68:4003410634 69:4134406264 70:4267514694 71:4402735924 72:4540069954 73:4679516784 74:4821076414 75:4964748844 76:5110534074 77:5258432104 78:5408442934 79:5560566564 80:5714802994 81:5871152224 82:6029614254 83:6190189084 84:6352876714 85:6517677144 86:6684590374 87:6853616404 88:7024755234 89:7198006864 90:7373371294 91:7550848524 92:7730438554
3	9	8	2	9:15588072 11:25933992 13:40105512 15:58102632 17:79925352 19:105573672 21:135047592 23:168347112 25:205472232 27:246422952 29:291199272 31:339801192 33:392228712 35:448481832 37:508560552 39:572464872 41:640194792 43:711750312 45:787131432 47:866338152 49:949370472 51:1036228392 53:1126911912 55:1221421032 57:1319755752 59:1421916072 61:1527901992 63:1637713512 65:1751350632 67:1868813352 69:1990101672 71:2115215592 73:2244155112 75:2376920232 77:2513510952 79:2653927272 81:2798169192

Table 11, continued.

e	r_1	q_1	t	$n{:}T$
3	9	8	3	10:21019416 11:25601734 13:36329566 14:42635604 16:56778116 17:64807874 19:82365066 20:92118544 22:113090416 23:124567614 25:148954166 26:162155084 28:189956316 29:204880954 31:236096866 32:252745224 34:287375816 35:305747894 37:343793166 38:363888964 40:405348916 41:427168434 43:472043066 44:495586304 46:543875616 47:569142574 49:620846566 50:647837244 52:702955916 53:731670314 55:790203666 56:820641784 58:882589816 59:914751654 61:980114366 62:1013999924 64:1082777316 65:1118386594 67:1190578666 68:1227911664
3	9	8	4	9:17679340 11:25106824 13:33568520 15:44087164 17:55508900 19:69118704 21:83500480 23:100201444 25:117543260 27:137335384 29:157637240 31:180520524 33:203782420 35:229756864 37:255978800 39:285044404 41:314226380 43:346383144 45:378525160 47:413773084 49:448875140 51:487214224 53:525276320 55:566706564 57:607728700
3	9	8	5	9:18267930 11:24234910 12:27800610 13:31607940 14:35718780 16:44238560 17:49144360 18:54282690 19:59733630 21:70806210 22:77052110 23:83521440 24:90312480 26:103937860 27:111523860 28:119324190 29:127455330 31:143633510 32:152559610 33:161690940 34:171162180 36:189893160 37:200159360 38:210621690 39:221433030 41:242716810 42:254323110 43:266116440 44:278267880 46:302104460 47:315050860
3	9	8	6	11:24290944 13:29090400 17:44892624 19:51485320 23:72171104 25:80557040 29:106126384 31:116305560 35:146758464
3	9	8	7	9:17841286 10:20098574 11:22465696 12:25003804 13:27718358 15:32651962 16:35757716 17:38966024 18:42271426 19:45760554 20:49438868 22:55994232 23:60063746 24:64223074
3	9	8	8	9:16309440 11:19939108 13:24131816
3	10	3	1	10:27229680
3	10	7	1	10:28360360 11:38369944 12:51048168 13:66395032 14:84410536 15:105094680 16:128447464 17:154468888 18:183158952 19:214517656 20:248545000 21:285240984
3	10	7	2	11:32821638 13:48999384 15:69838410 17:95338716
3	10	7	3	10:26098960 11:31022502 13:42673310 14:49603688 16:65225580
3	10	7	4	11:29083018 13:37745138
3	10	7	5	11:26725766
3	10	9	1	10:27881280 11:37399746 12:49220292 13:63342918 14:79767624 15:98494410 16:119523276 17:142854222 18:168487248 19:196422354 20:226659540 21:259198806 22:294040152 23:331183578 24:370629084 25:412376670 26:456426336 27:502778082 28:551431908 29:602387814 30:655645800 31:711205866 32:769068012 33:829232238 34:891698544 35:956466930 36:1023537396 37:1092909942 38:1164584568 39:1238561274 40:1314840060 41:1393420926 42:1474303872 43:1557488898 44:1642976004 45:1730765190
3	10	9	2	11:33931624 13:49850122 15:69888780 17:94047598 19:122326576 21:154725714 23:191245012 25:231884470 27:276644088 29:325523866 31:378523804 33:435643902 35:496884160 37:562244578 39:631725156
3	10	9	3	10:28360360 11:33576932 13:45663264 14:52709928 16:68420408 17:77297164 19:96631792 20:107338640 22:130297416 23:142834356 25:169417280 26:183784312 28:213991384 29:230188508 31:264019728 32:282046944 34:319502312
3	10	9	4	11:33017964 13:42567000 15:54291990 17:66921070 19:81870336 21:97579460 23:115753002 25:134542170 27:155939988
3	10	9	5	11:32018860 12:36051316 13:40327474 14:44915584 16:54352440 17:59749006 18:65379264 19:71331484 21:83356420 22:90117096 23:97101454
3	10	9	6	11:32020046 13:37448950 17:54879372
3	10	9	7	10:27229680 11:29921682 12:32779896
3	11	5	1	11:43102908
3	11	6	1	11:42991290 12:57152200 13:74501590
3	11	6	2	11:34860924
3	11	7	1	11:43948304 12:57897844 13:74866584 14:94854524 15:117861664
3	11	7	2	11:36853984 13:54042254
3	11	7	3	11:34084260
3	11	8	1	11:44398050 12:58065088 13:74578046 14:93936924 15:116141722 16:141192440 17:169089078 18:199831636
3	11	8	2	11:38350960 13:55716280 15:77989440
3	11	8	3	11:36310596 13:48744500 14:56109802
3	11	8	4	11:34084260
3	11	9	1	11:44597896 12:57932808 13:73936360 14:92608552 15:113949384 16:137958856 17:164636968 18:193983720 19:225999112 20:260683144 21:298035816 22:338057128 23:380747080
3	11	9	2	11:39630000 13:57069122 15:79169524 17:105931206 19:137354168
3	11	9	3	11:38350960 13:51210052 14:58746210 16:75576386 17:85119380

Table 11, continued.

e	r_1	q_1	t	n:T
3	11	9	4	11:36853984 13:46666924 15:58832718
3	11	9	5	11:34860924 12:38885496
3	11	10	1	11:48168728 12:61425070 13:77168772 14:95399834 15:116118256 16:139324038 17:165017180 18:193197682 19:223865544 20:257020766 21:292663348 22:330793290 23:371410592 24:414515254 25:460107276 26:508186658 27:558753400 28:611807502 29:667348964 30:725377786 31:785893968 32:848897510
3	11	10	2	11:44597896 13:62613932 15:85028688 17:111842164 19:143054360 21:178665276 23:218674912 25:263083268 27:311890344
3	11	10	3	11:44398050 13:58185442 14:66138012 16:83735020 17:93612054 19:115018678 20:126820176 22:152036416 23:165762378
3	11	10	4	11:43948304 13:54919240 15:68169136 17:82308120 19:98883408
3	11	10	5	11:42991290 12:47655782 13:52562048 14:57784708 16:68441200
3	11	10	6	11:43102908
3	12	7	1	12:62614008
3	12	11	1	12:76298488 13:93840546 14:114051244 15:136930582 16:162478560 17:190695178 18:221580436 19:255134334 20:291356872 21:330248050 22:371807868 23:416036326 24:462933424 25:512499162
3	12	11	2	13:78396942 15:103522356 17:133309050 19:167757024 21:206866278
3	12	11	3	13:73896100 14:82919856 16:102721952 17:113752544 19:137525724
3	12	11	4	13:70625240 15:85718602
3	12	11	5	12:62614008 13:68311662
3	13	9	1	13:89685000
3	13	10	1	13:92795238 14:114882856 15:140158954
3	13	10	2	13:71046504
3	13	11	1	13:97199152 14:119020562 15:143861172 16:171720982 17:202599992 18:236498202
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4	5	1	3	5:3907610
4	5	1	4	5:3471120
4	5	2	1	5:4501980 6:6295066 7:7543528 8:7608102 9:10341316 10:15743170
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4	5	4	6	7:5406740
4	6	1	1	6:5200560

Table 11, continued.

e	r_1	q_1	t	n:T
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4	6	5	4	7:6894500
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4	7	4	1	7:7718004
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4	7	6	3	7:7718004
4	8	7	1	8:11619144
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5	4	1	3	4:1911096
5	4	3	1	4:1809976 5:3195334 6:4300772 7:4924730 8:5981968
5	4	3	2	5:3270434 7:4508496
5	4	3	3	4:2400208 5:3138156
5	4	3	4	5:2771200
5	5	3	1	5:3684060
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5	5	4	2	5:3684060
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8	2	1	2	3:670330
9	2	1	1	2:407056
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11	1	0	1	2:459256 3:560846 4:720680 5:1163290 6:1665580 7:1988646 8:2771200 9:4633930 10:8609460
11	1	0	2	3:700264 5:1049824 7:1748376 9:3858896
11	1	0	3	4:590358 5:896656 7:1484896
11	1	0	4	5:684676
12	1	0	1	2:215110 3:282268 4:361710
12	1	0	2	3:168770
13	1	0	1	2:111540

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